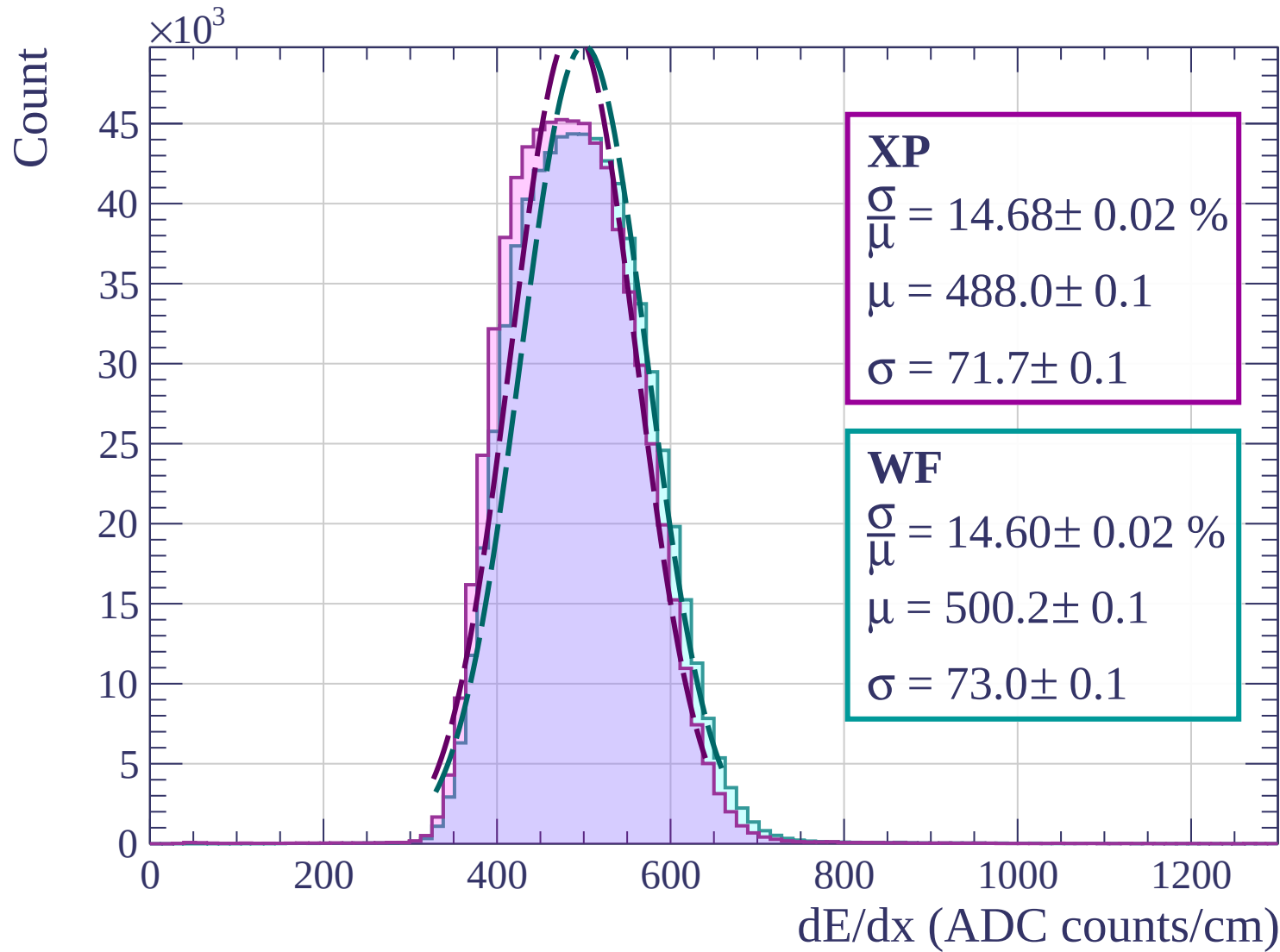
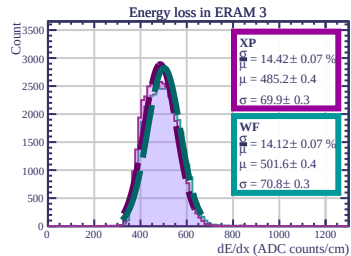
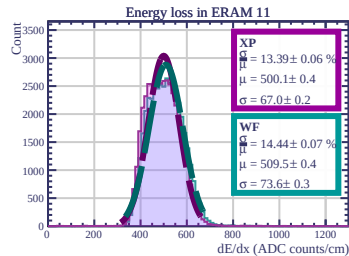
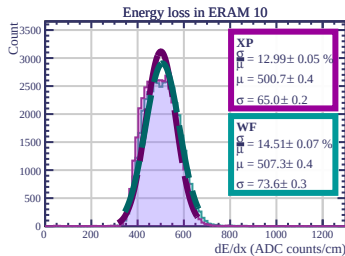
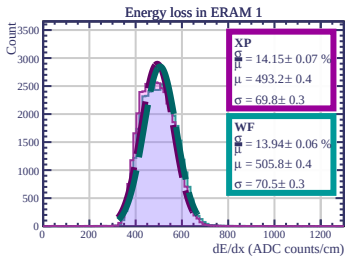
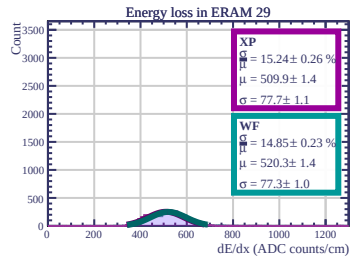
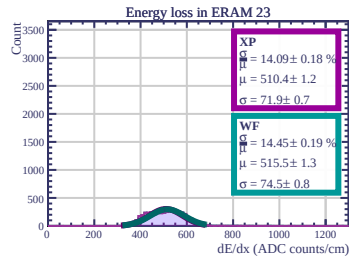
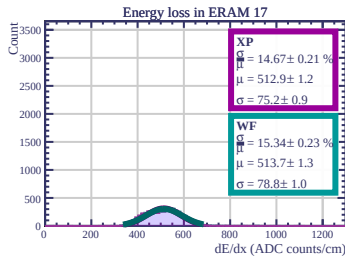
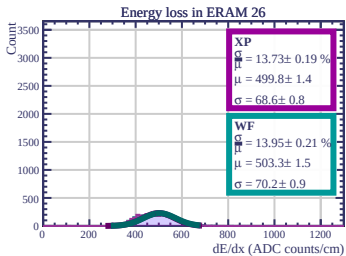
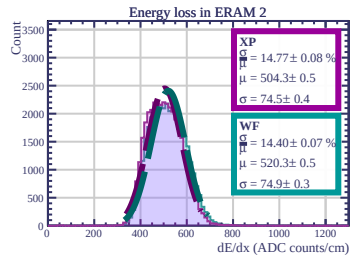
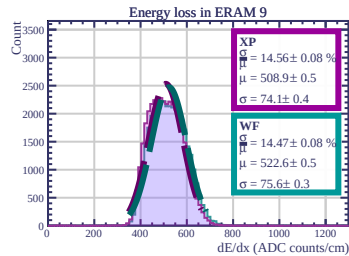
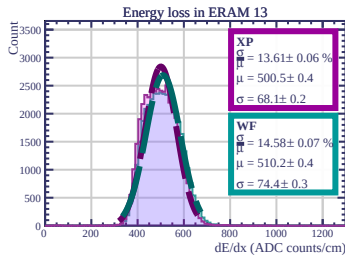
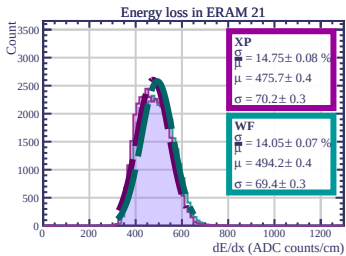
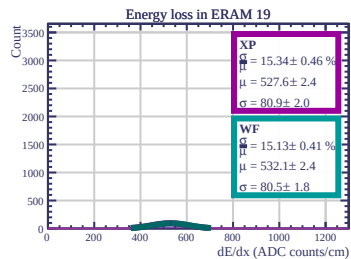
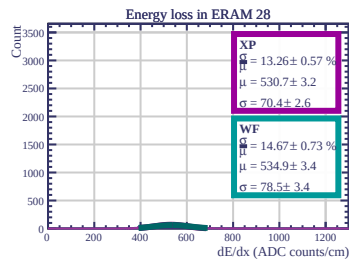
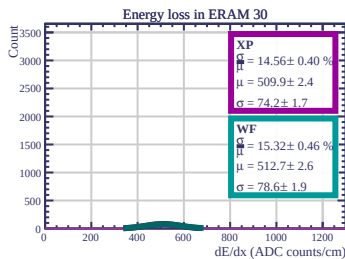
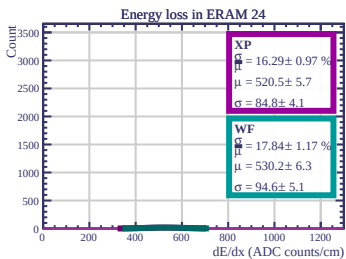
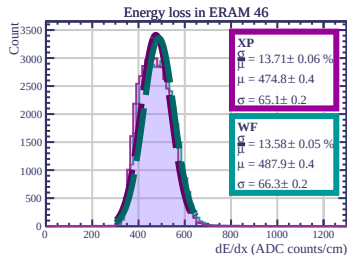
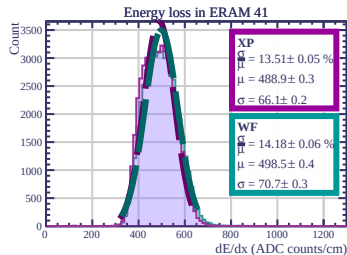
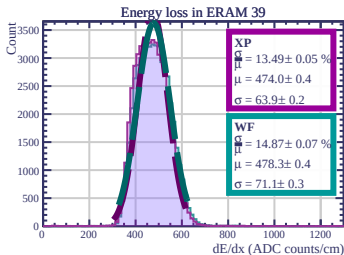
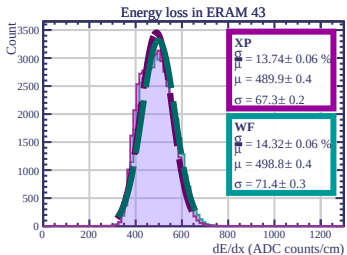
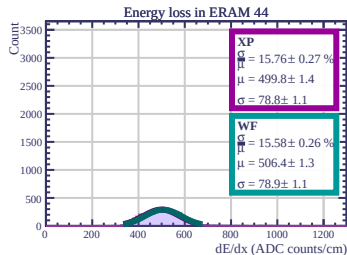
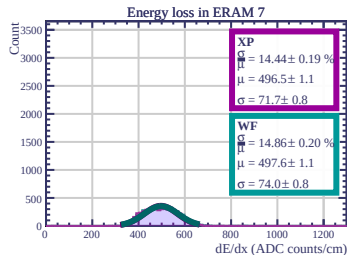
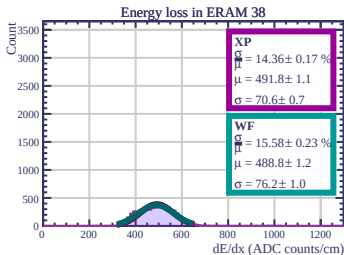
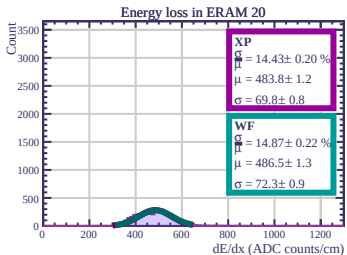
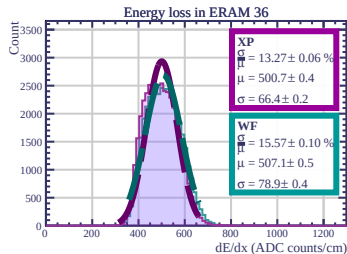
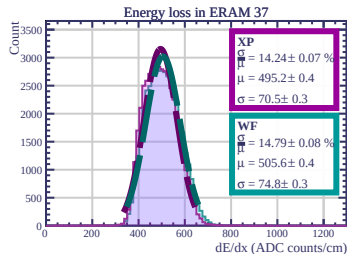
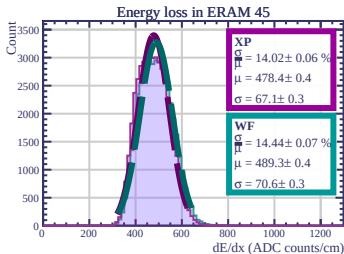
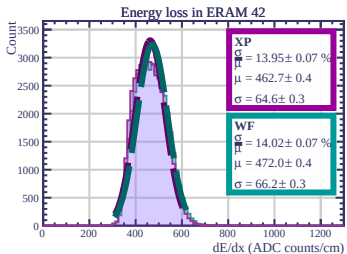
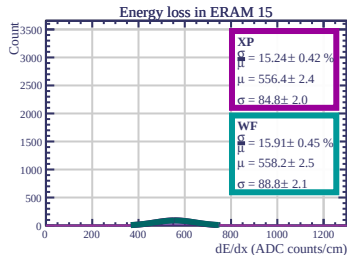
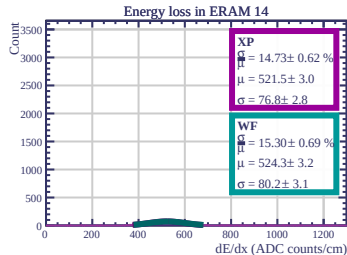
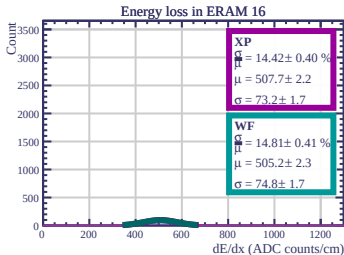
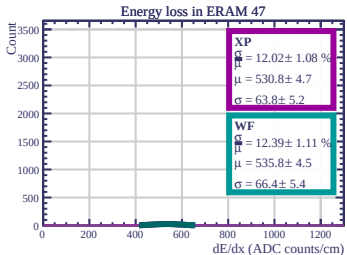
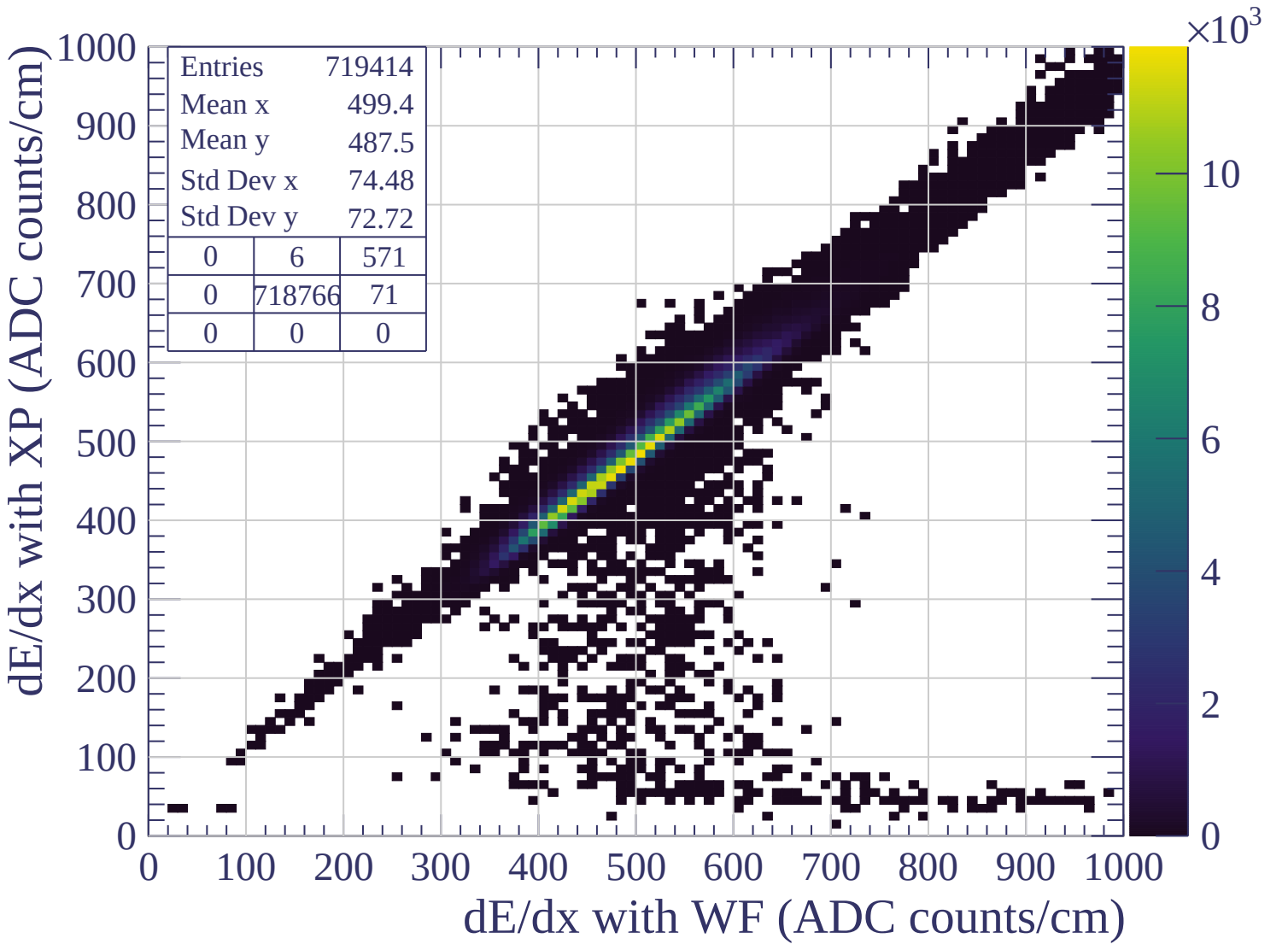
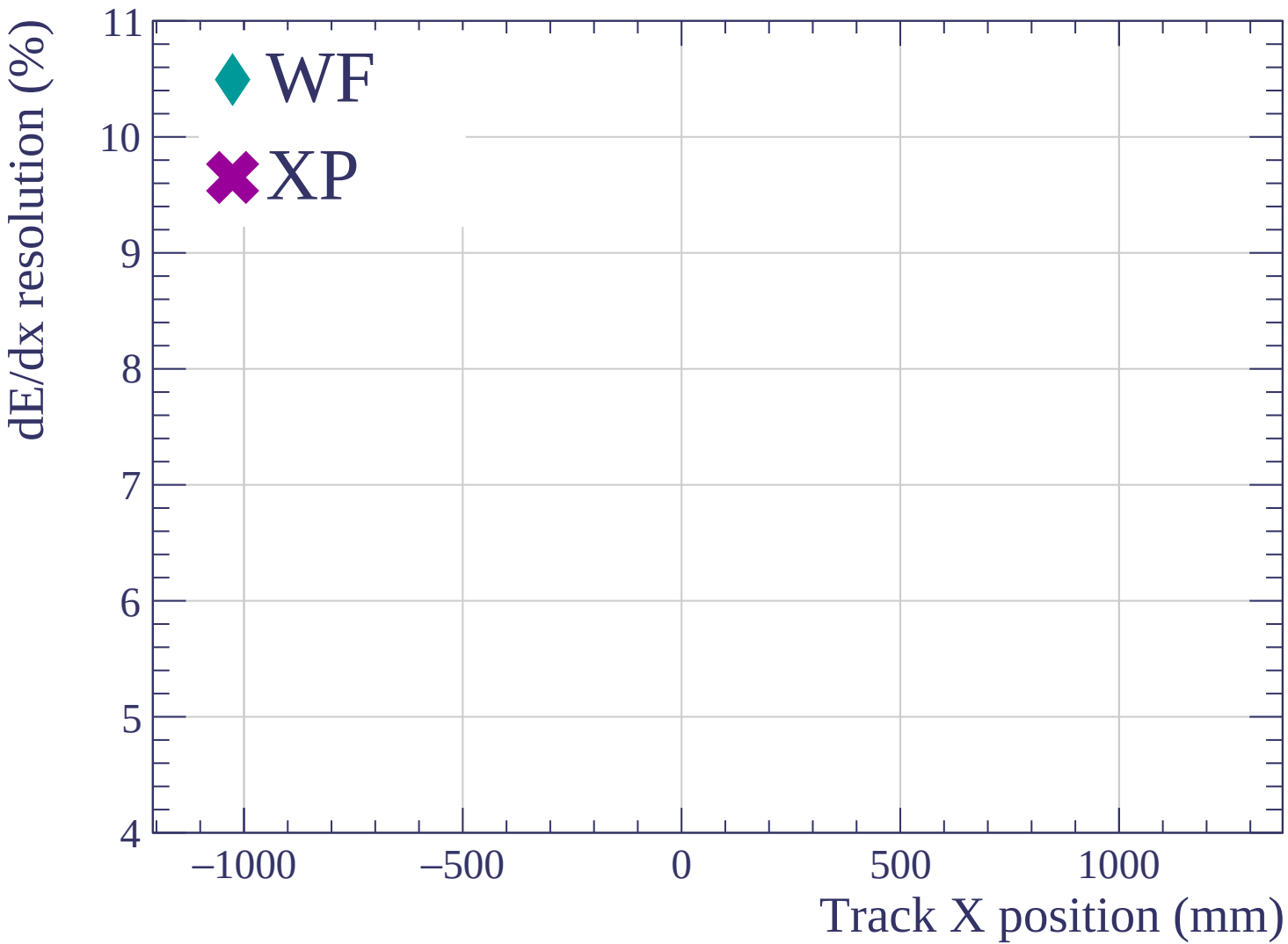


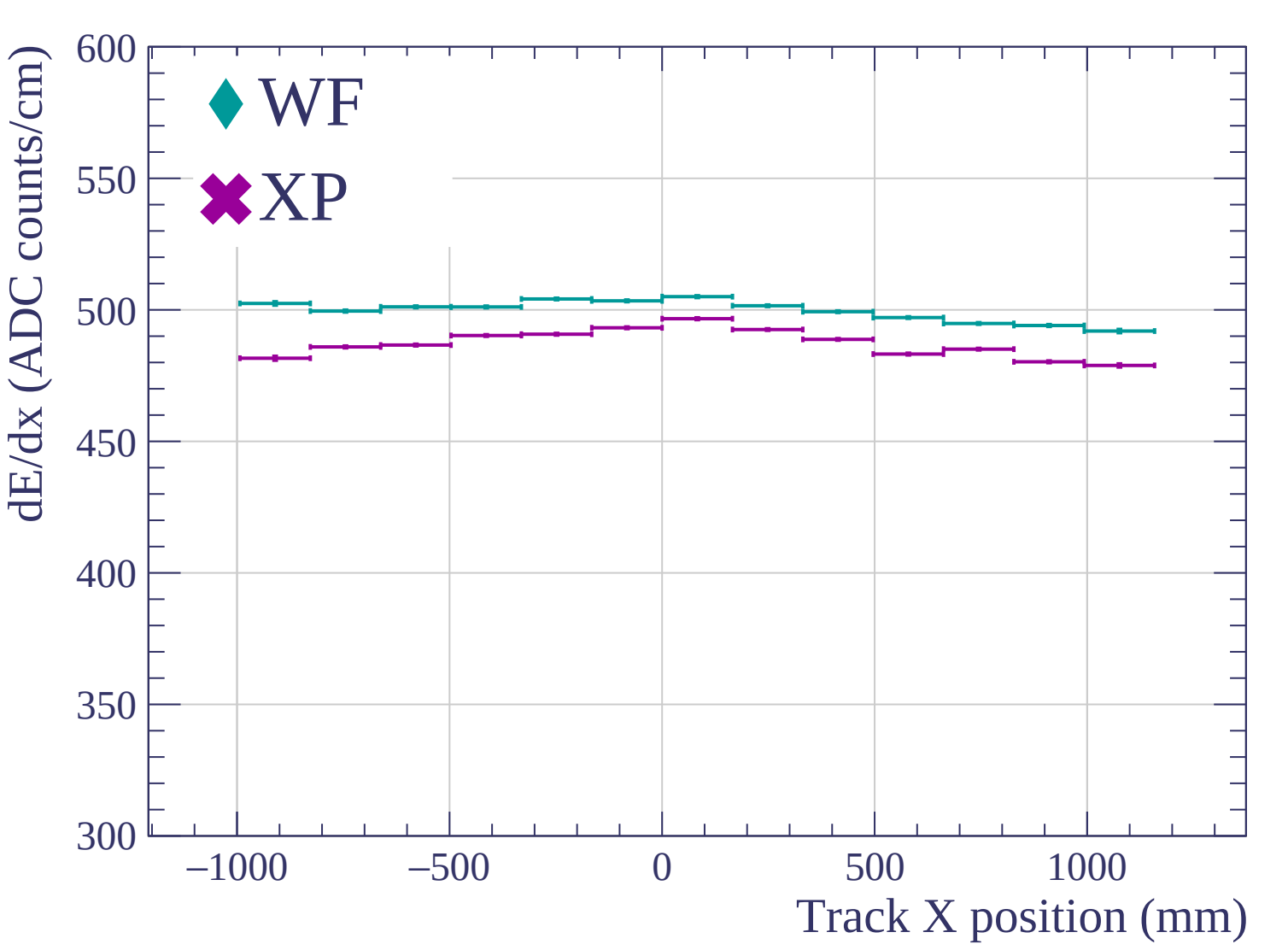


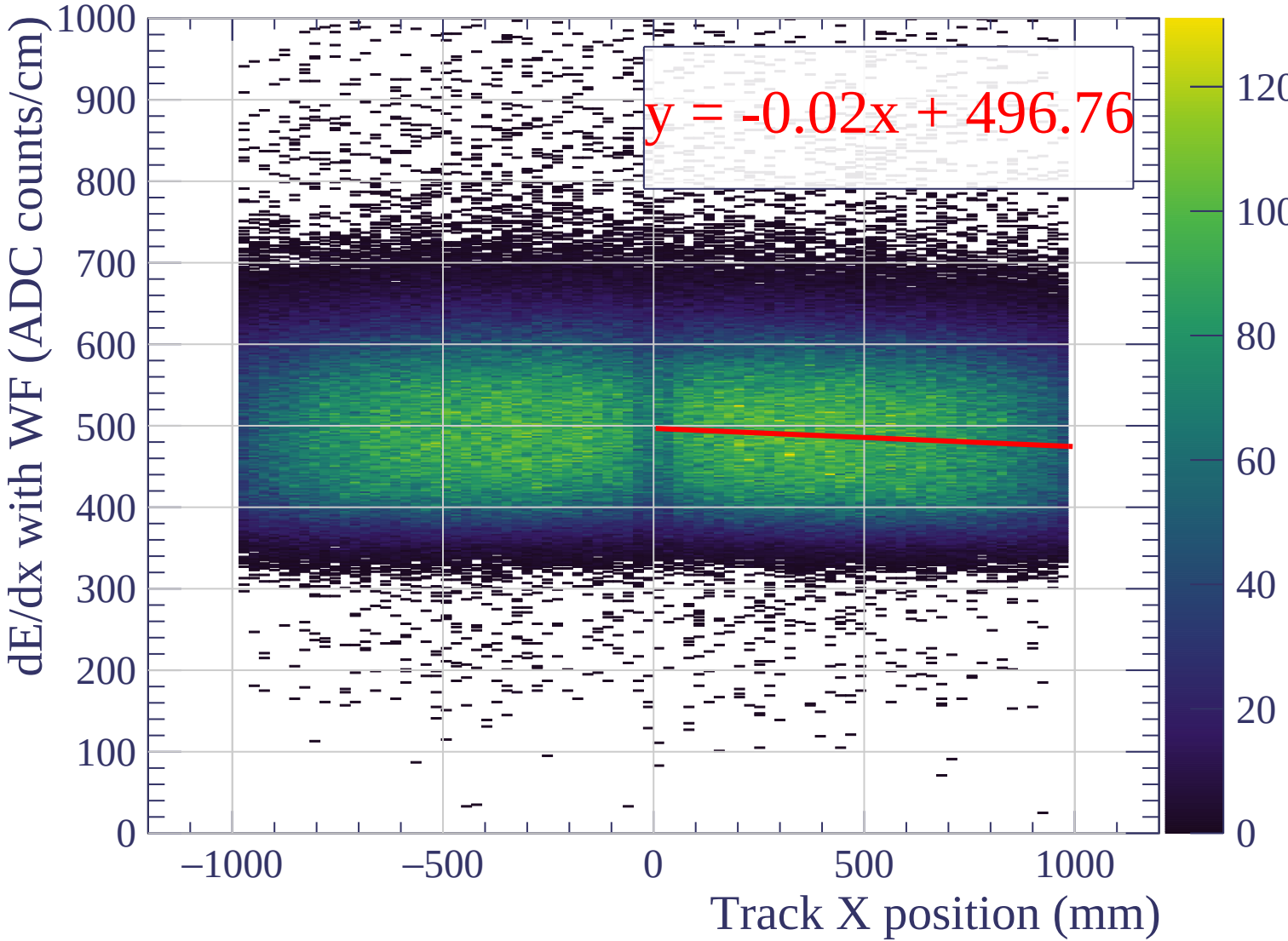


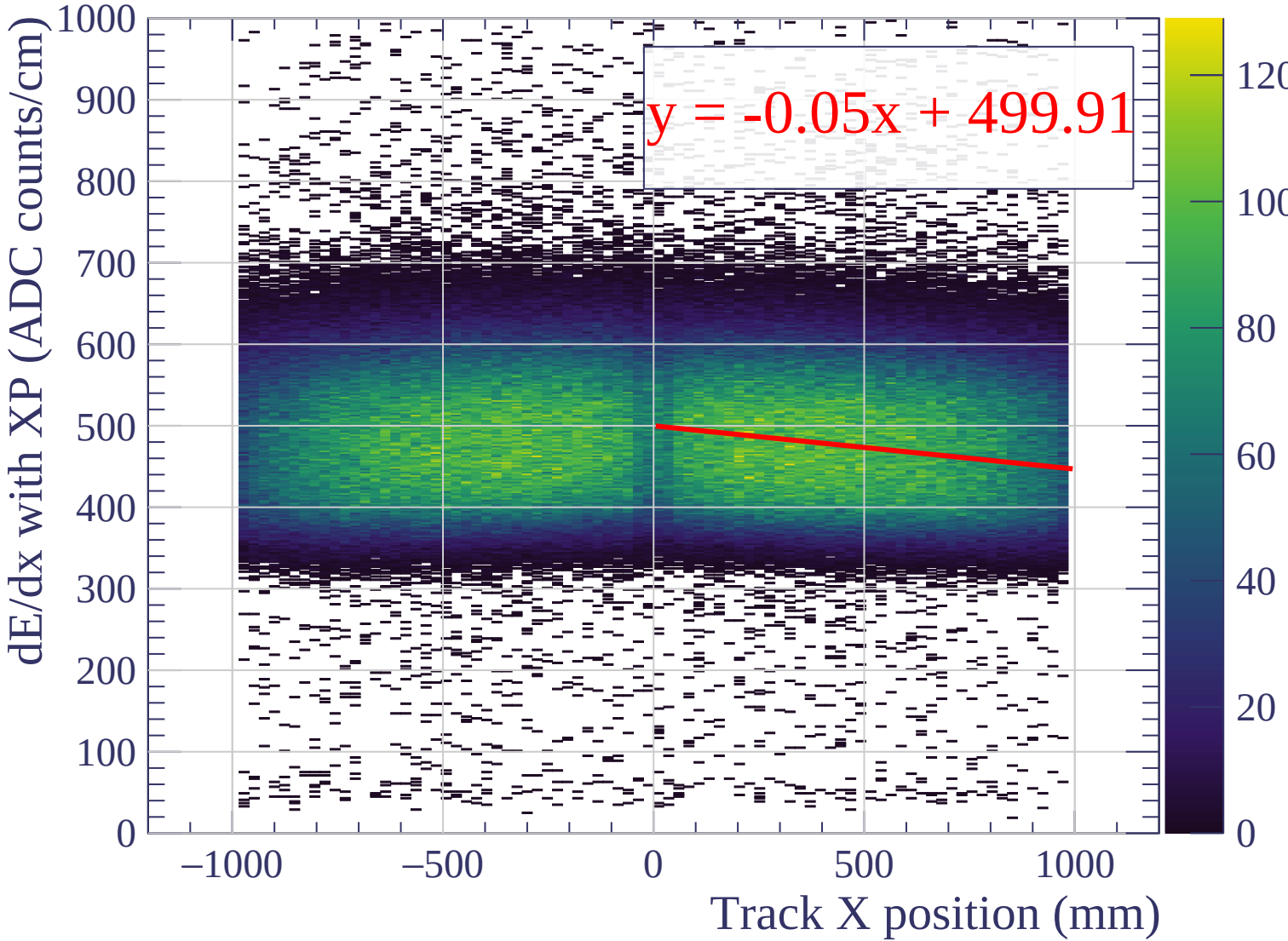


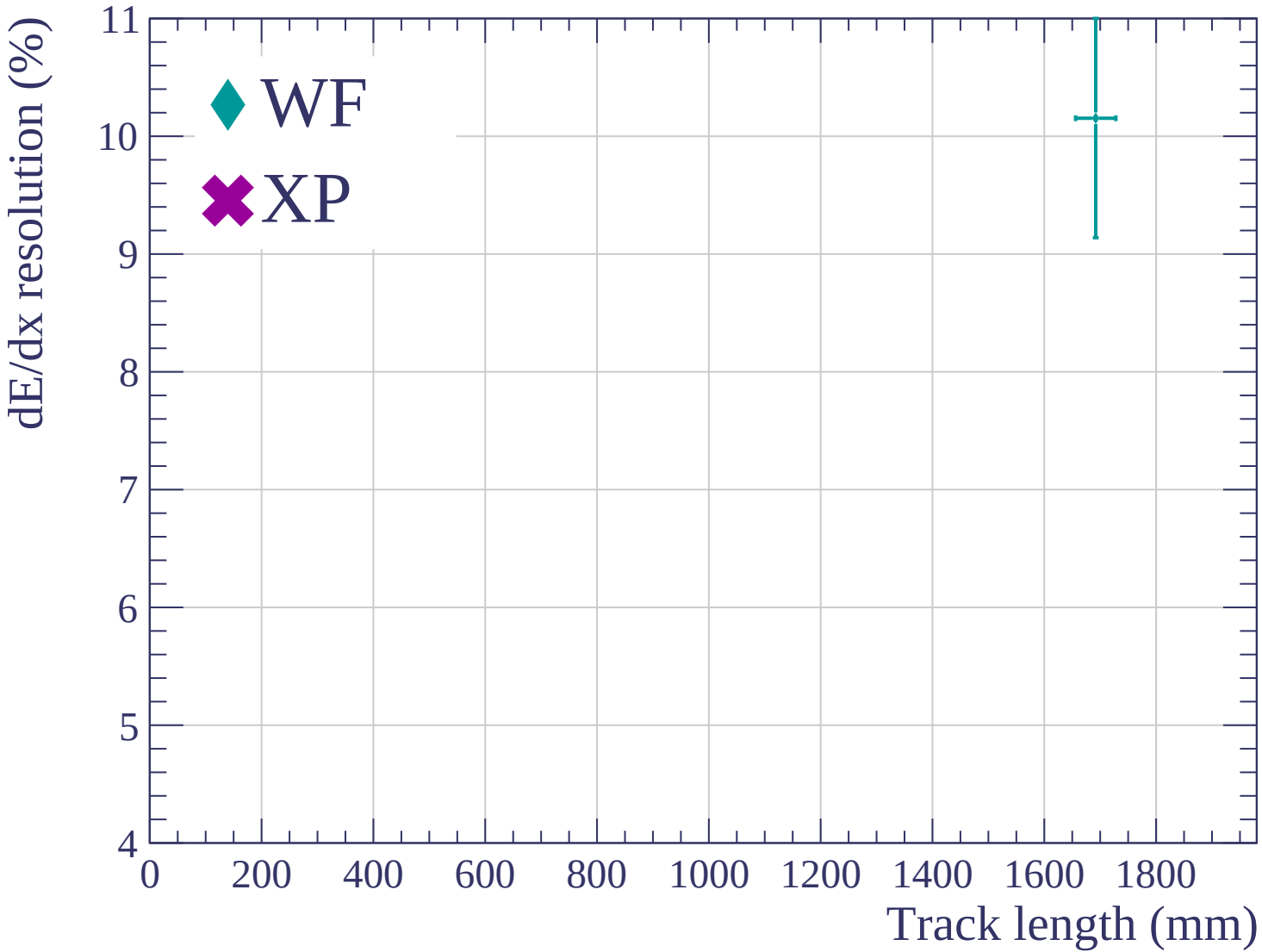


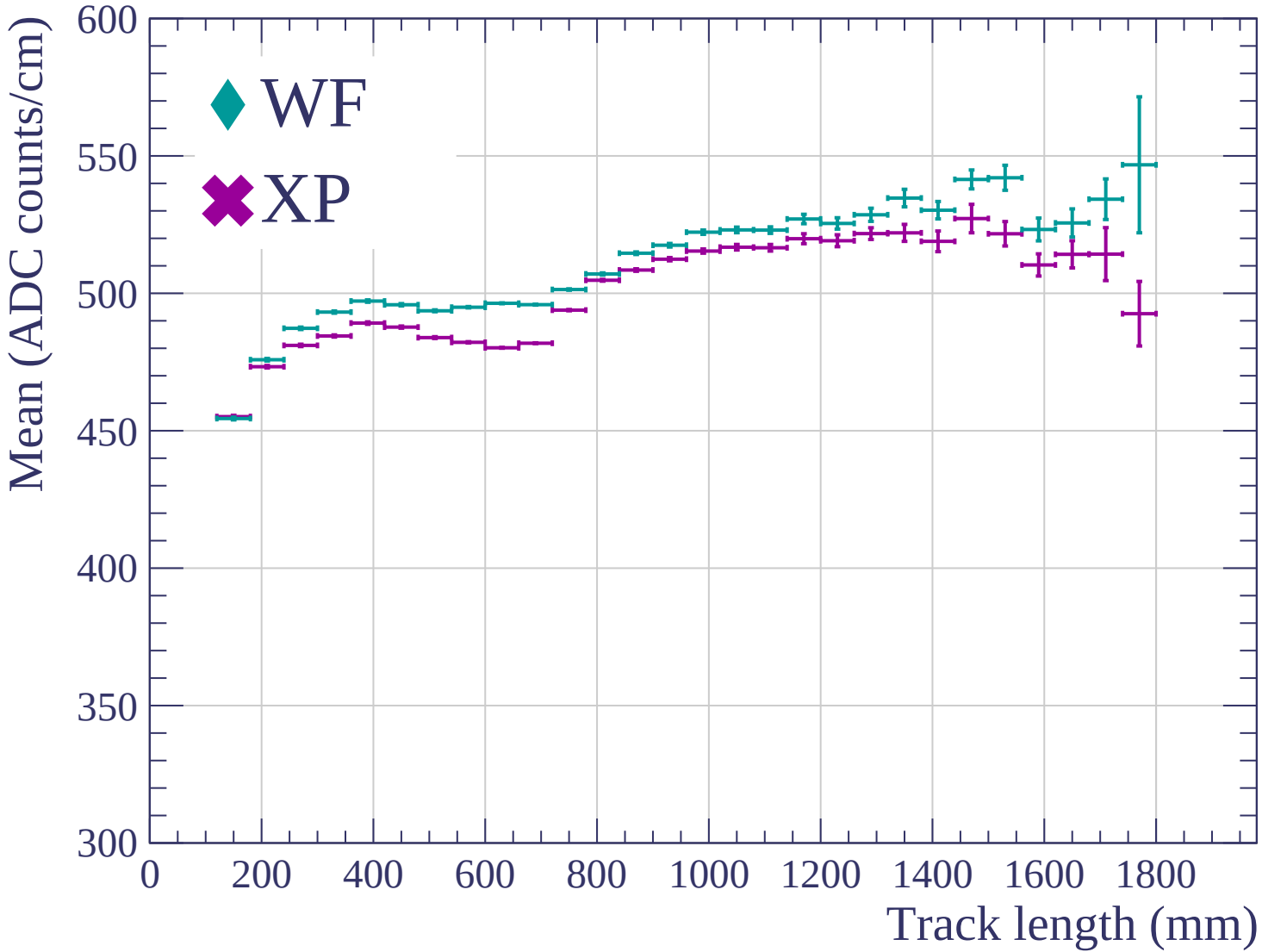


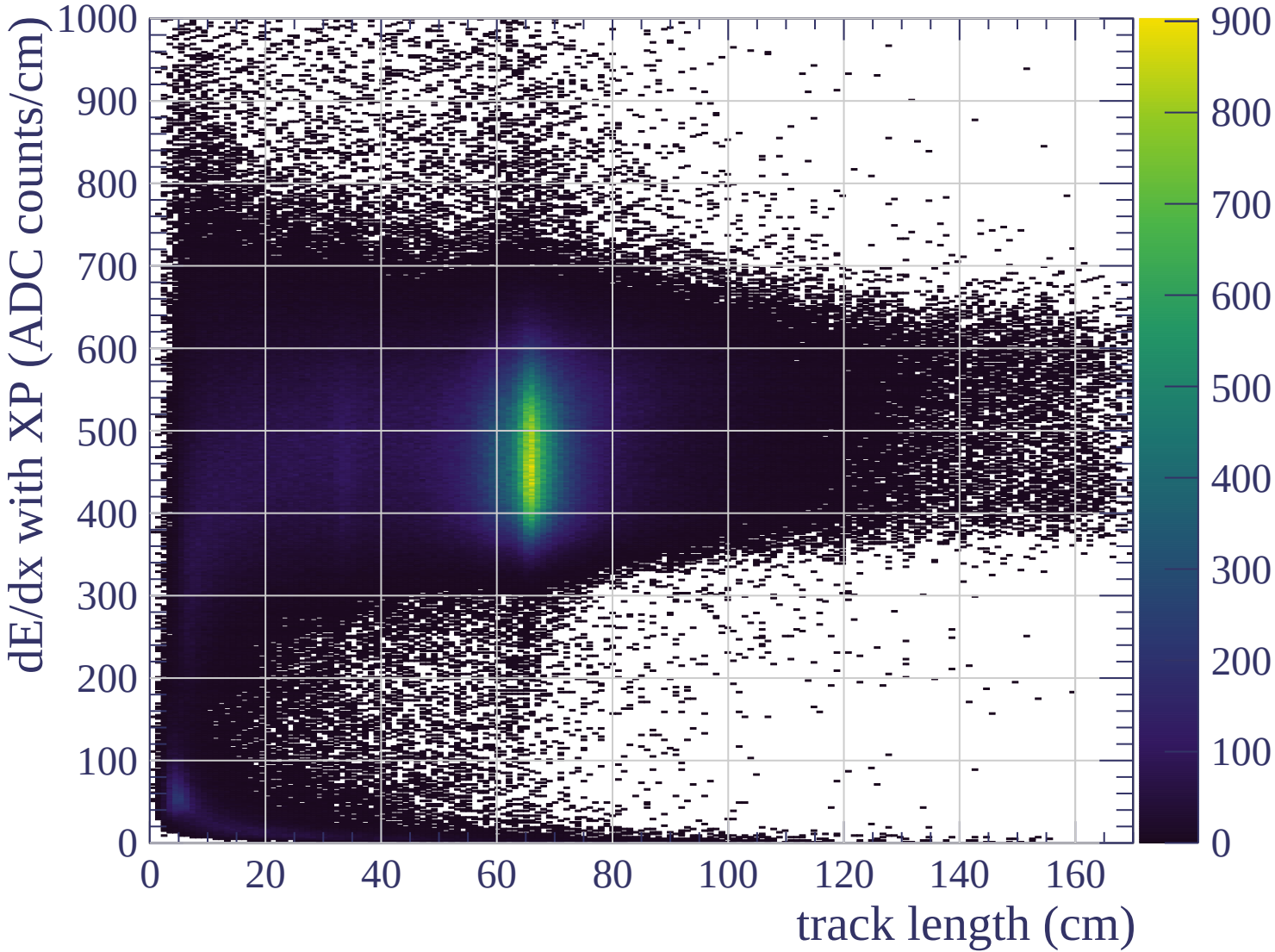


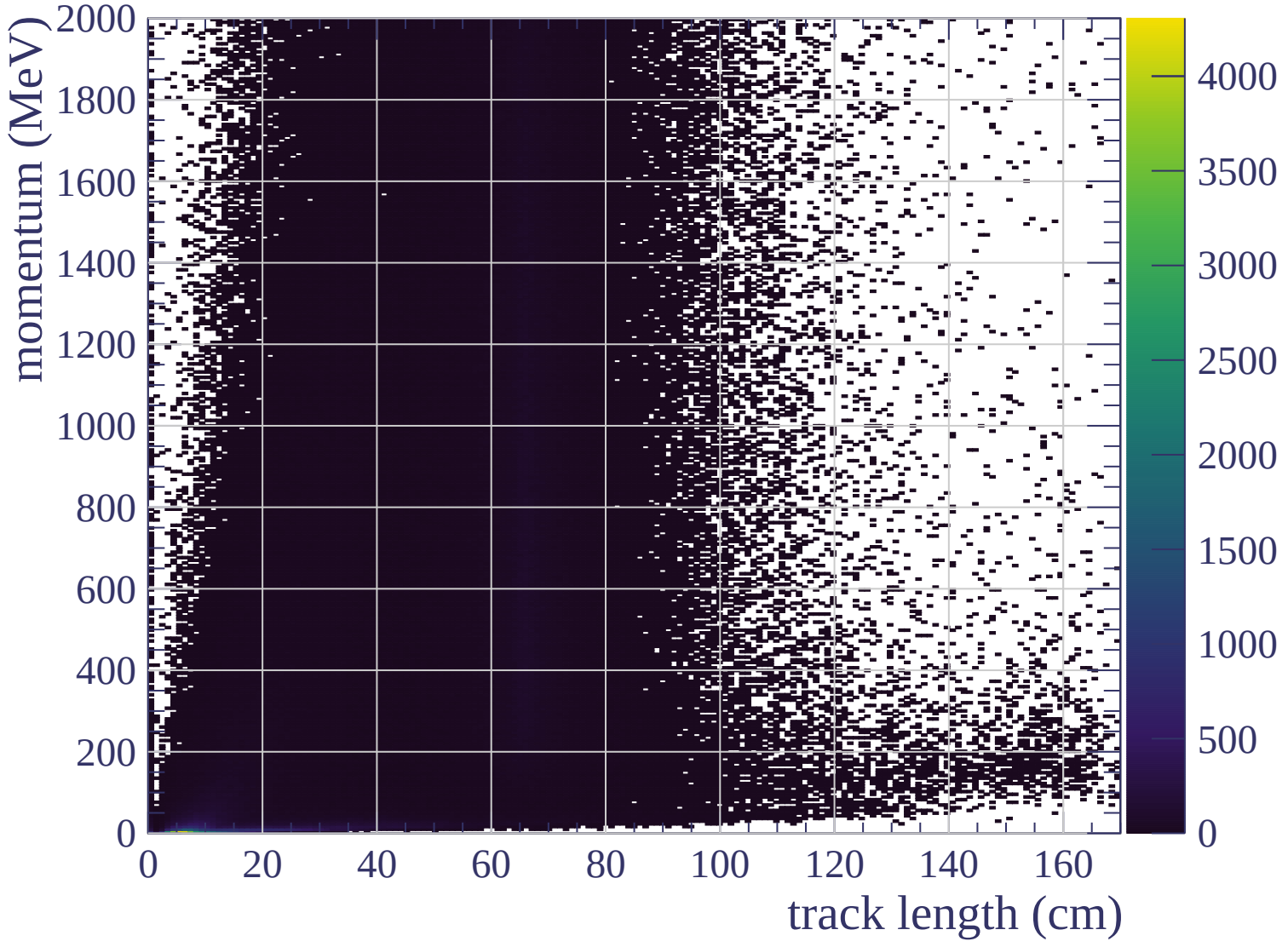


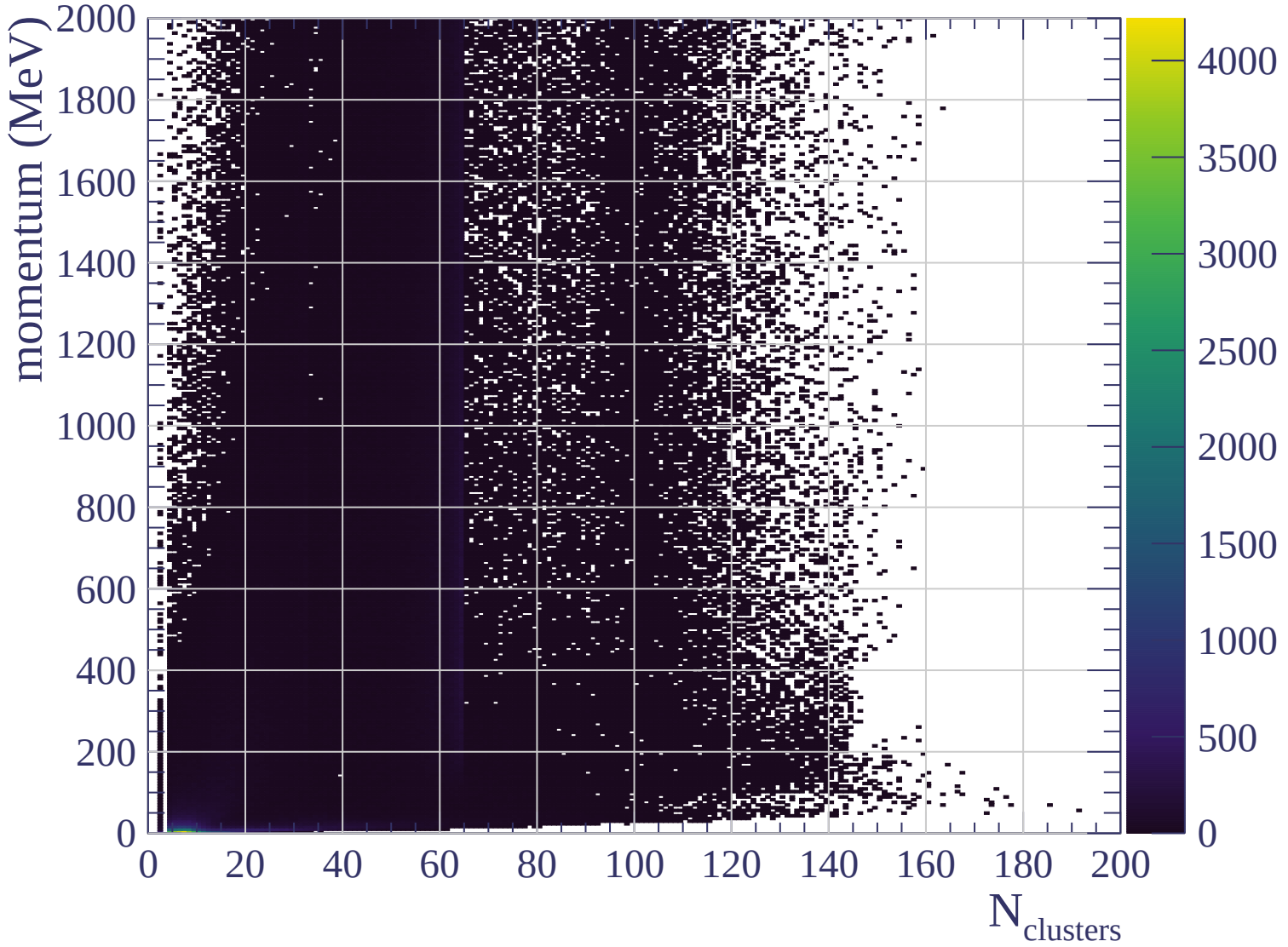


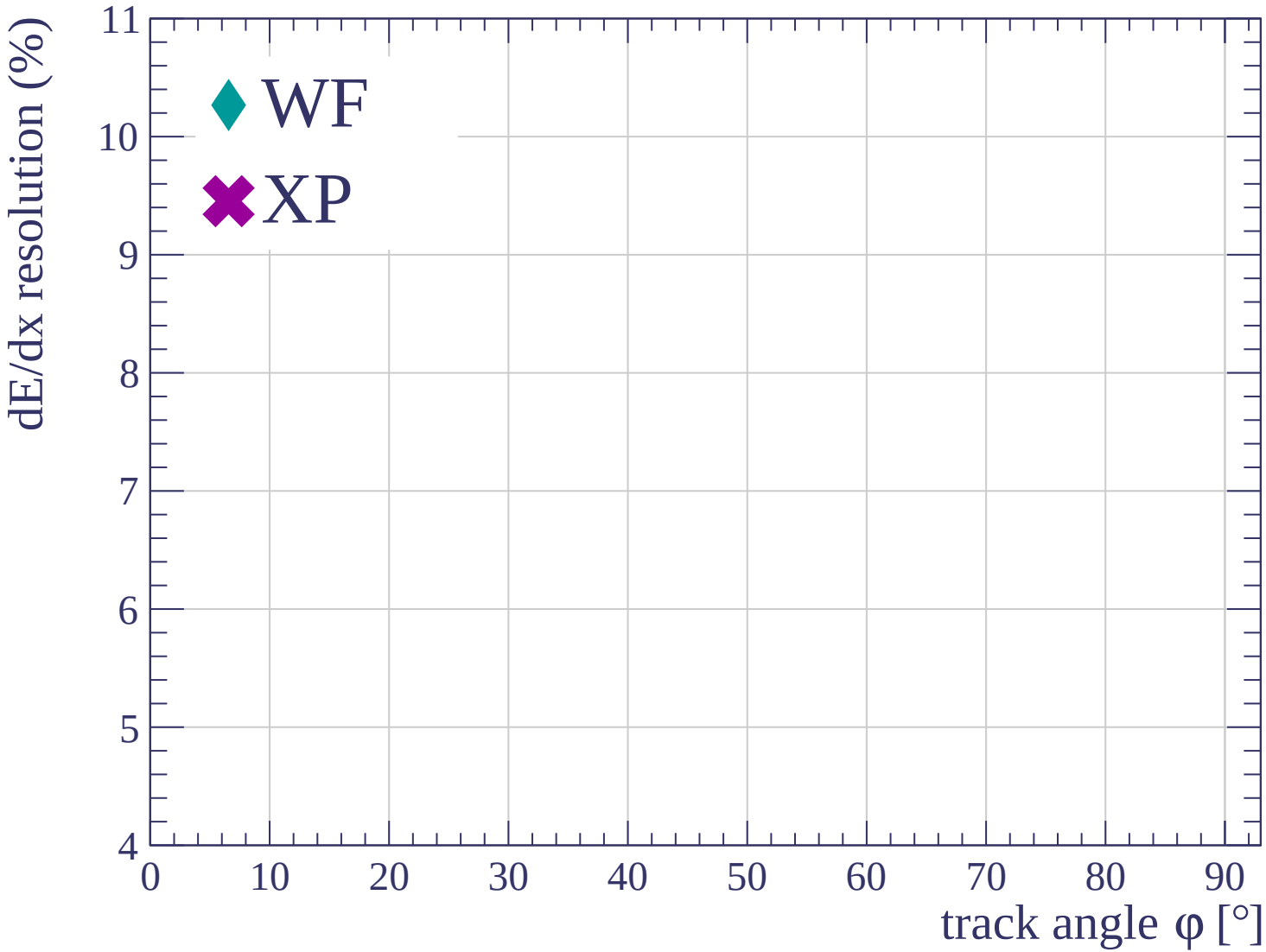


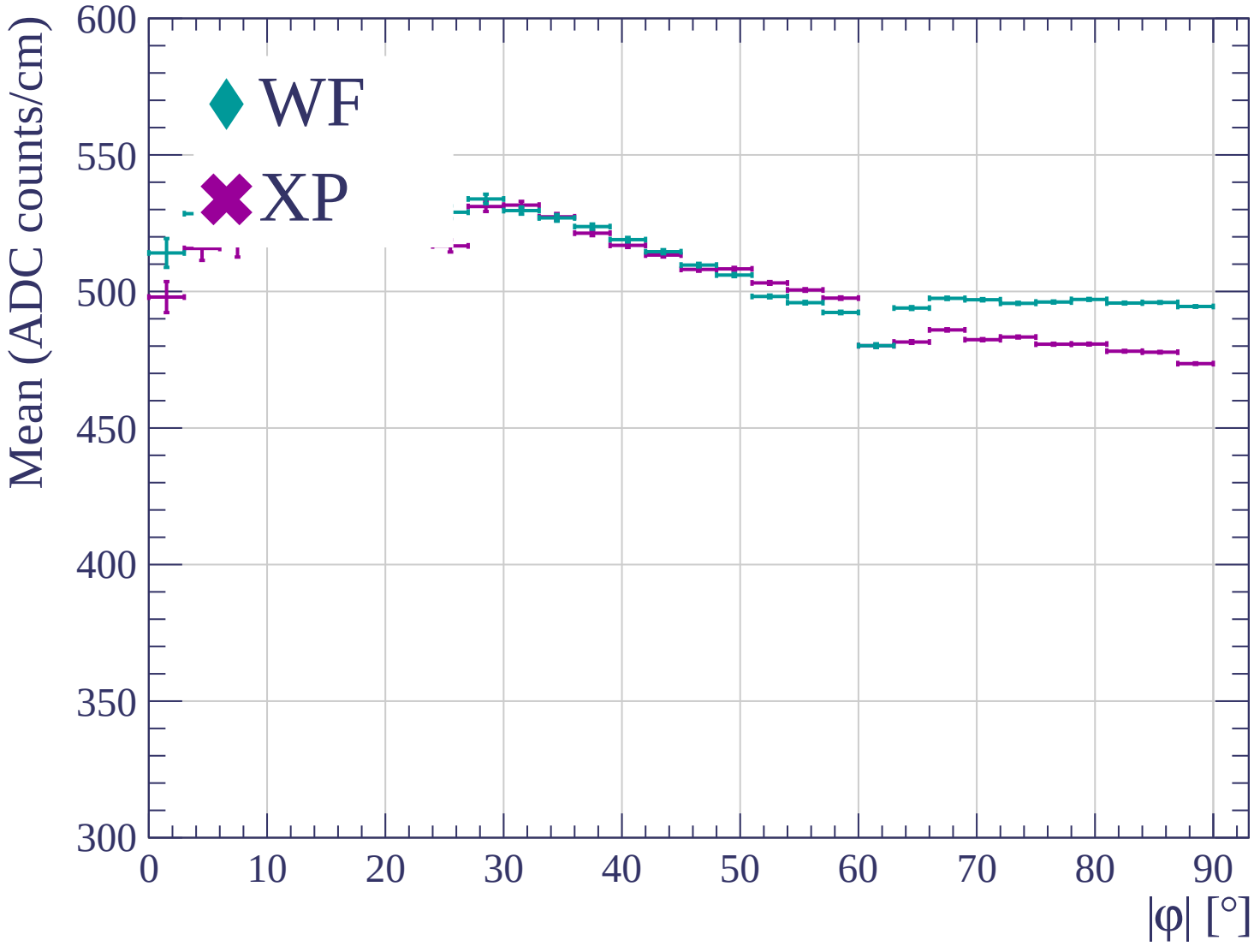


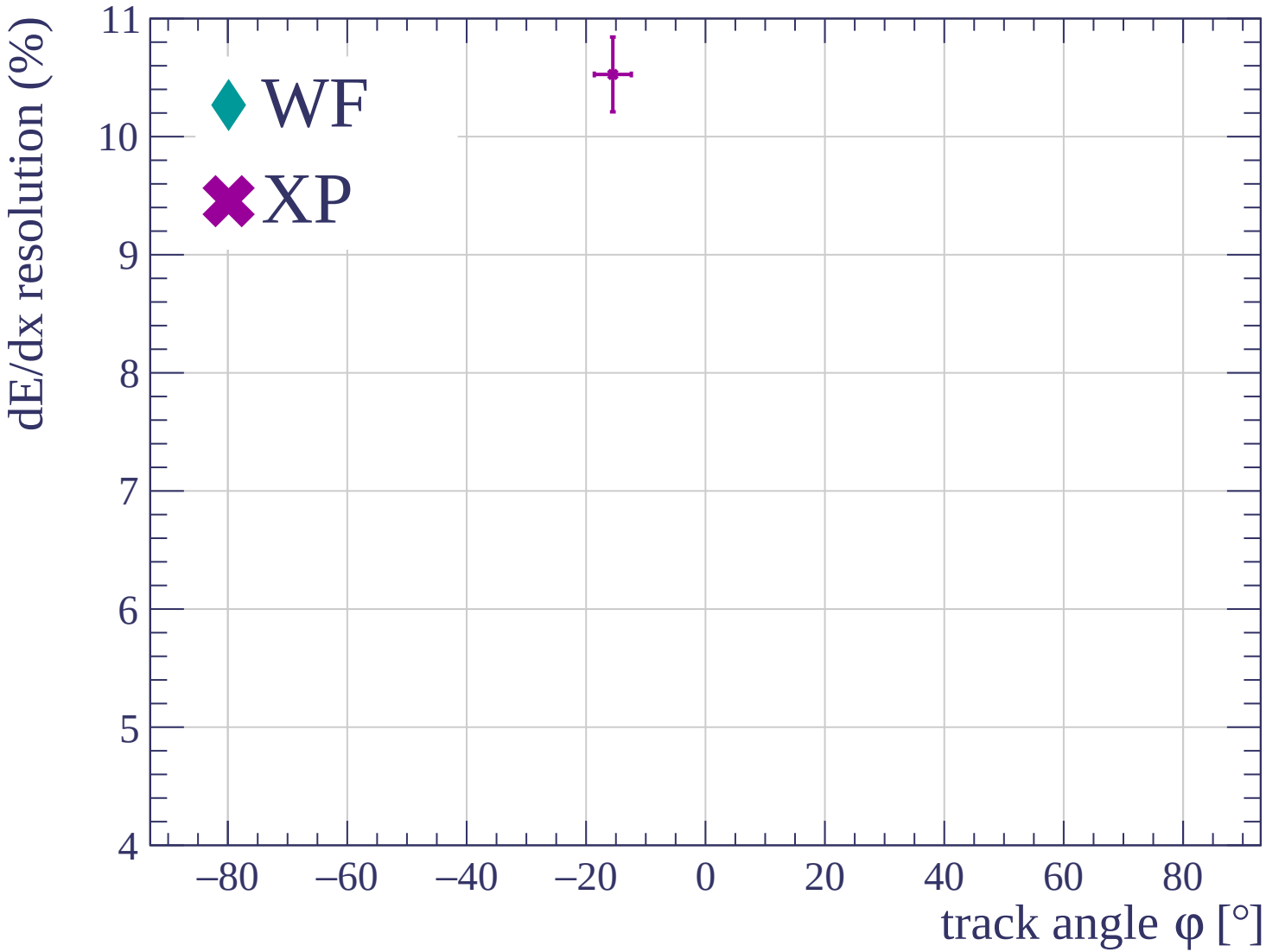


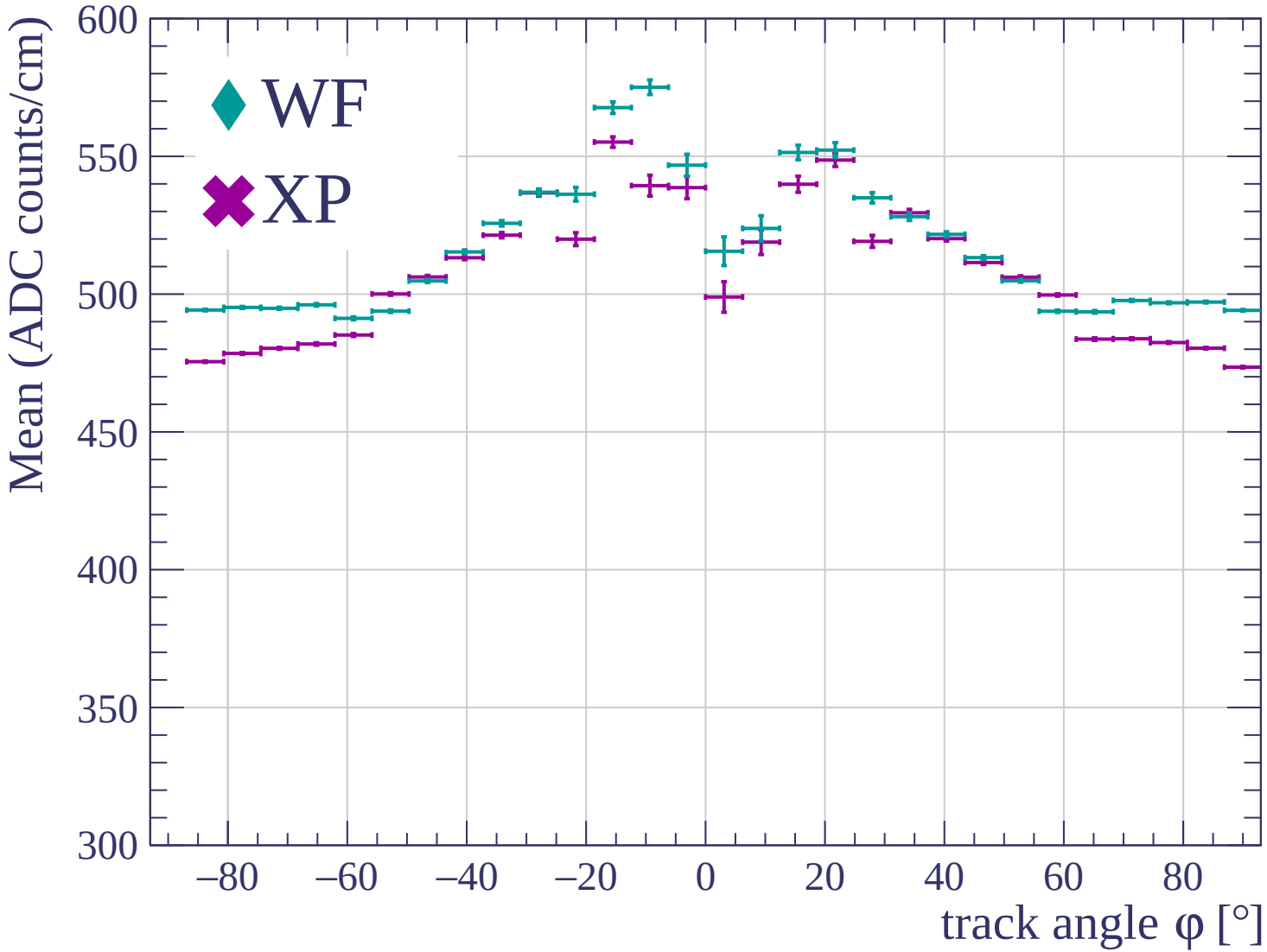


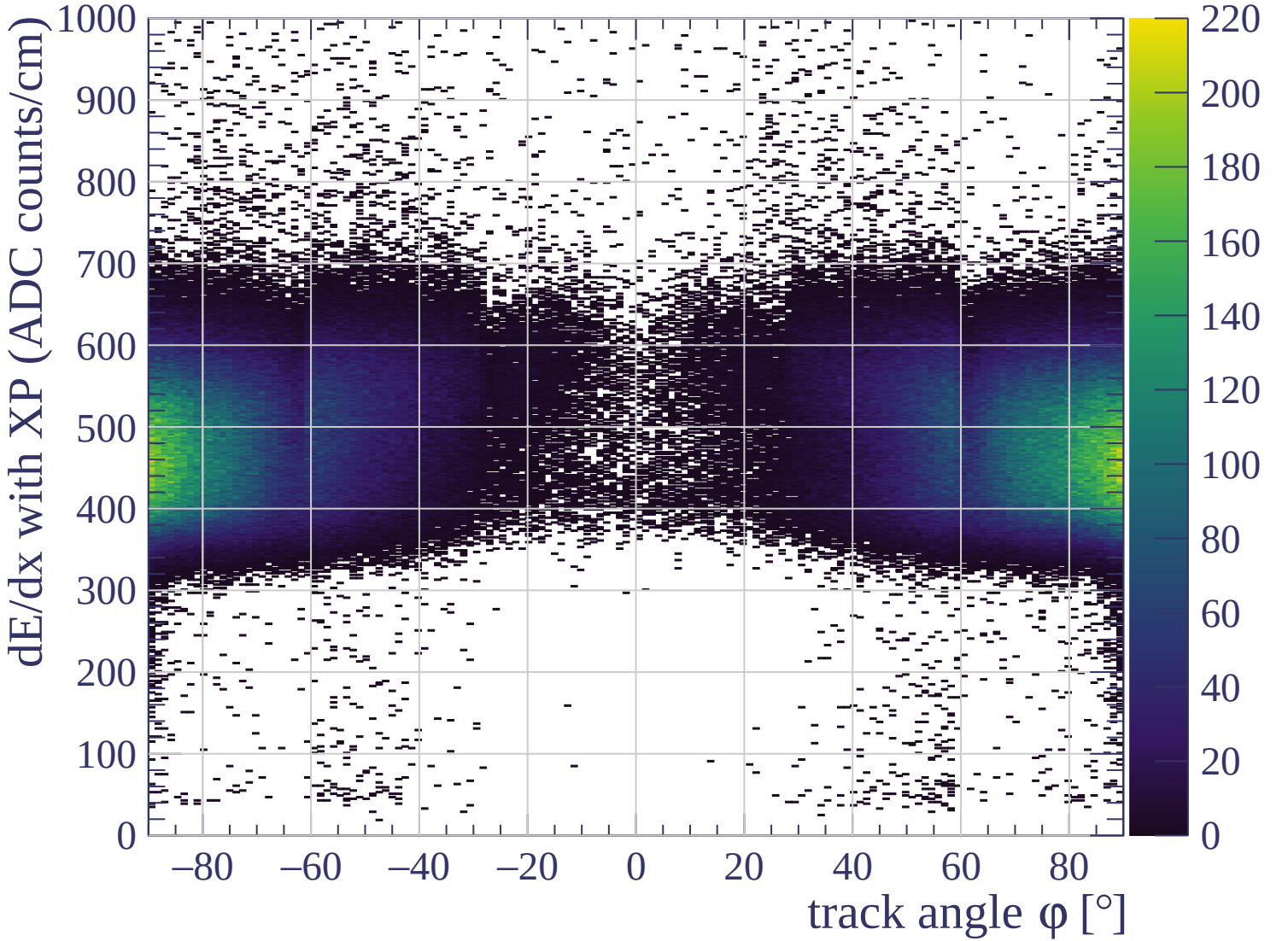


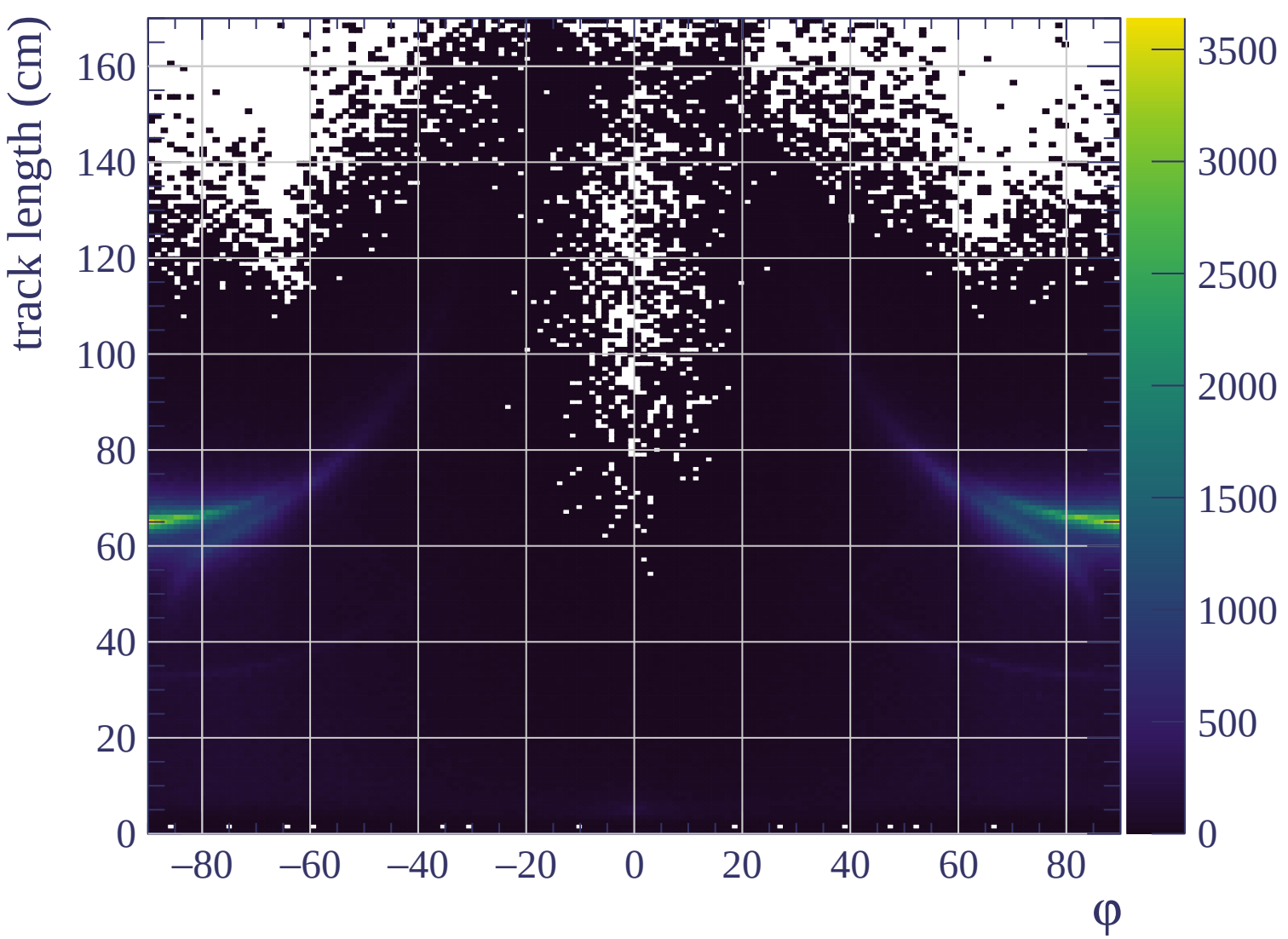


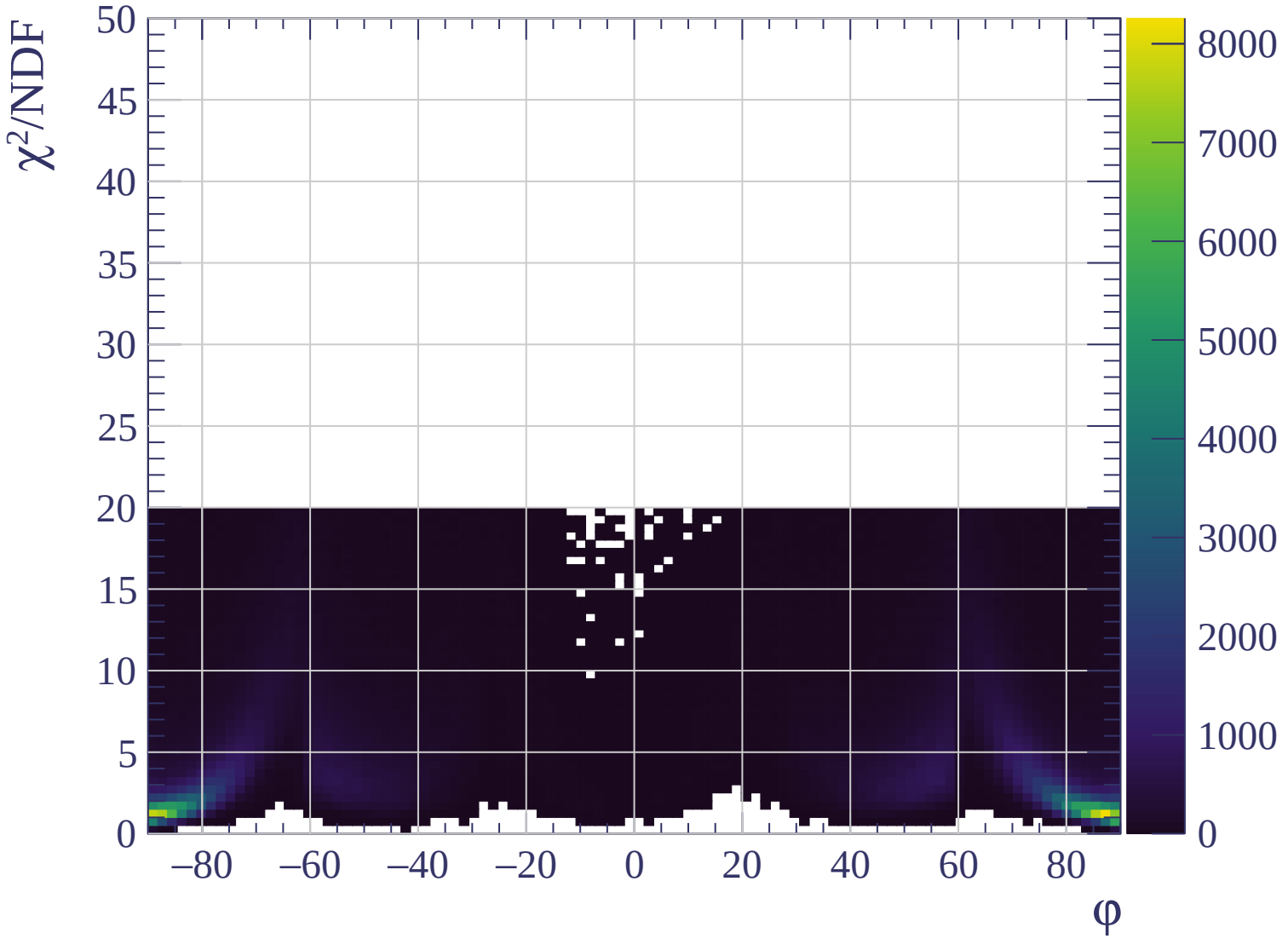


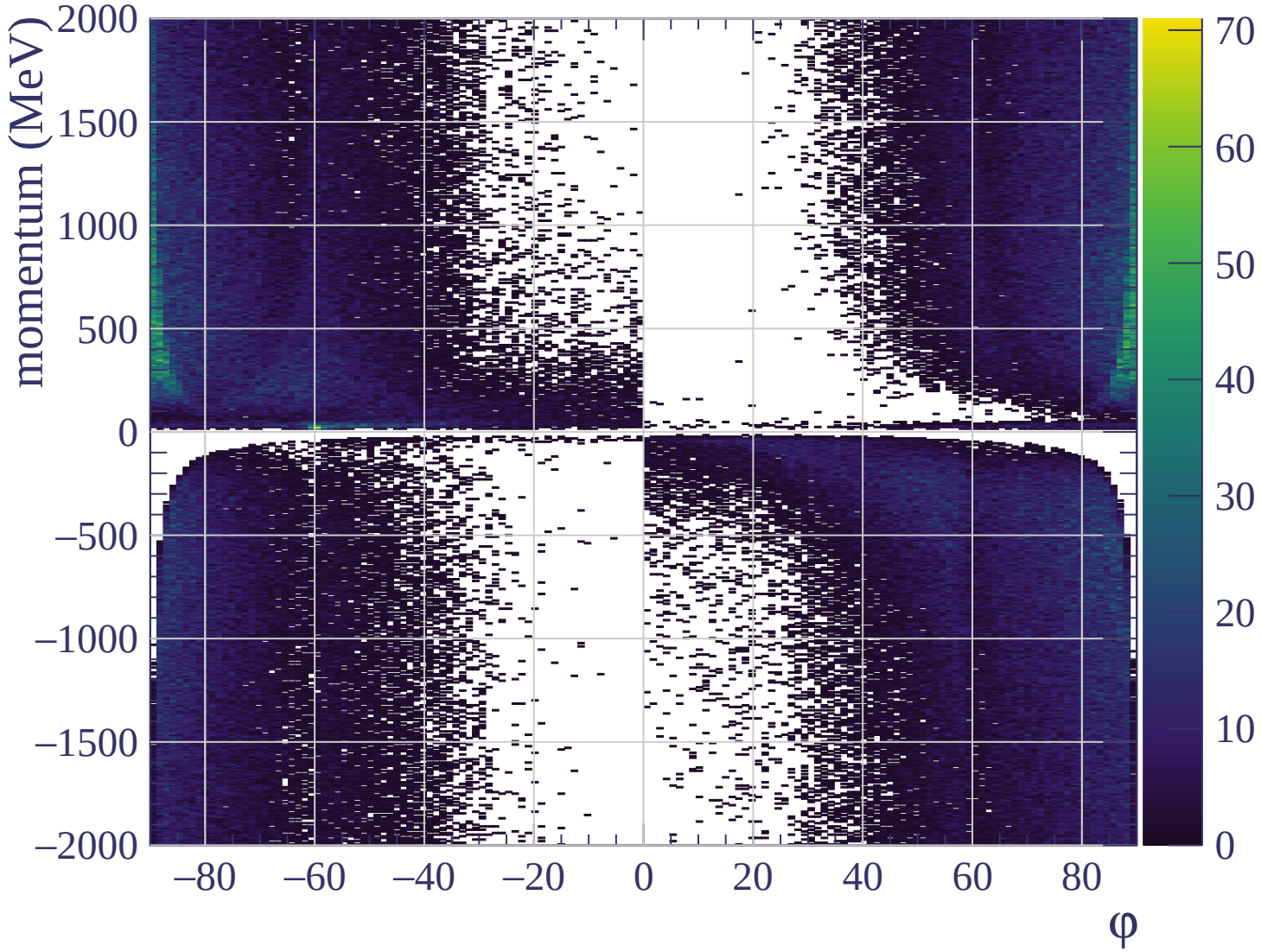


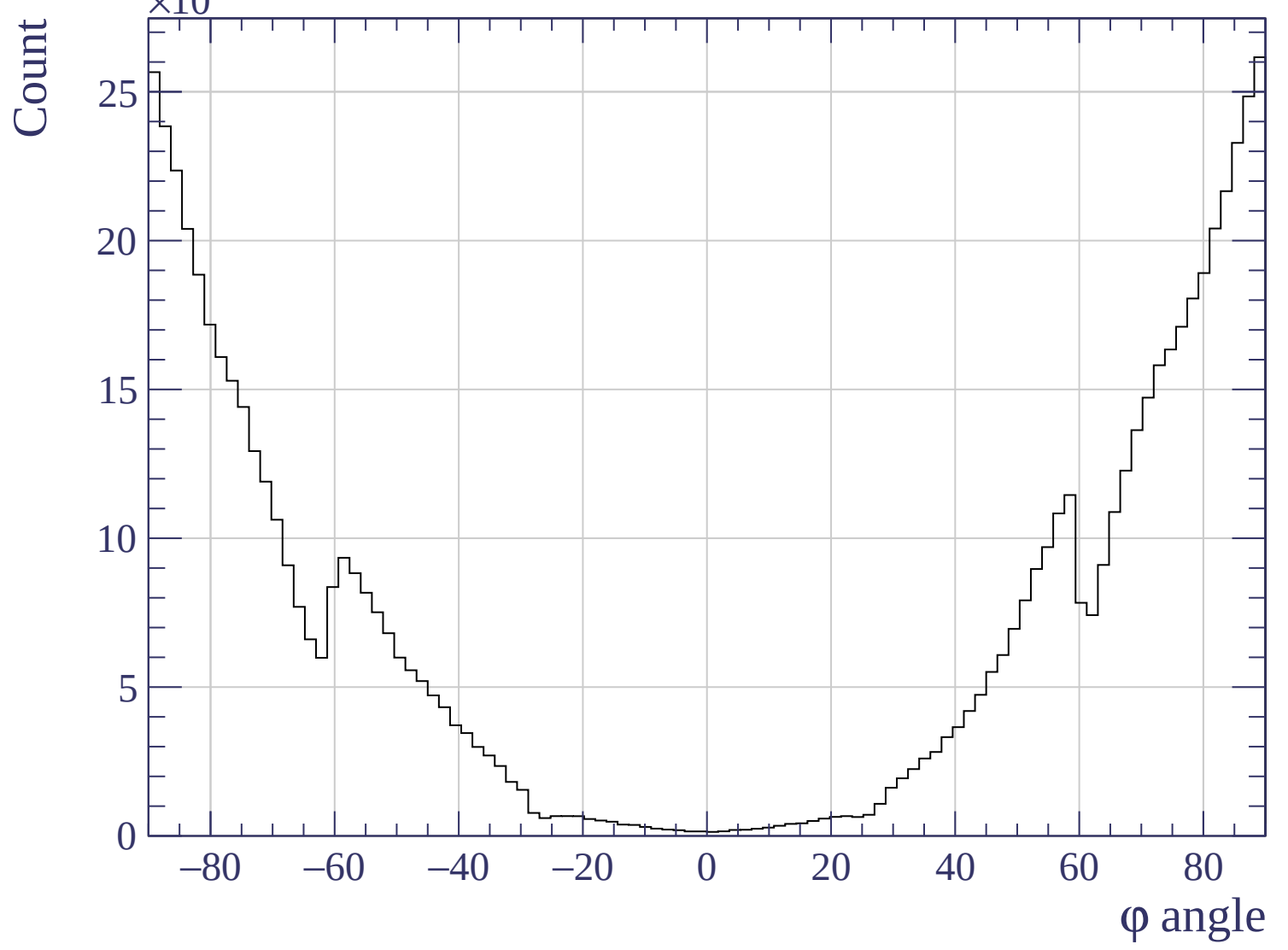


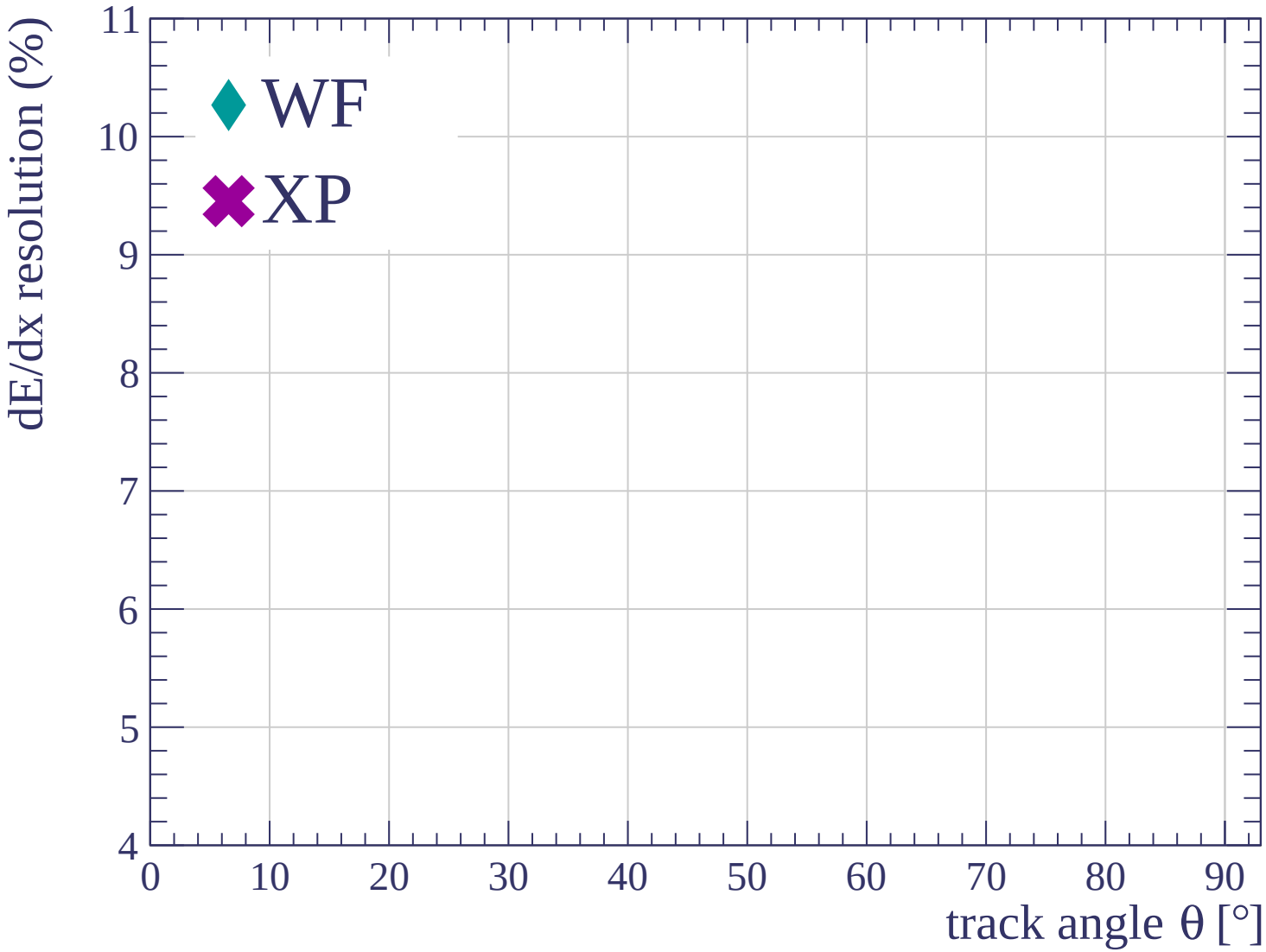


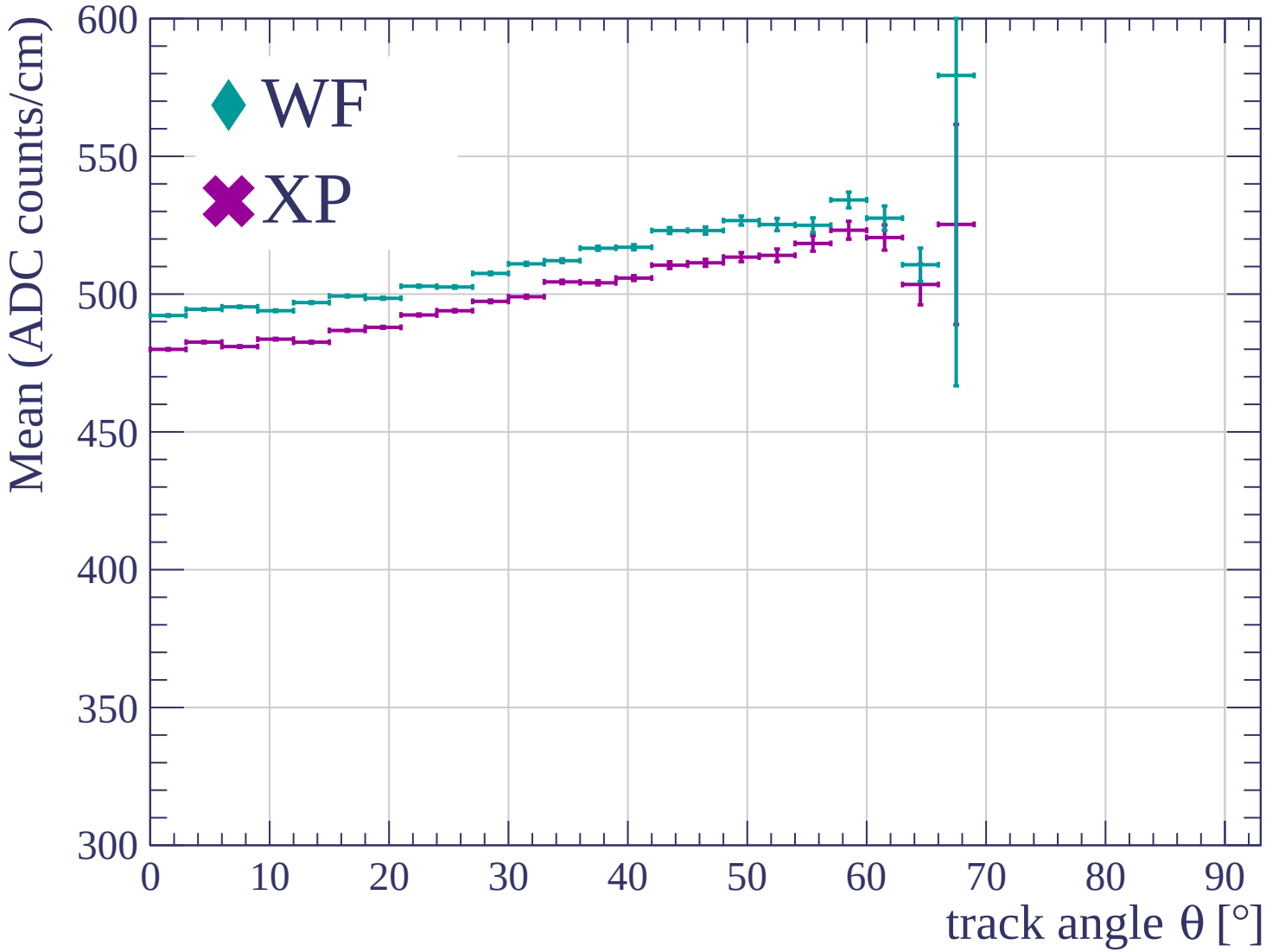


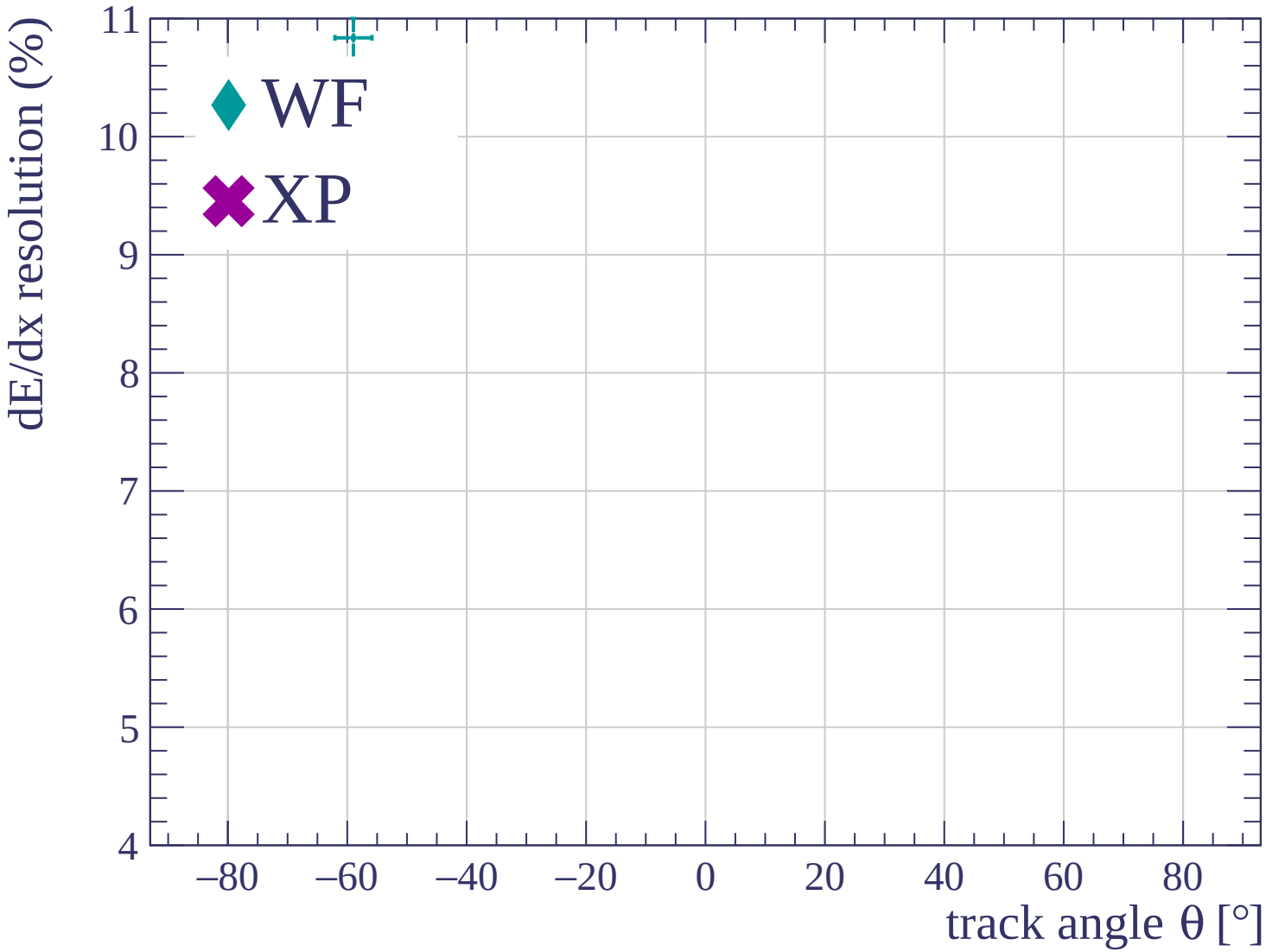


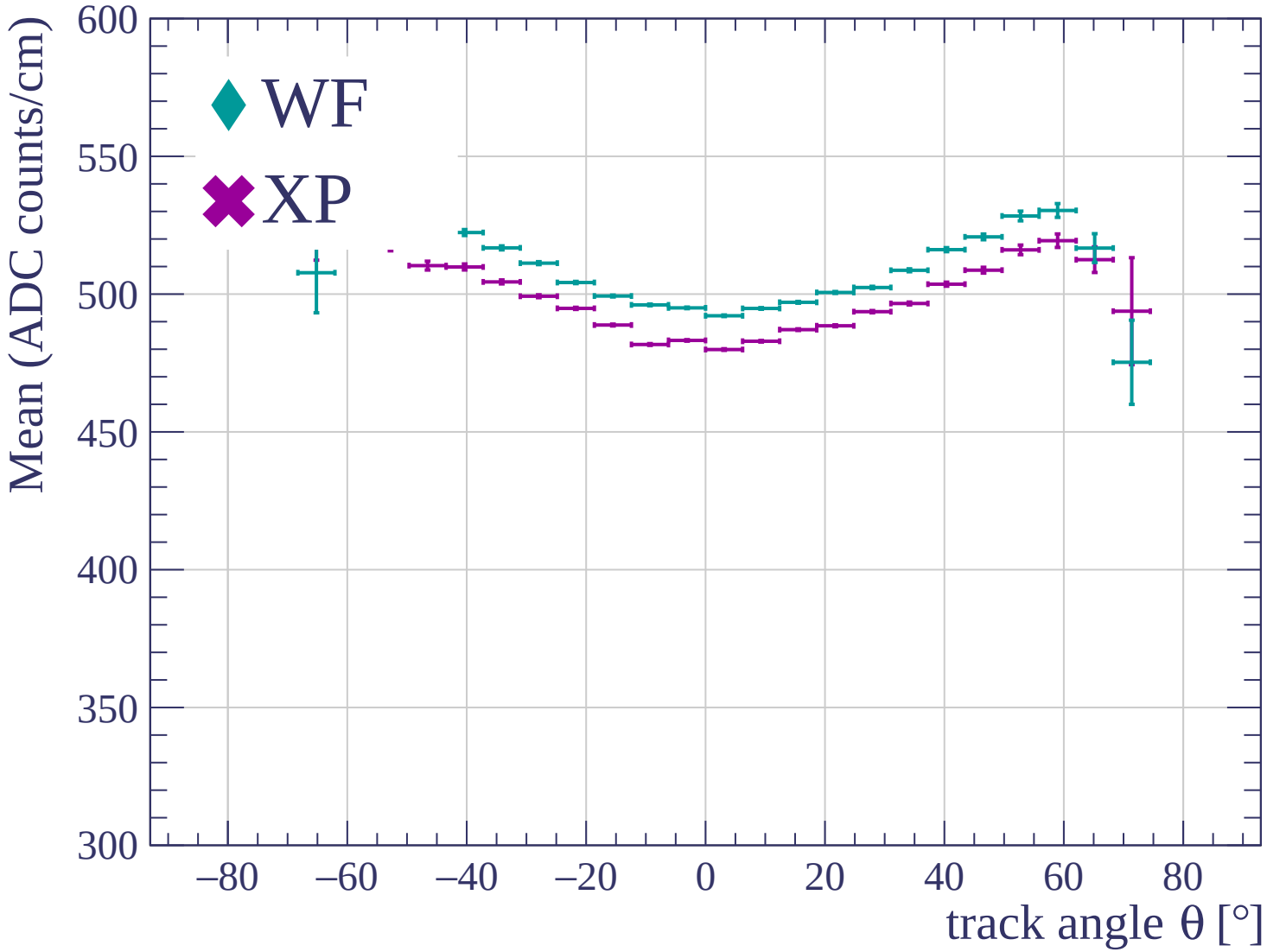


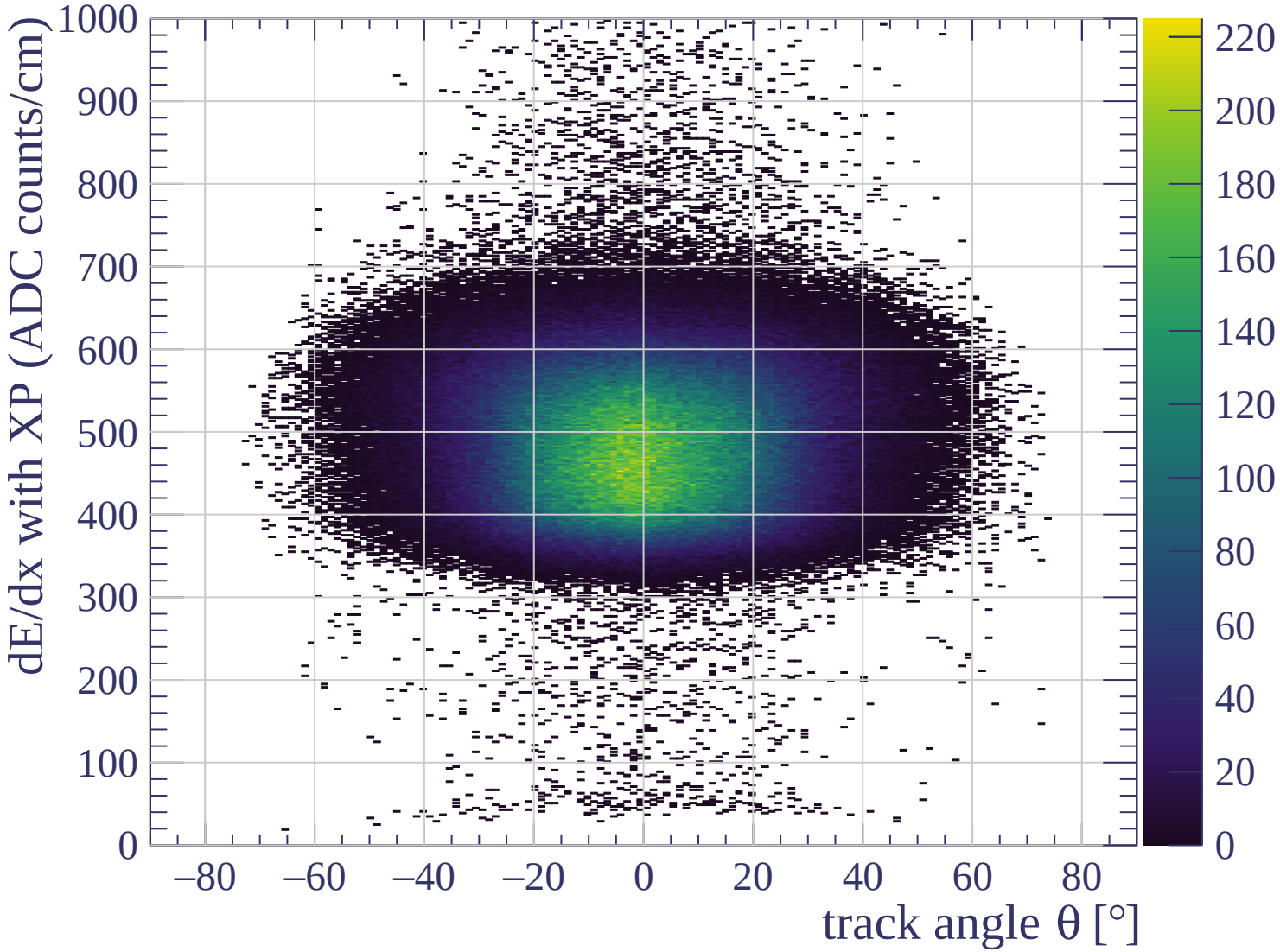


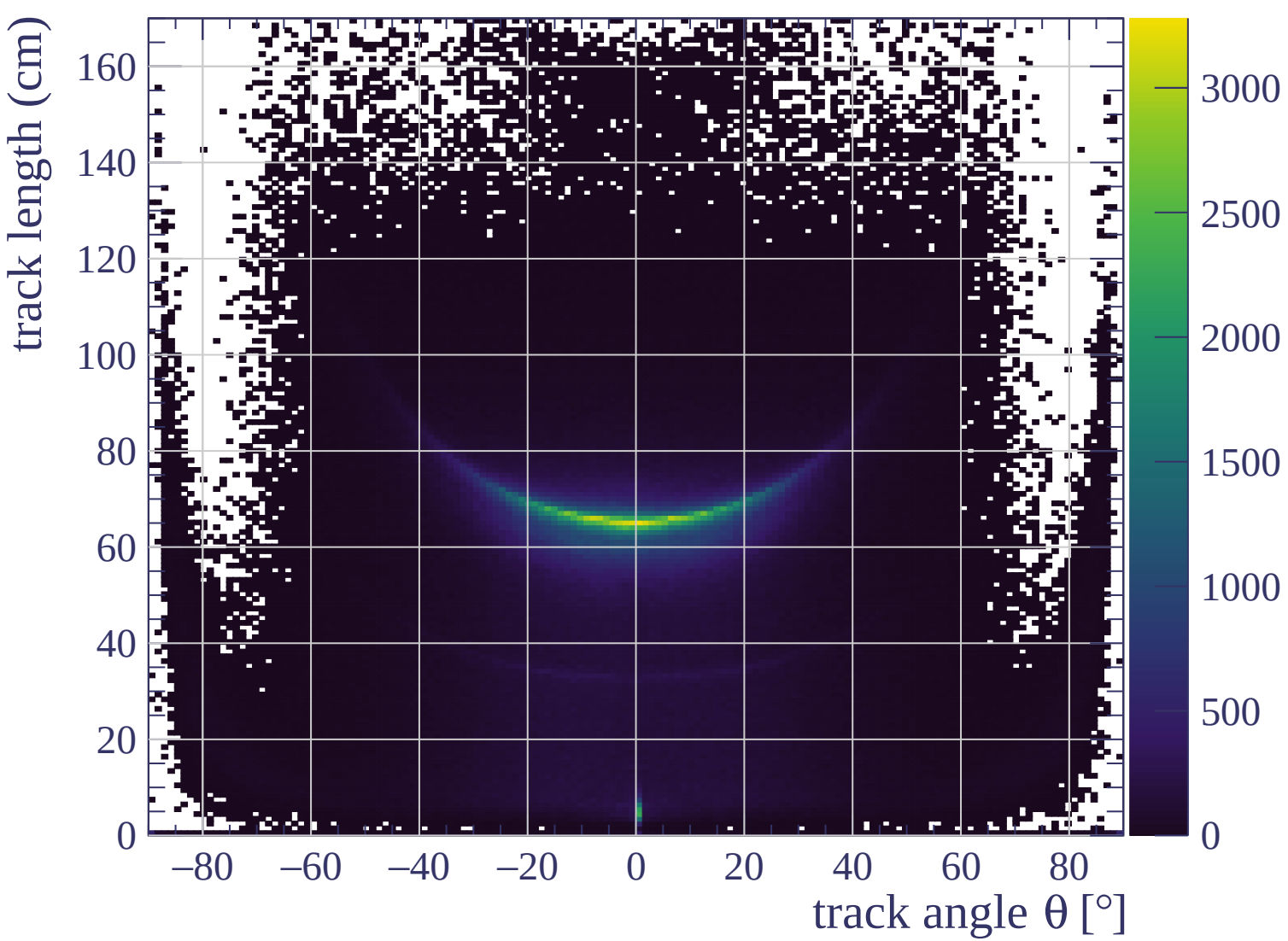


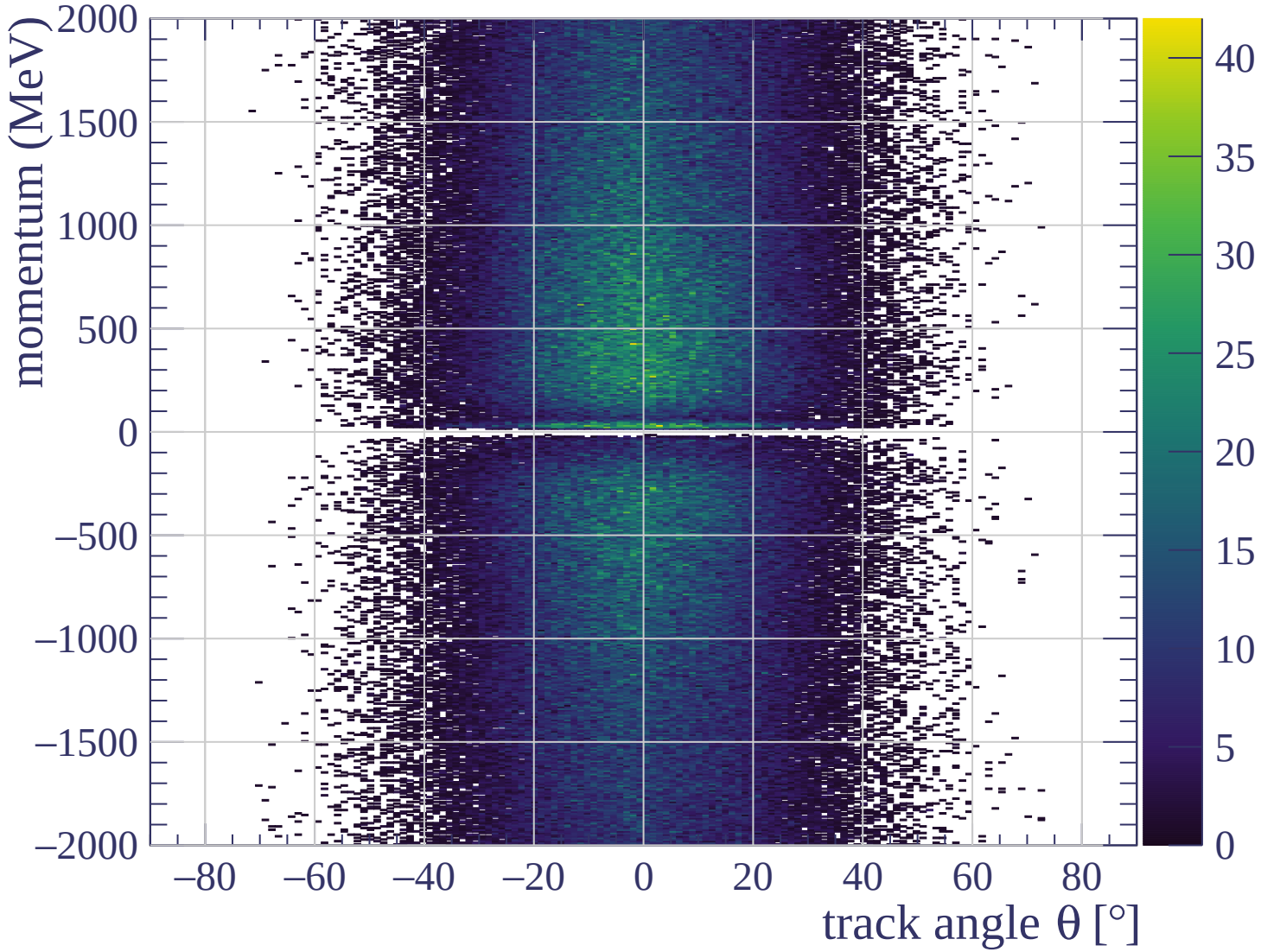


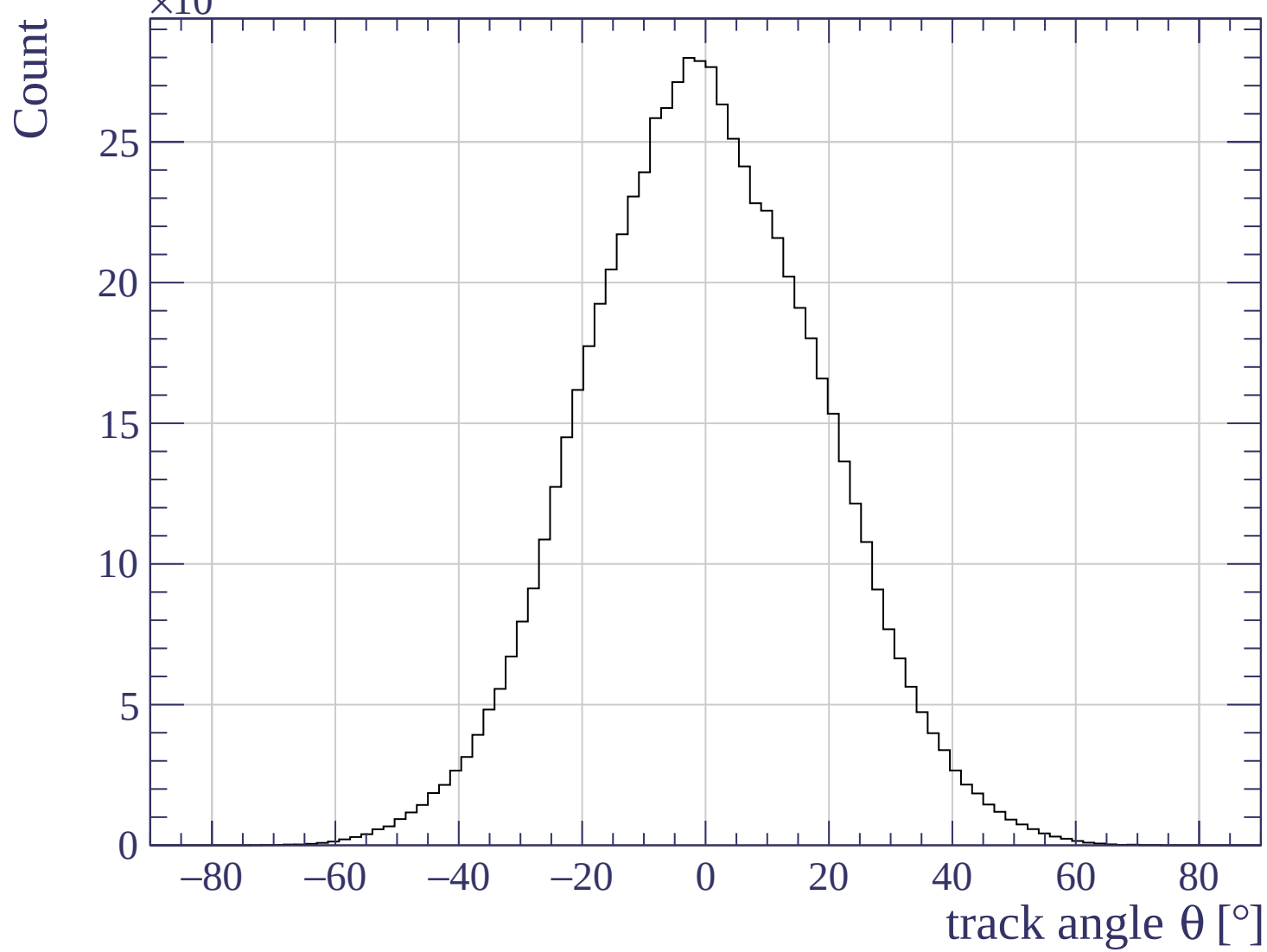


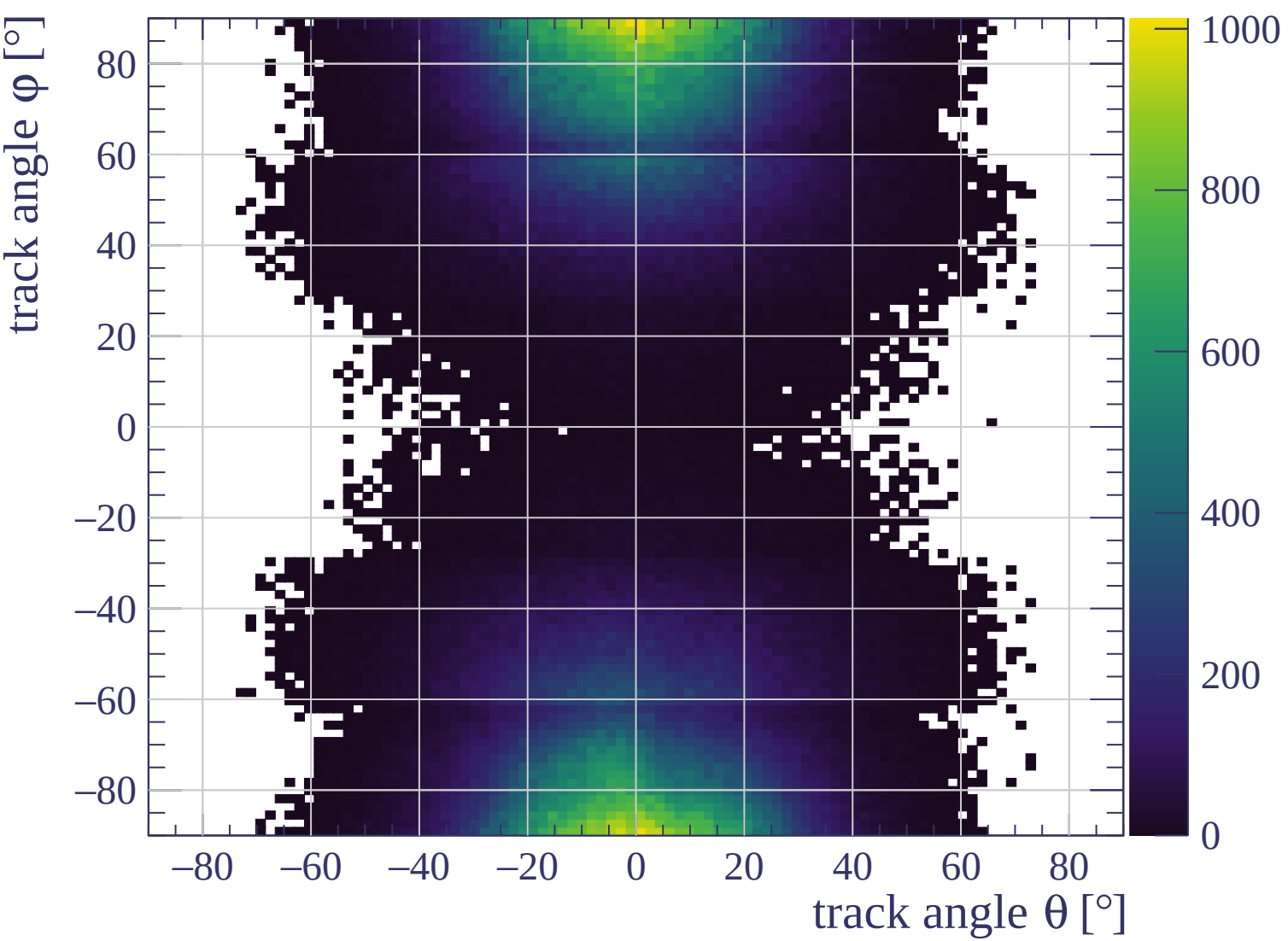


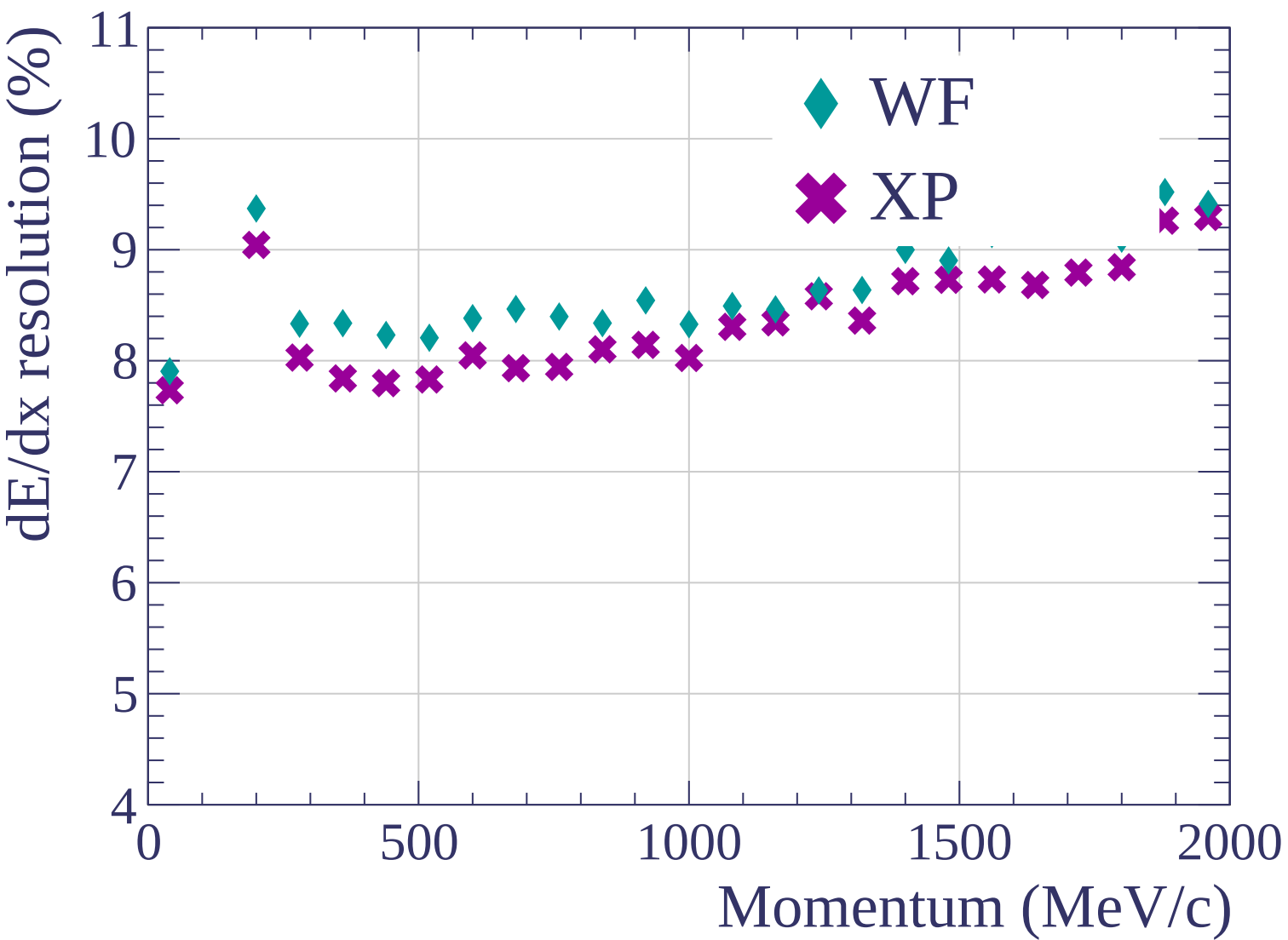


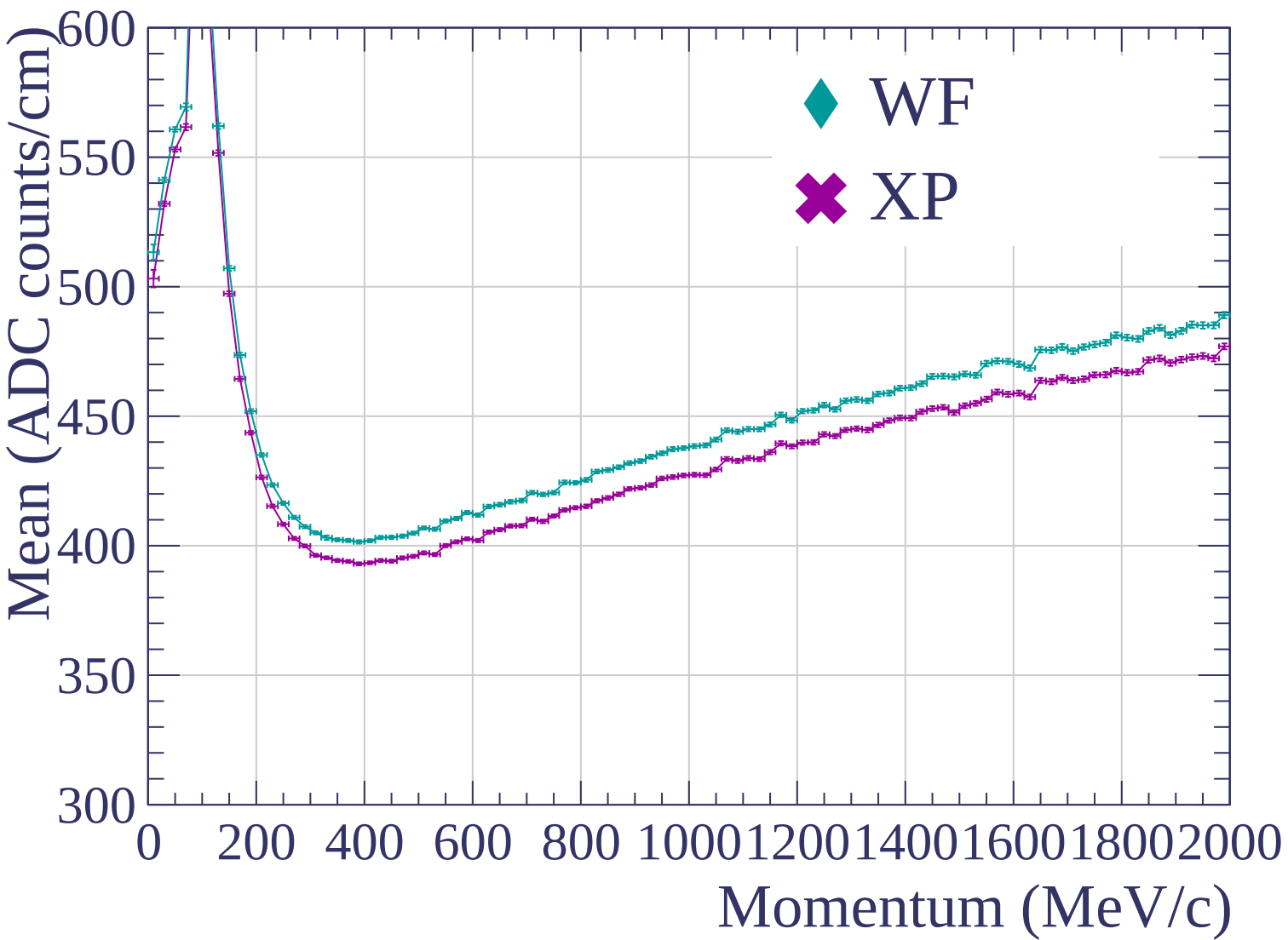


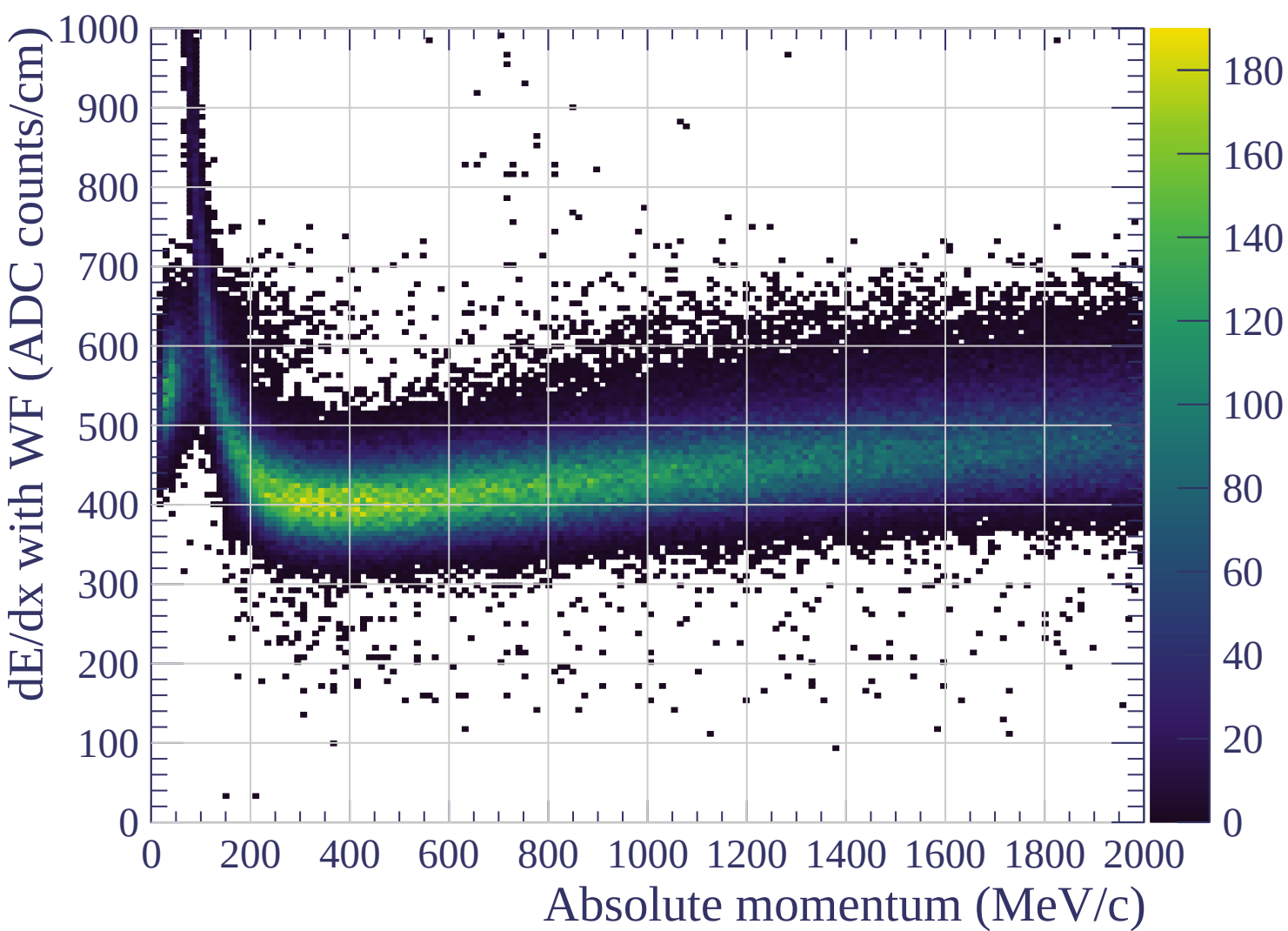


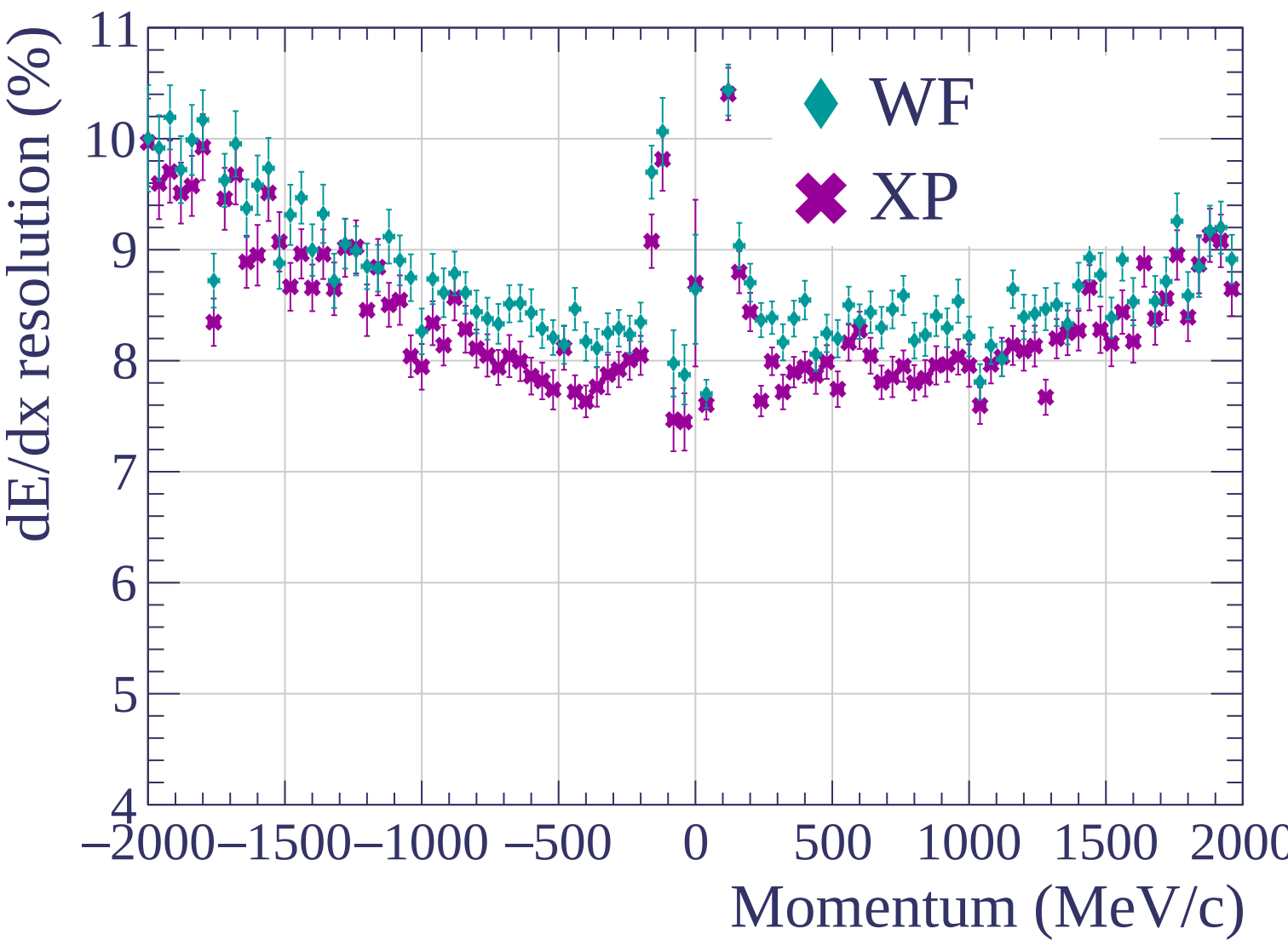


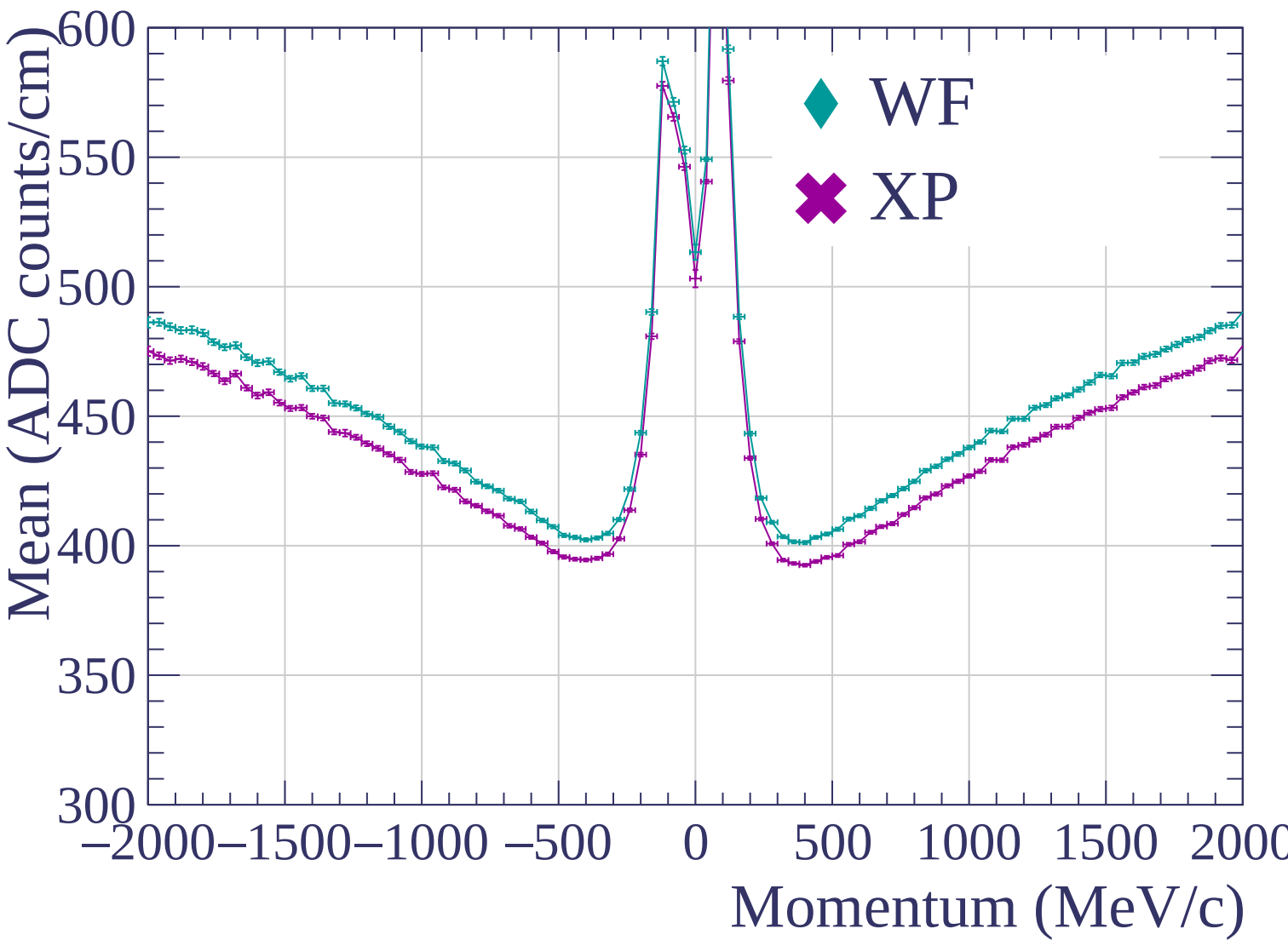


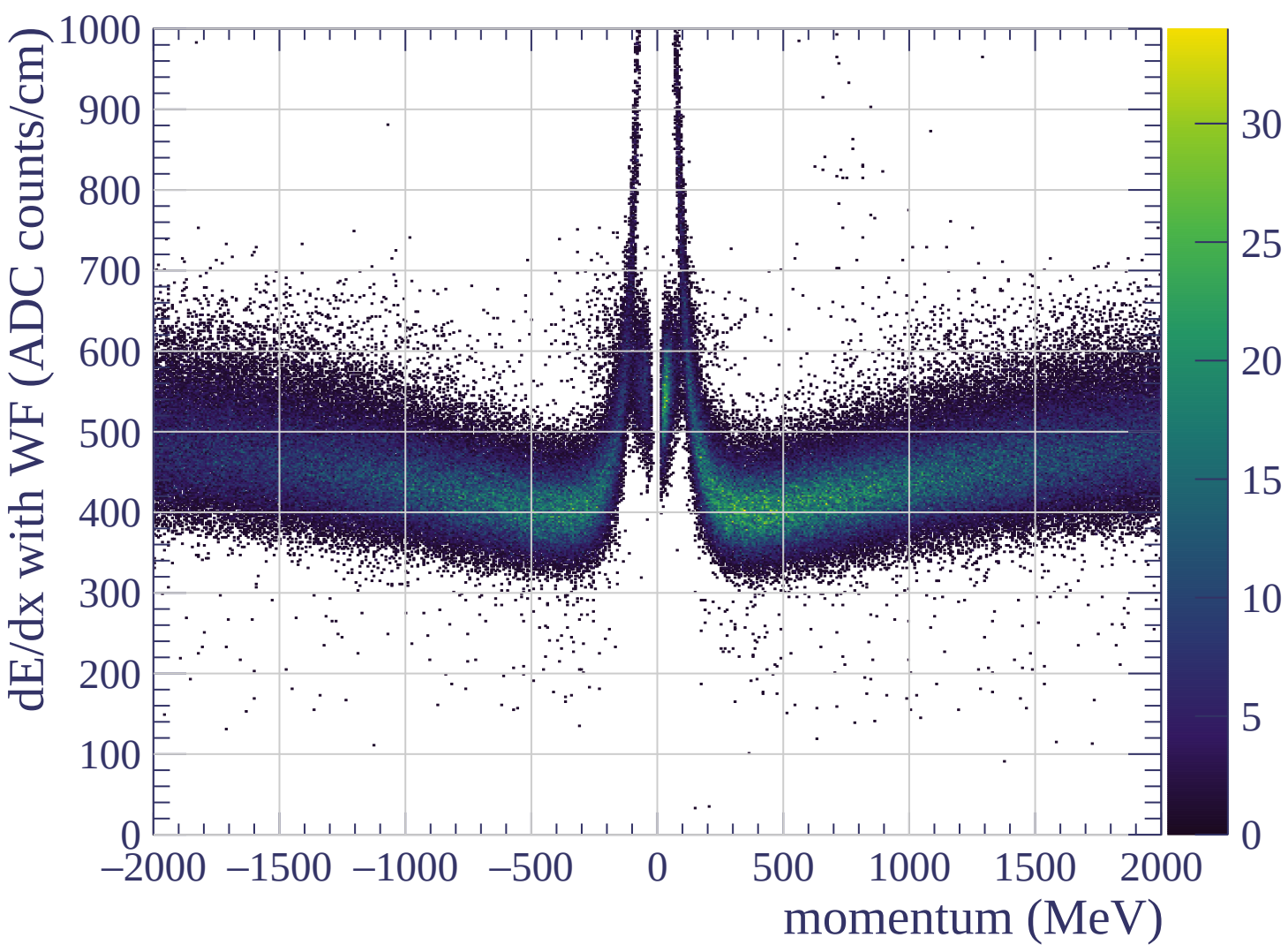


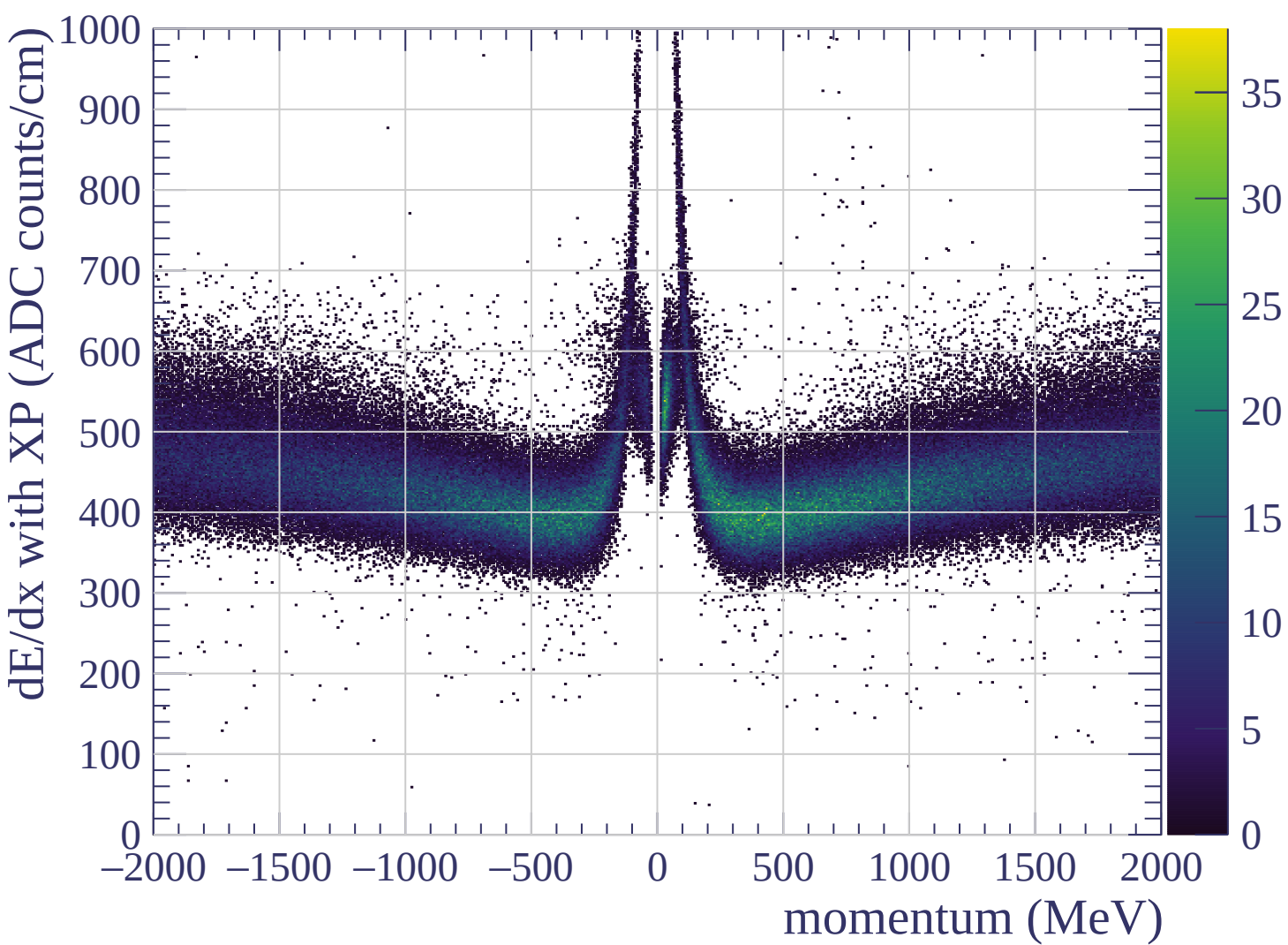


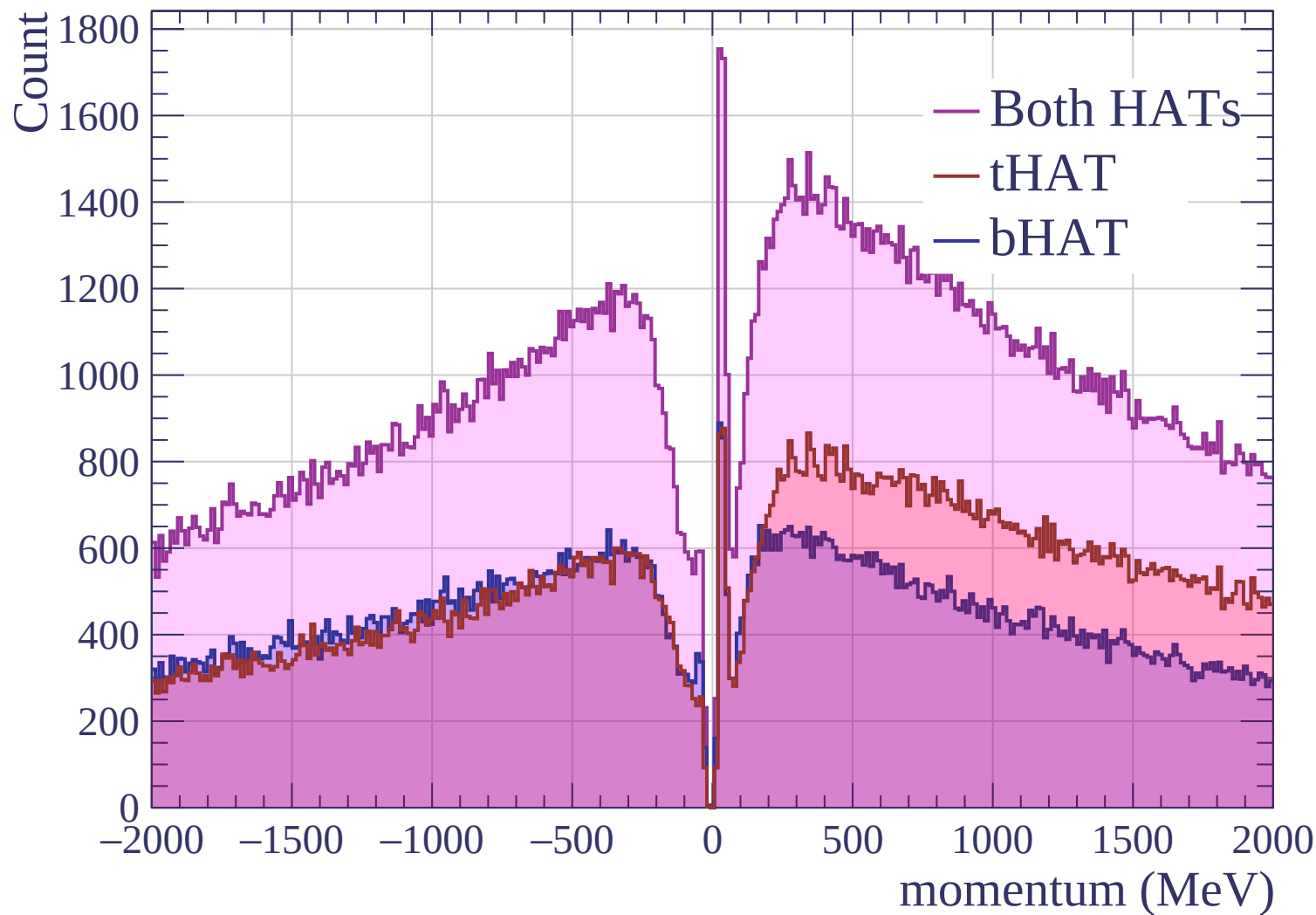


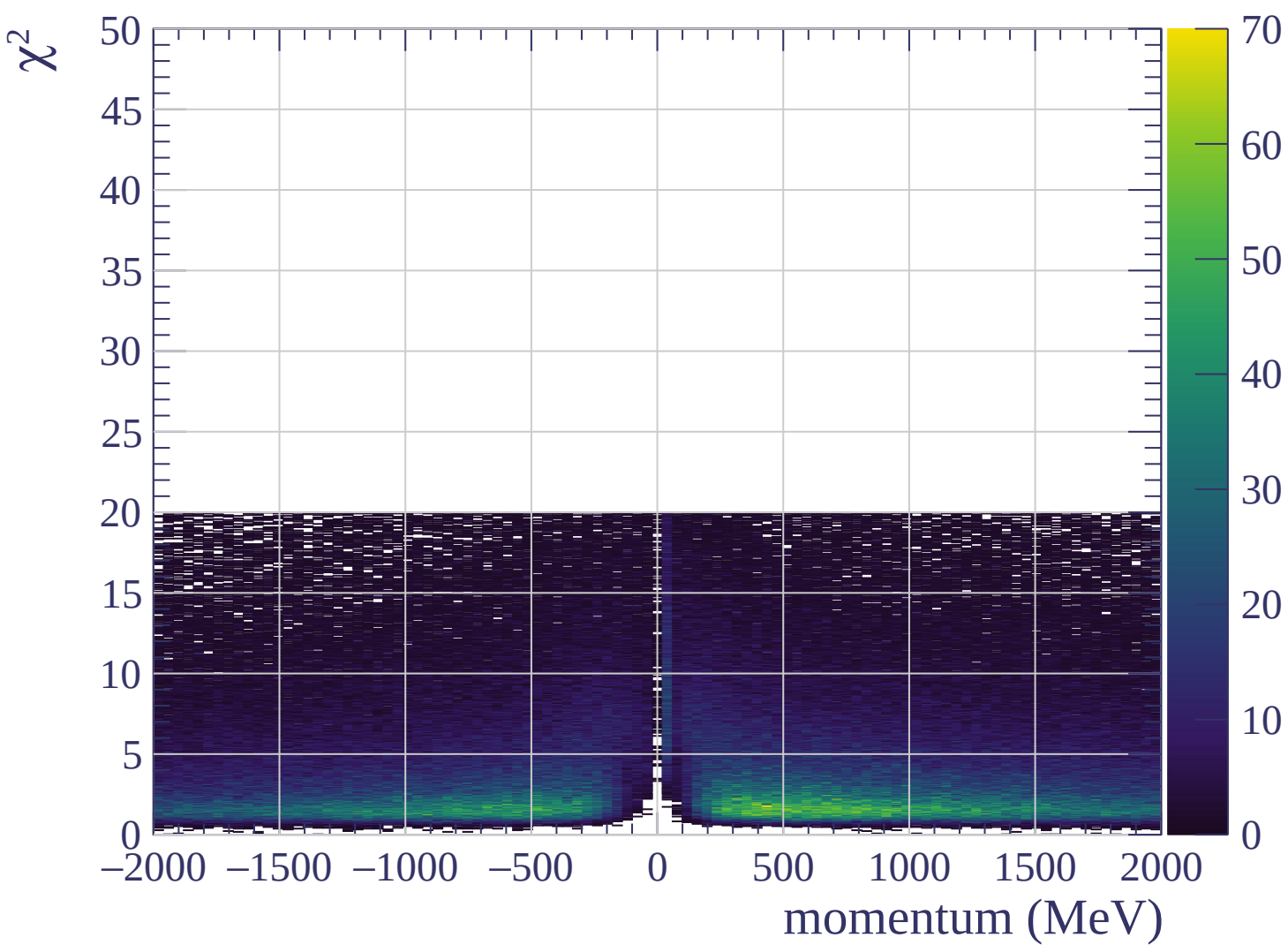


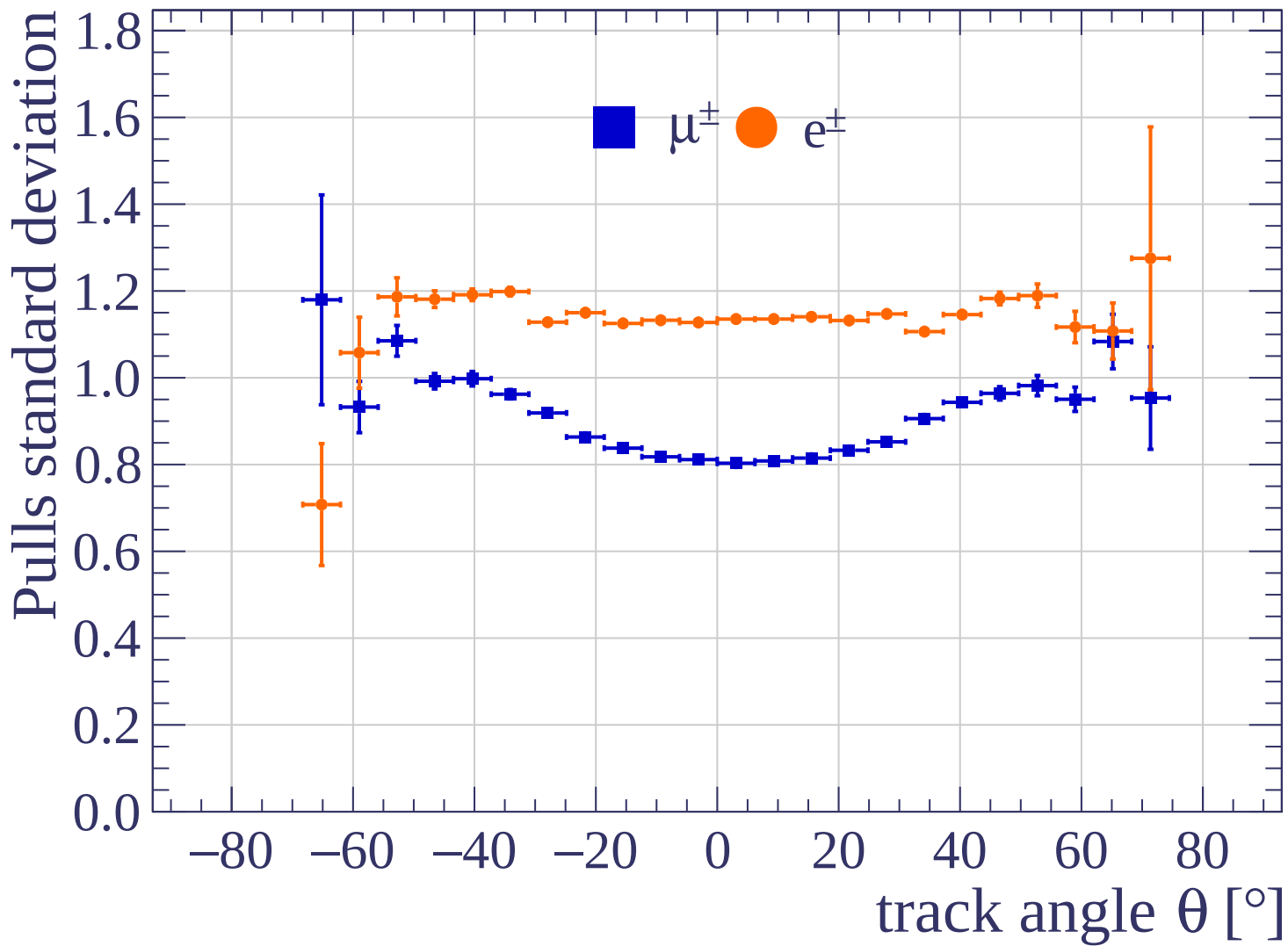


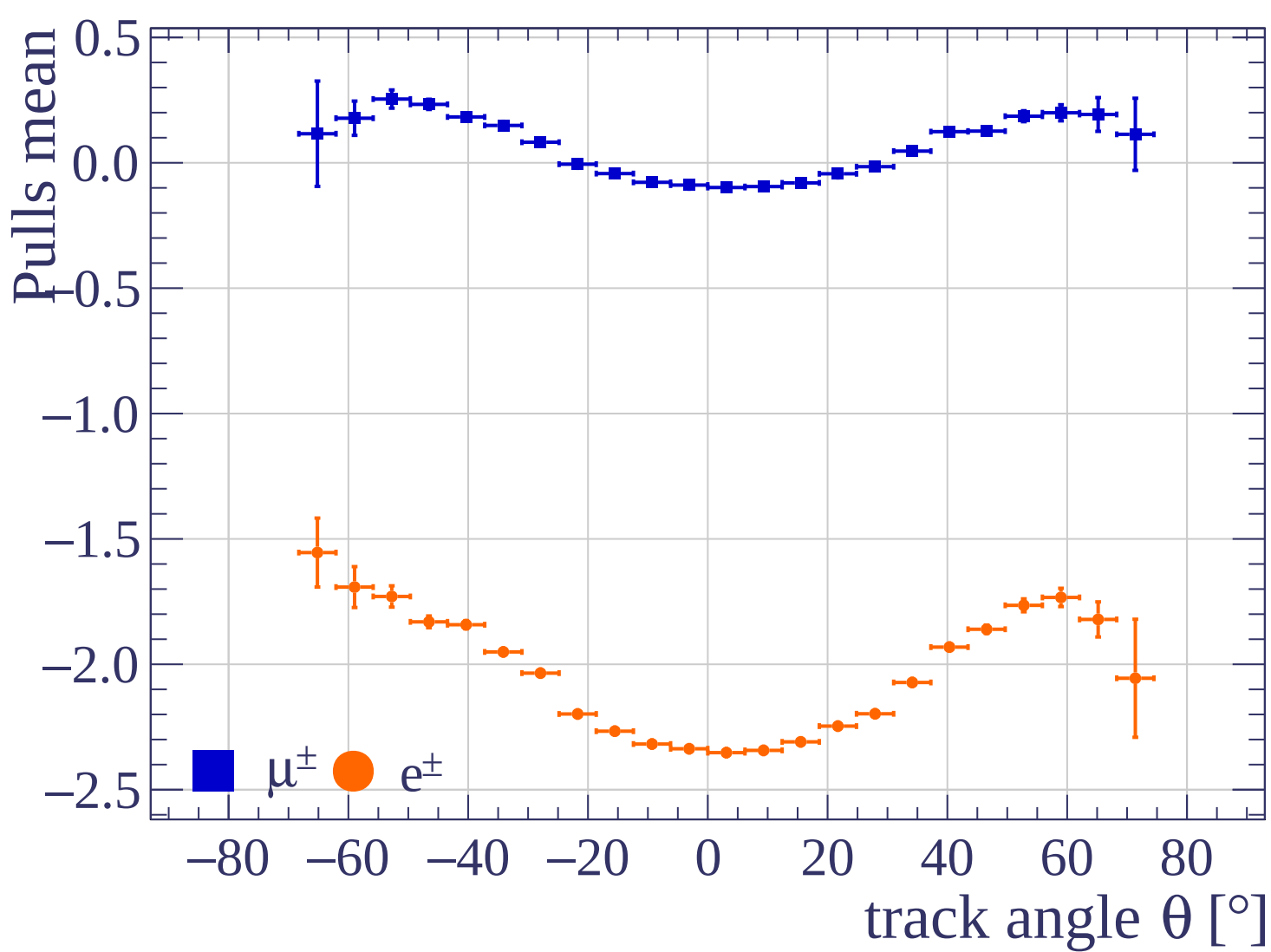


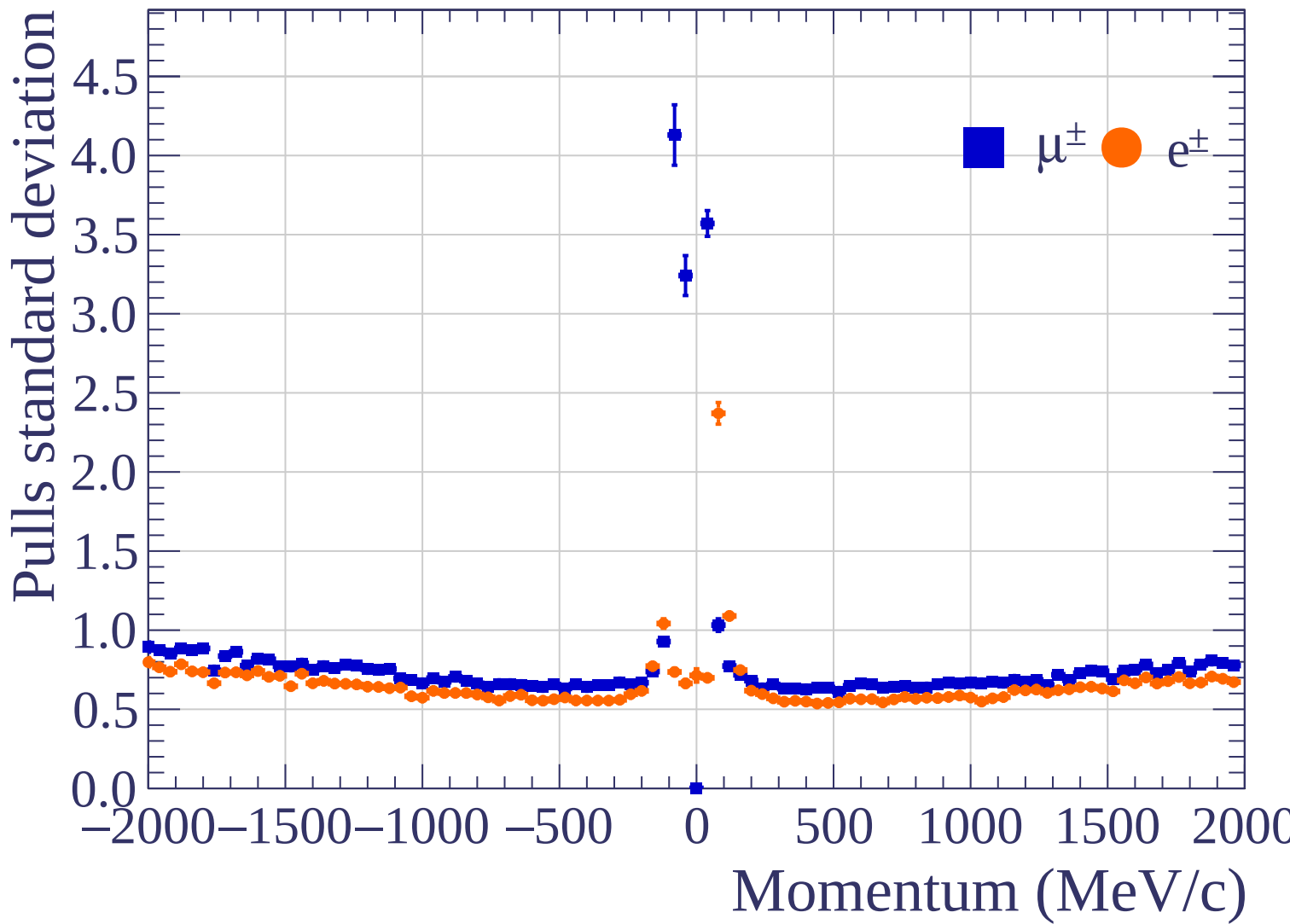


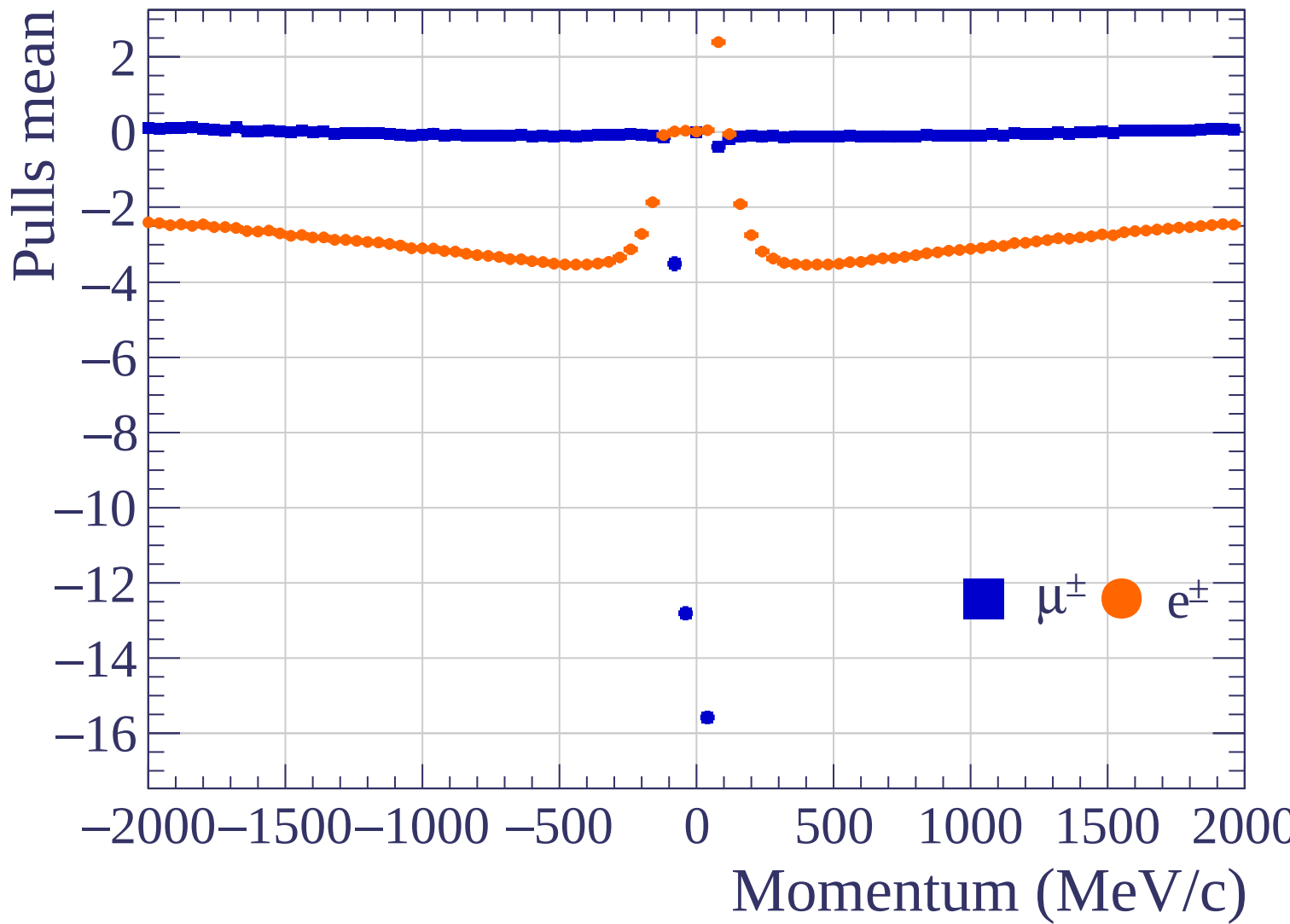


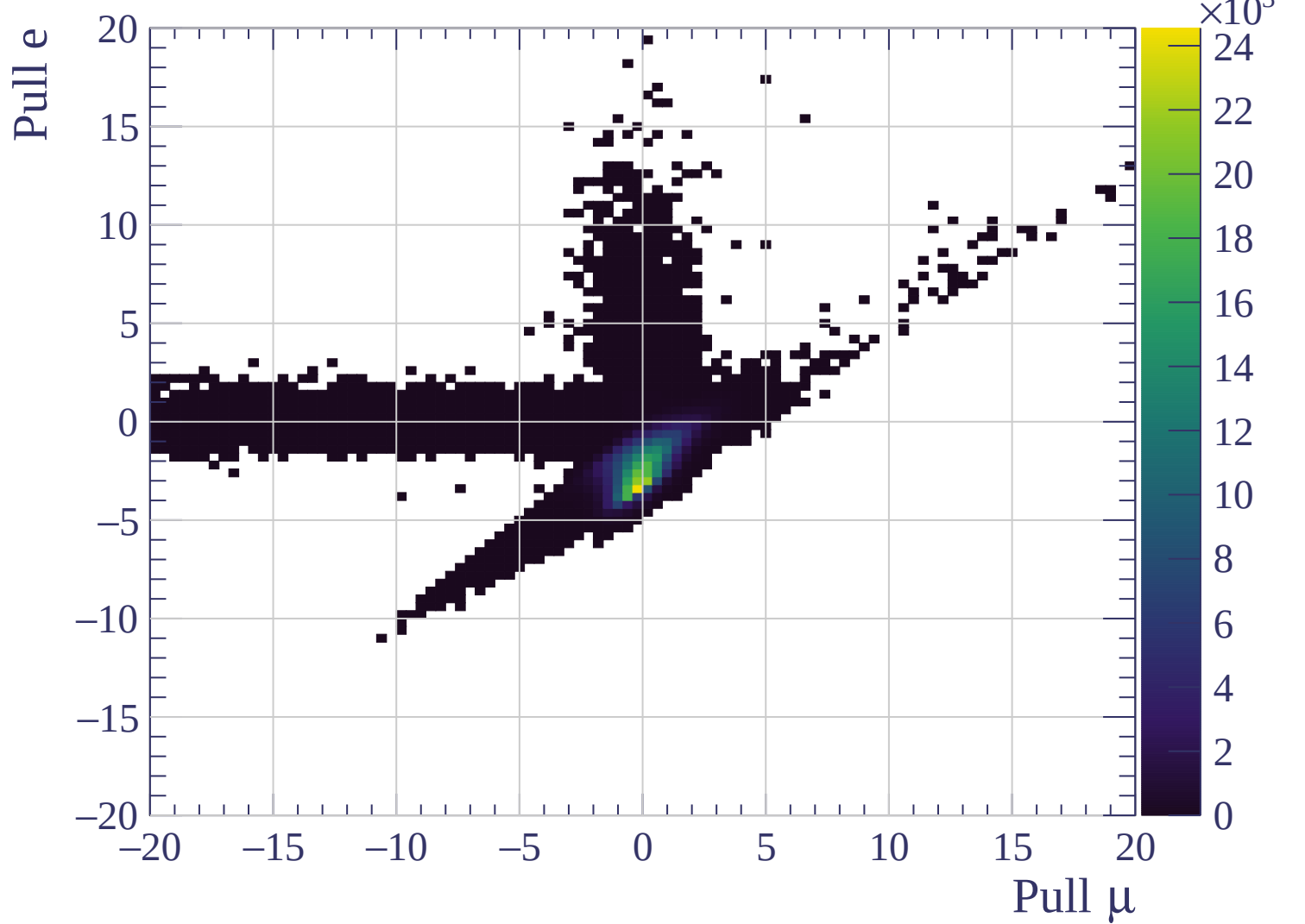


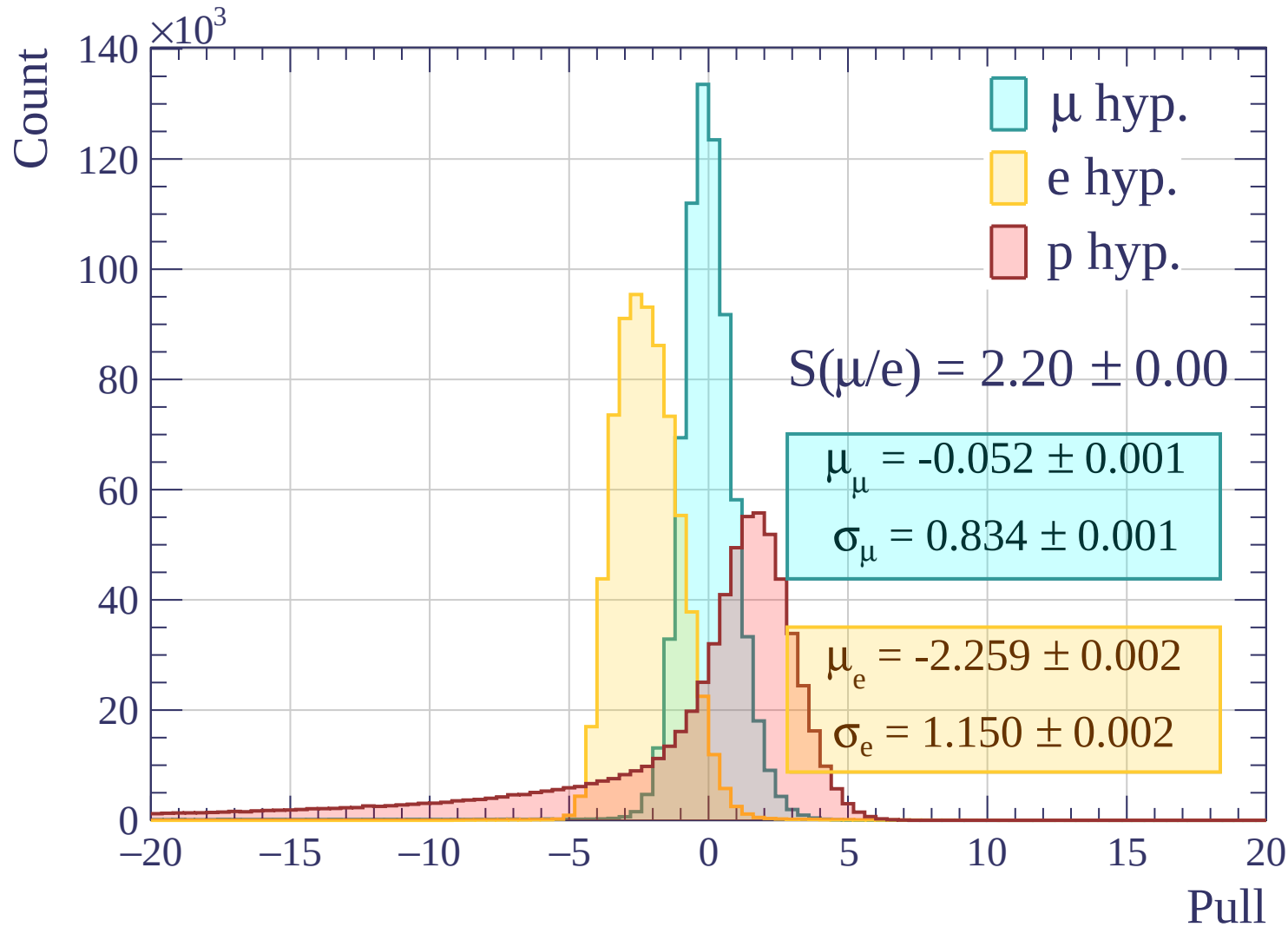


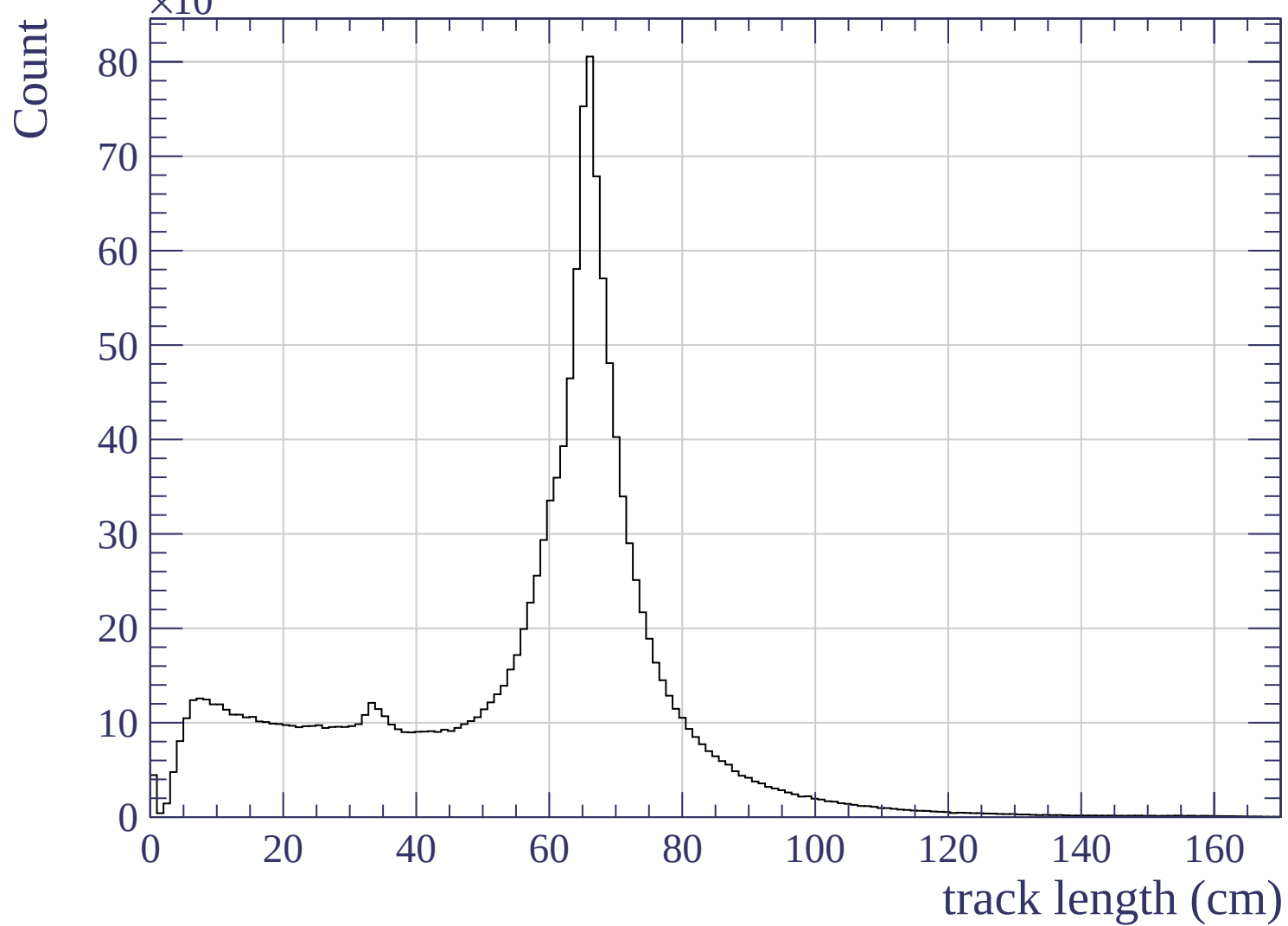


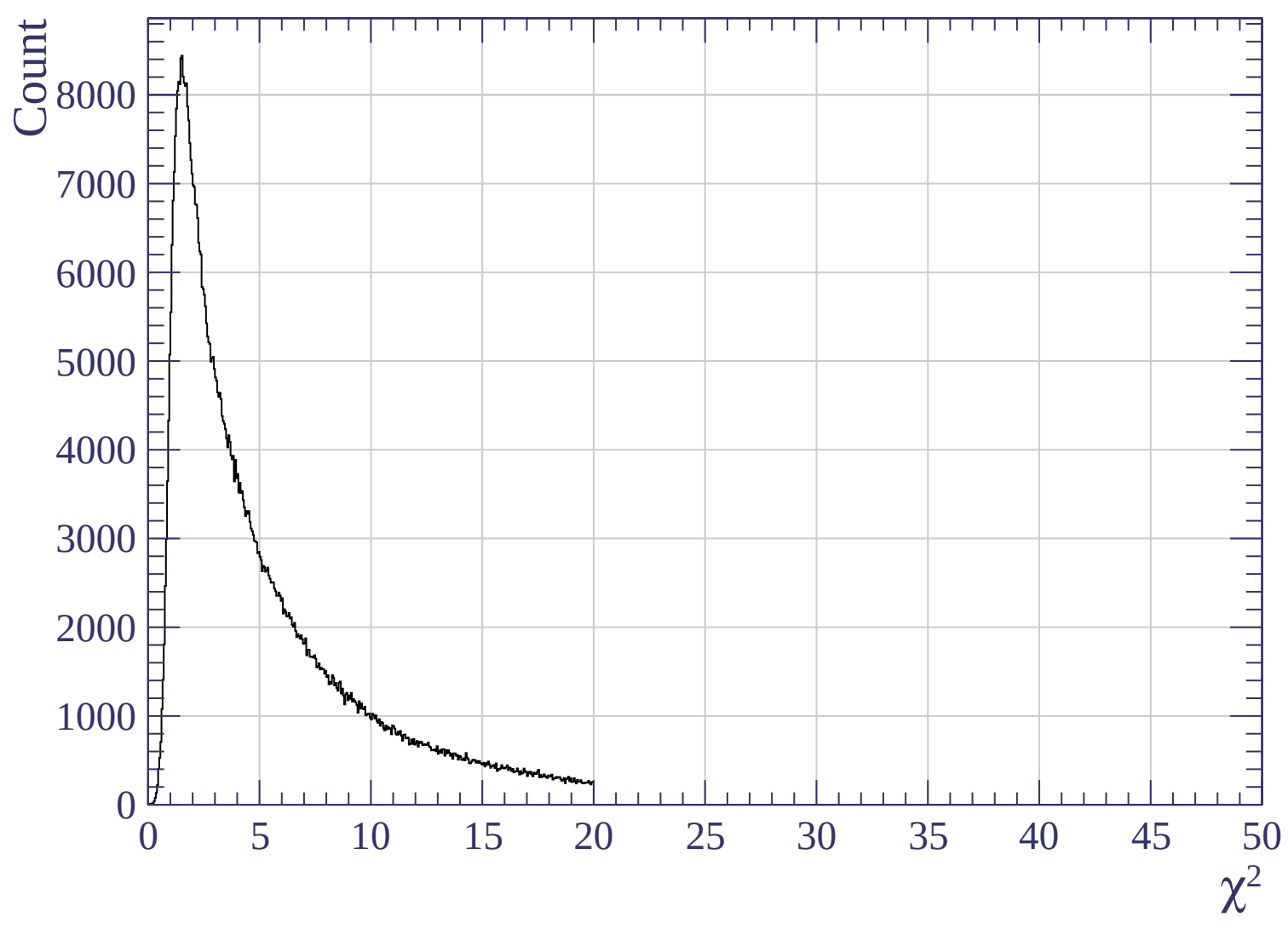


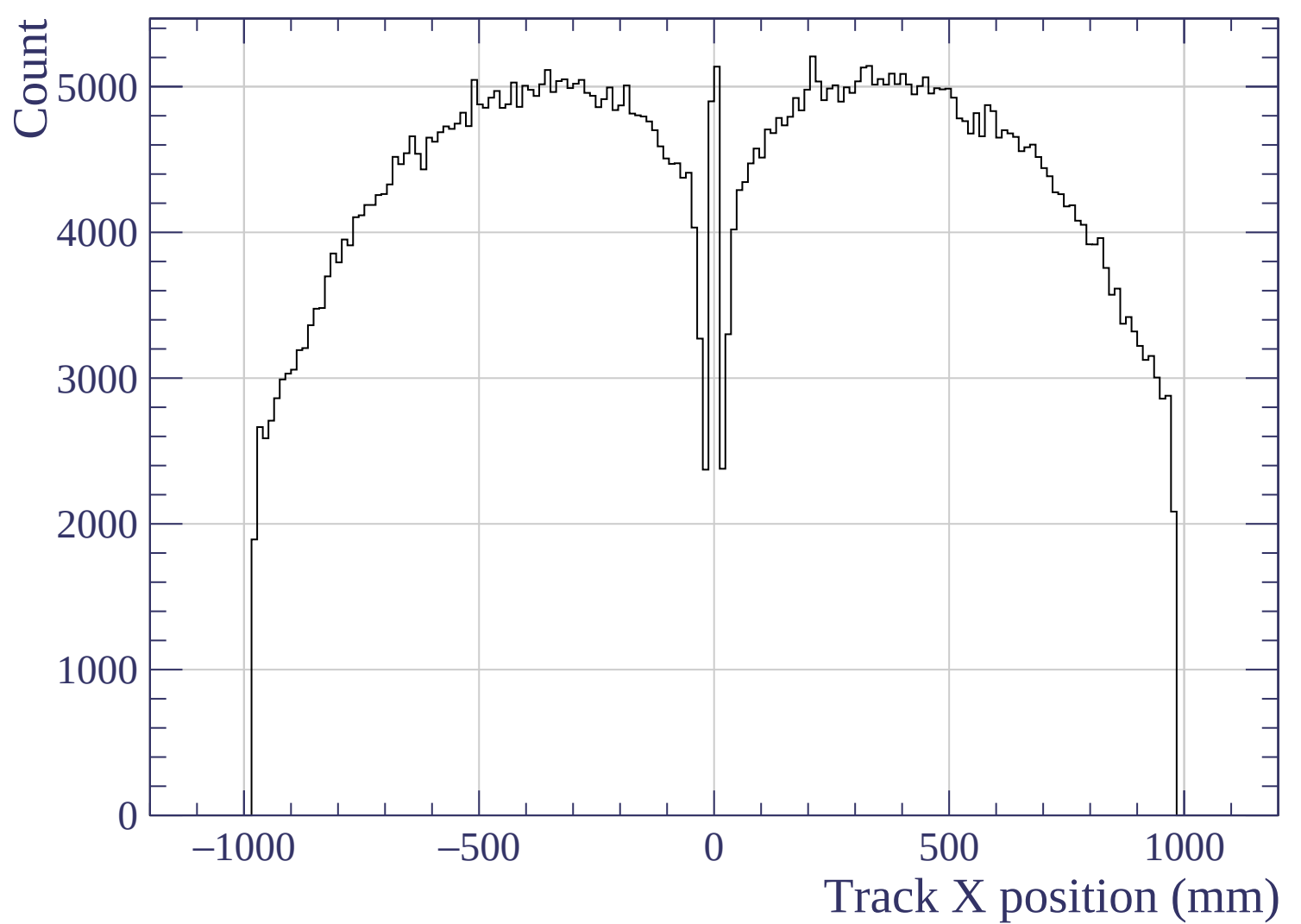


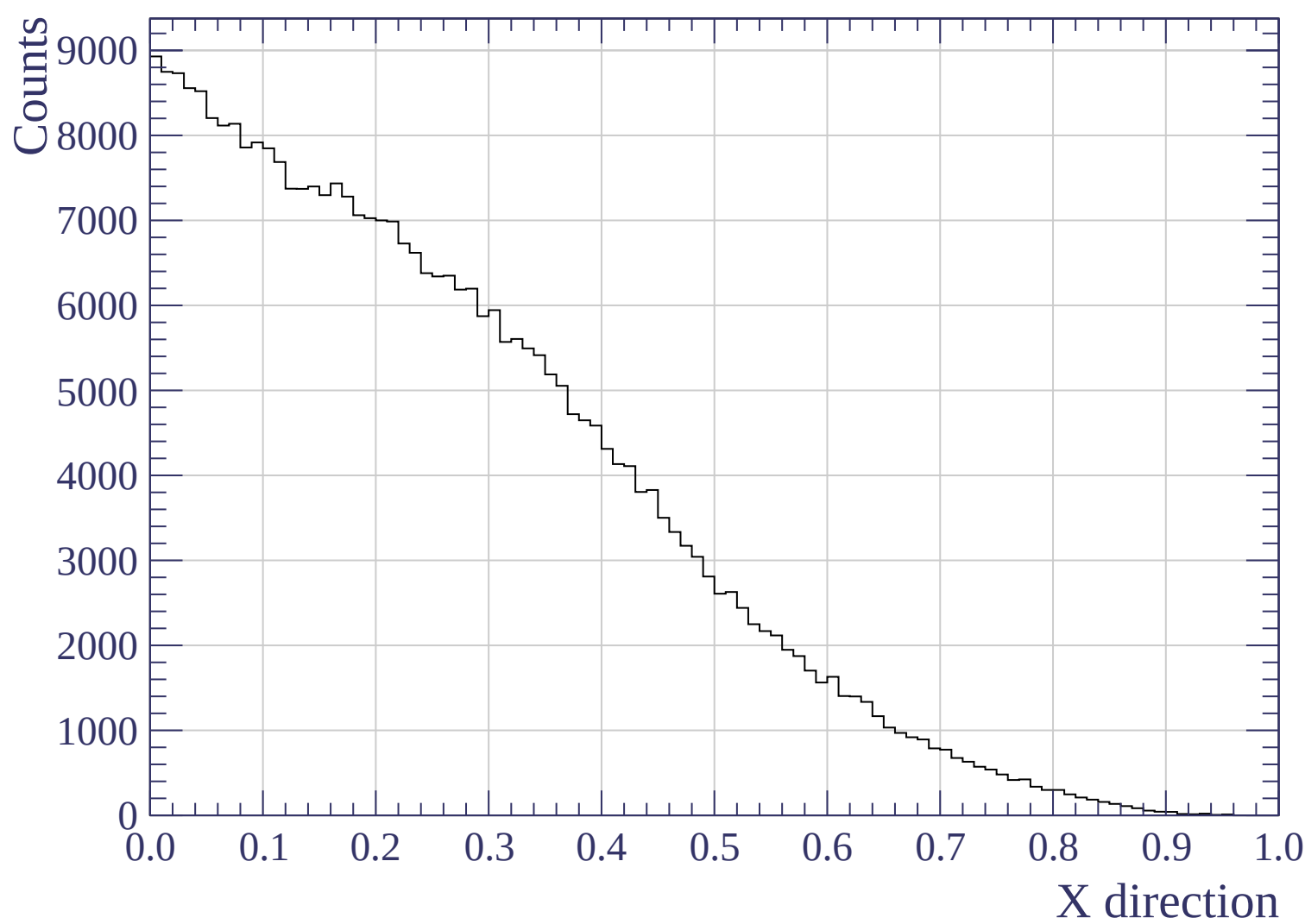


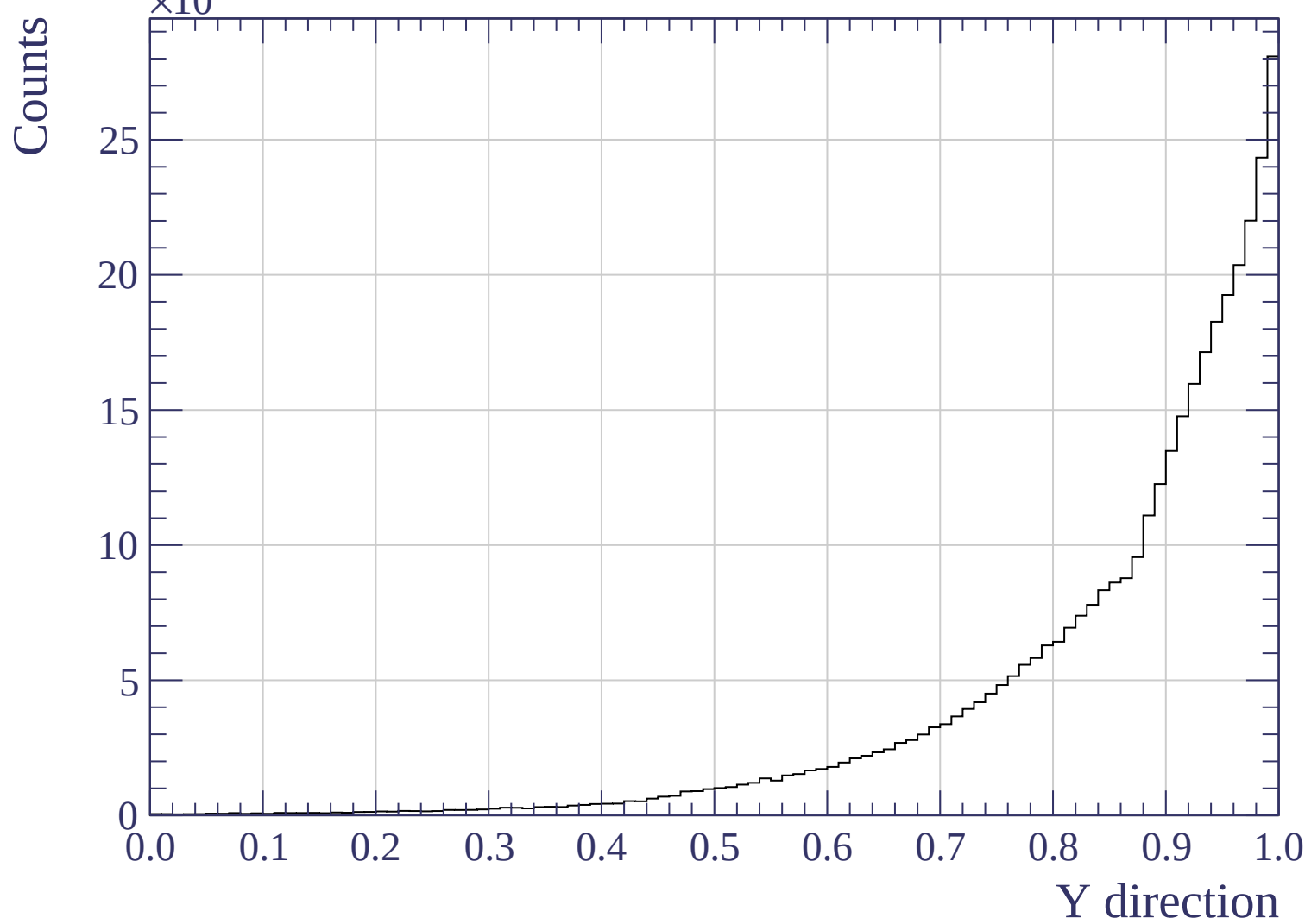


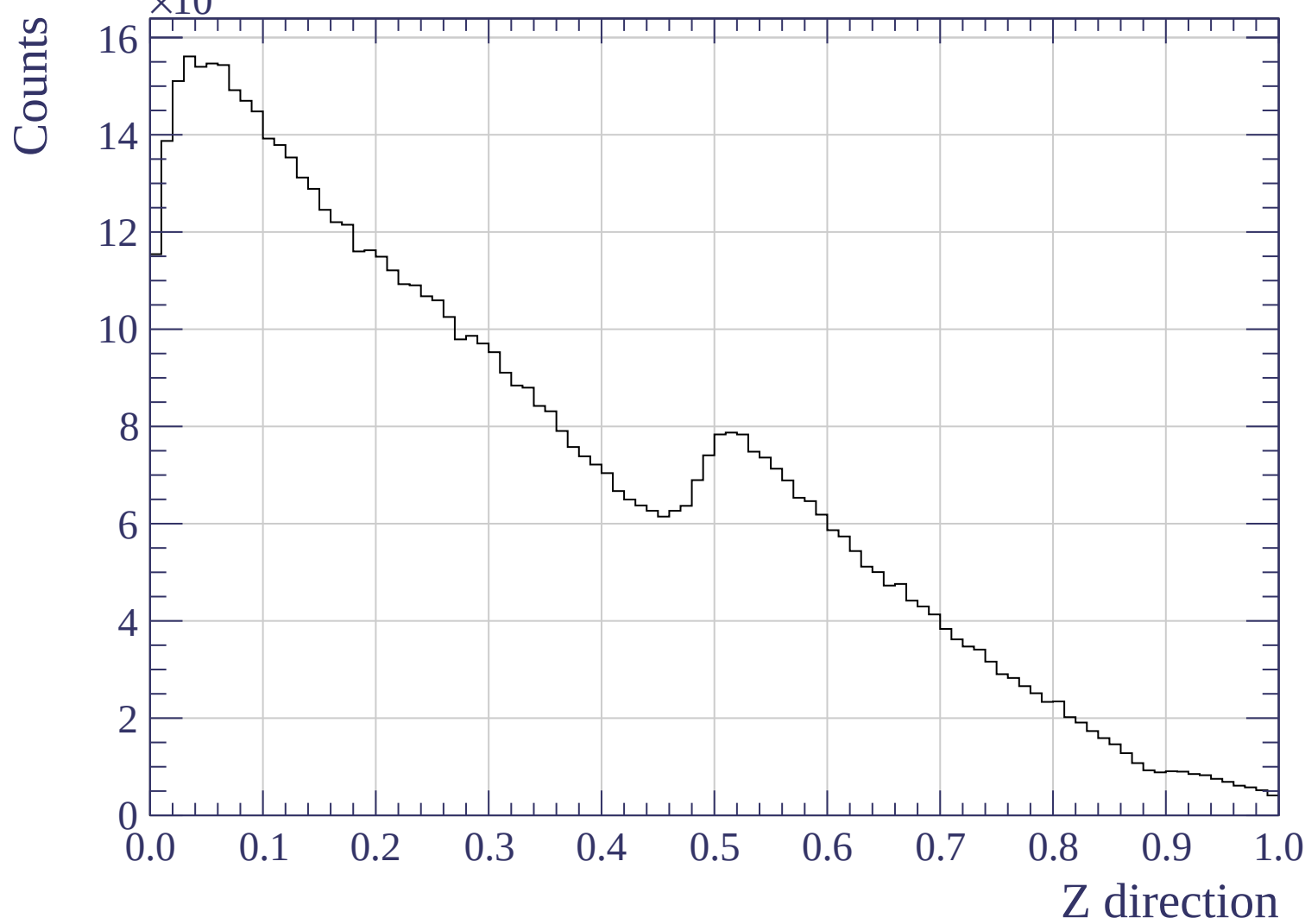


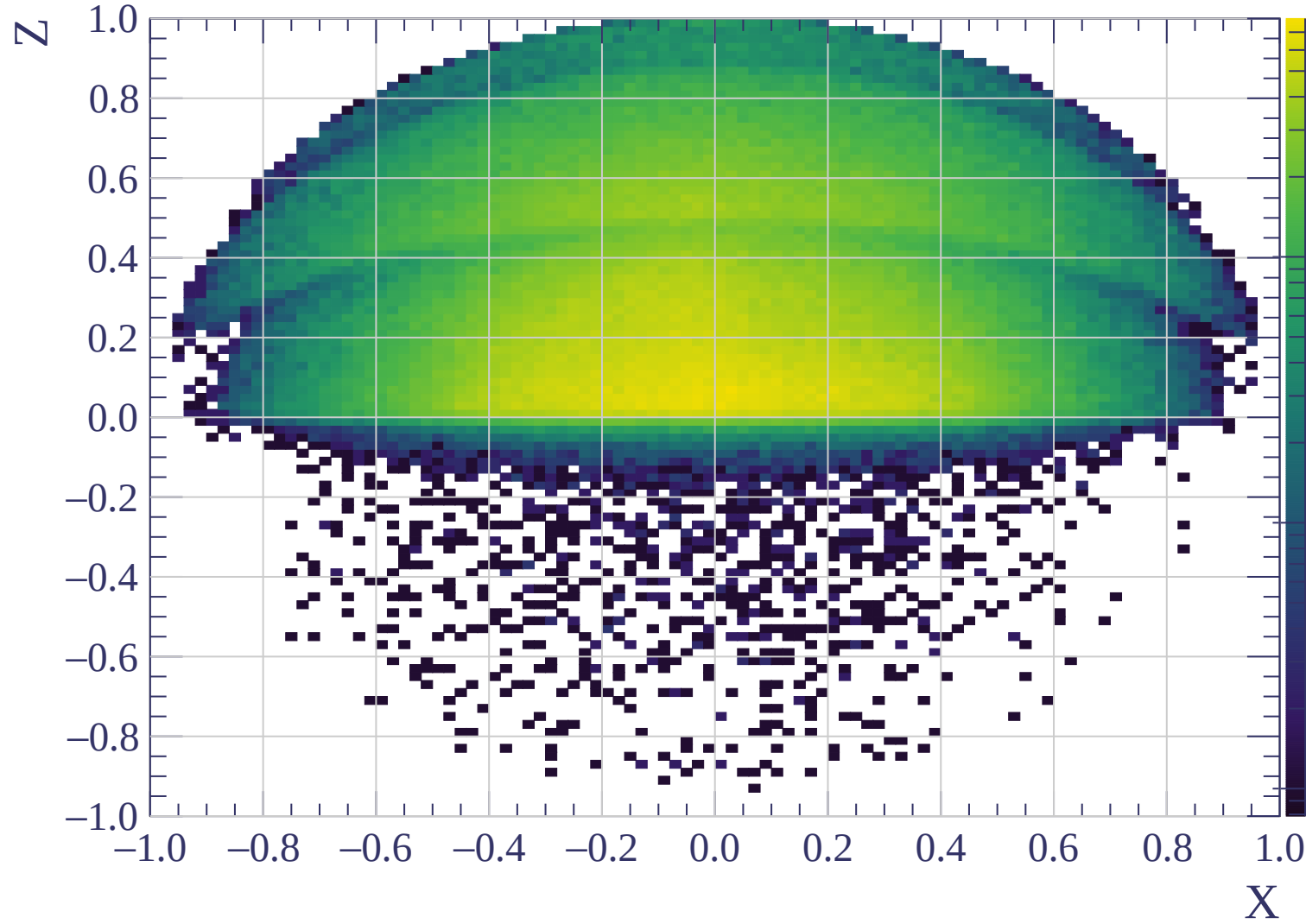


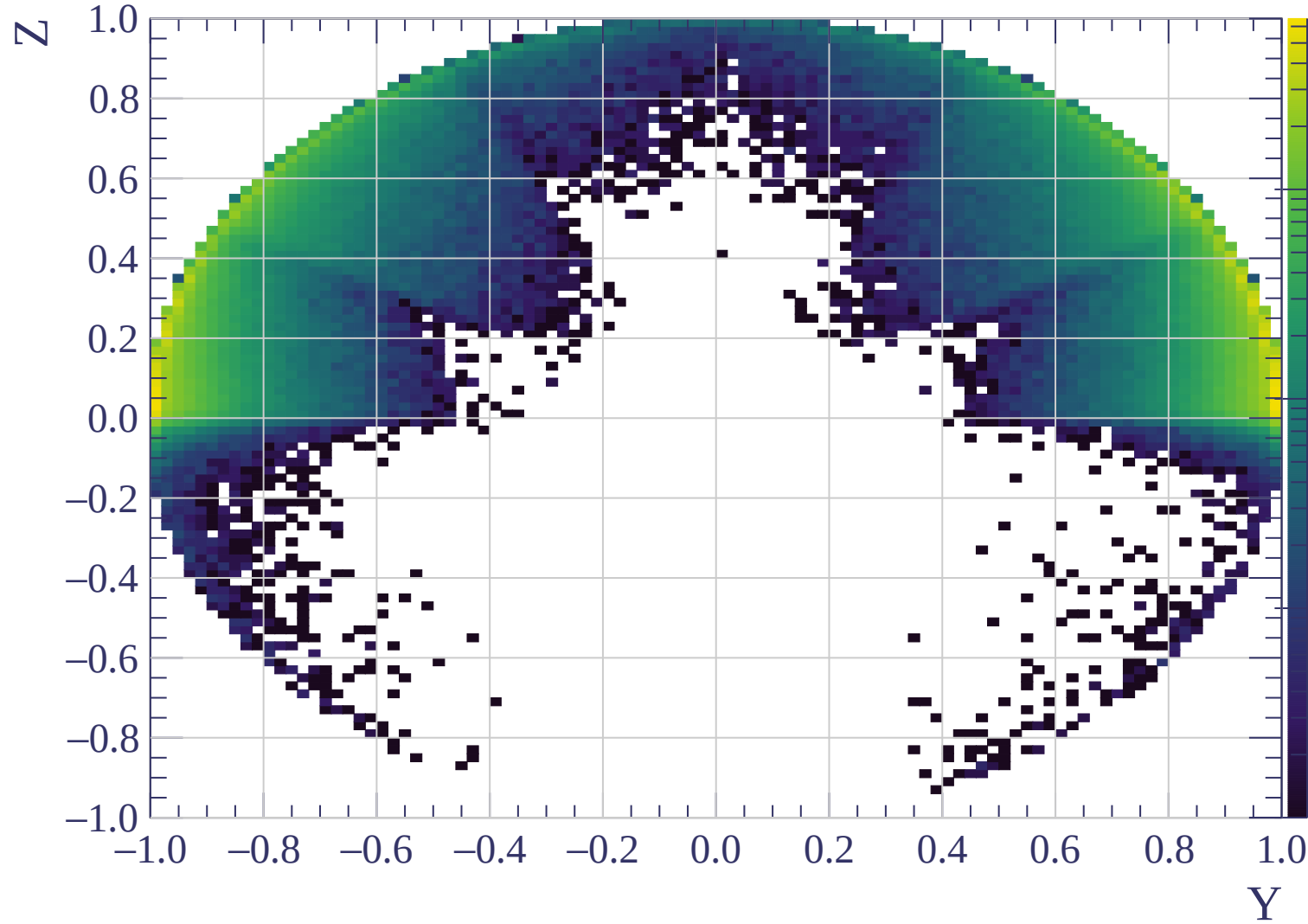


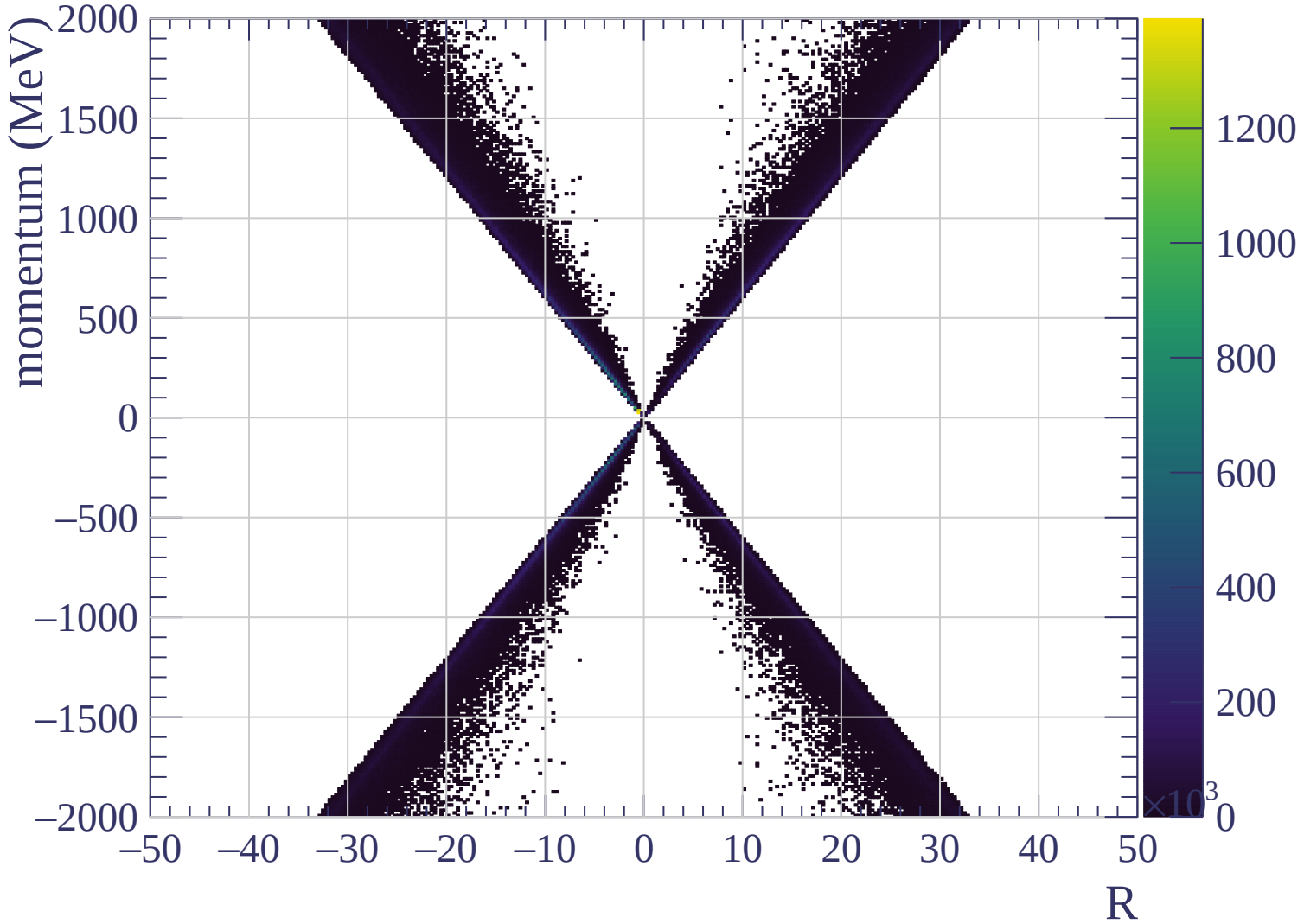


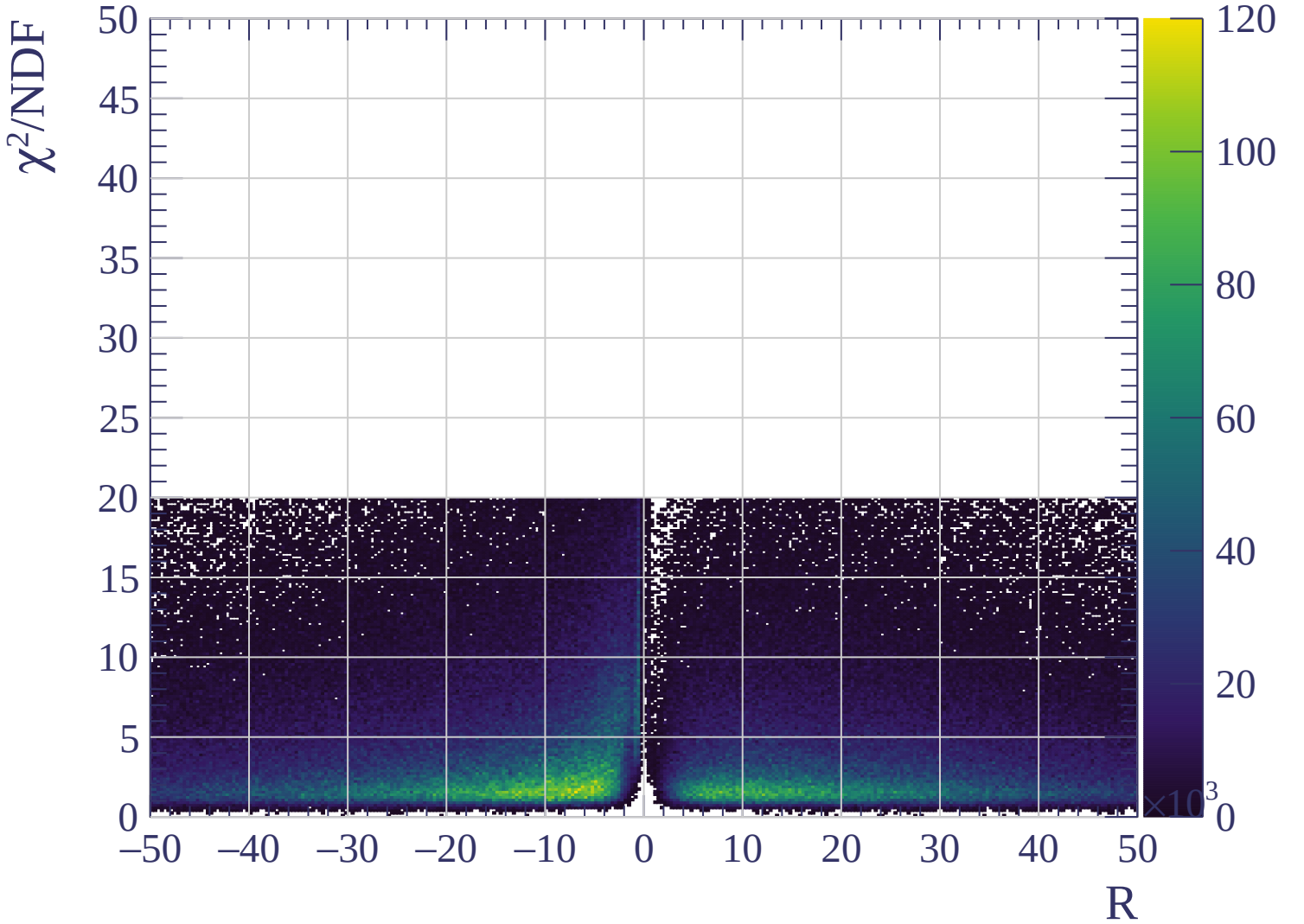




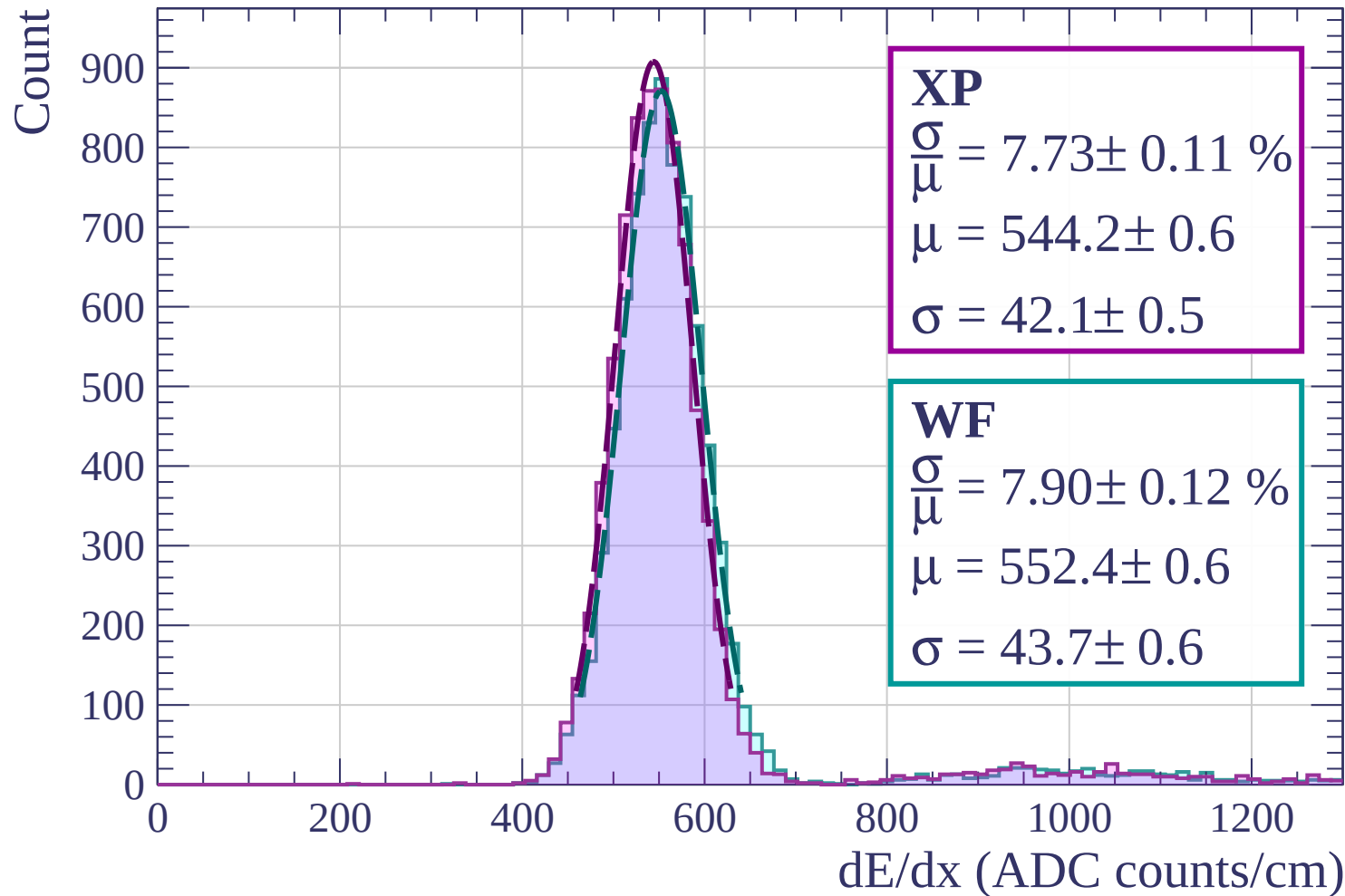






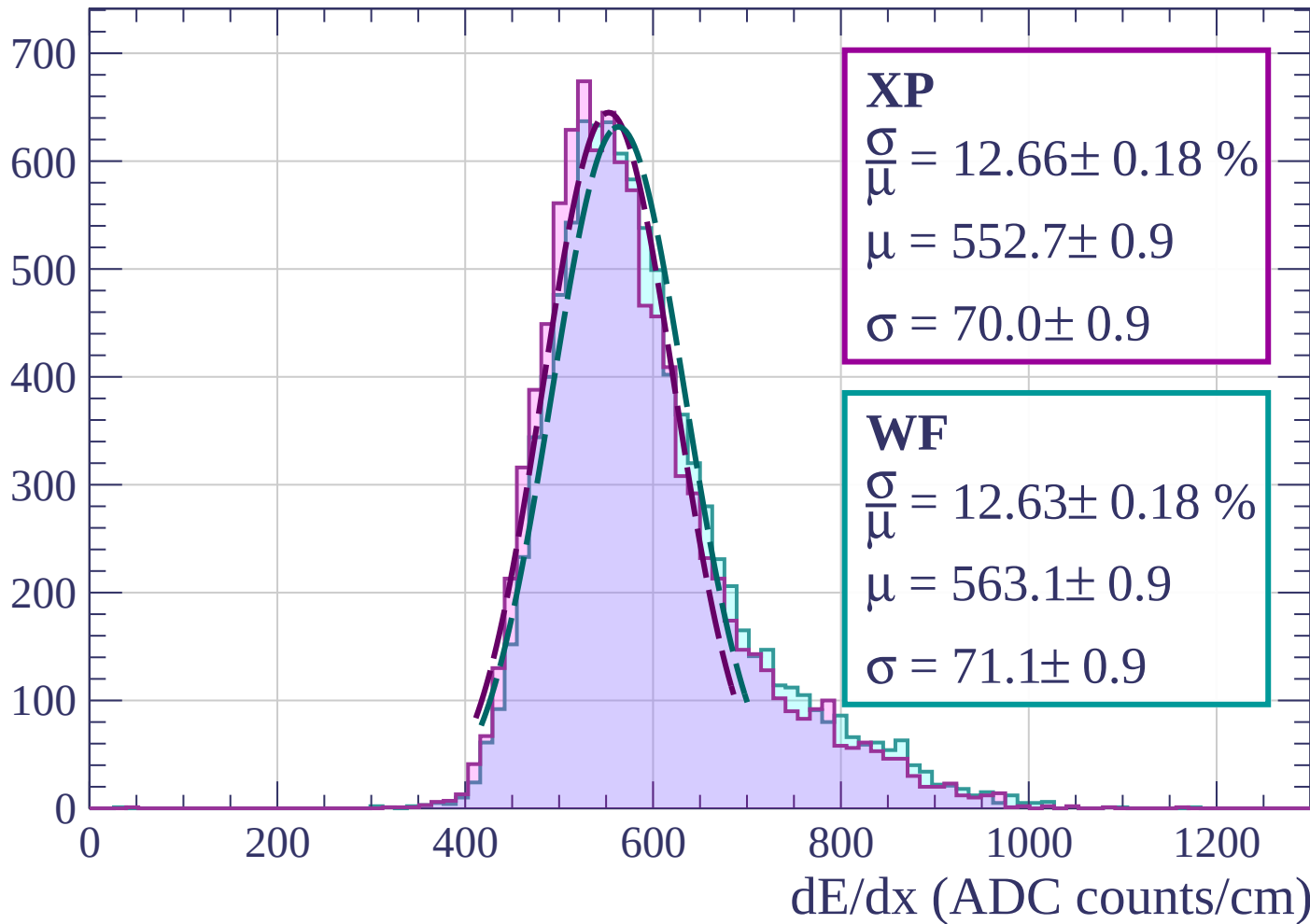


Energy loss | $0 < p < 80$

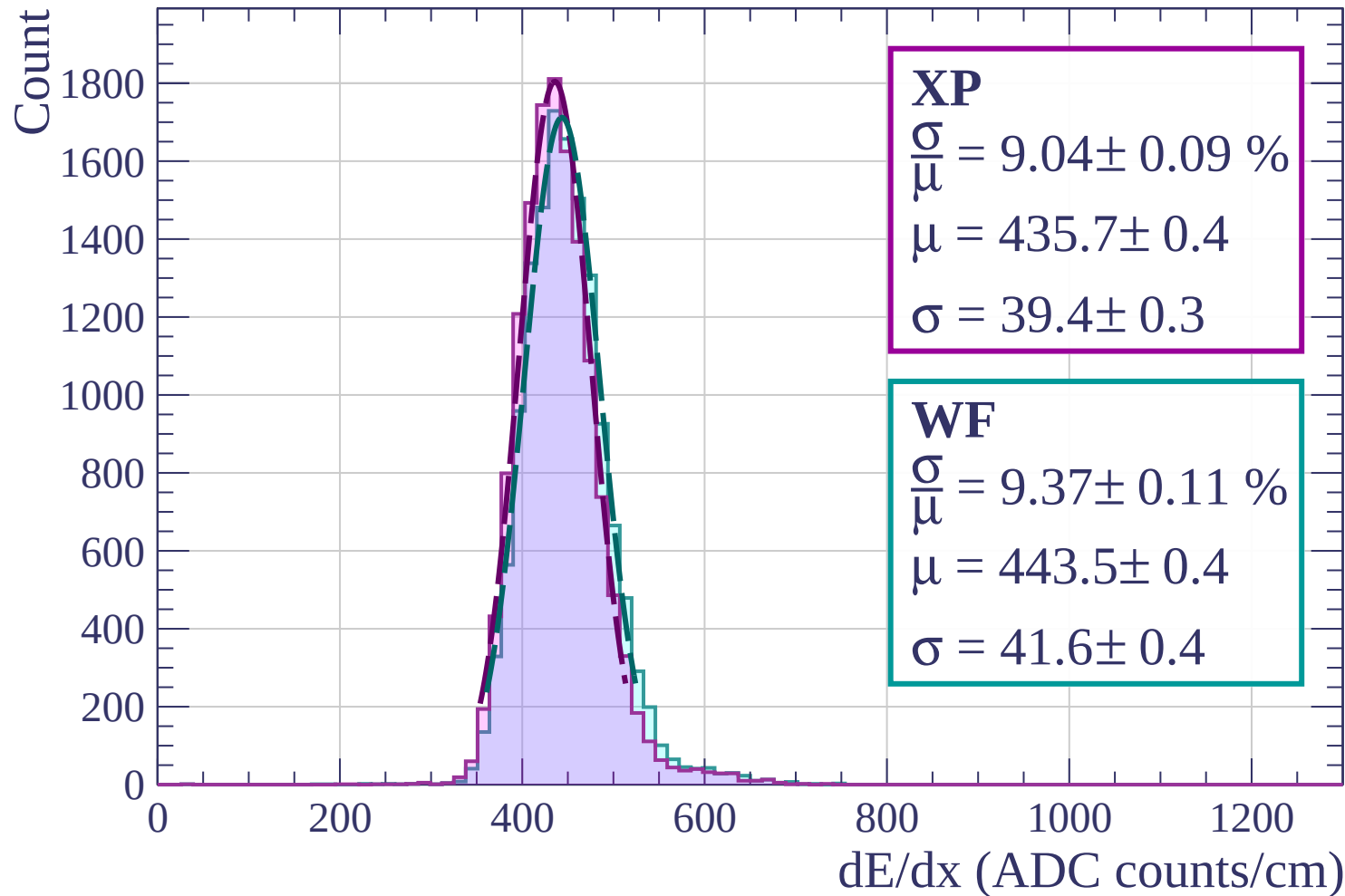


Energy loss | $80 < p < 160$

Count



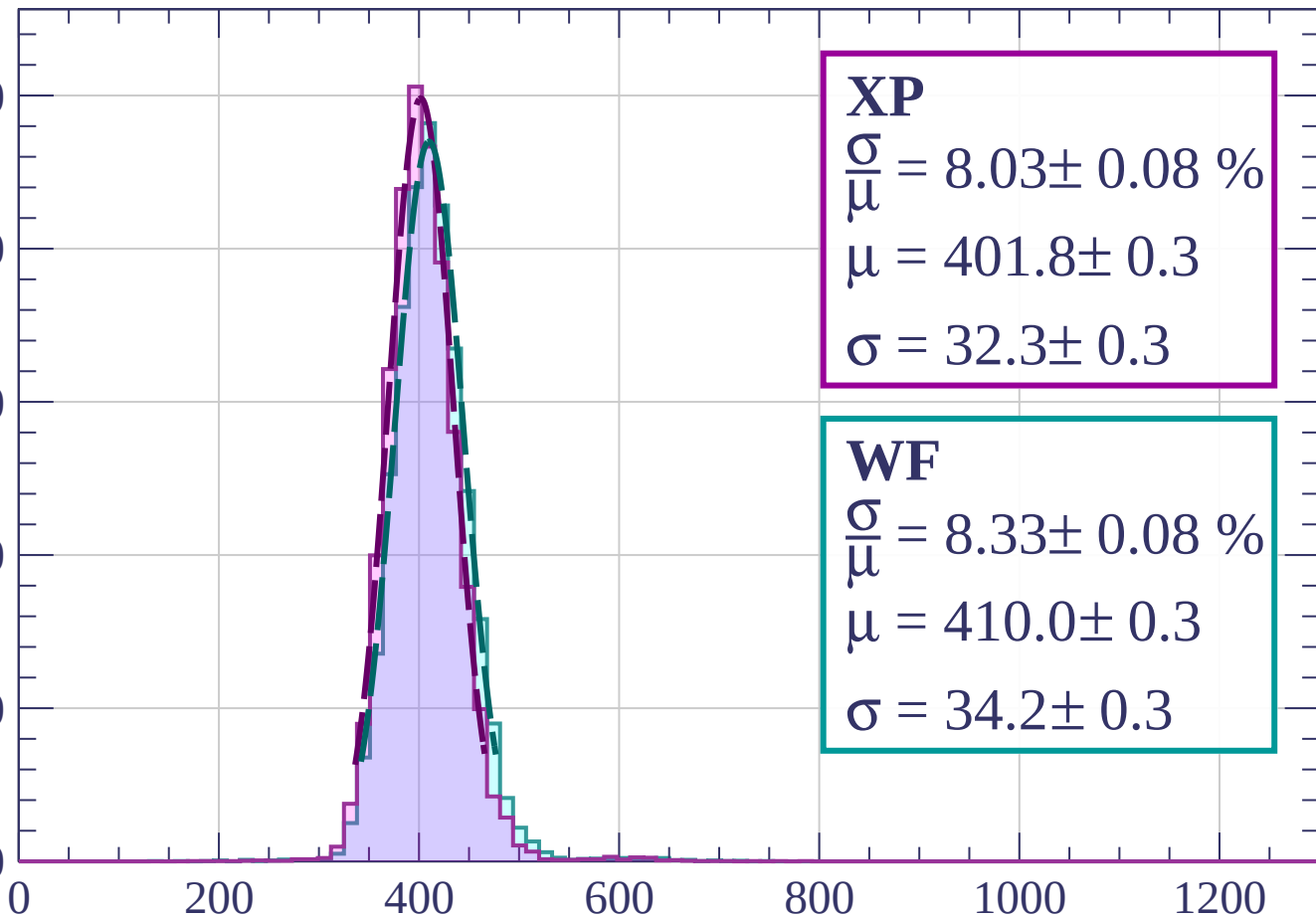
Energy loss | $160 < p < 240$



Energy loss | $240 < p < 320$

Count

2500
2000
1500
1000
500
0



XP

$$\frac{\sigma}{\mu} = 8.03 \pm 0.08 \%$$

$$\mu = 401.8 \pm 0.3$$

$$\sigma = 32.3 \pm 0.3$$

WF

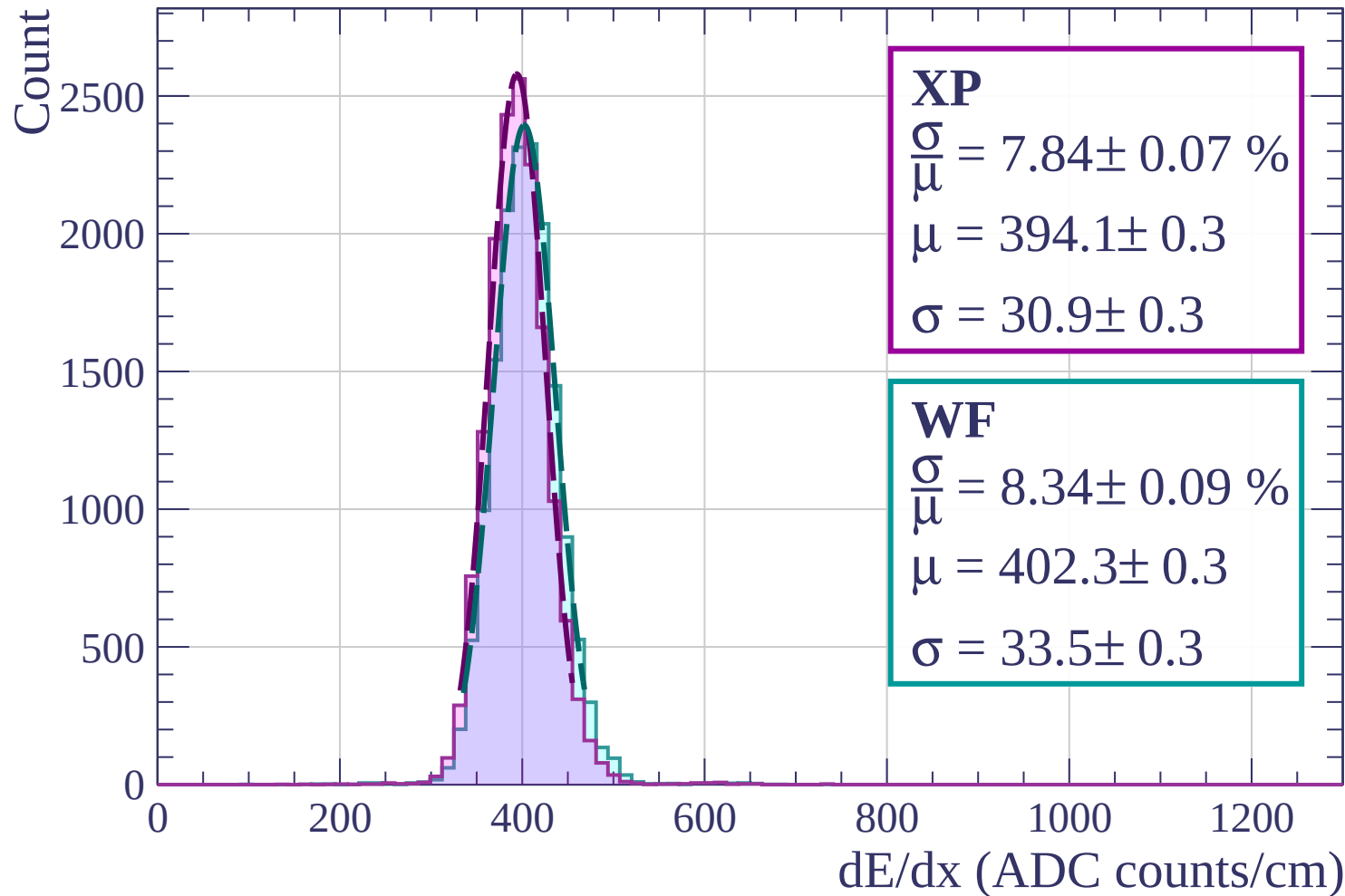
$$\frac{\sigma}{\mu} = 8.33 \pm 0.08 \%$$

$$\mu = 410.0 \pm 0.3$$

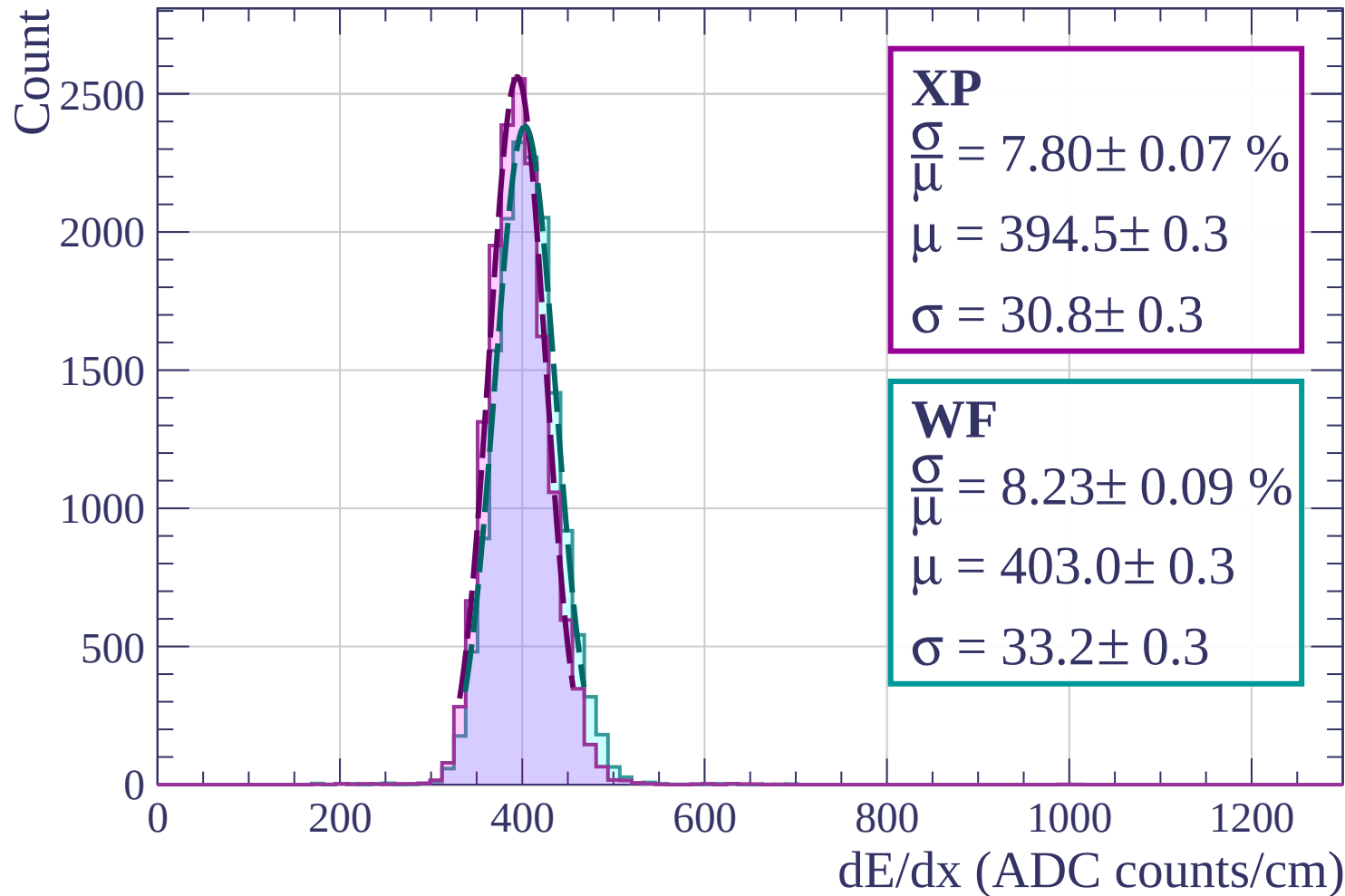
$$\sigma = 34.2 \pm 0.3$$

dE/dx (ADC counts/cm)

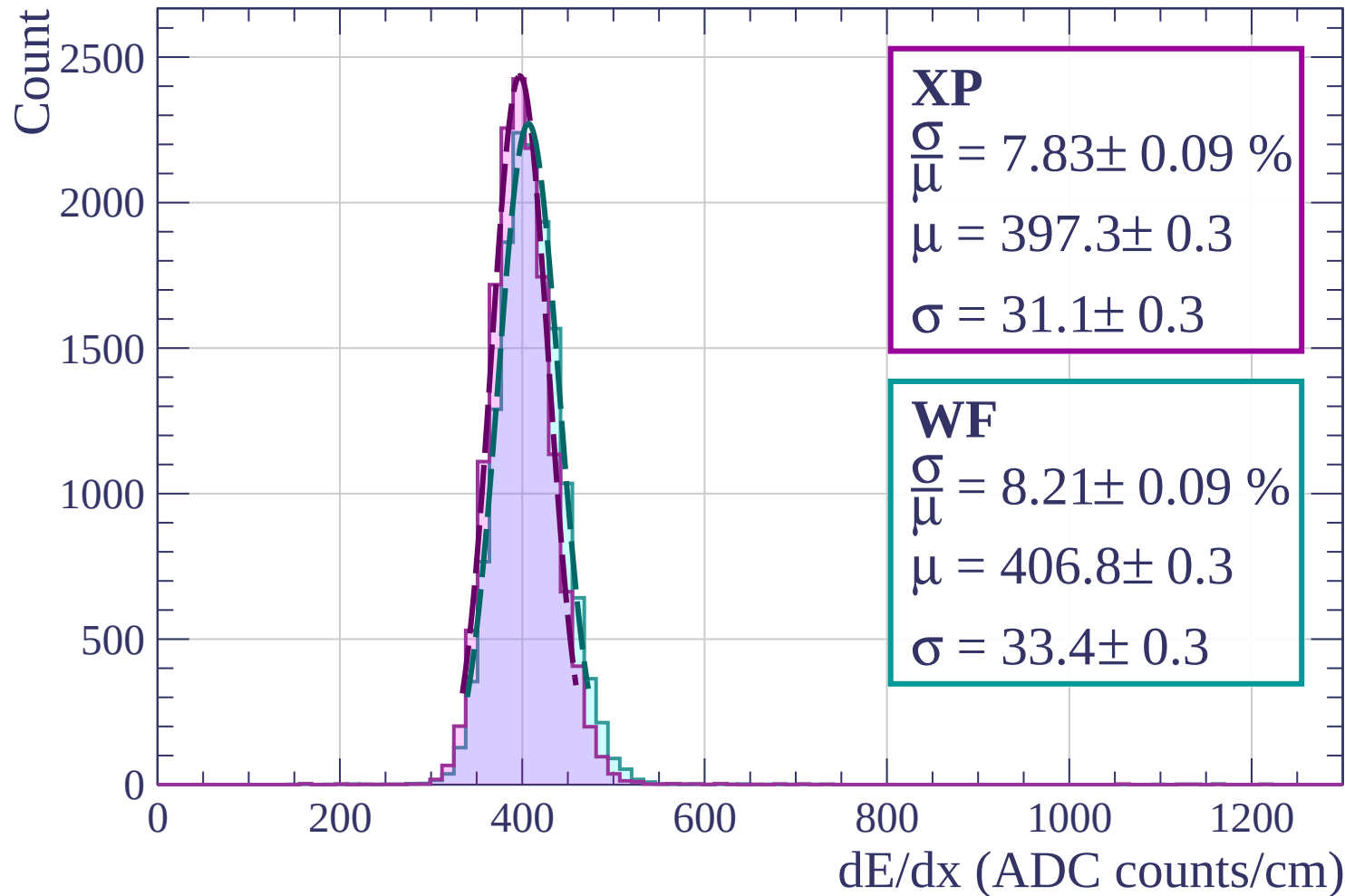
Energy loss | $320 < p < 400$



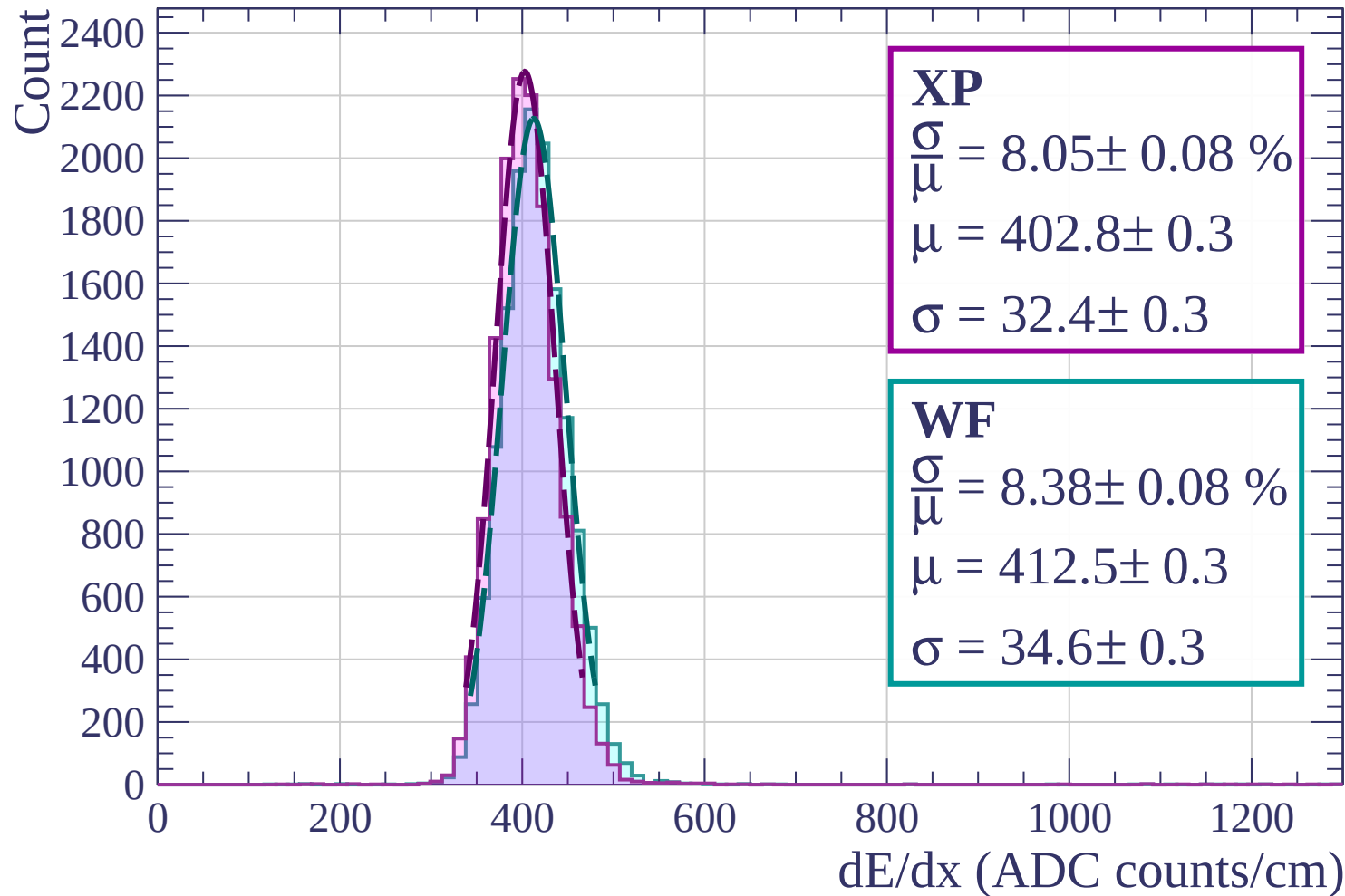
Energy loss | $400 < p < 480$



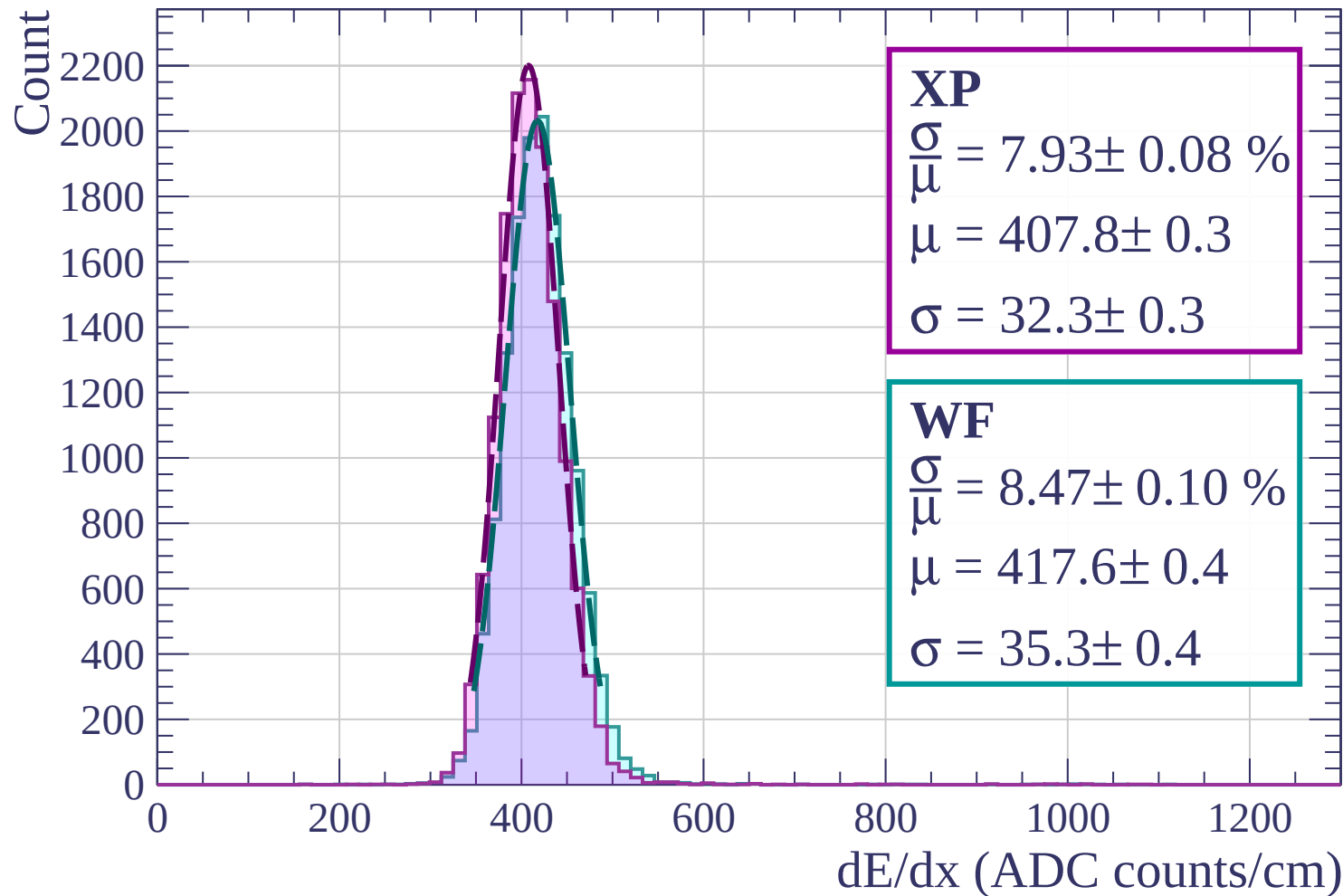
Energy loss | $480 < p < 560$



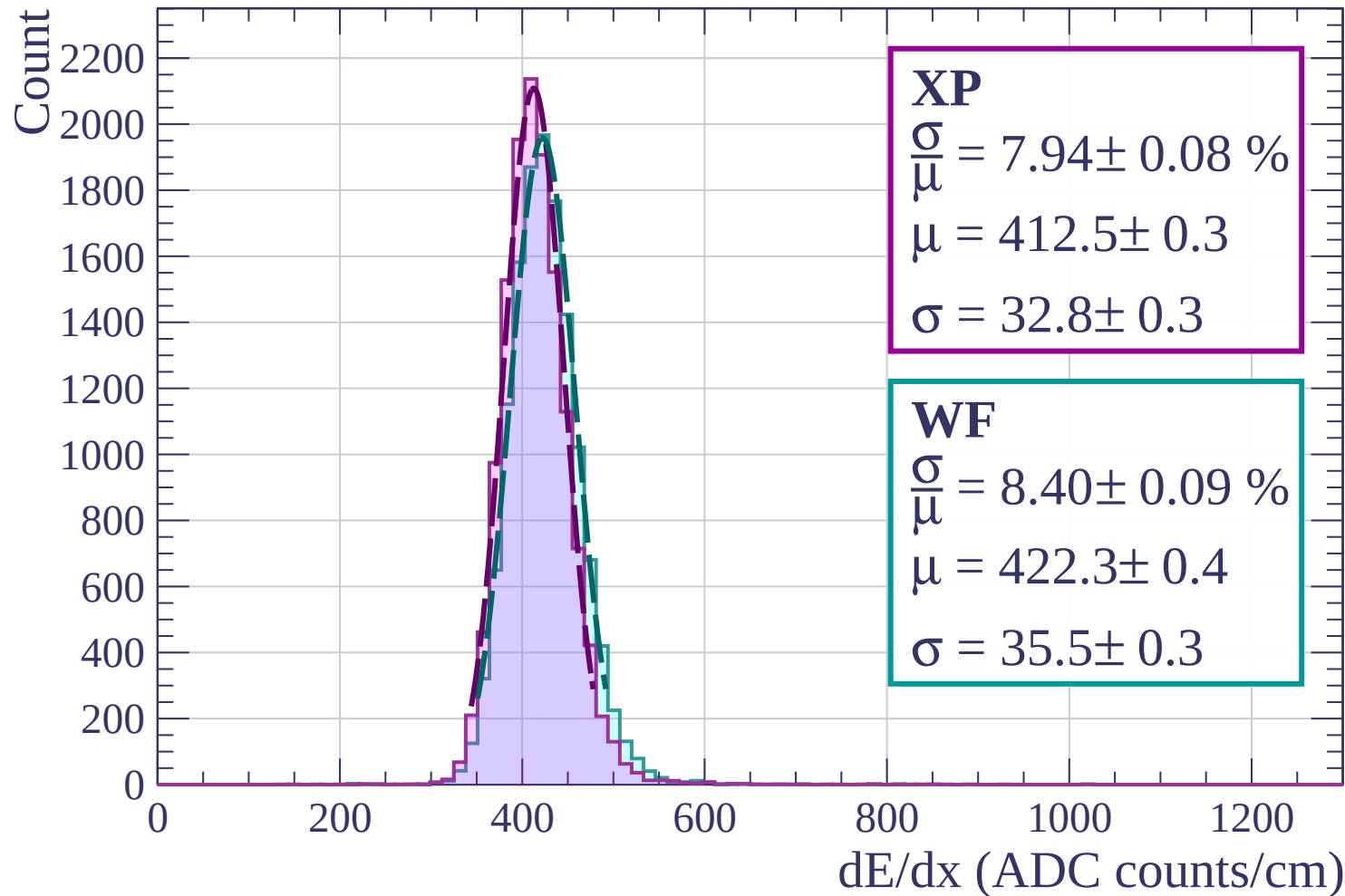
Energy loss | $560 < p < 640$



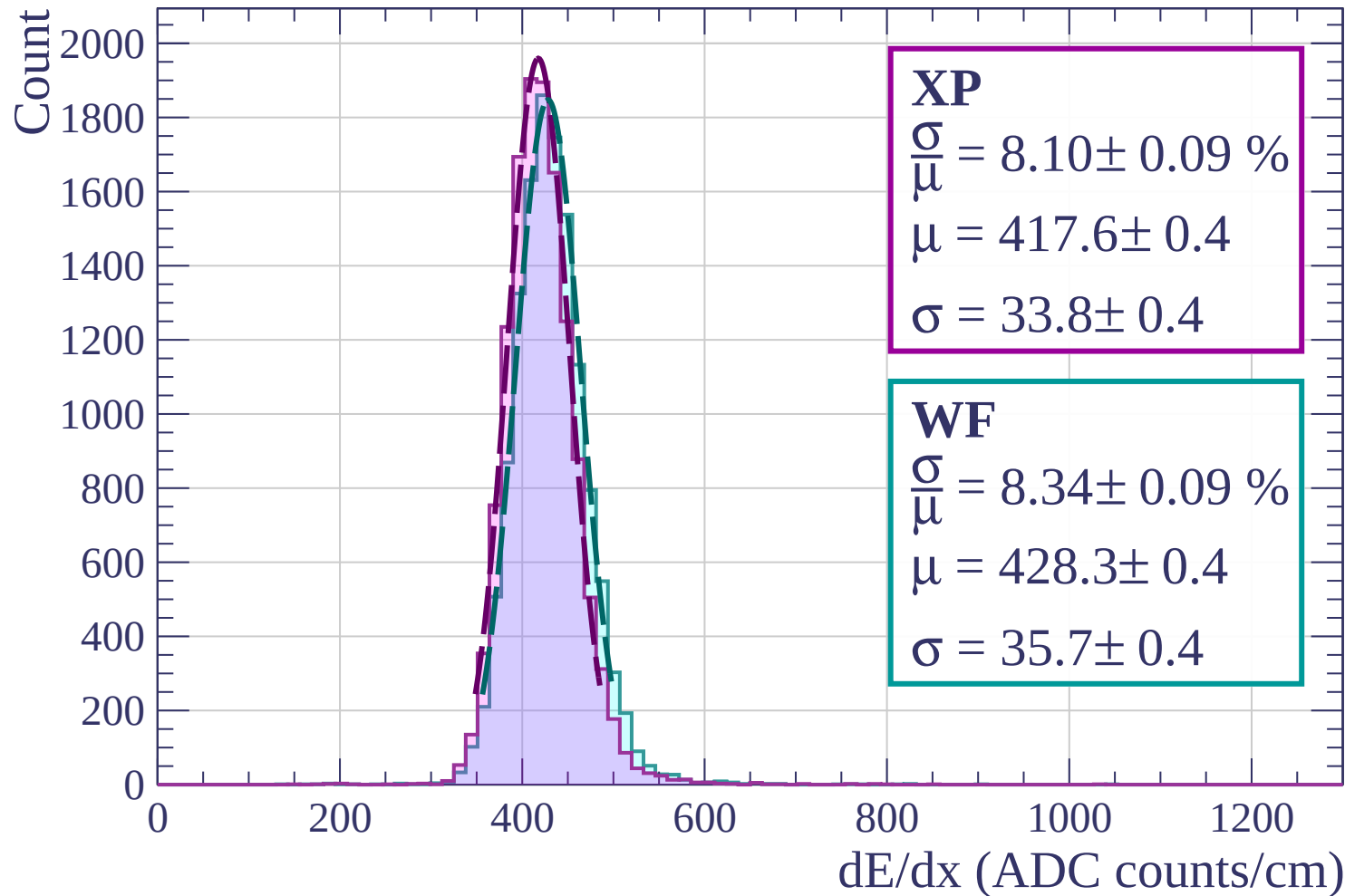
Energy loss | $640 < p < 720$



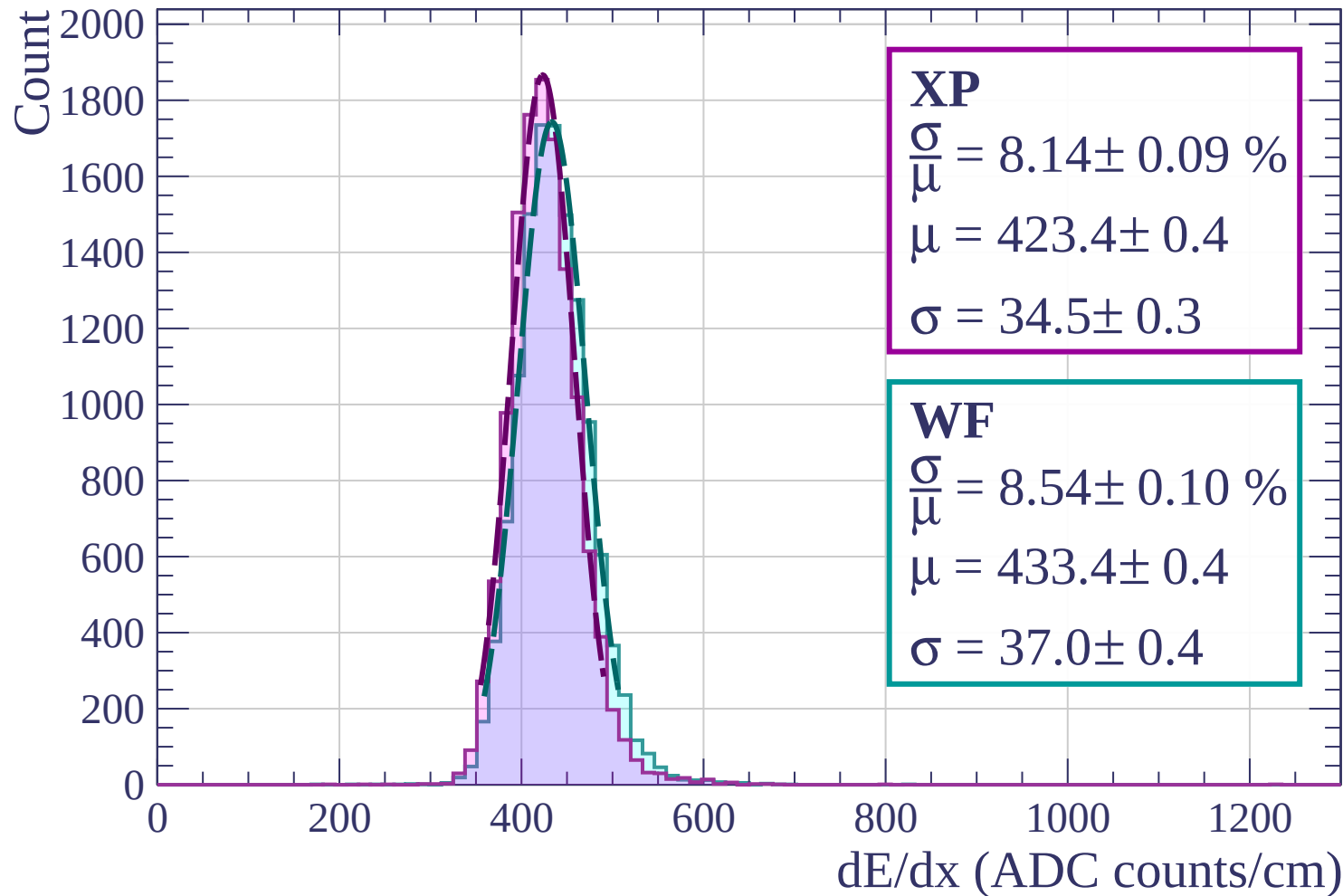
Energy loss | $720 < p < 800$



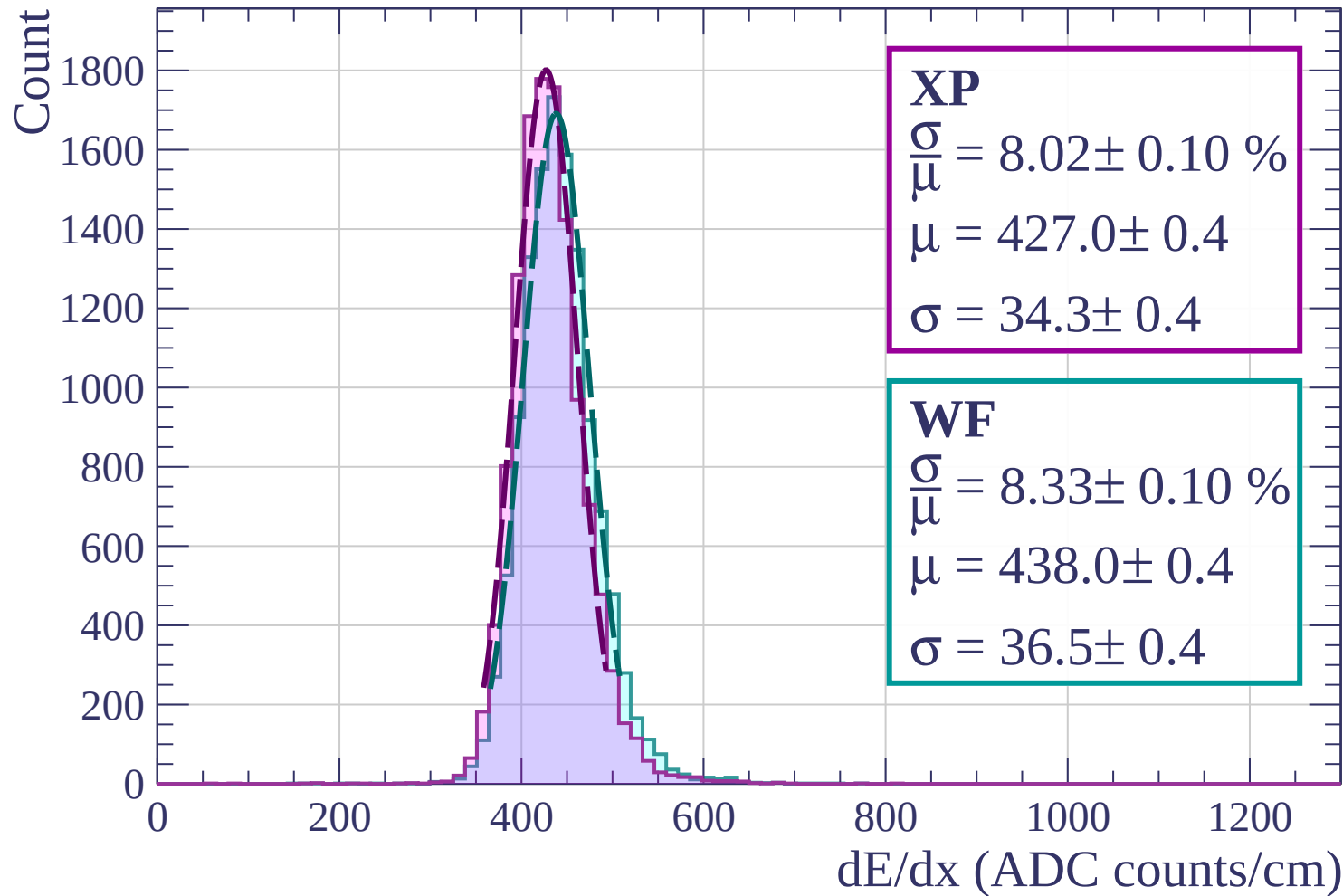
Energy loss | $800 < p < 880$



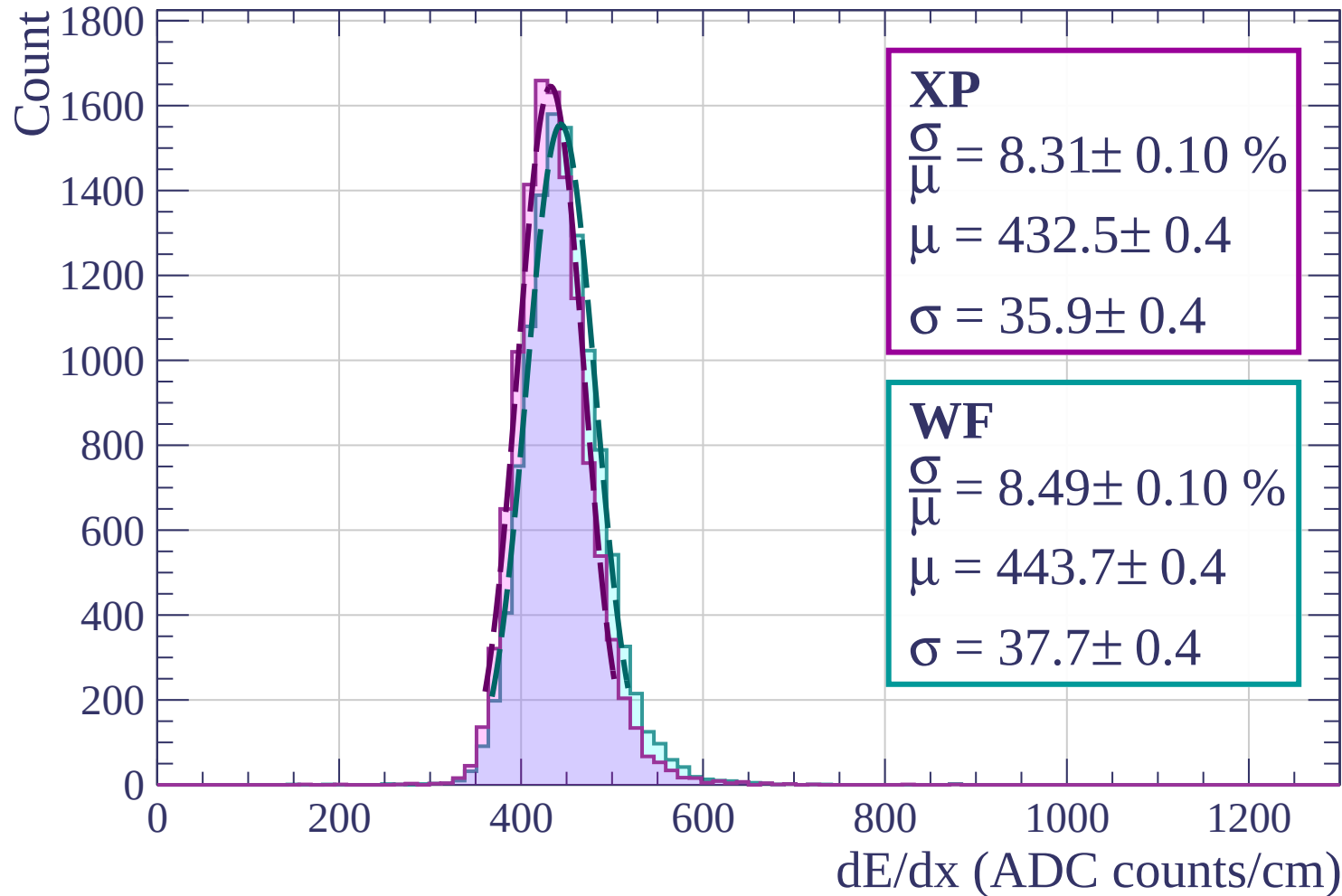
Energy loss | $880 < p < 960$



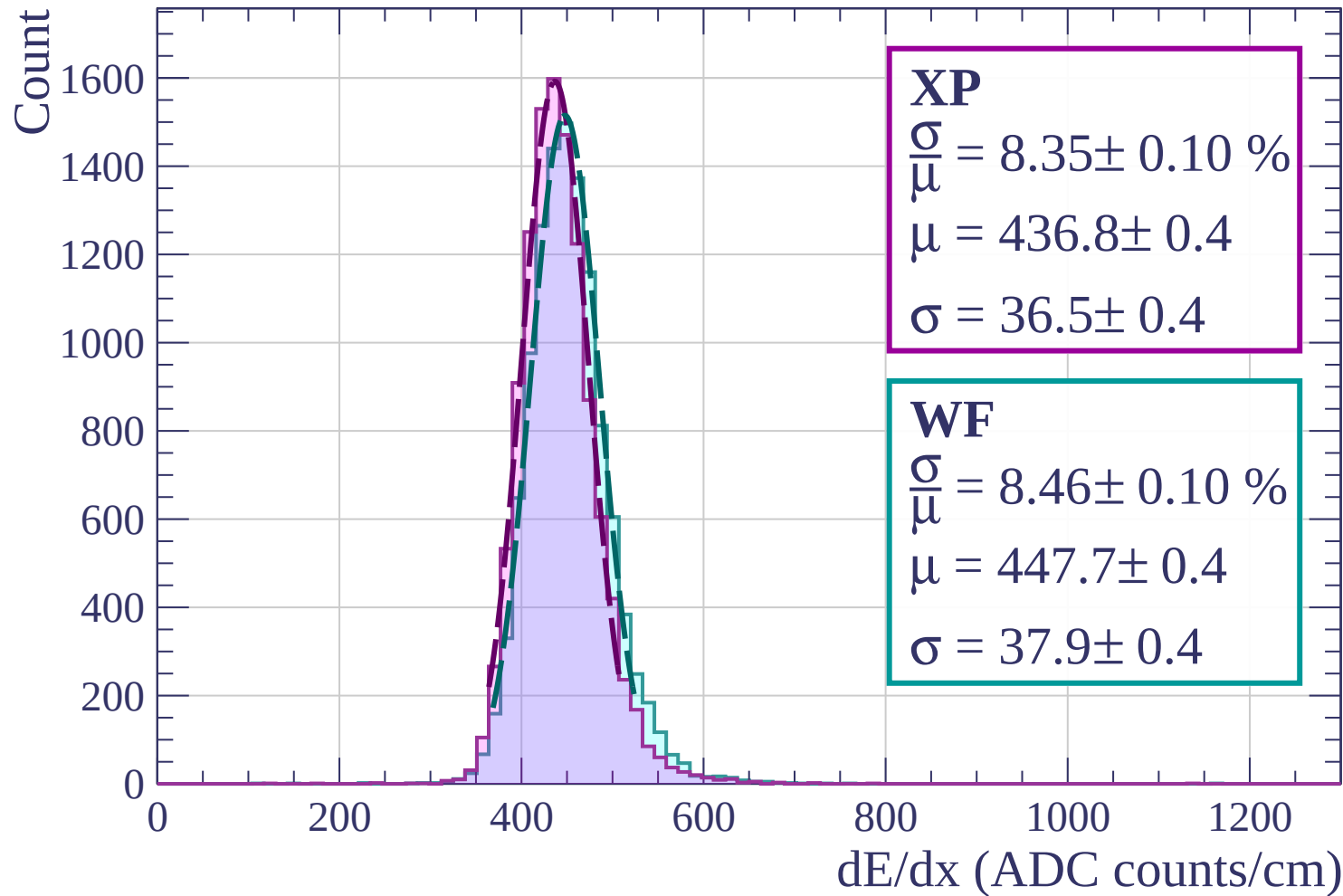
Energy loss | $960 < p < 1040$



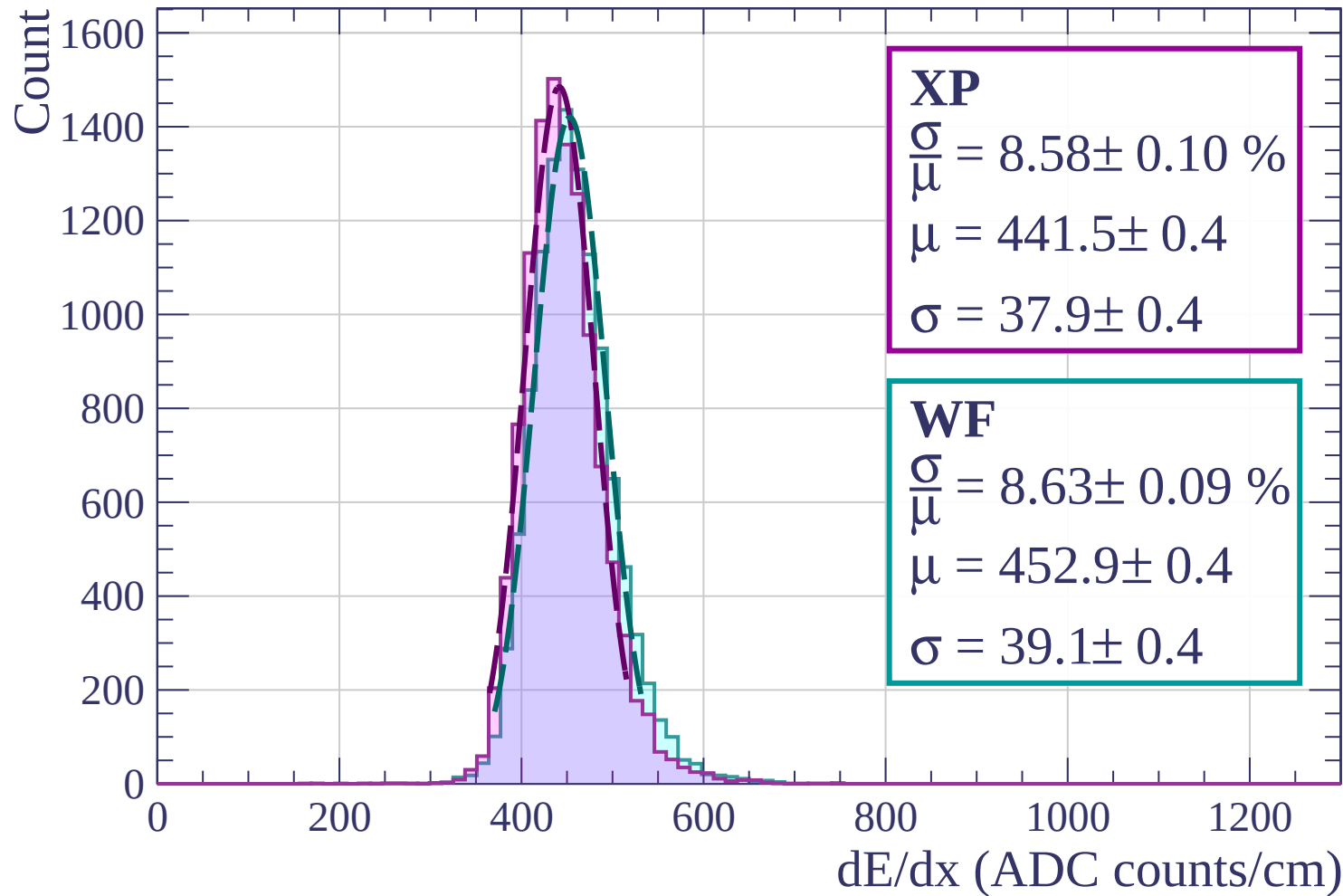
Energy loss | $1040 < p < 1120$



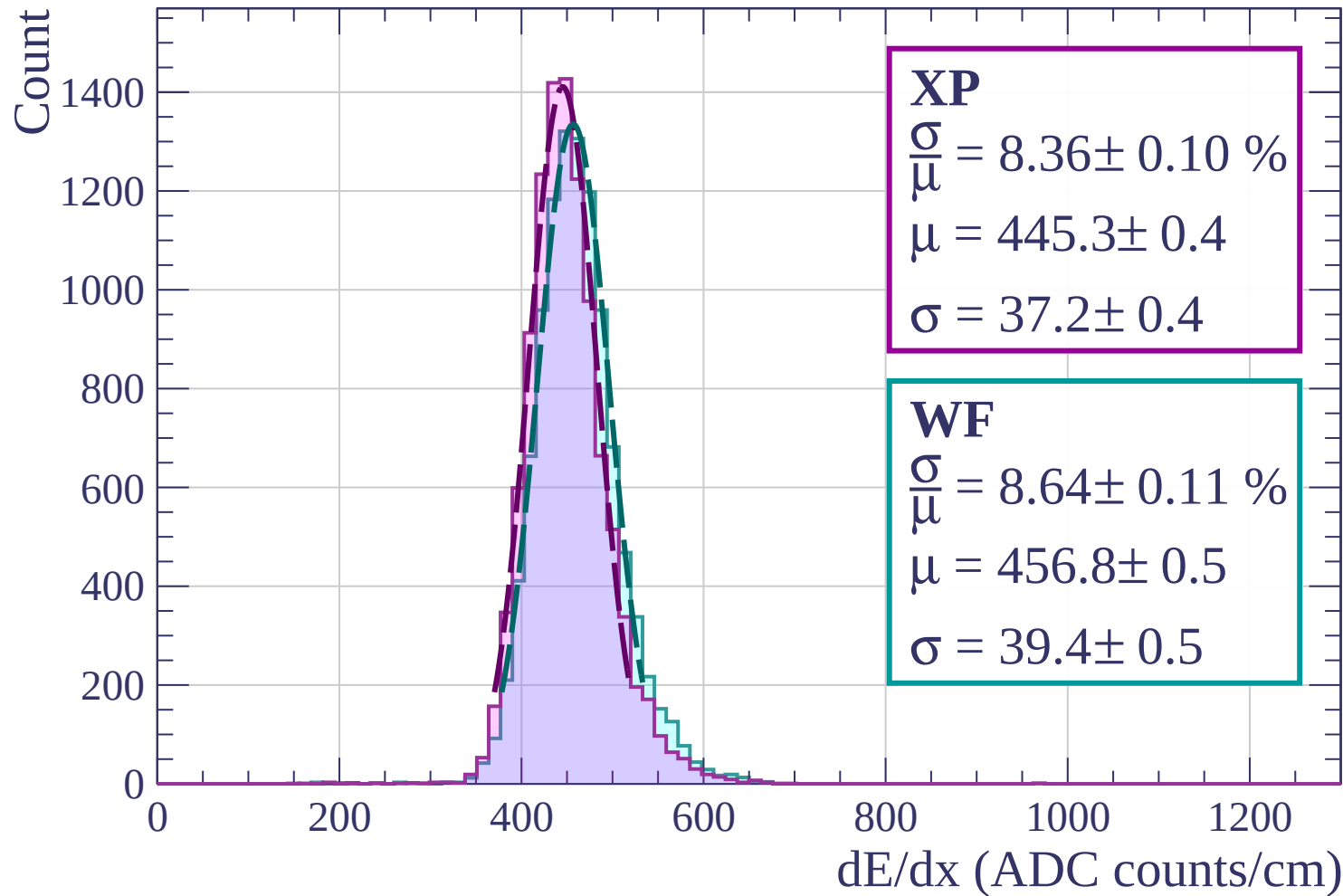
Energy loss | $1120 < p < 1200$



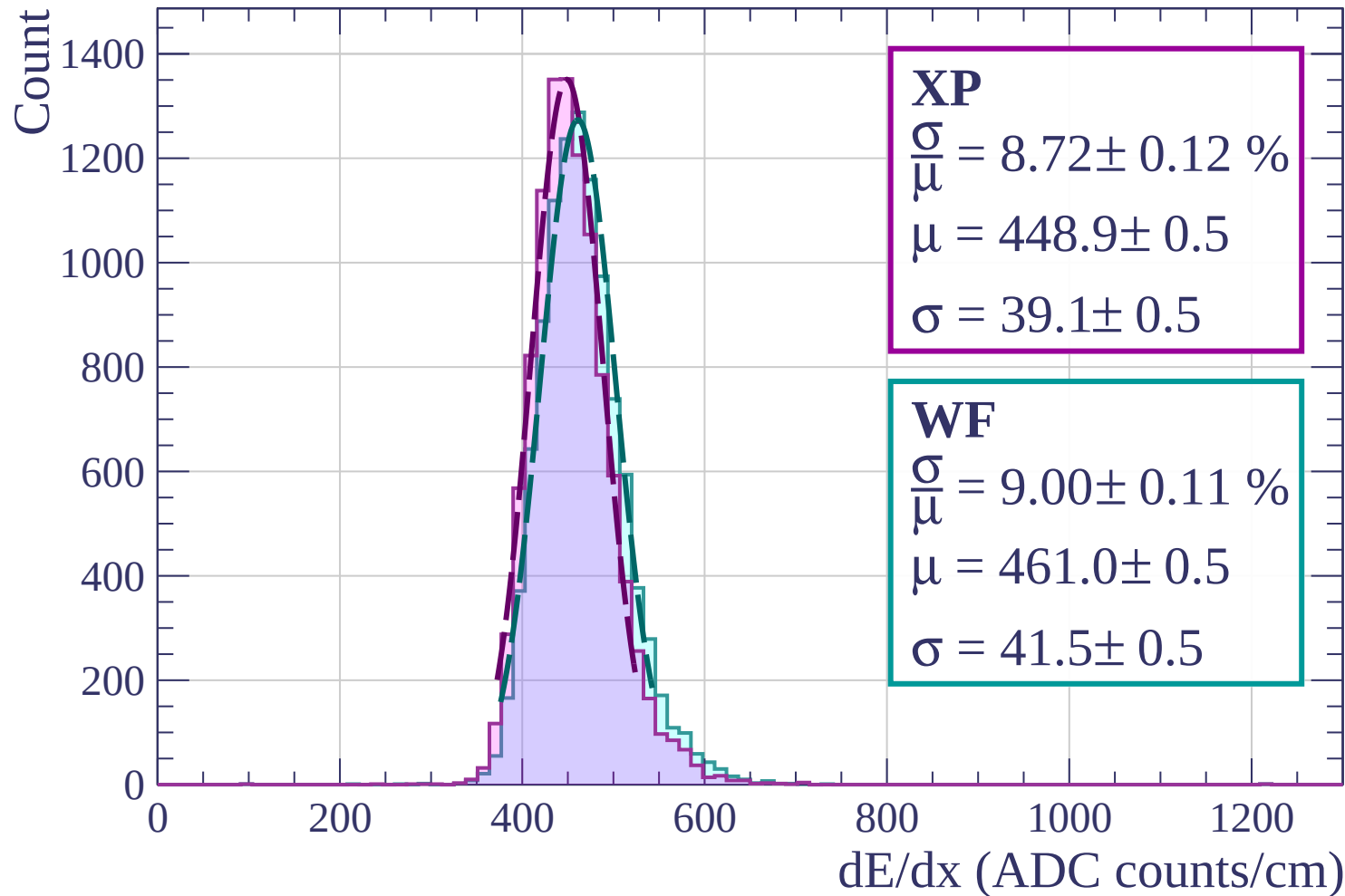
Energy loss | $1200 < p < 1280$



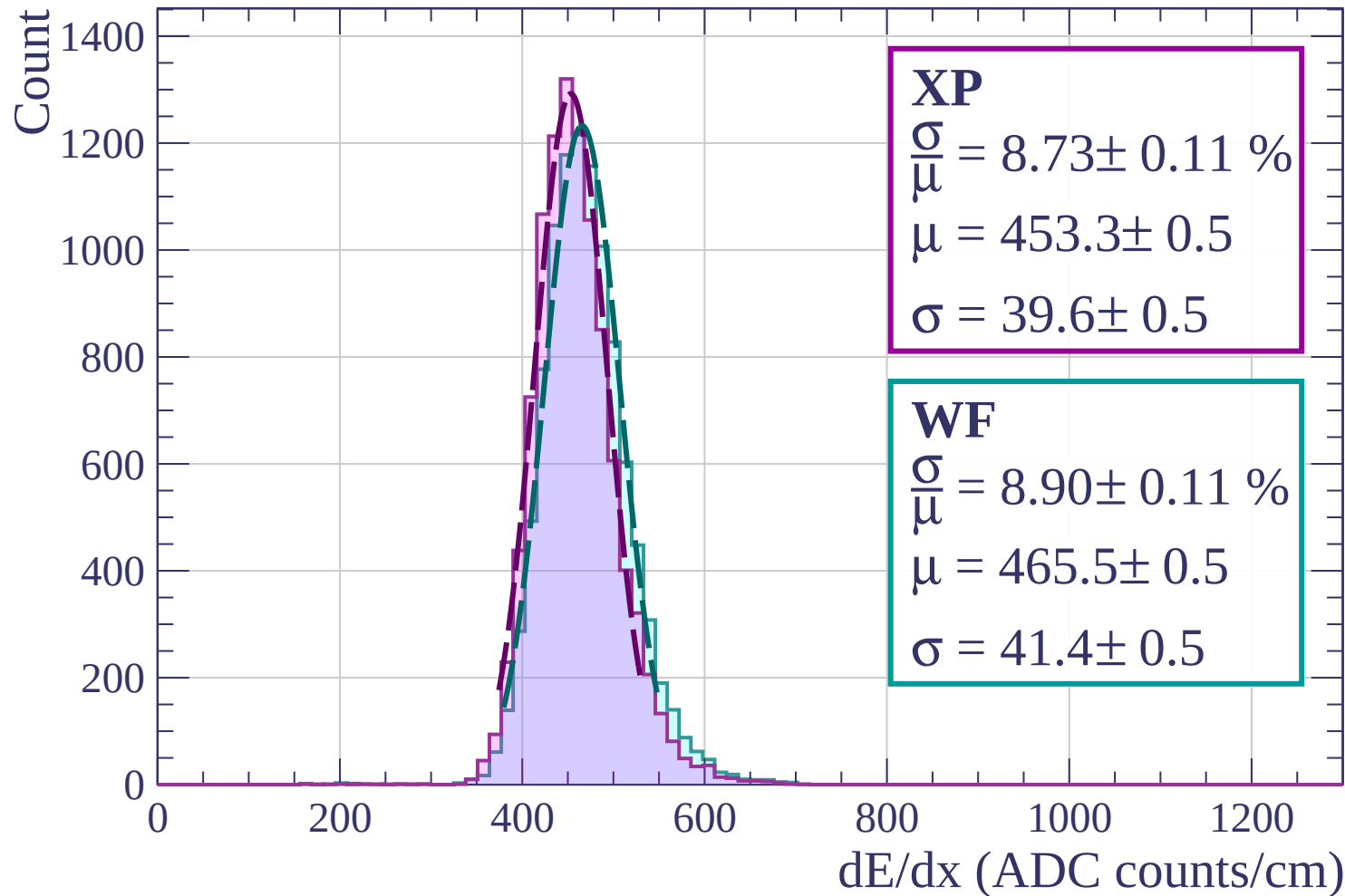
Energy loss | $1280 < p < 1360$



Energy loss | $1360 < p < 1440$

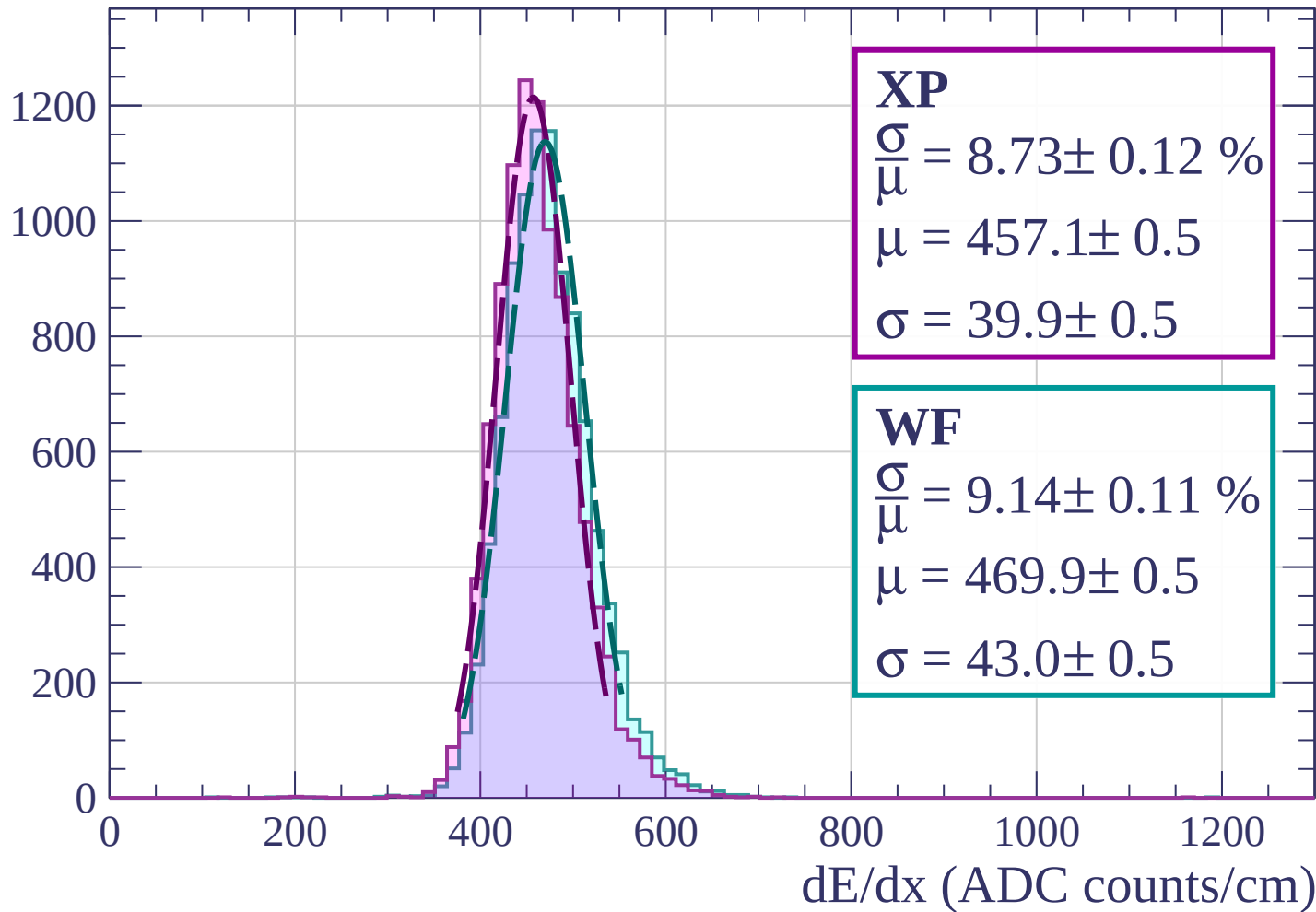


Energy loss | $1440 < p < 1520$

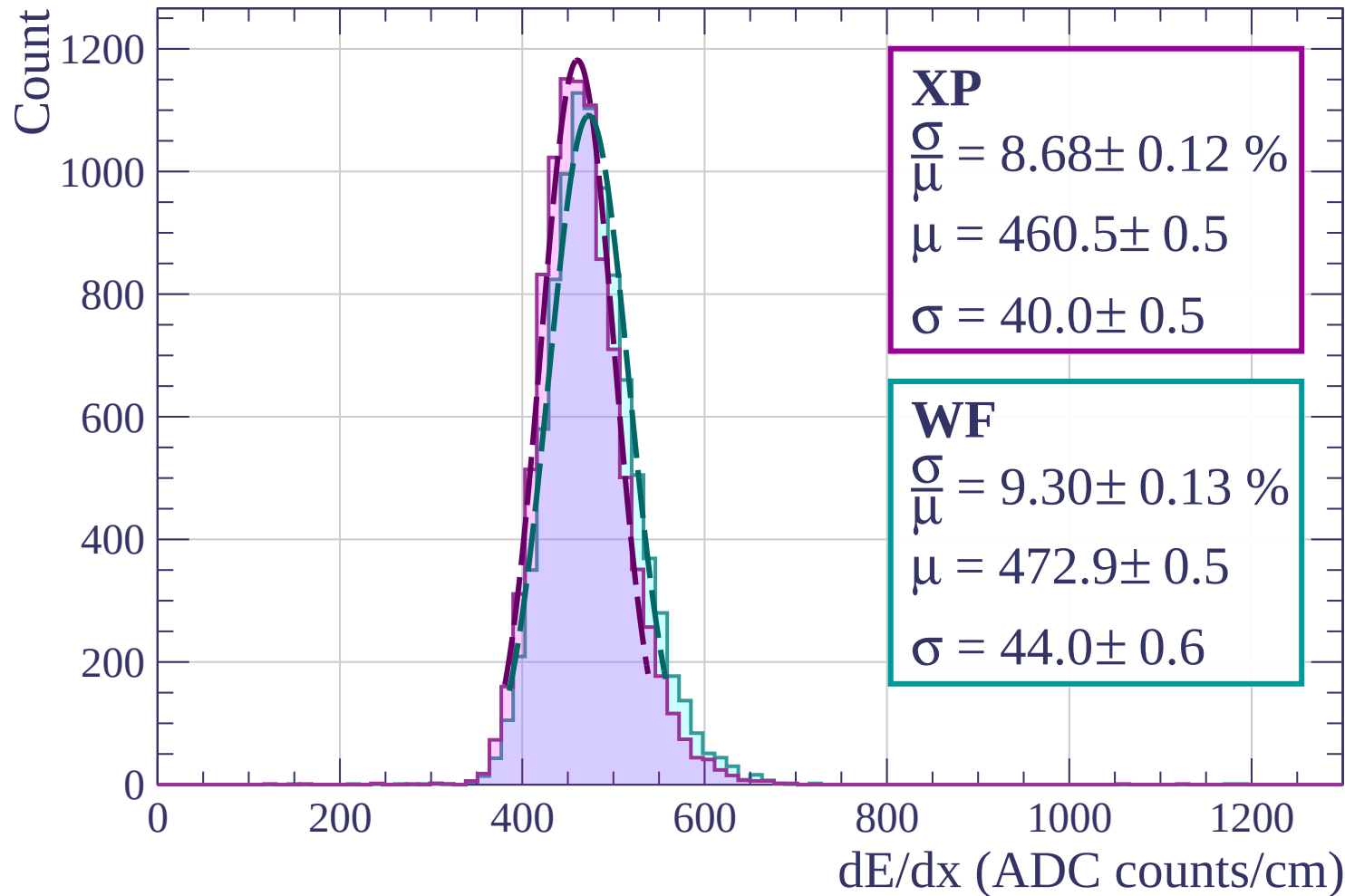


Energy loss | $1520 < p < 1600$

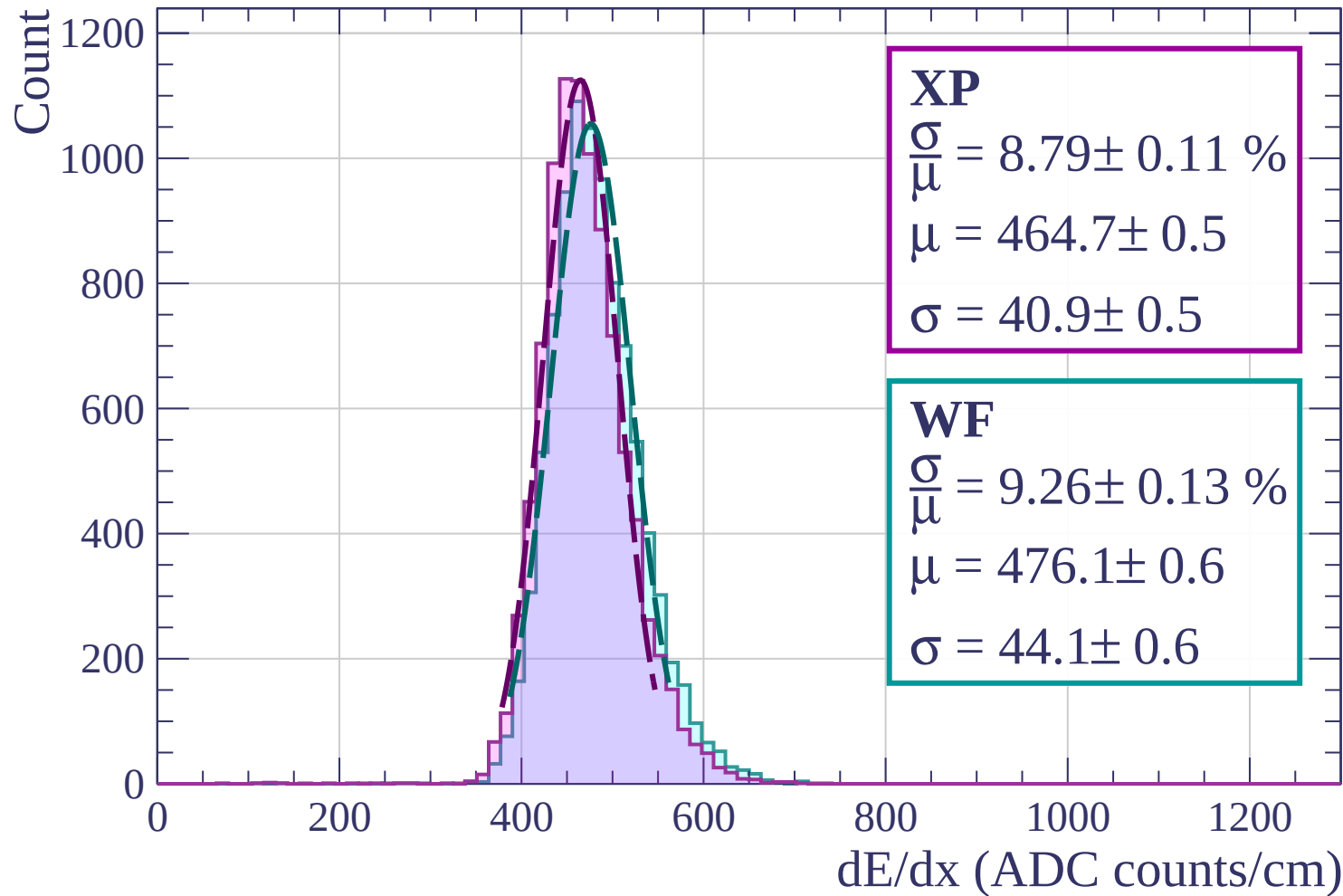
Count



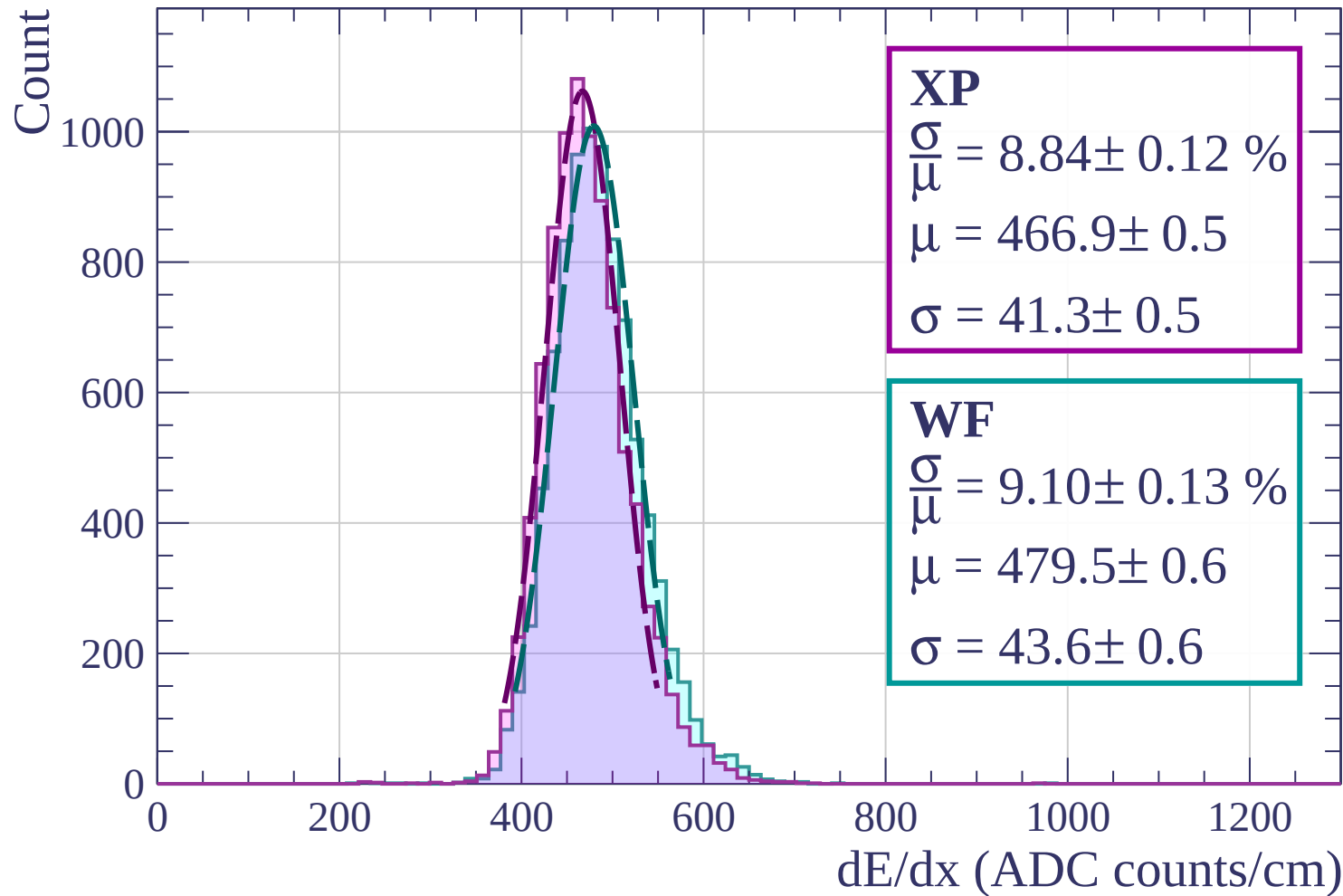
Energy loss | $1600 < p < 1680$



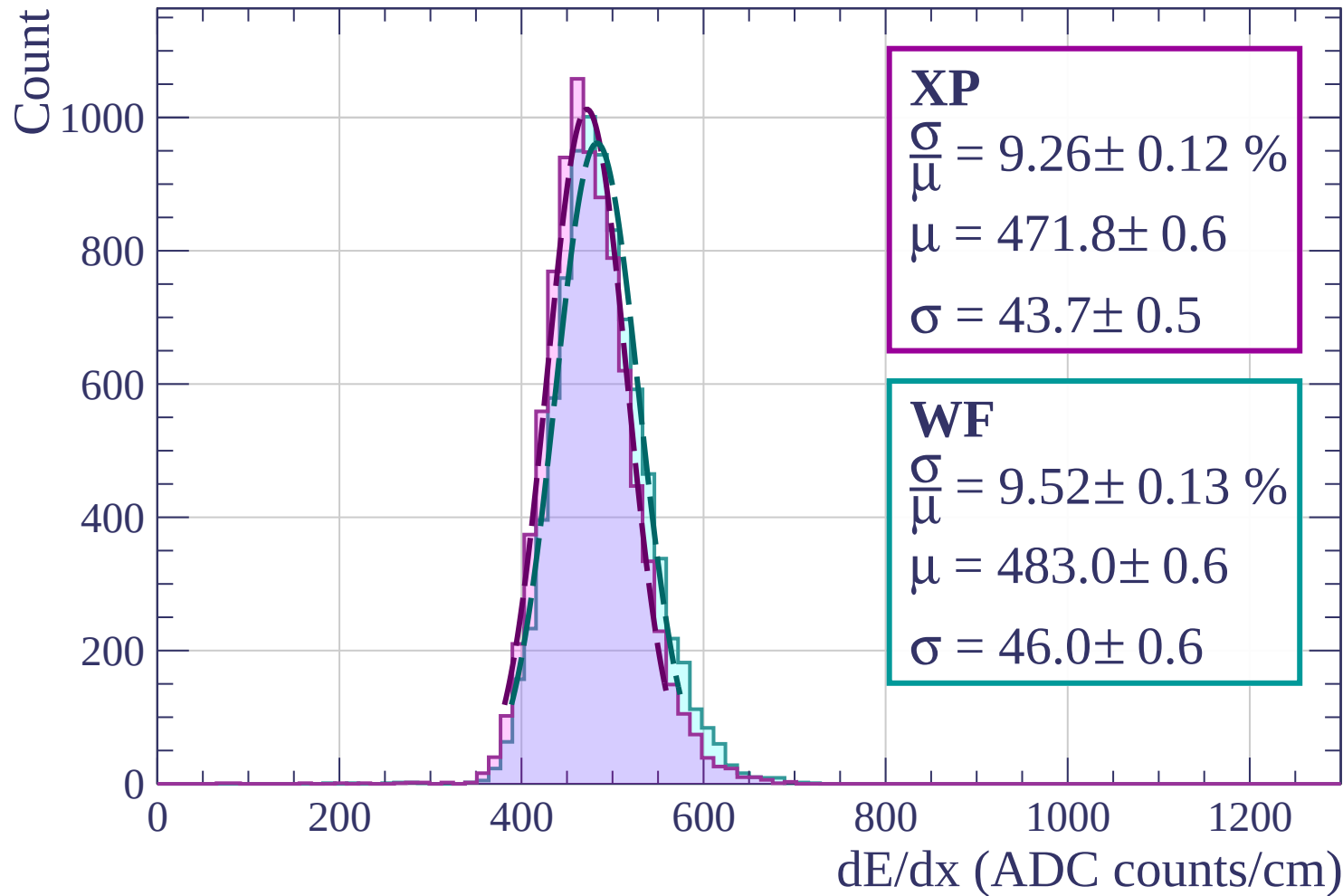
Energy loss | $1680 < p < 1760$



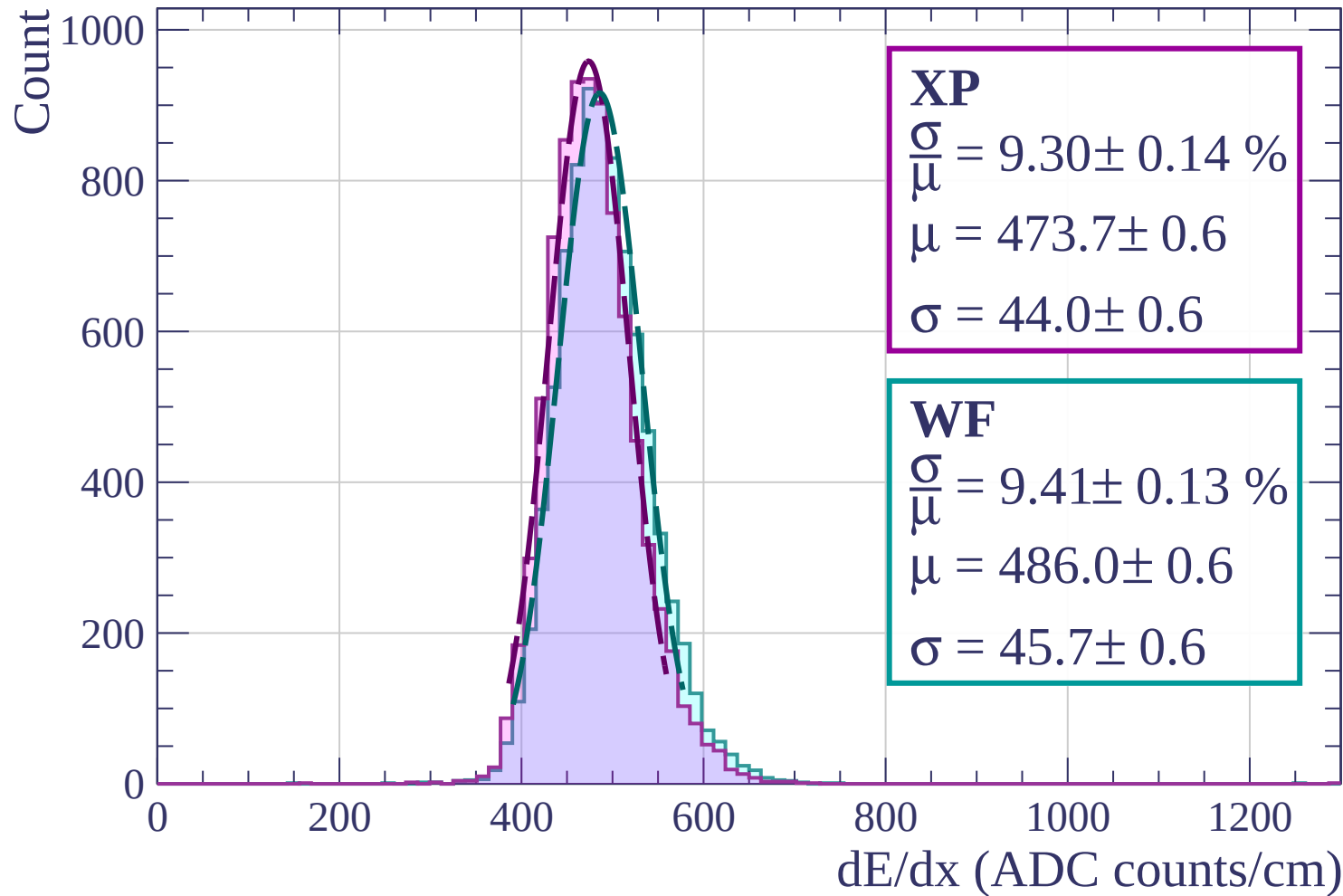
Energy loss | $1760 < p < 1840$



Energy loss | $1840 < p < 1920$

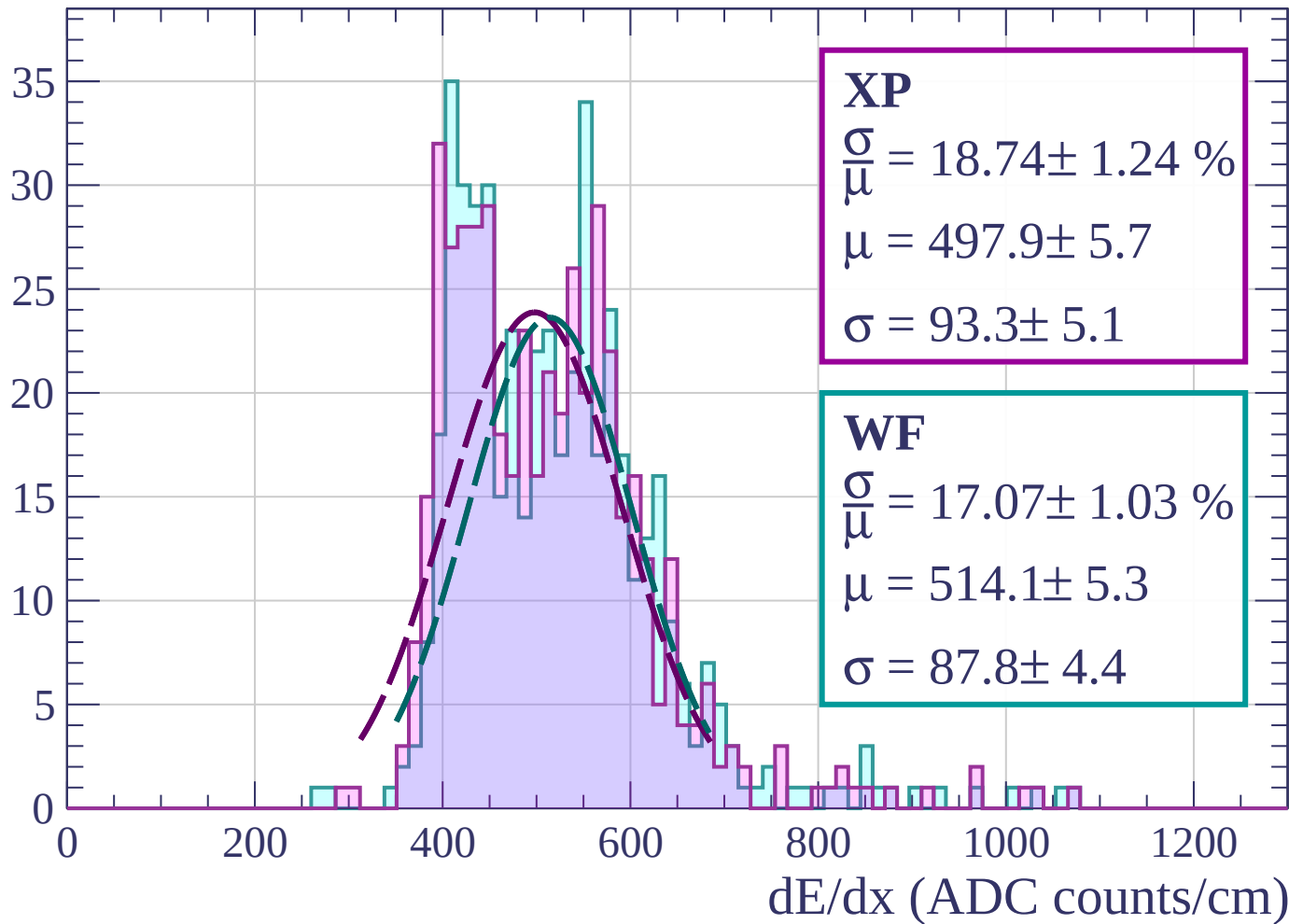


Energy loss | $1920 < p < 2000$



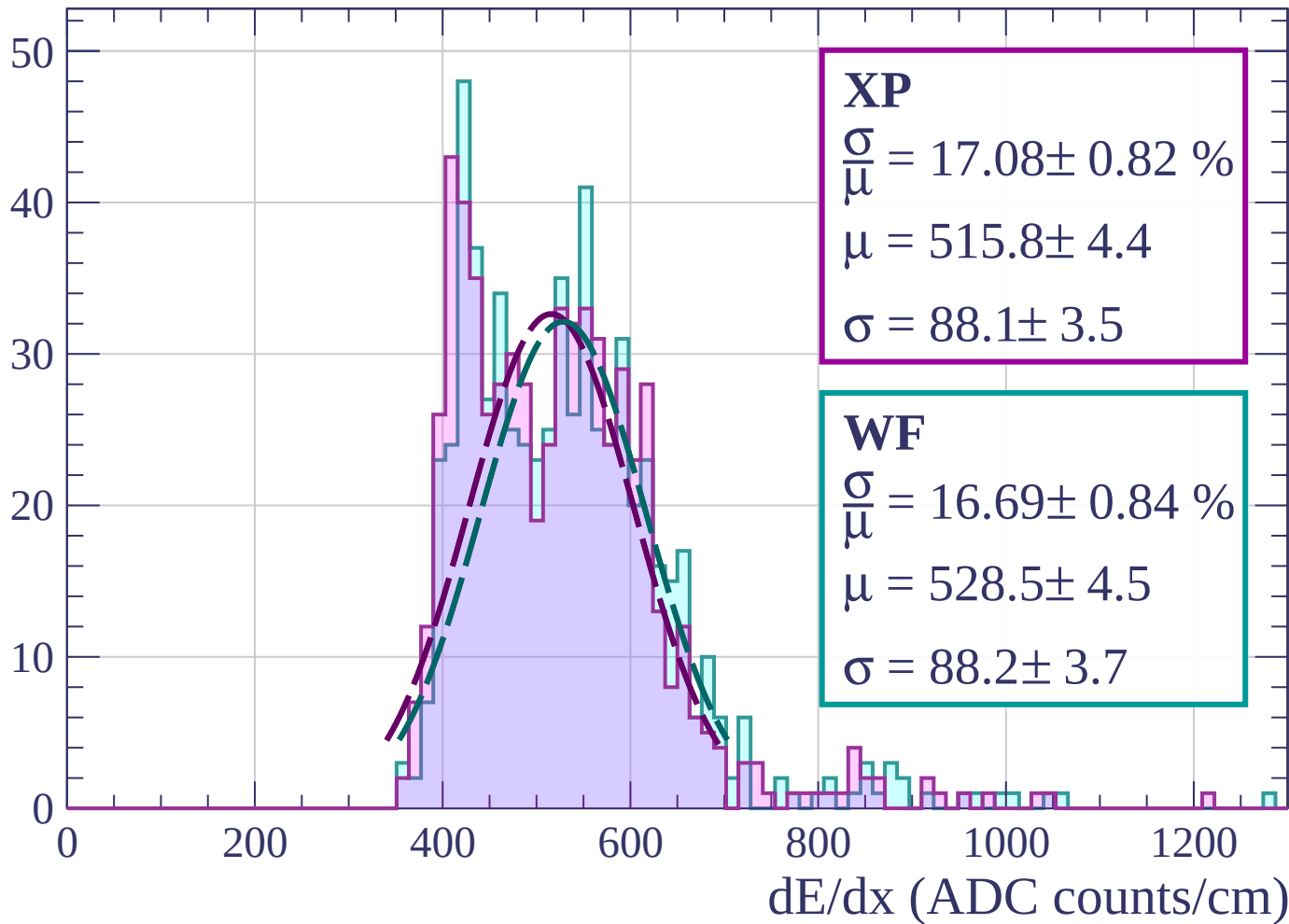
Energy loss | $0 < \phi < 3$

Count



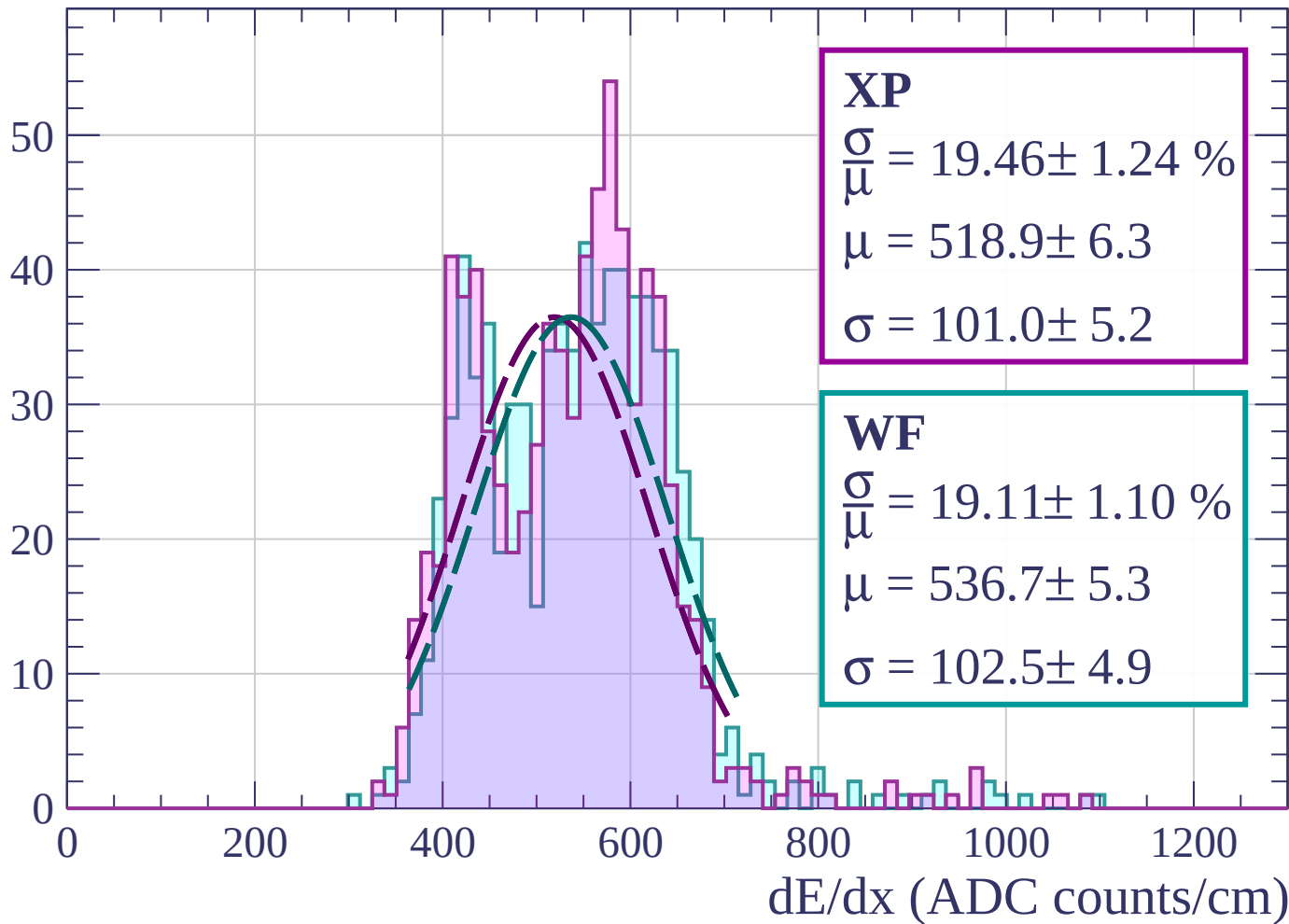
Energy loss | $3 < \phi < 6$

Count



Energy loss | $6 < \phi < 9$

Count



Energy loss | $9 < \phi < 12$

Count

XP

$$\frac{\sigma}{\mu} = 15.21 \pm 0.43 \%$$

$$\mu = 541.5 \pm 2.8$$

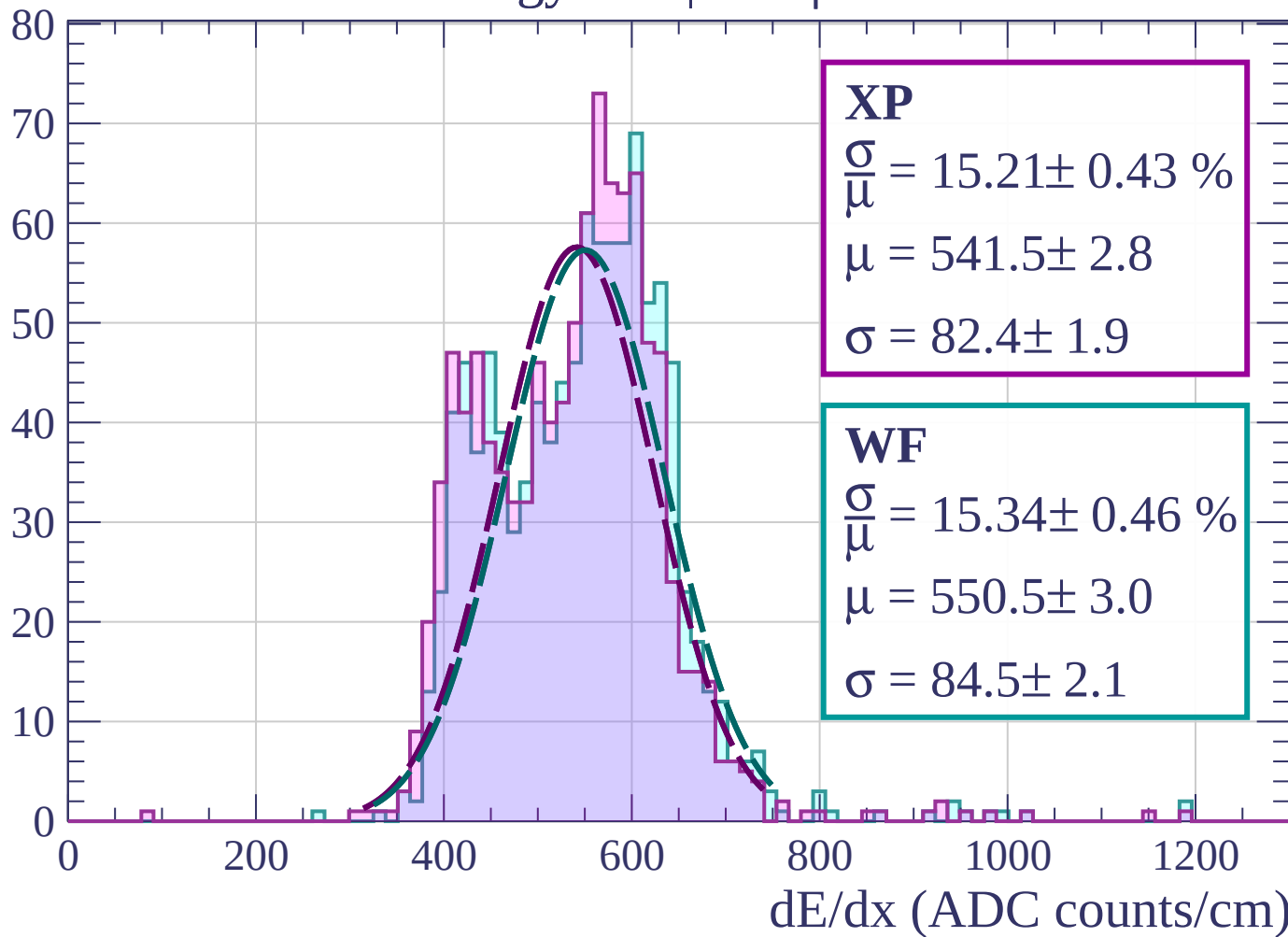
$$\sigma = 82.4 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 15.34 \pm 0.46 \%$$

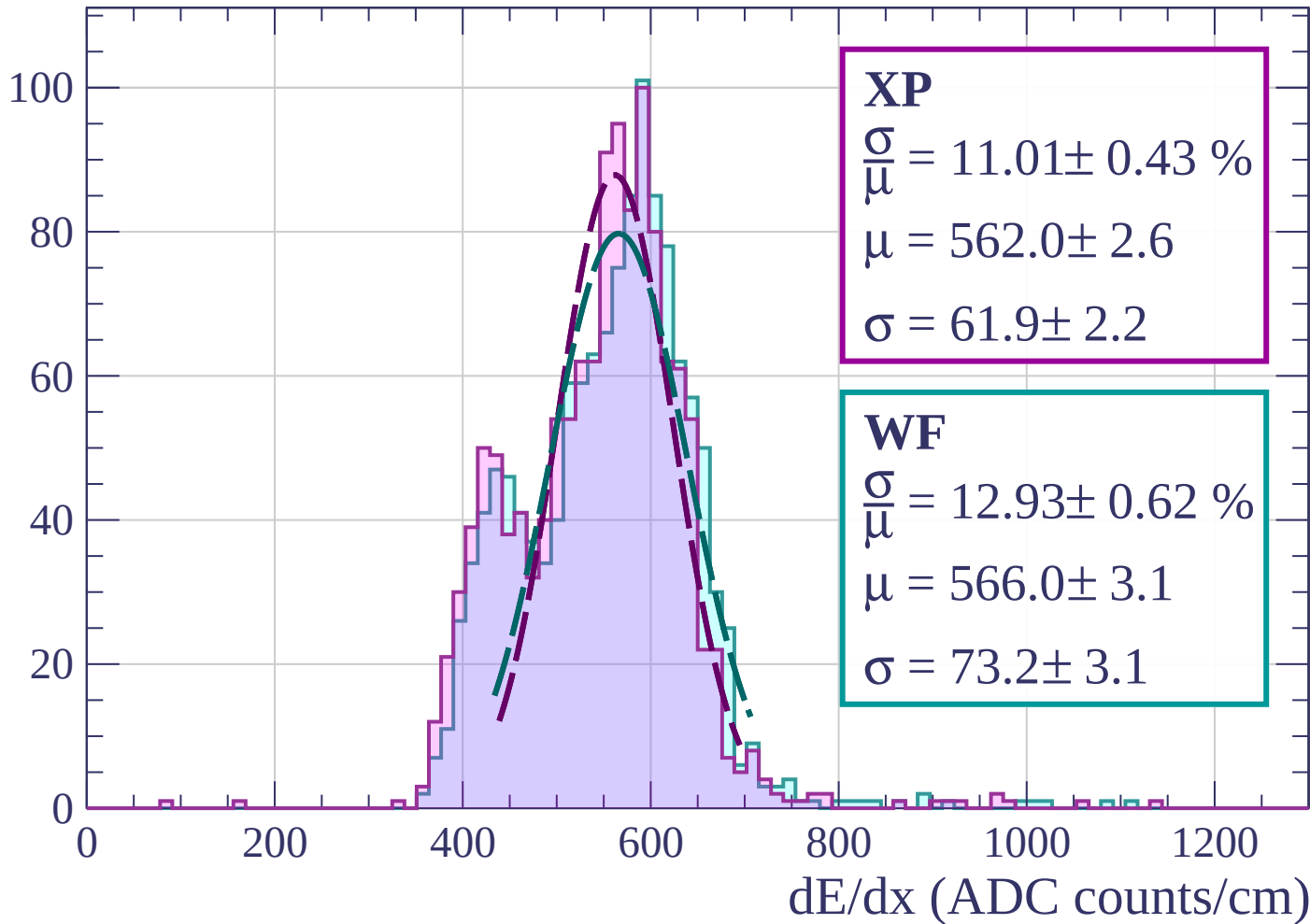
$$\mu = 550.5 \pm 3.0$$

$$\sigma = 84.5 \pm 2.1$$



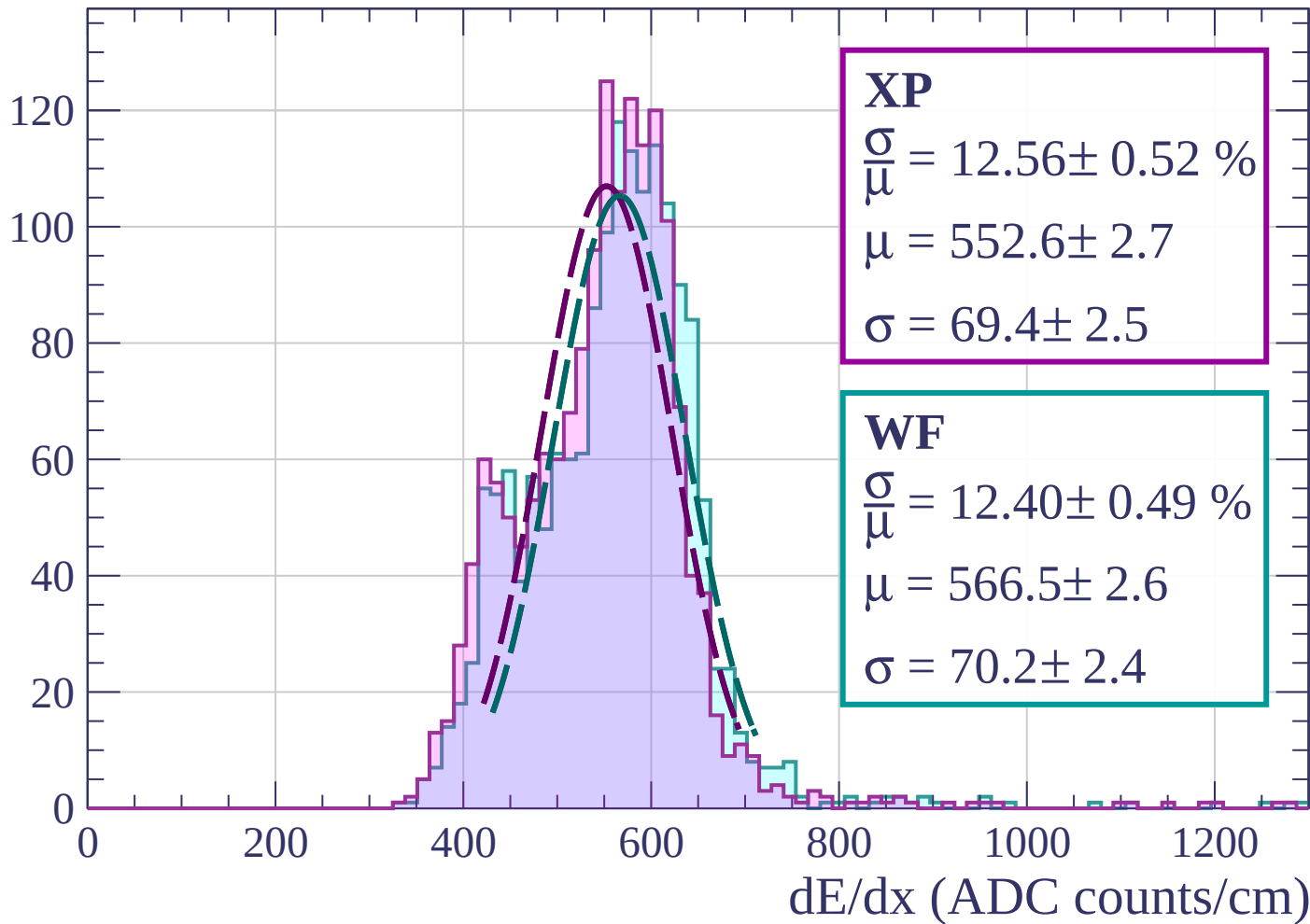
Energy loss | $12 < \phi < 15$

Count

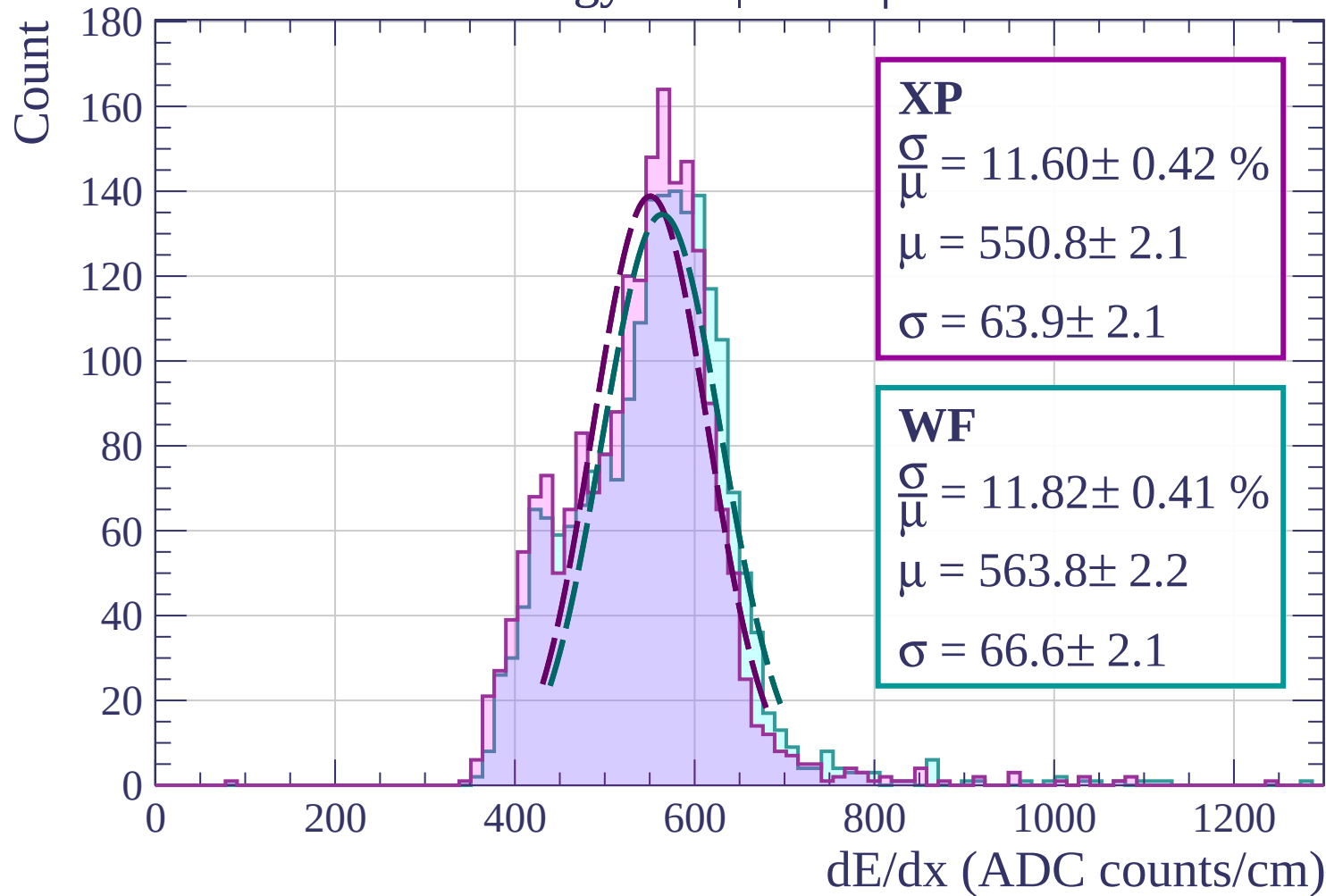


Energy loss | $15 < \phi < 18$

Count

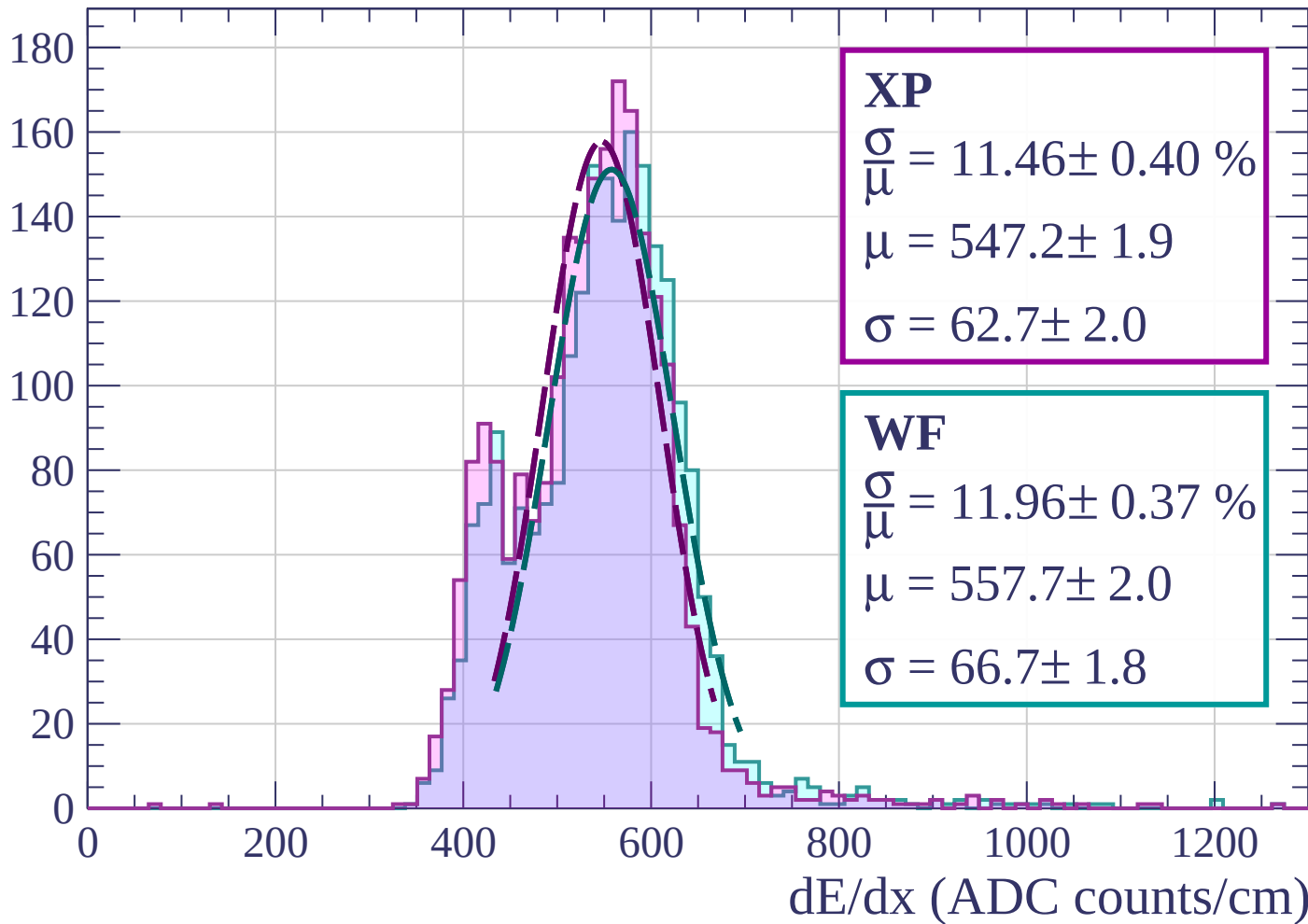


Energy loss | $18 < \phi < 21$



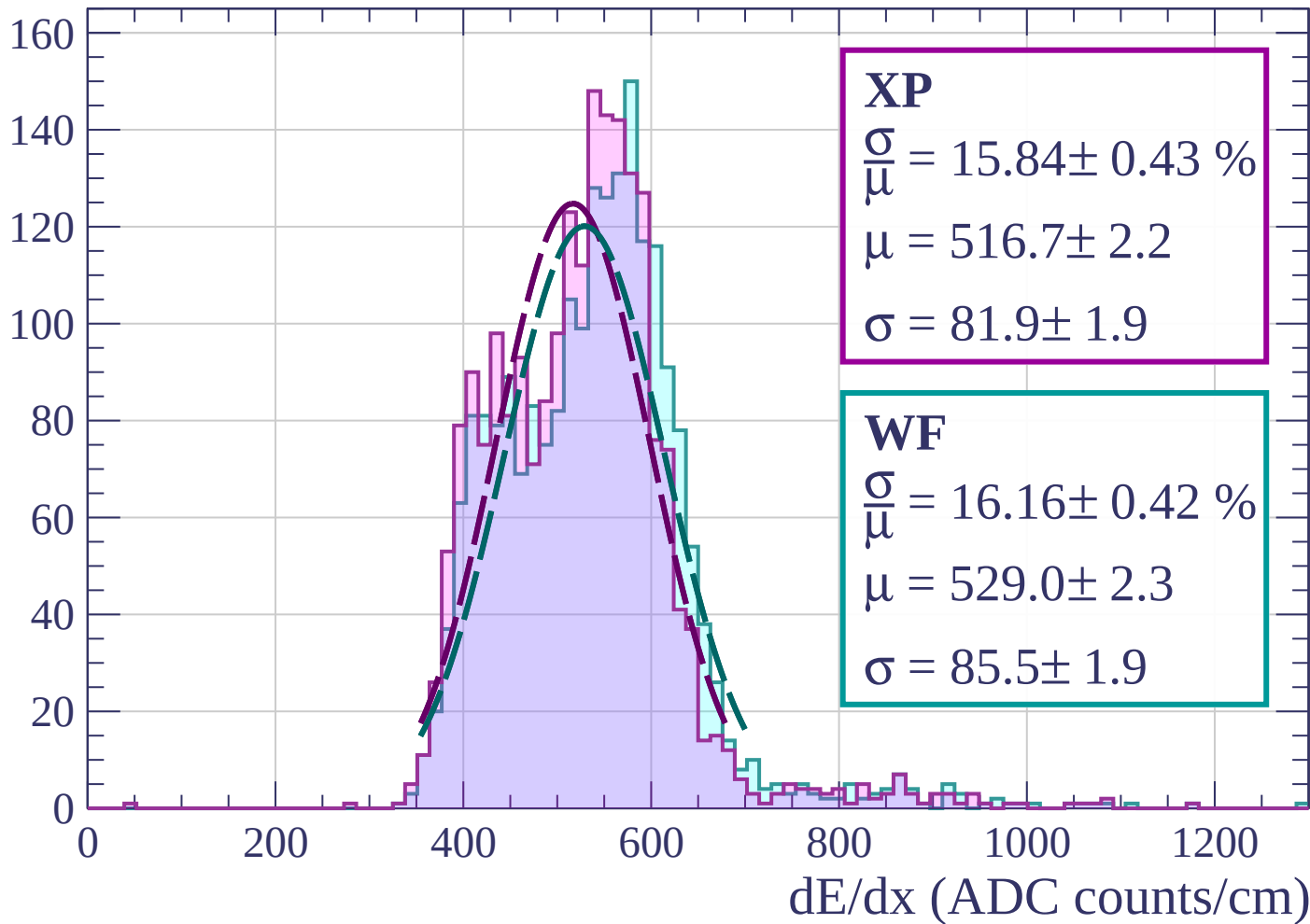
Energy loss | $21 < \phi < 24$

Count

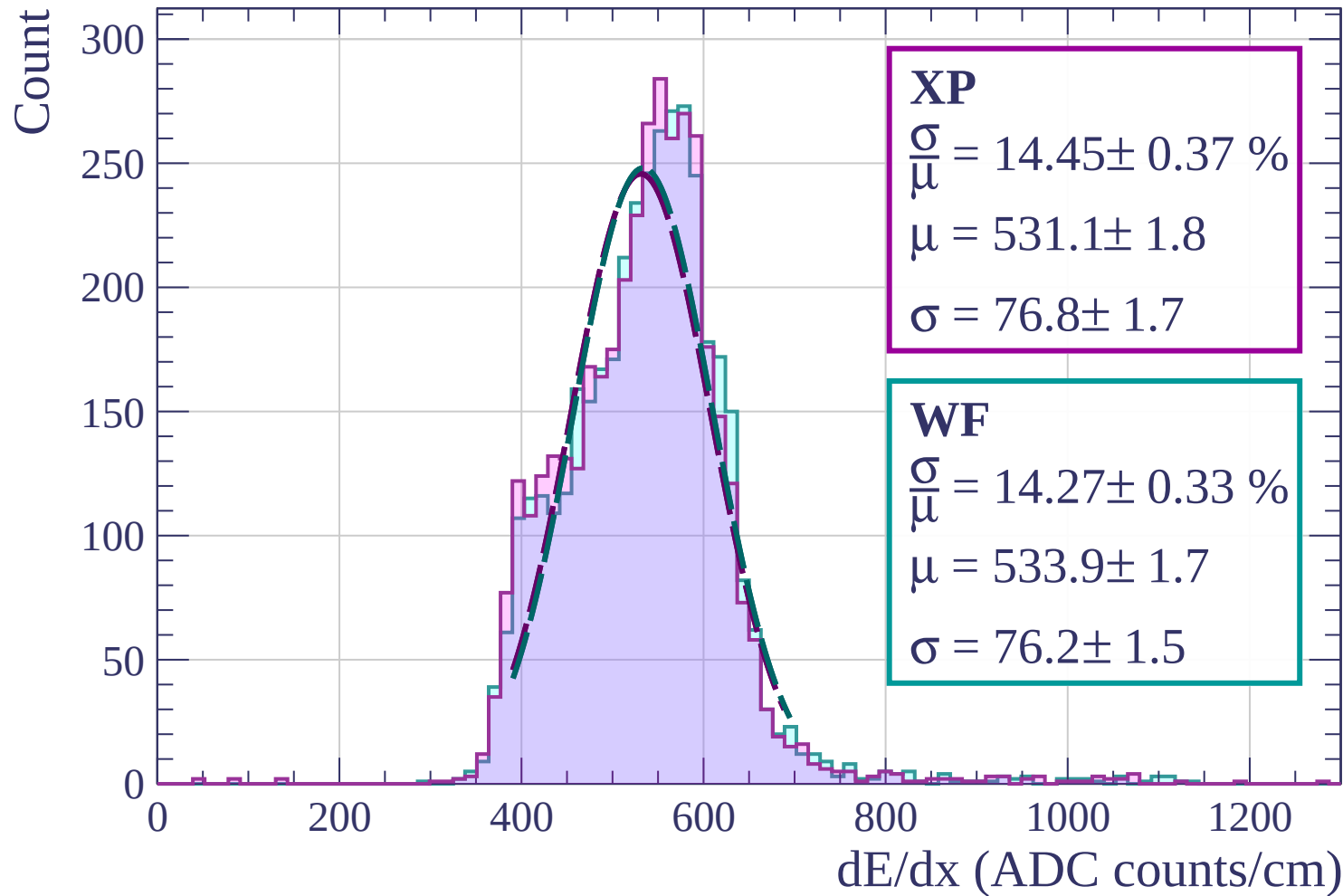


Energy loss | $24 < \phi < 27$

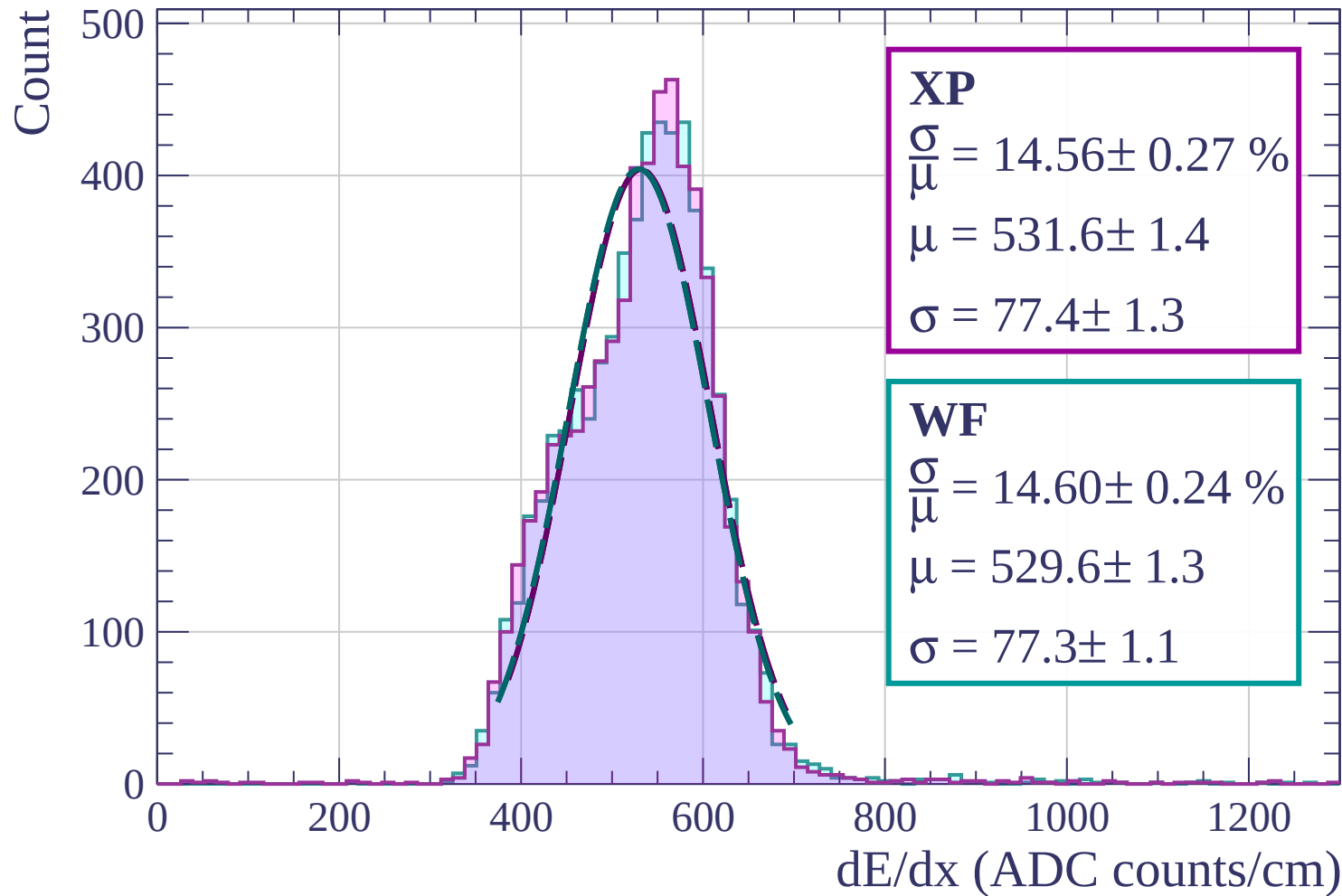
Count



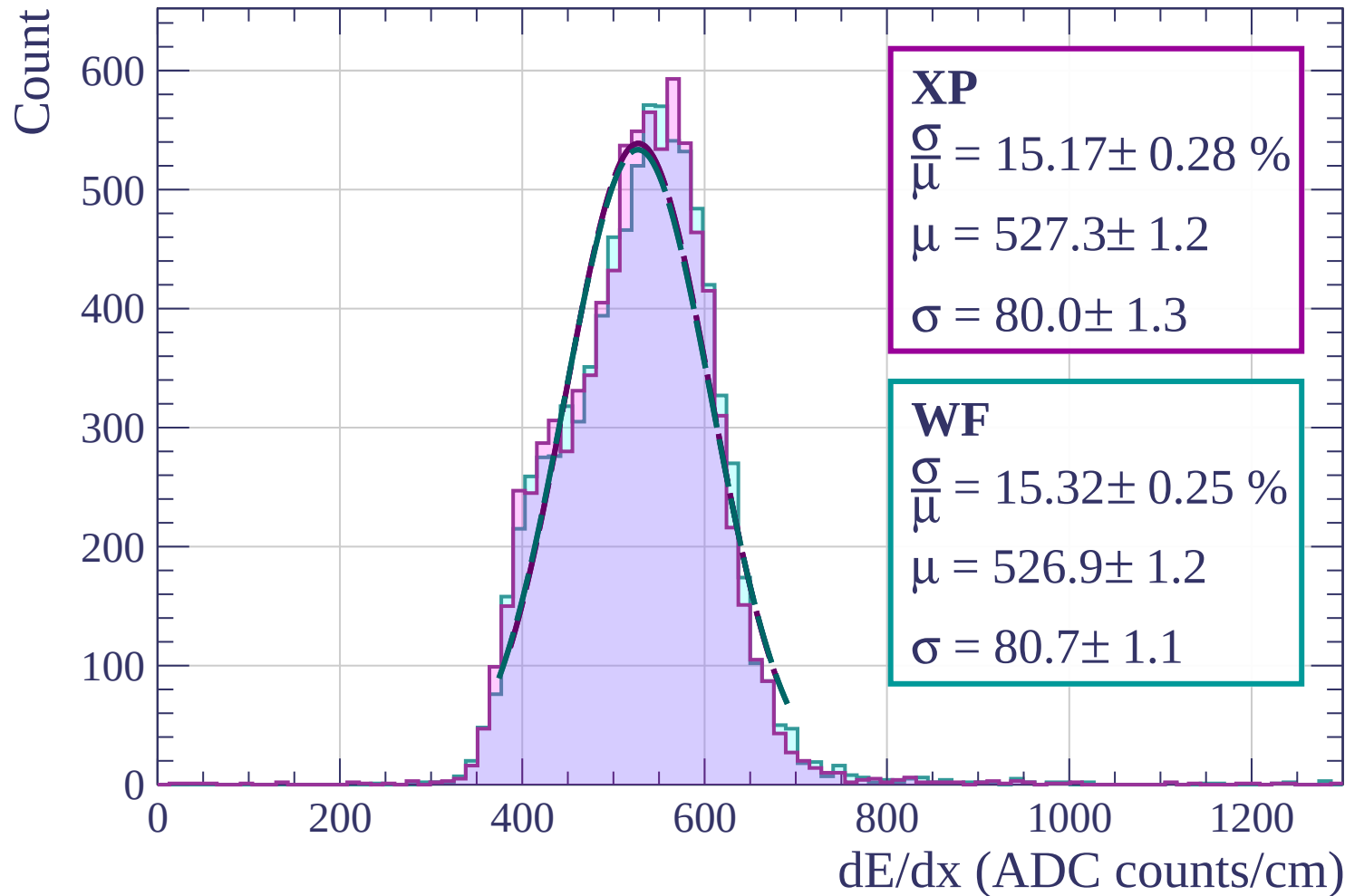
Energy loss | $27 < \phi < 30$



Energy loss | $30 < \phi < 33$

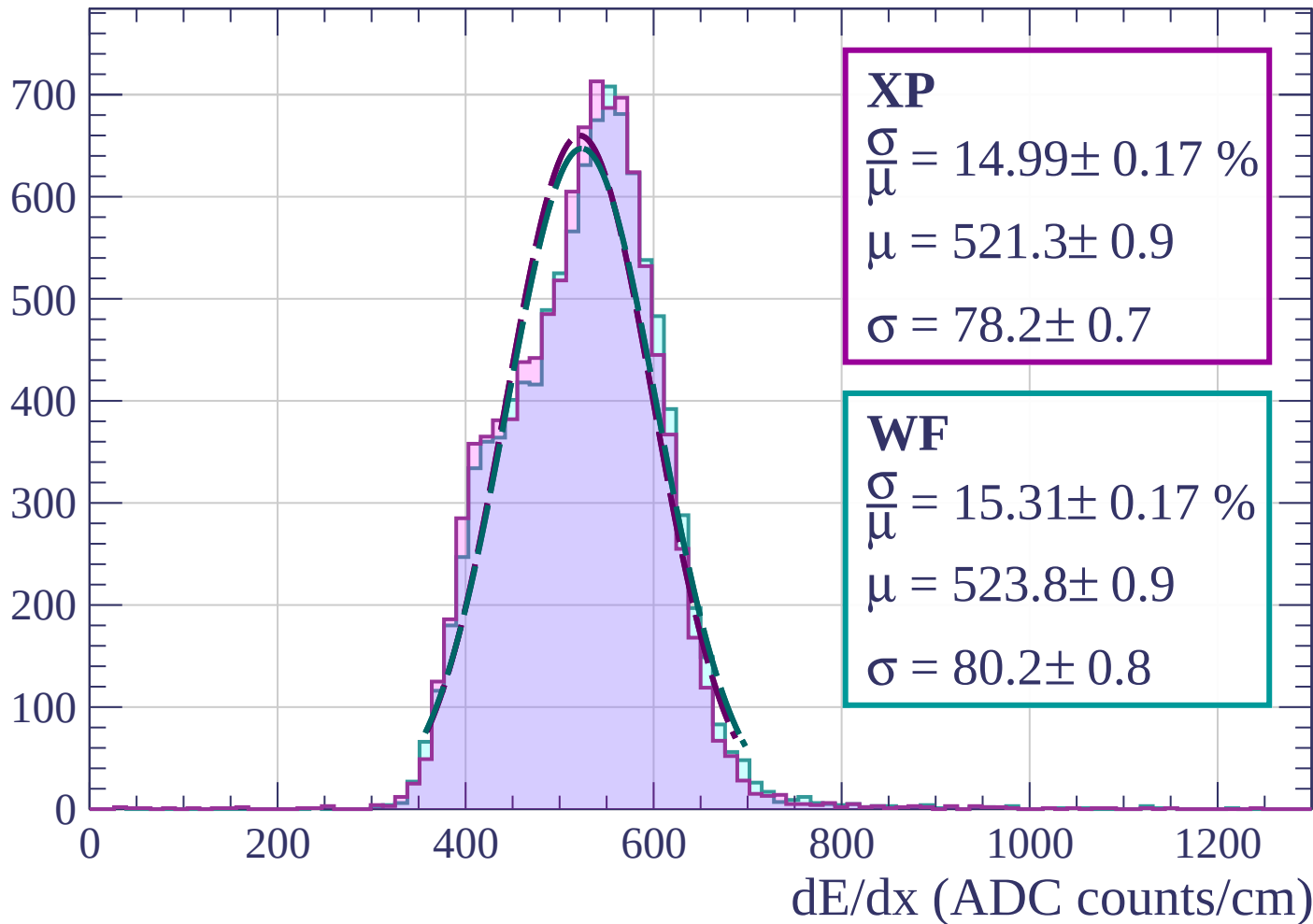


Energy loss | $33 < \phi < 36$

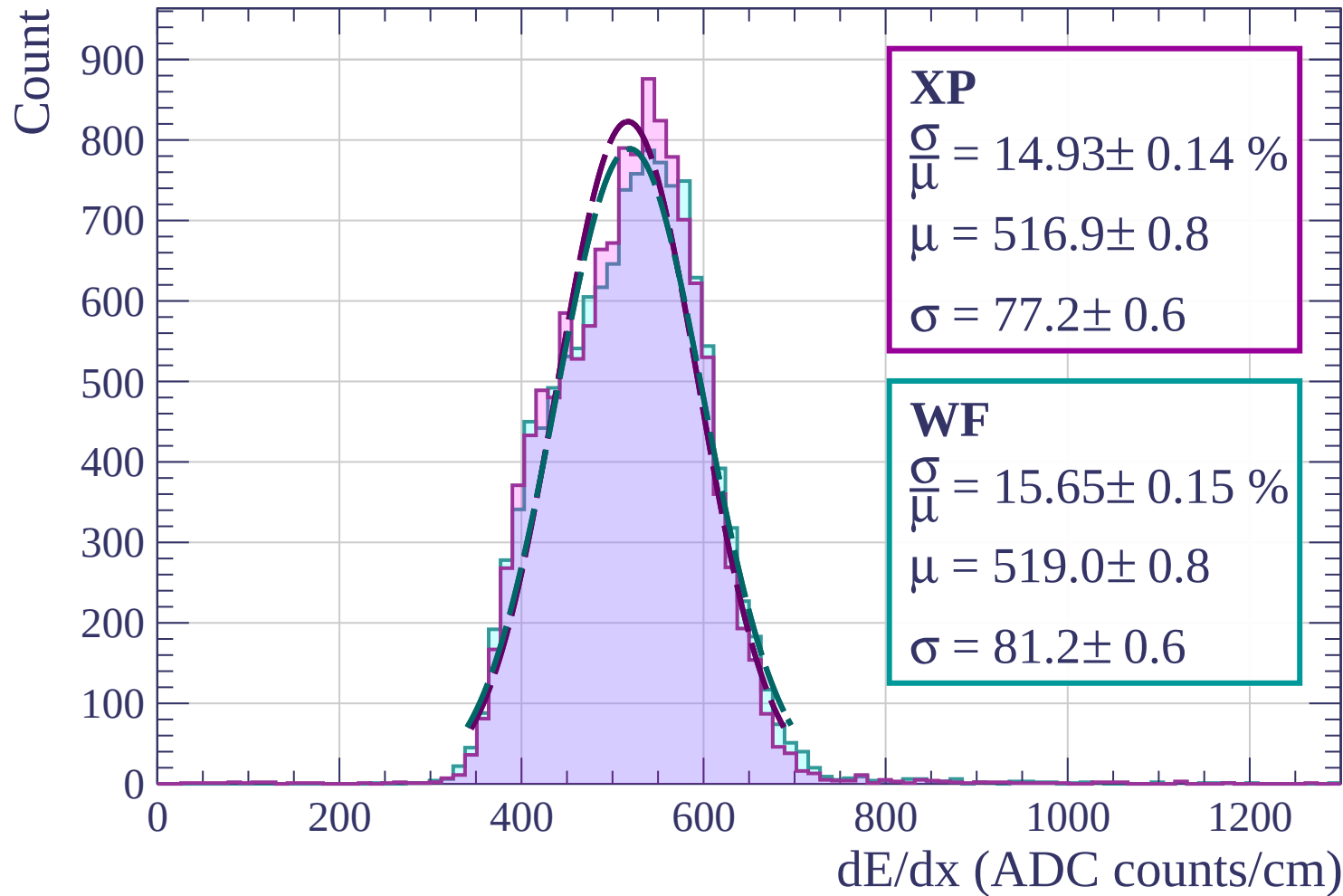


Energy loss | $36 < \phi < 39$

Count



Energy loss | $39 < \phi < 42$



Energy loss | $42 < \phi < 45$

Count

1000

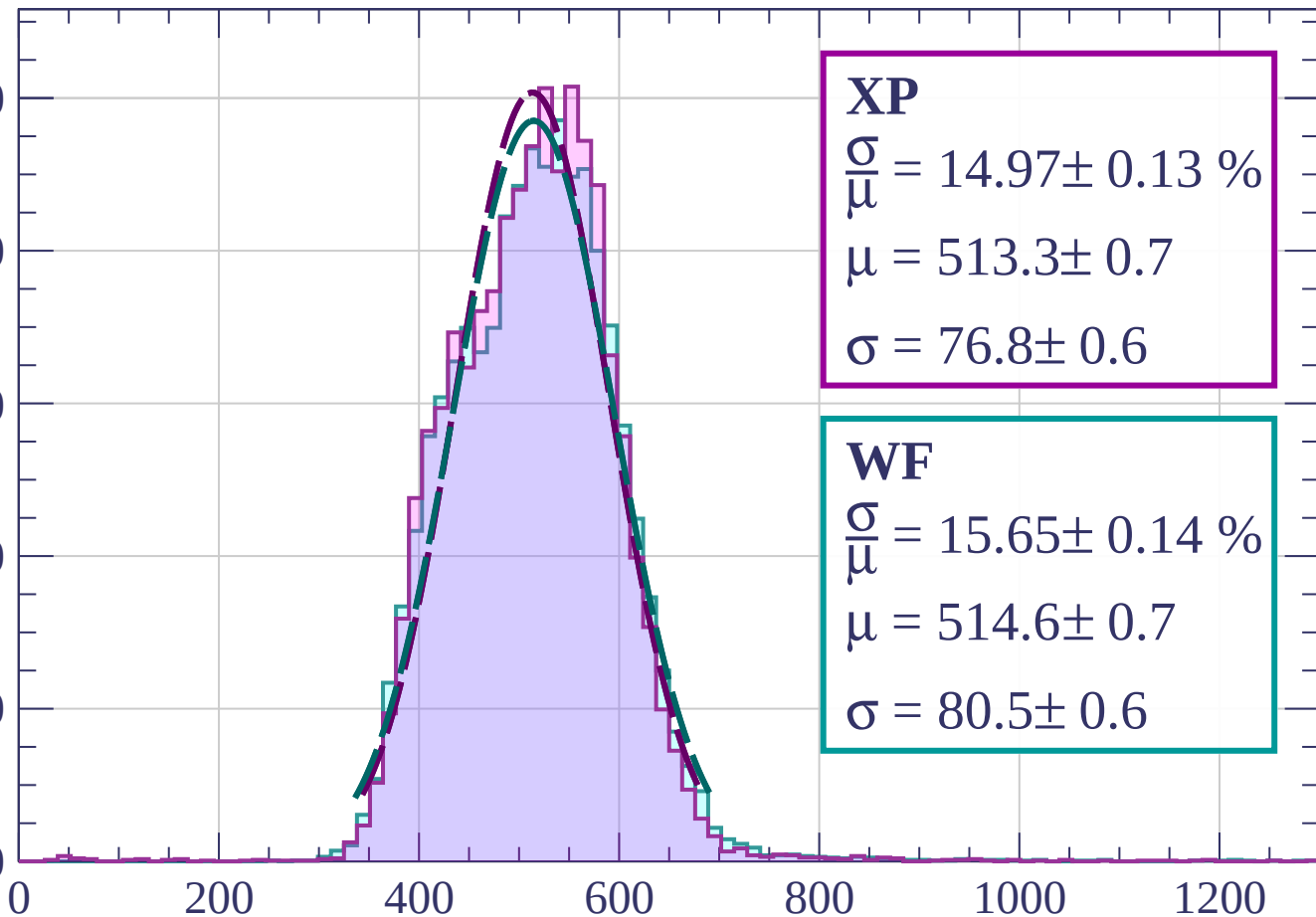
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 14.97 \pm 0.13 \%$$

$$\mu = 513.3 \pm 0.7$$

$$\sigma = 76.8 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 15.65 \pm 0.14 \%$$

$$\mu = 514.6 \pm 0.7$$

$$\sigma = 80.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $45 < \phi < 48$

Count

1200

1000

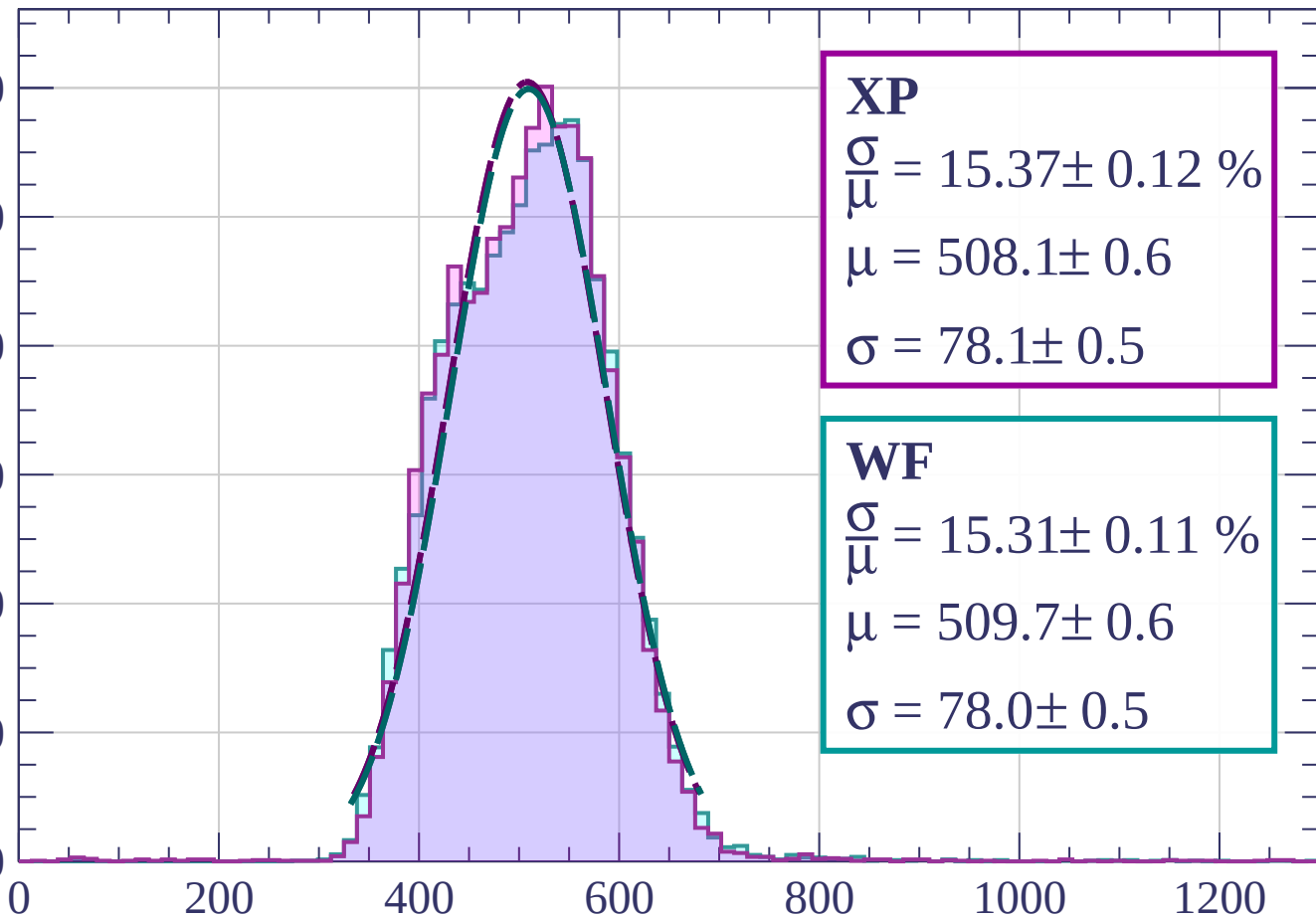
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 15.37 \pm 0.12 \%$$

$$\mu = 508.1 \pm 0.6$$

$$\sigma = 78.1 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 15.31 \pm 0.11 \%$$

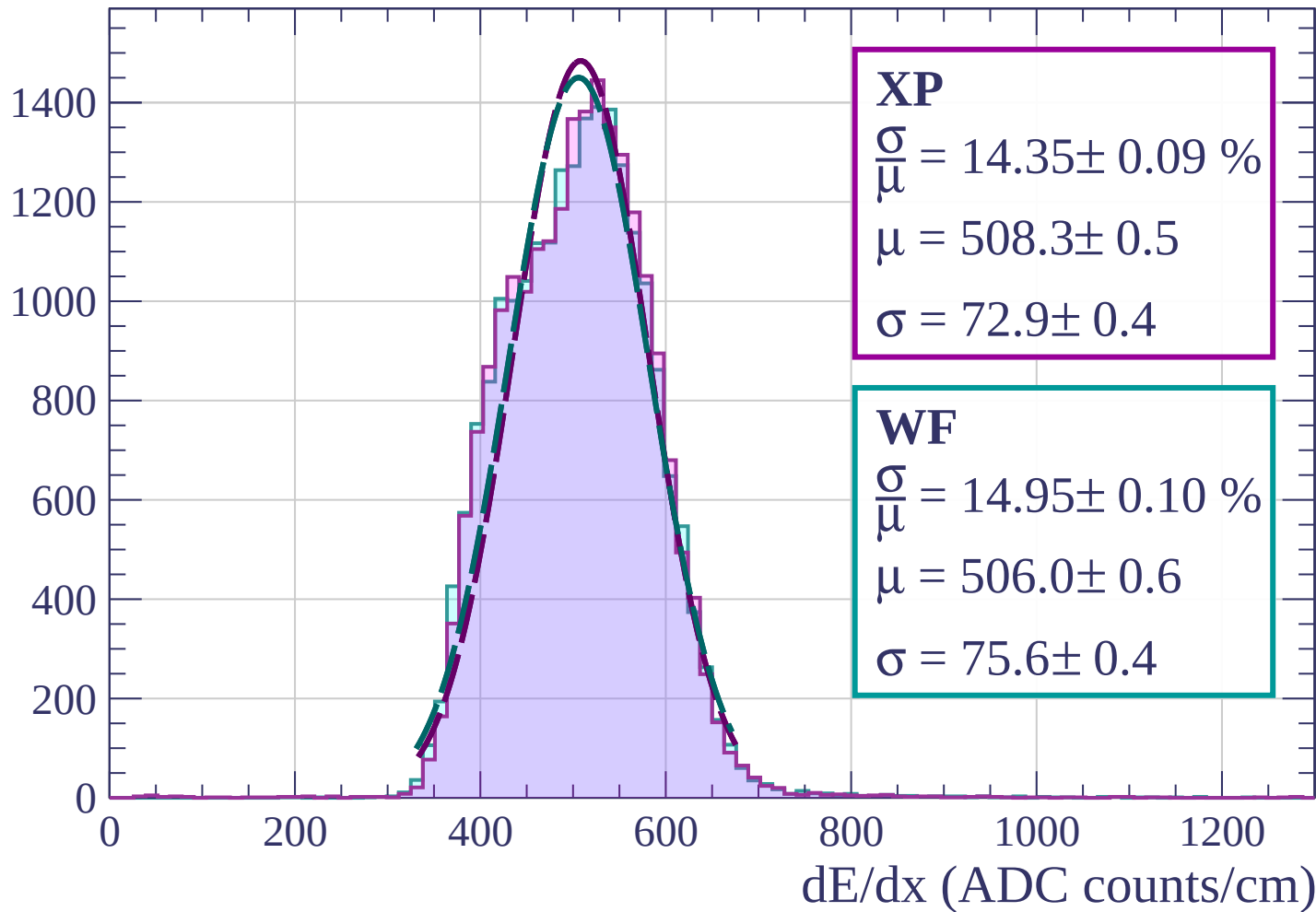
$$\mu = 509.7 \pm 0.6$$

$$\sigma = 78.0 \pm 0.5$$

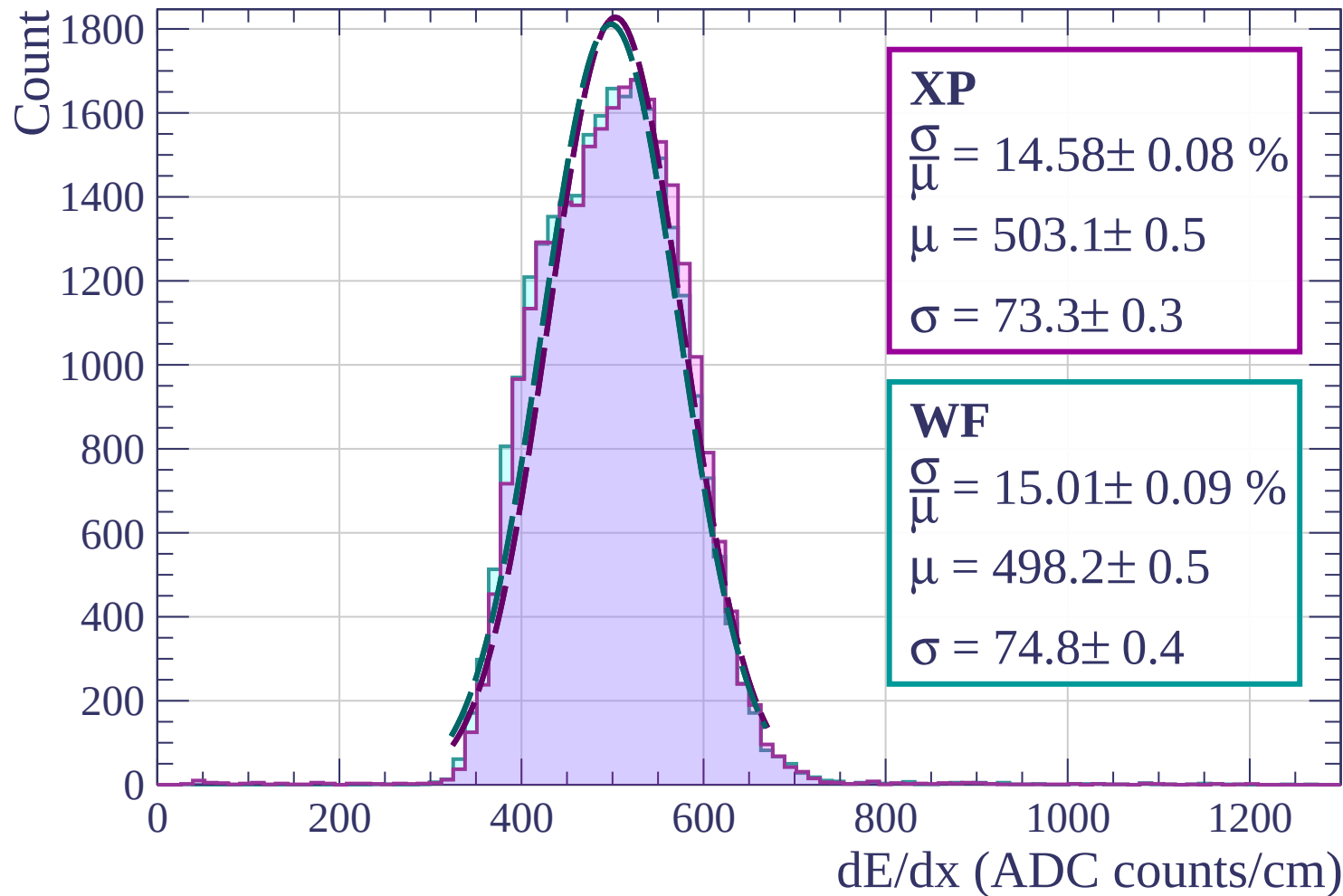
dE/dx (ADC counts/cm)

Energy loss | $48 < \phi < 51$

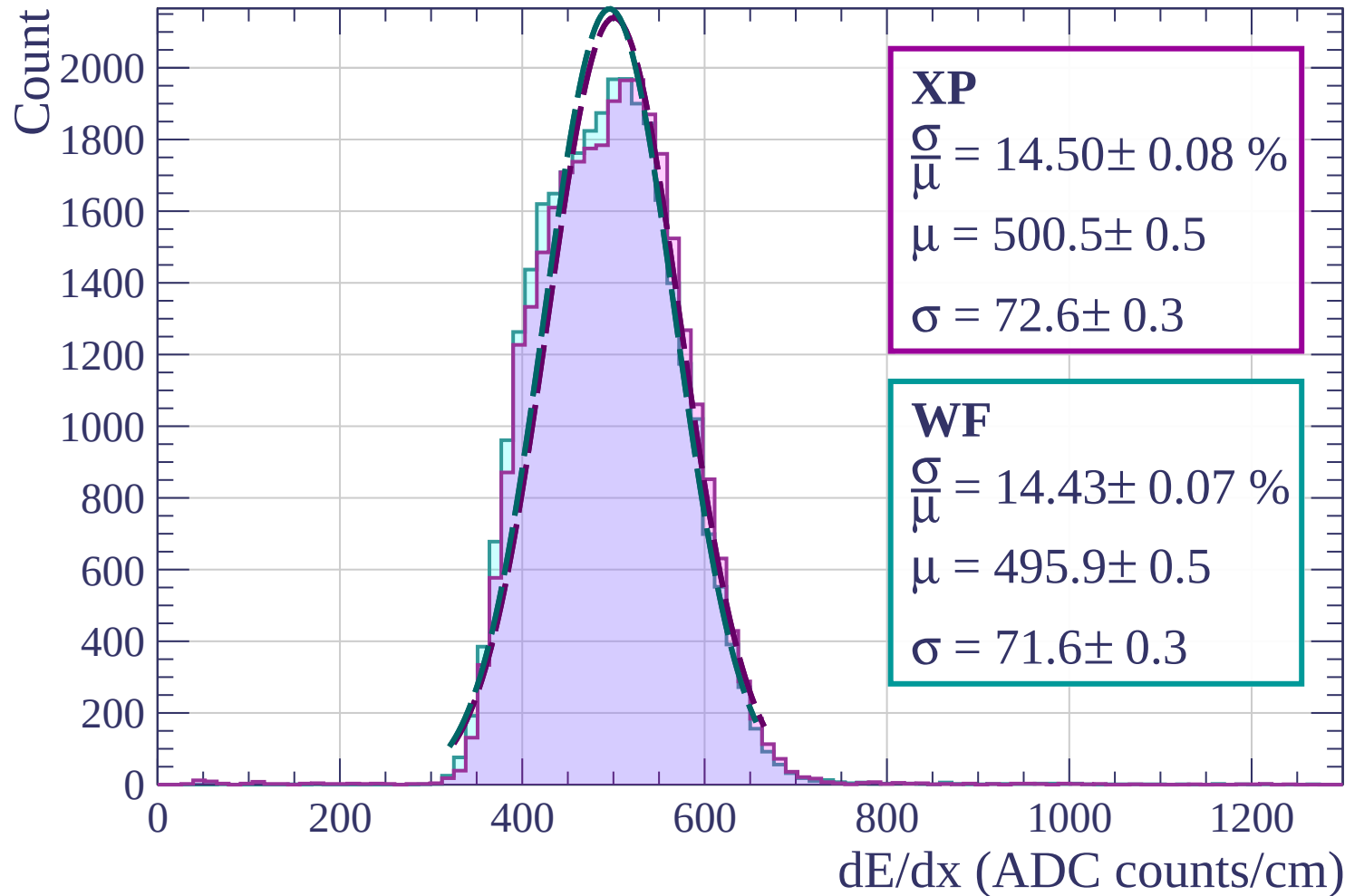
Count



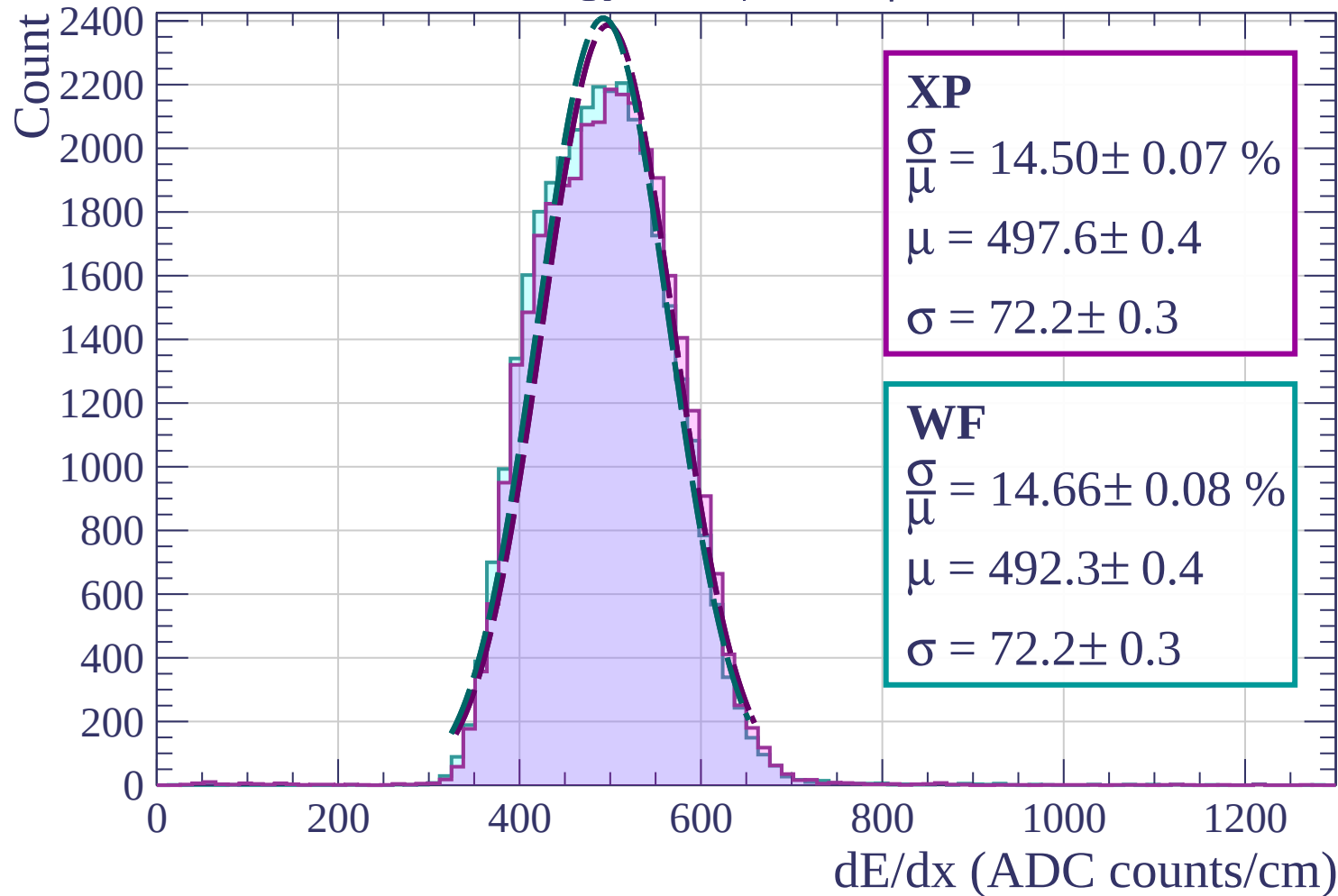
Energy loss | $51 < \phi < 54$



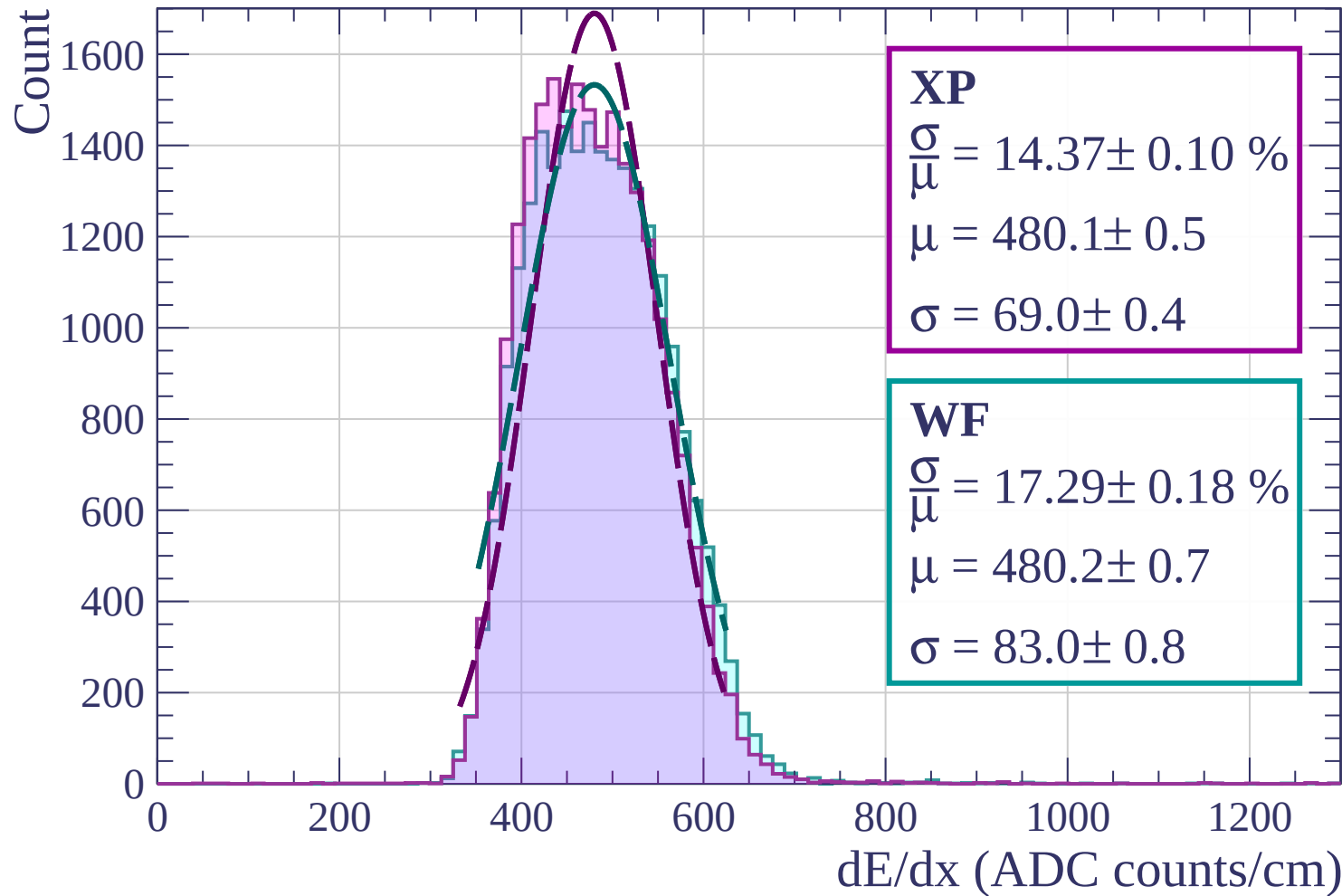
Energy loss | $54 < \phi < 57$



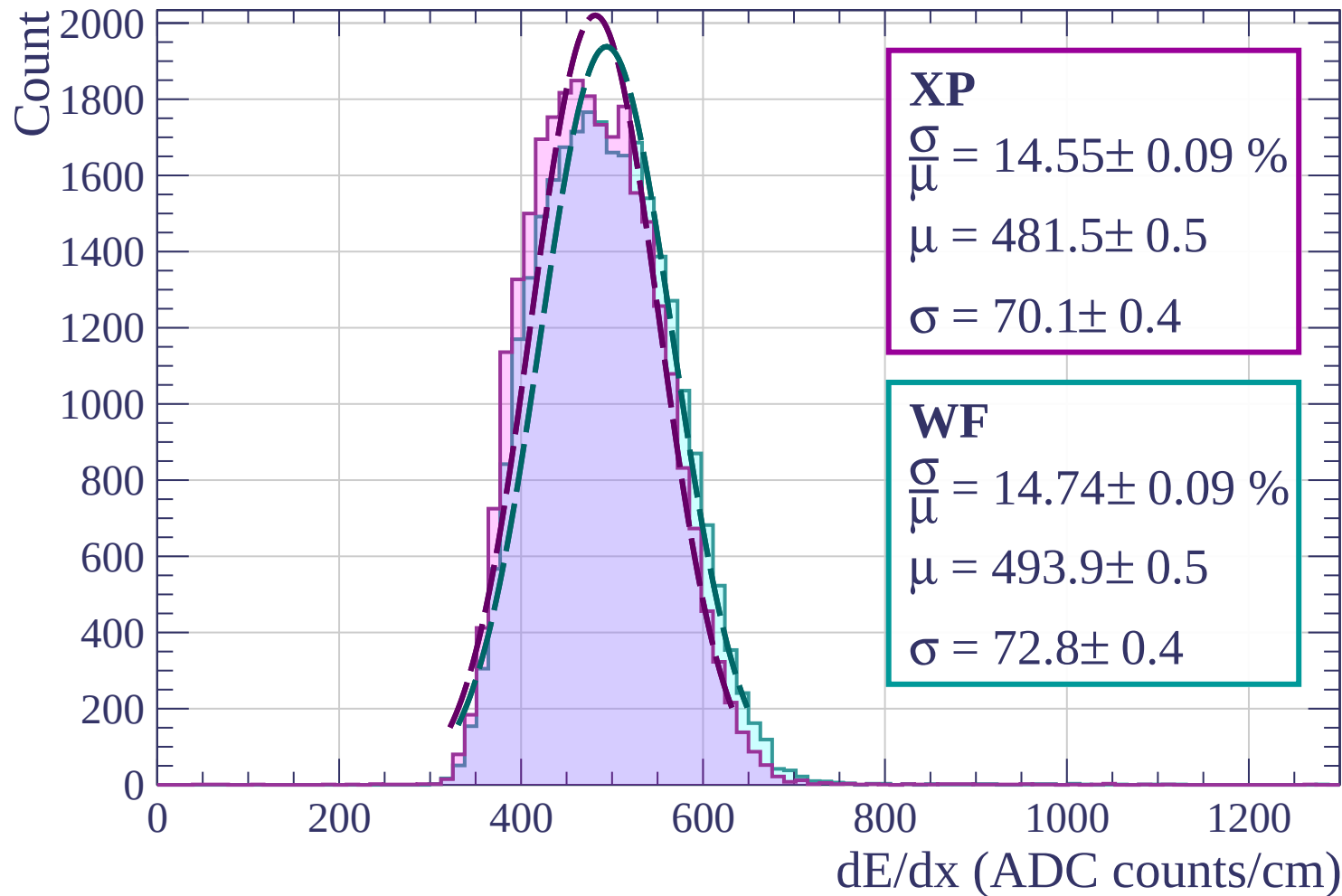
Energy loss | $57 < \phi < 60$



Energy loss | $60 < \phi < 63$

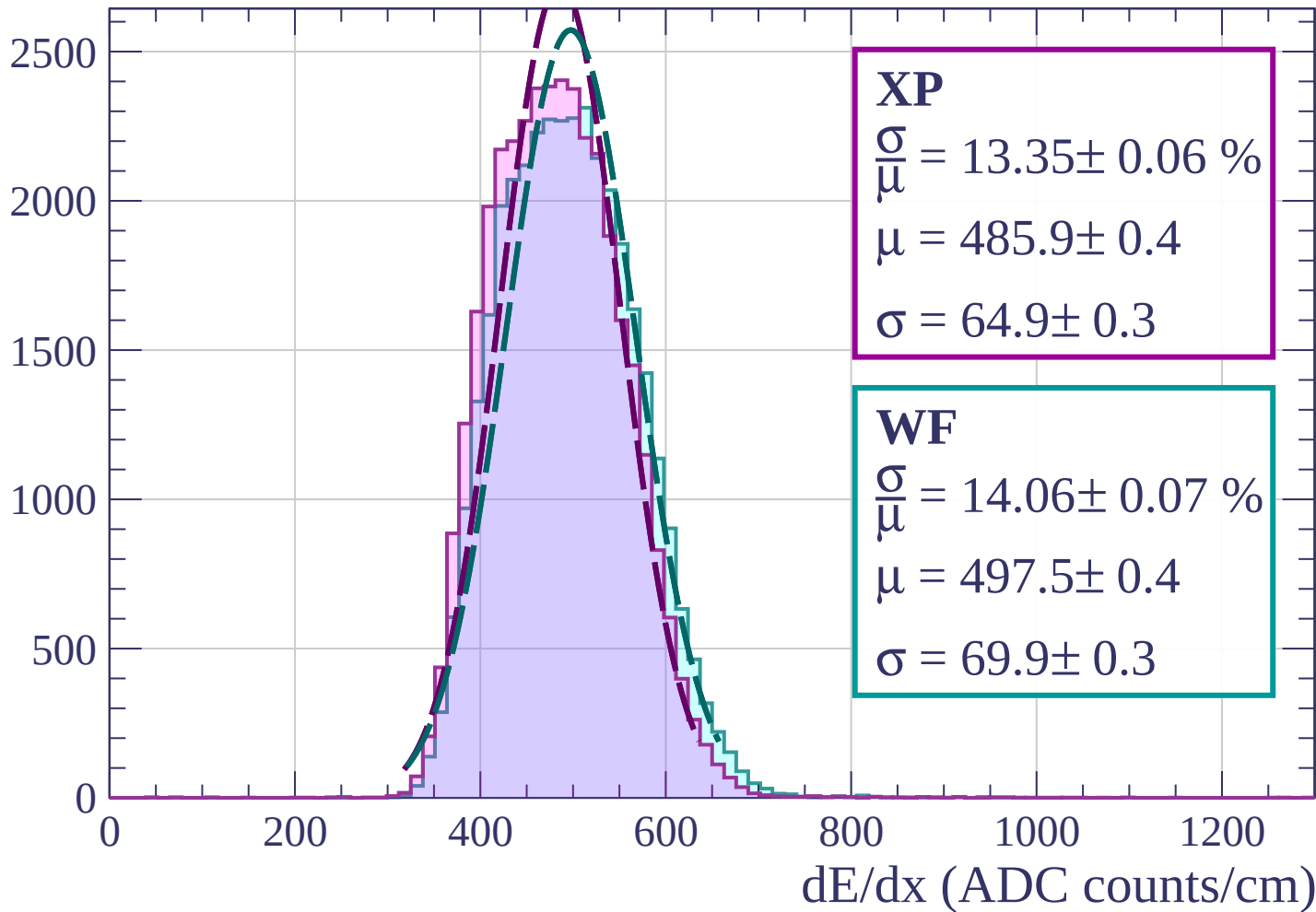


Energy loss | $63 < \phi < 66$

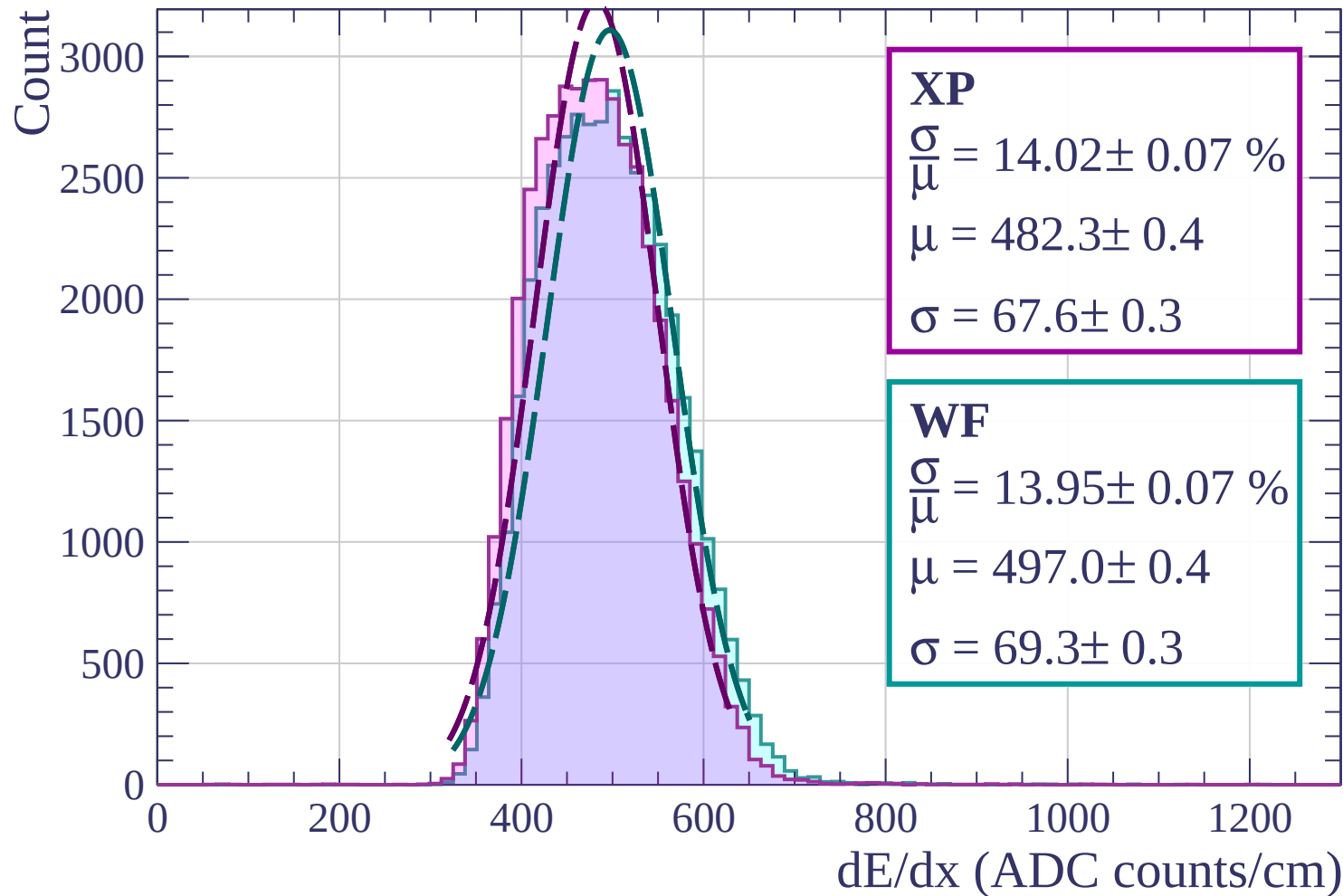


Energy loss | $66 < \phi < 69$

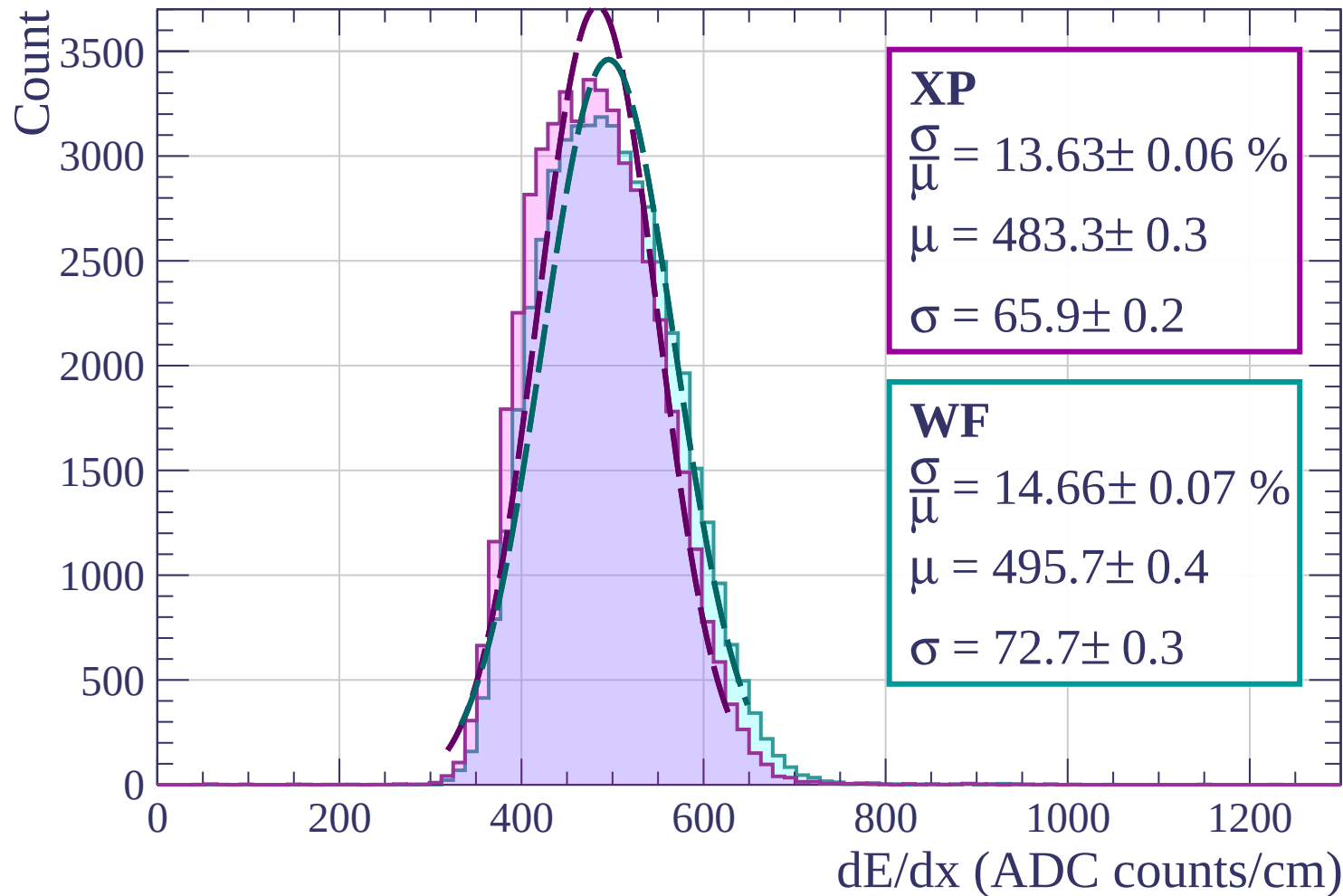
Count



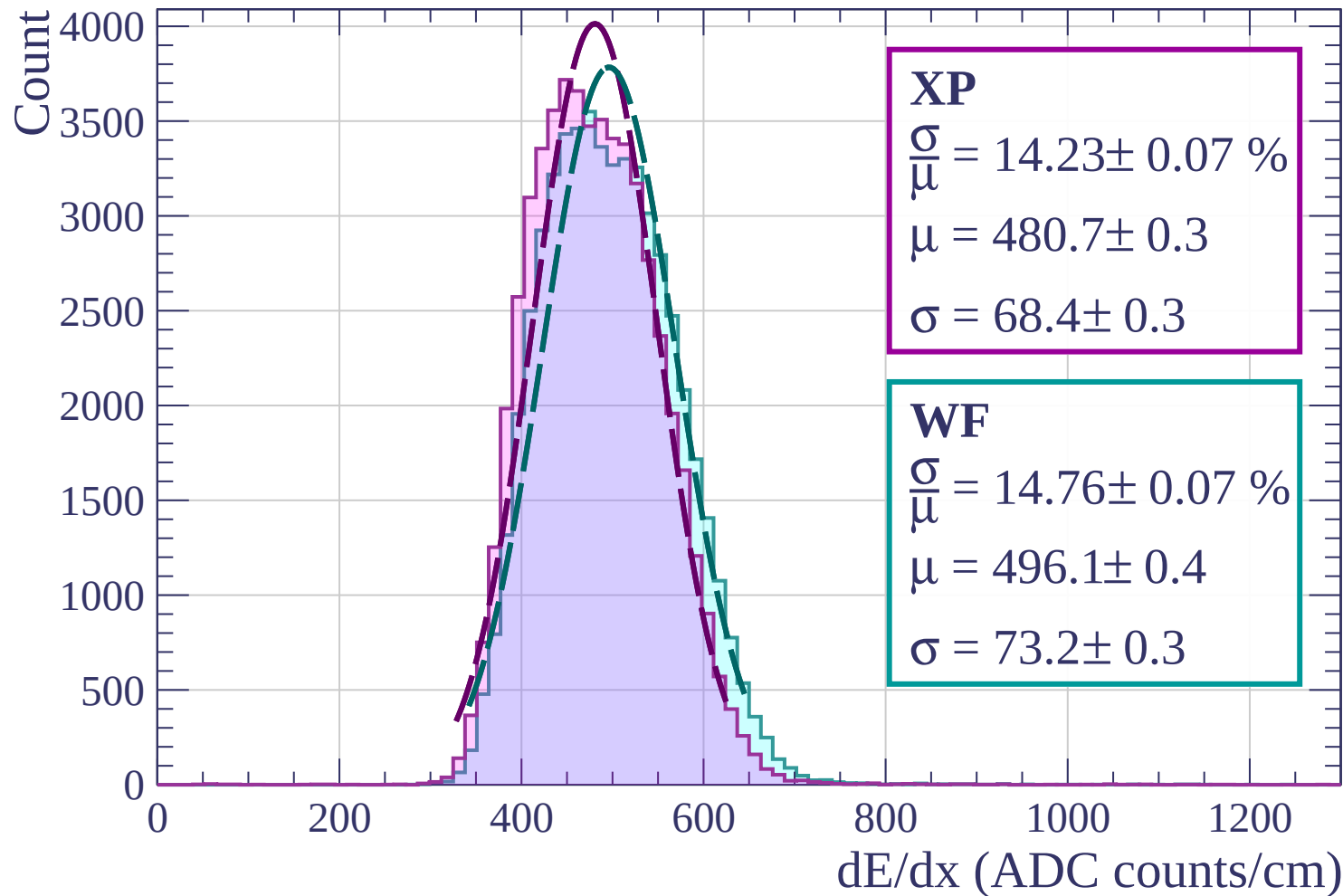
Energy loss | $69 < \phi < 72$



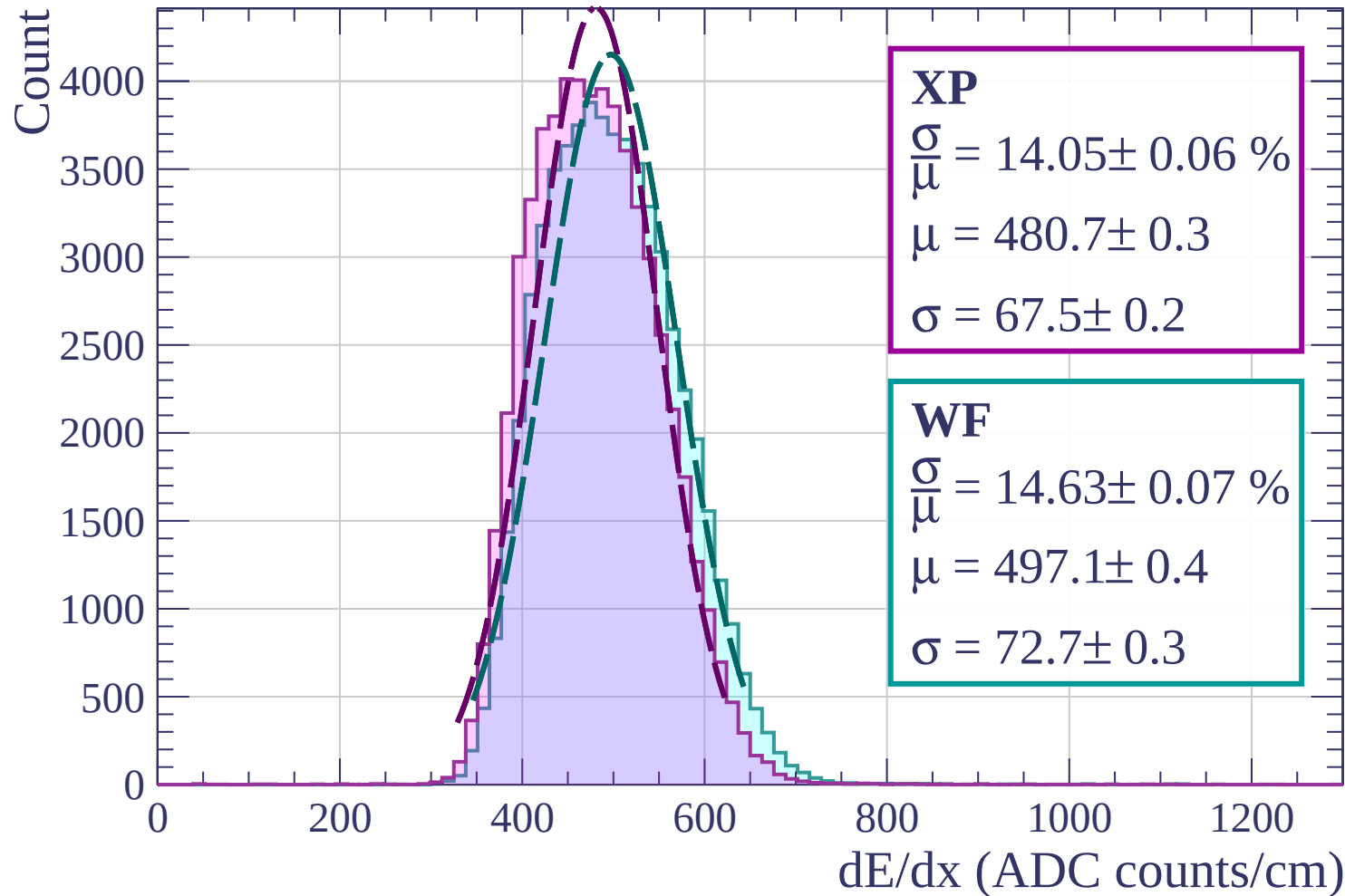
Energy loss | $72 < \phi < 75$



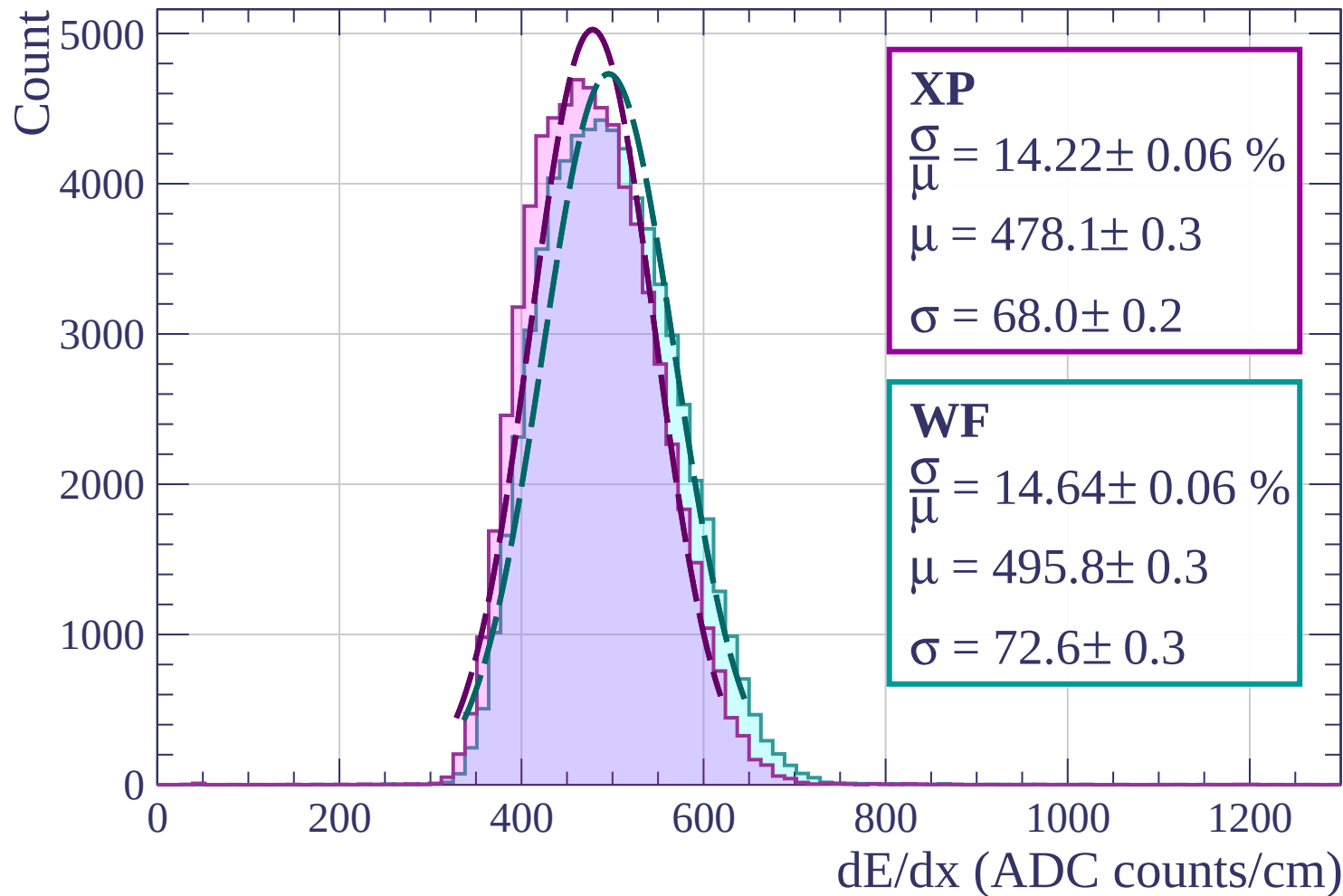
Energy loss | $75 < \phi < 78$



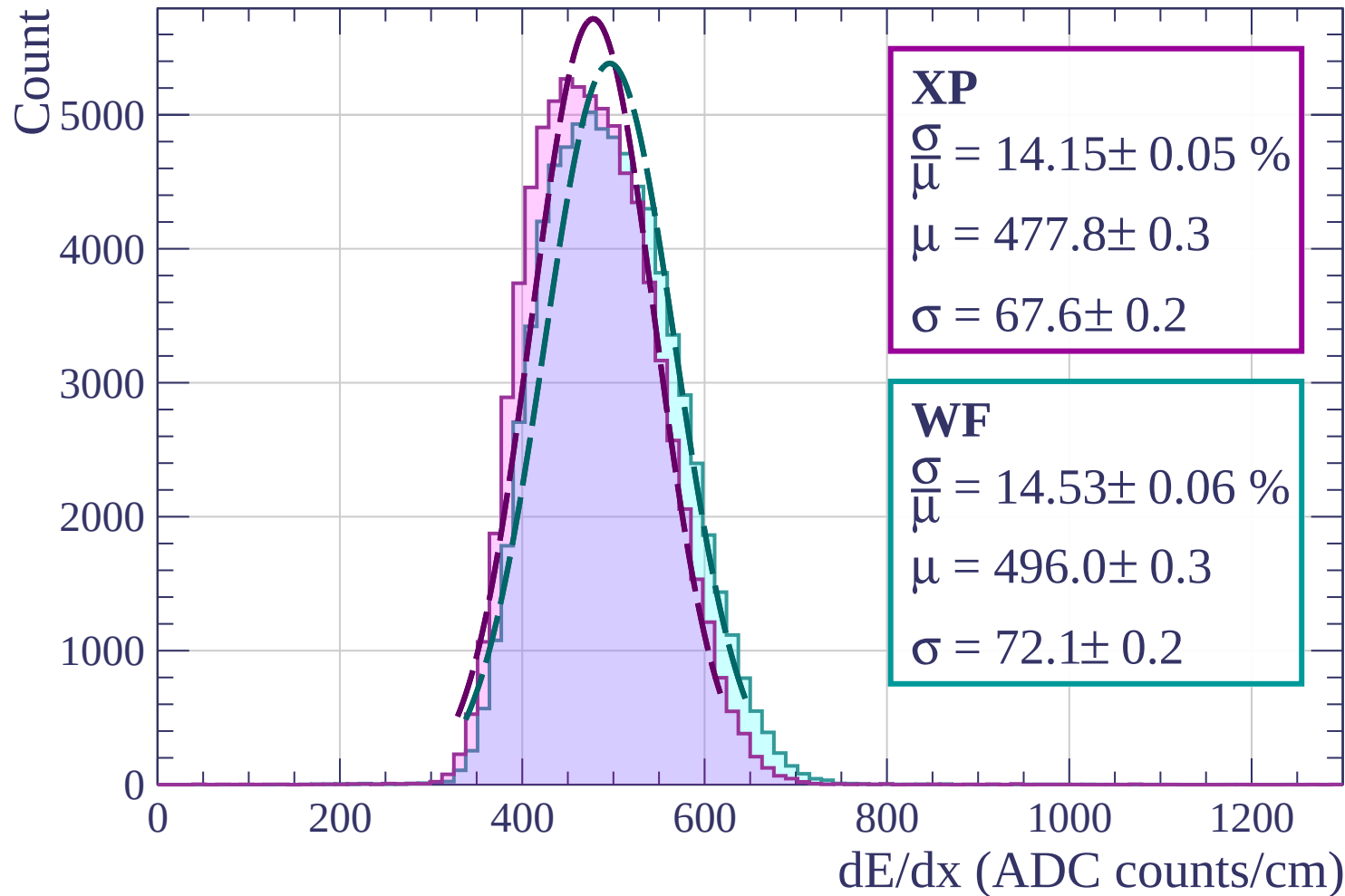
Energy loss | $78 < \phi < 81$



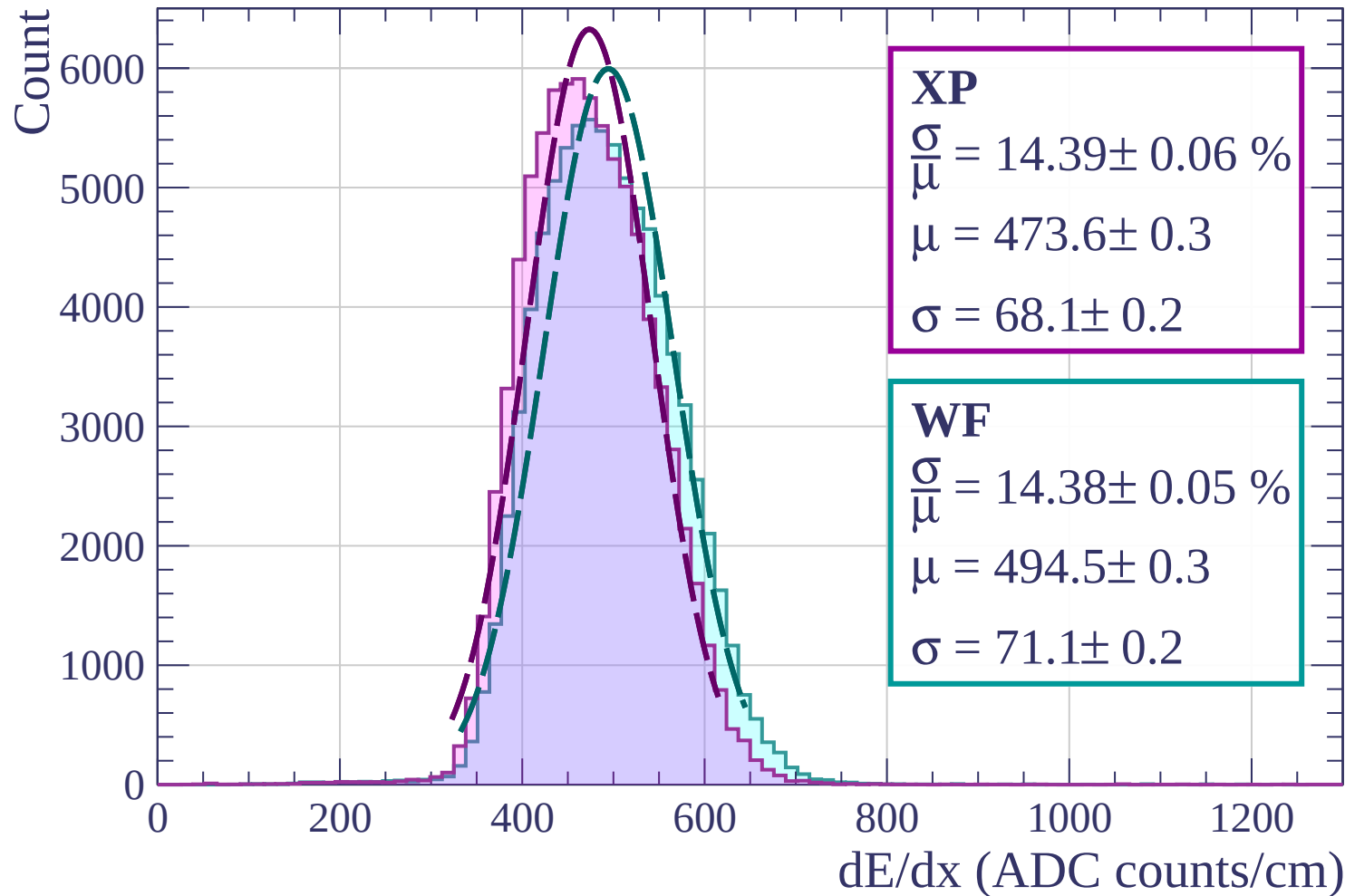
Energy loss | $81 < \phi < 84$



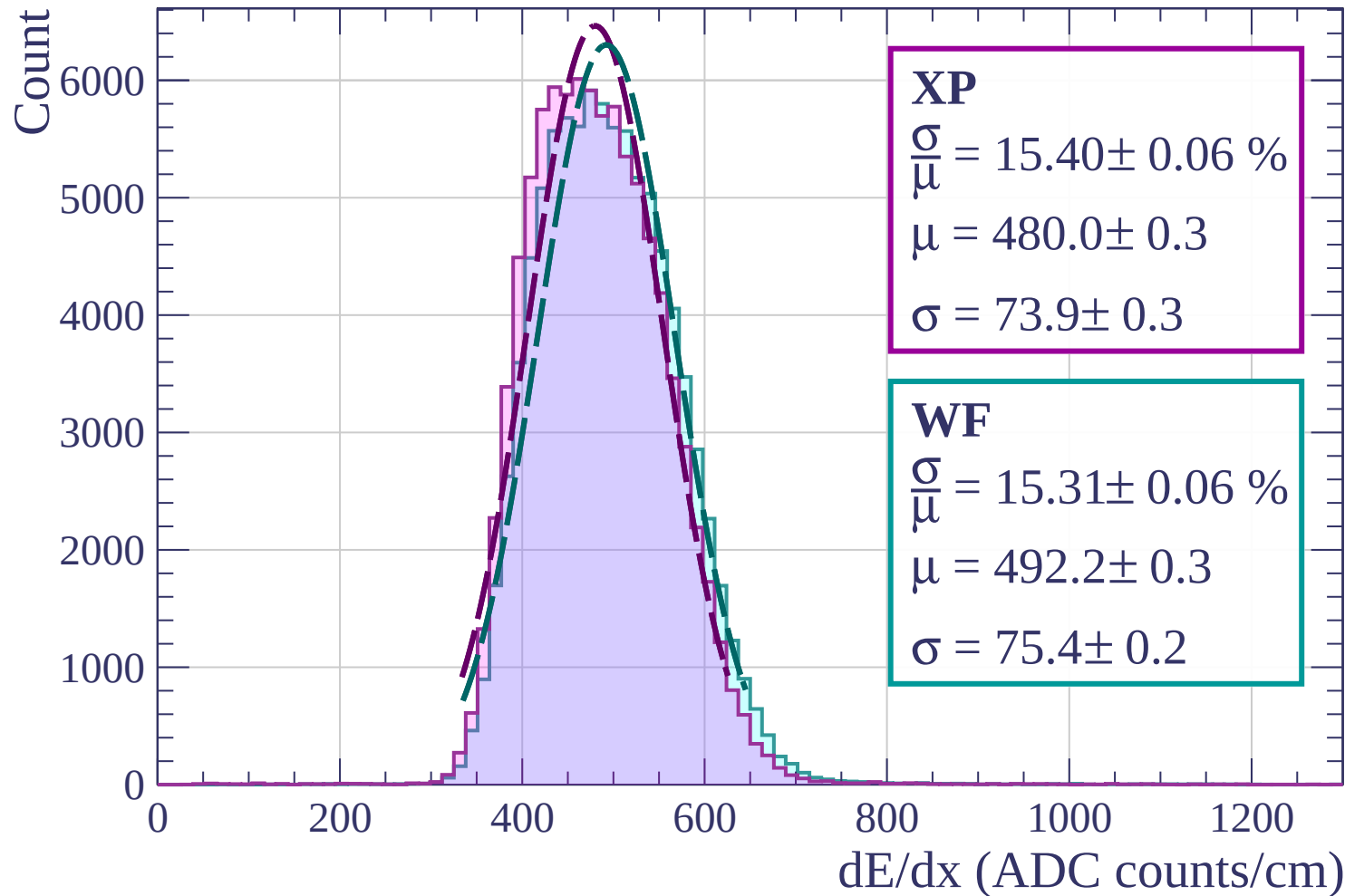
Energy loss | $84 < \phi < 87$



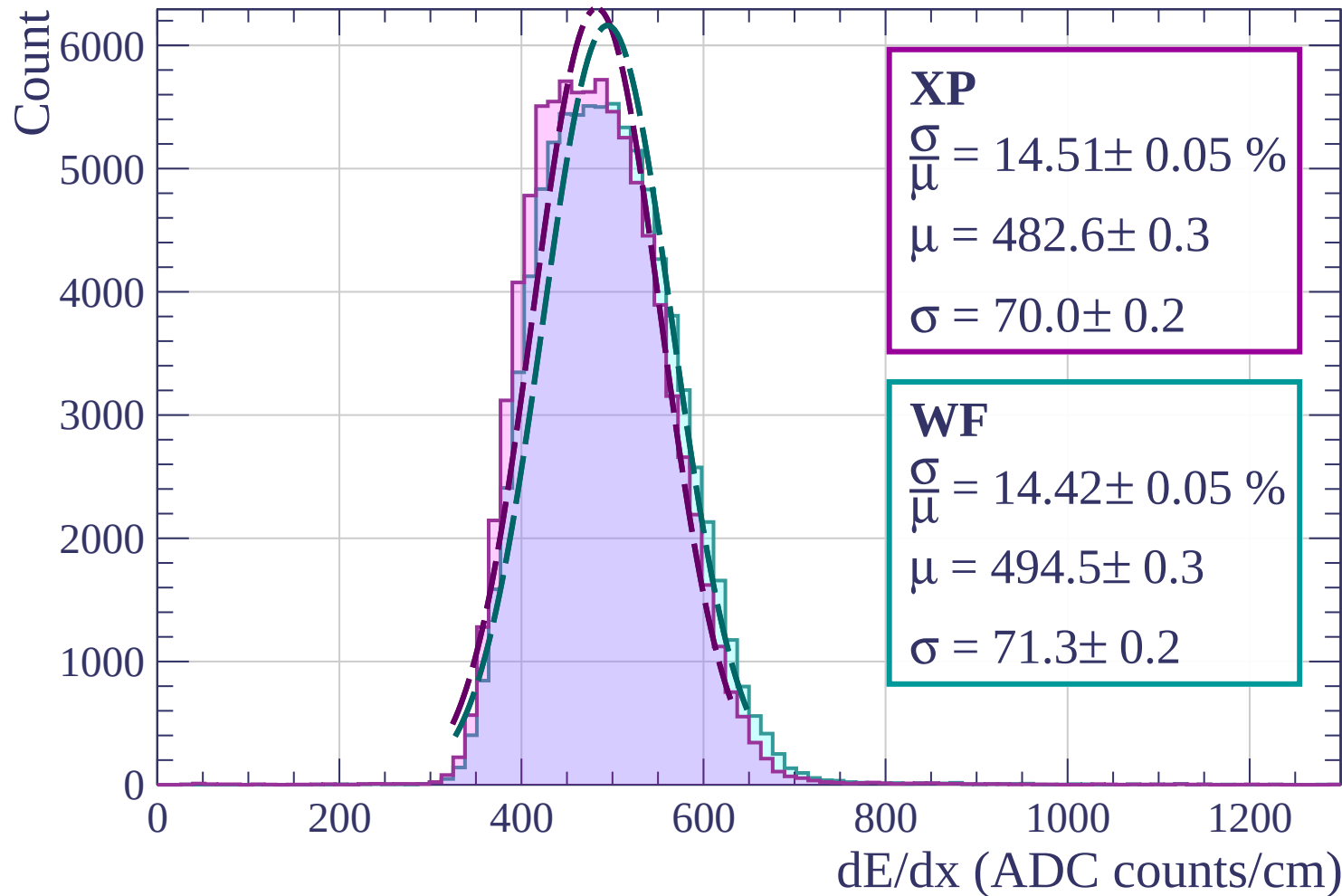
Energy loss | $87 < \phi < 90$



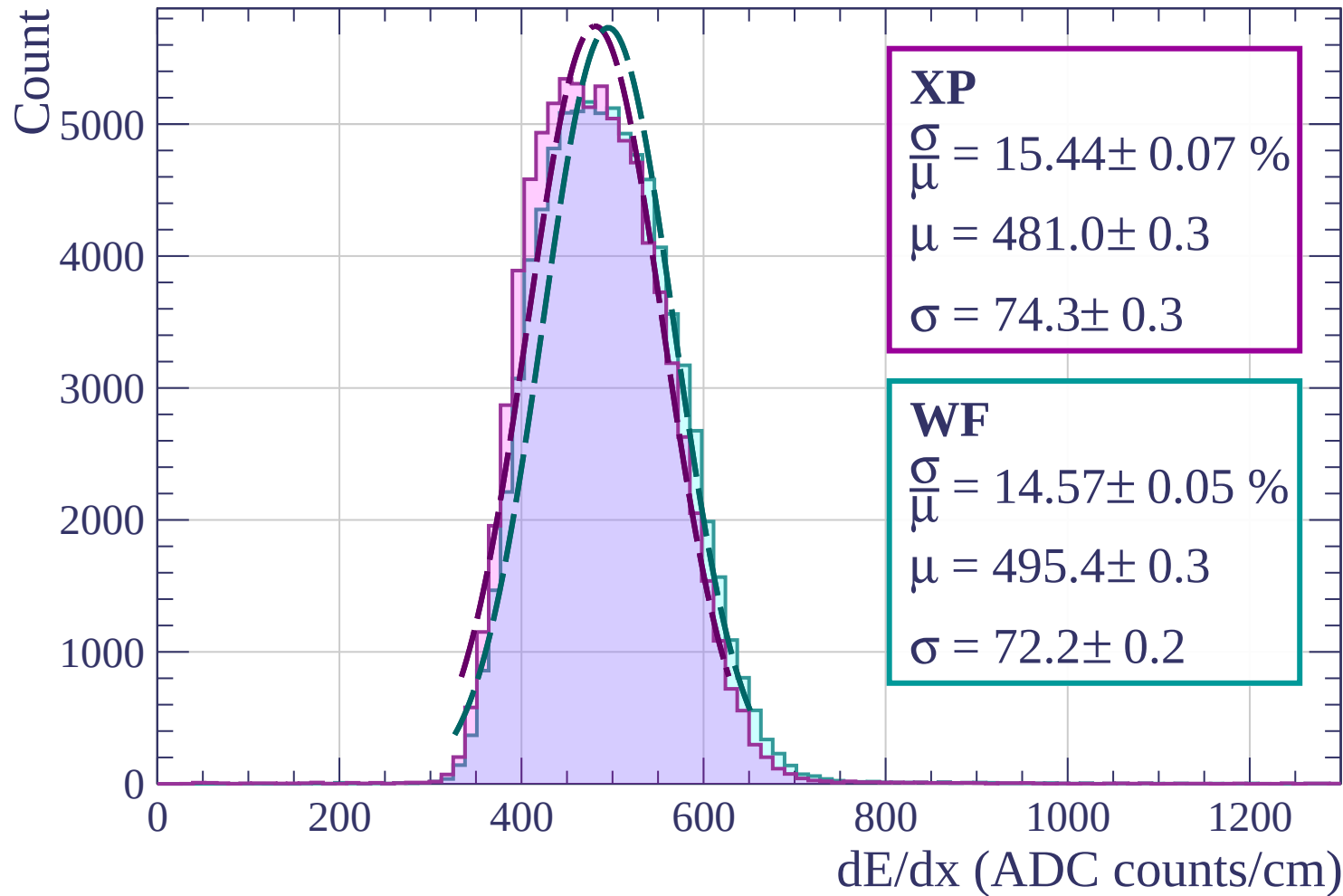
Energy loss | $0 < \theta < 3$



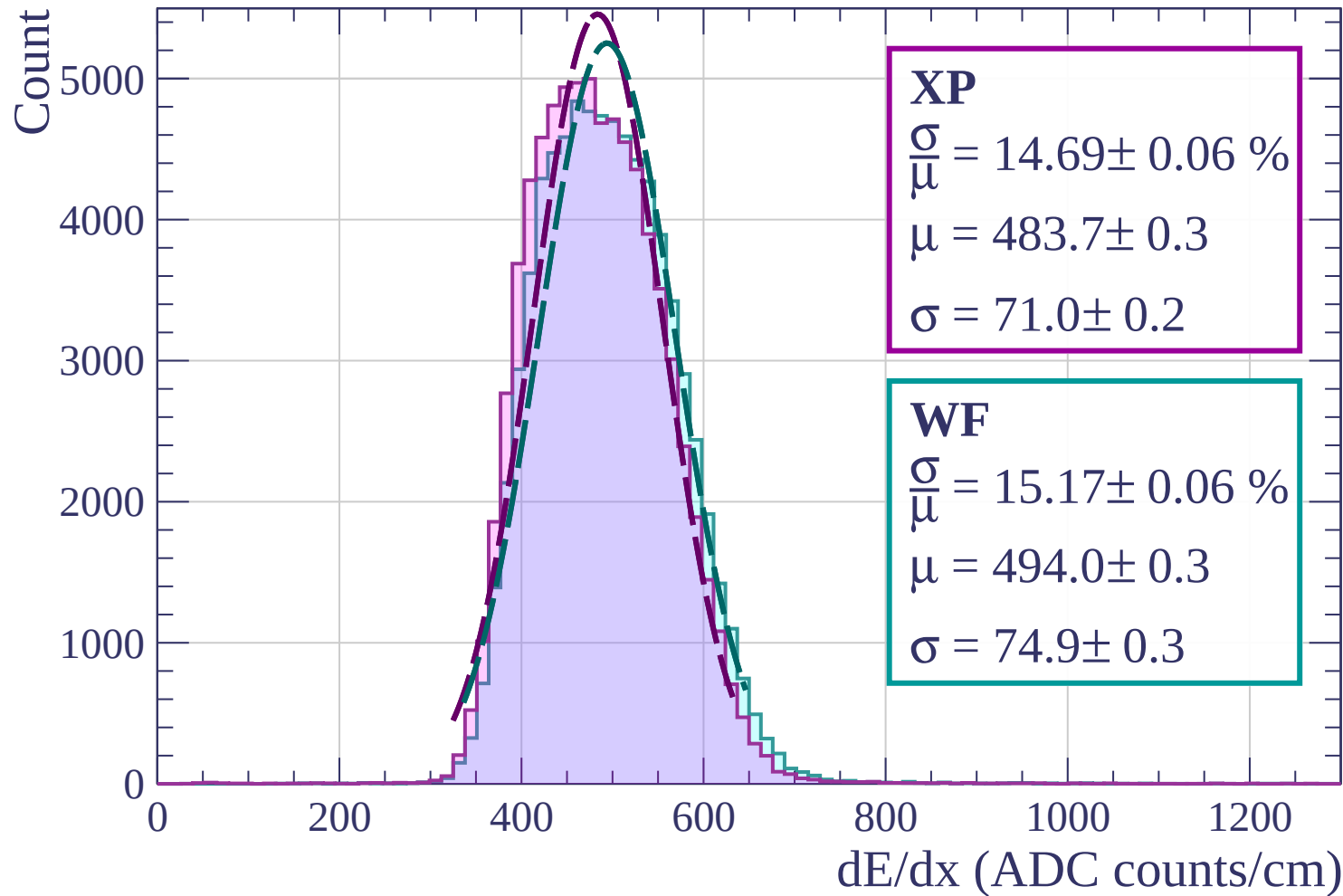
Energy loss | $3 < \theta < 6$



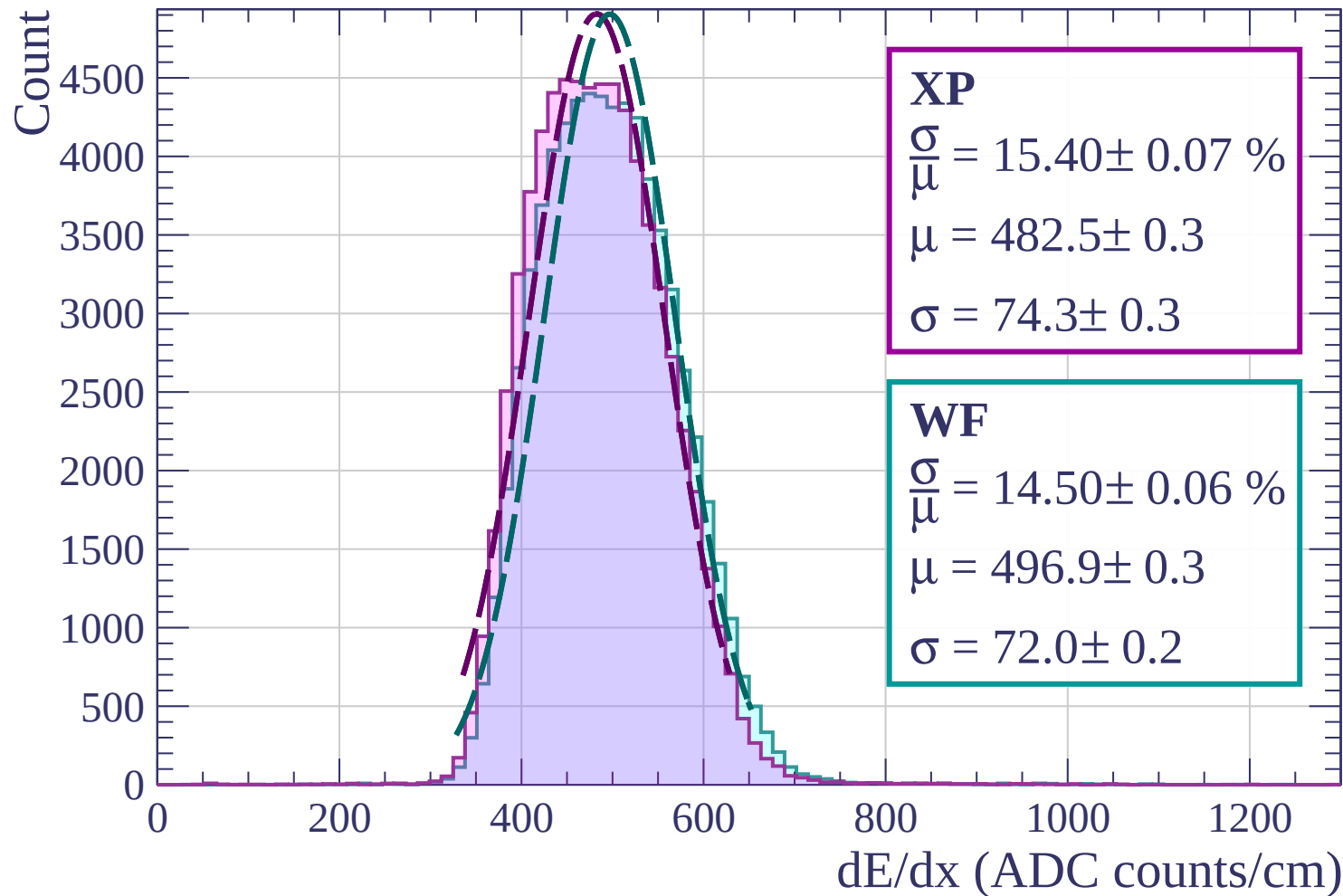
Energy loss | $6 < \theta < 9$



Energy loss | $9 < \theta < 12$

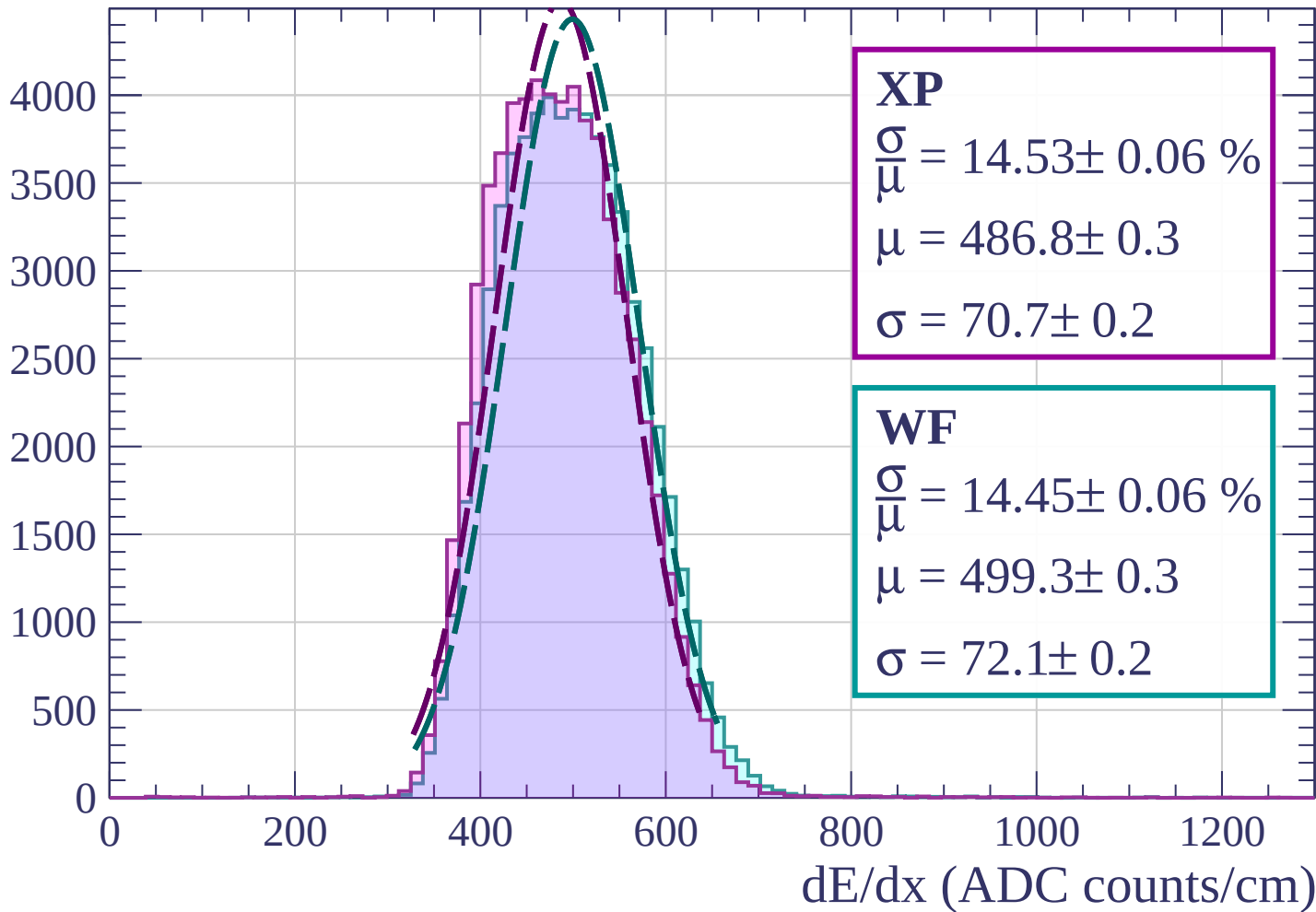


Energy loss | $12 < \theta < 15$

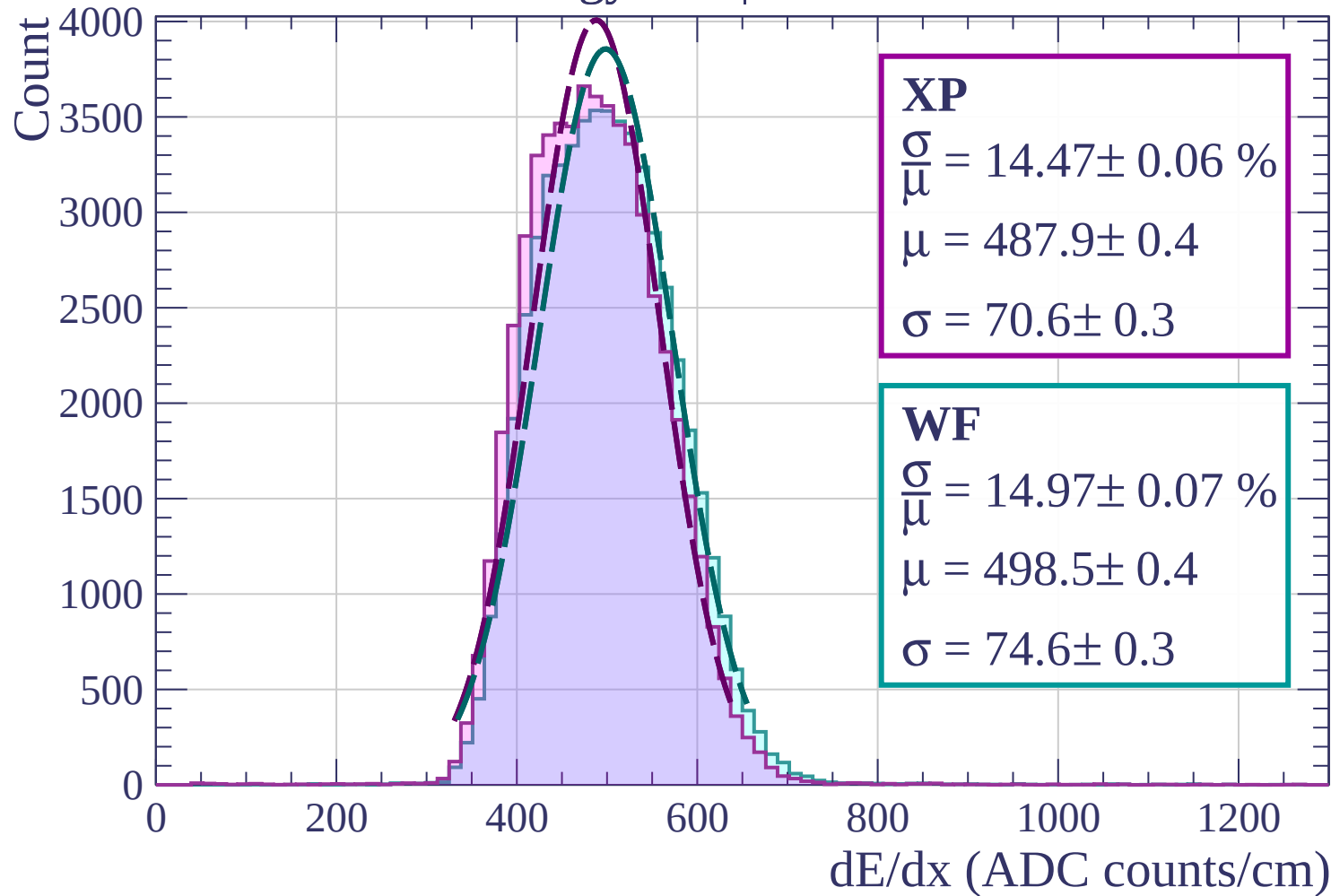


Energy loss | $15 < \theta < 18$

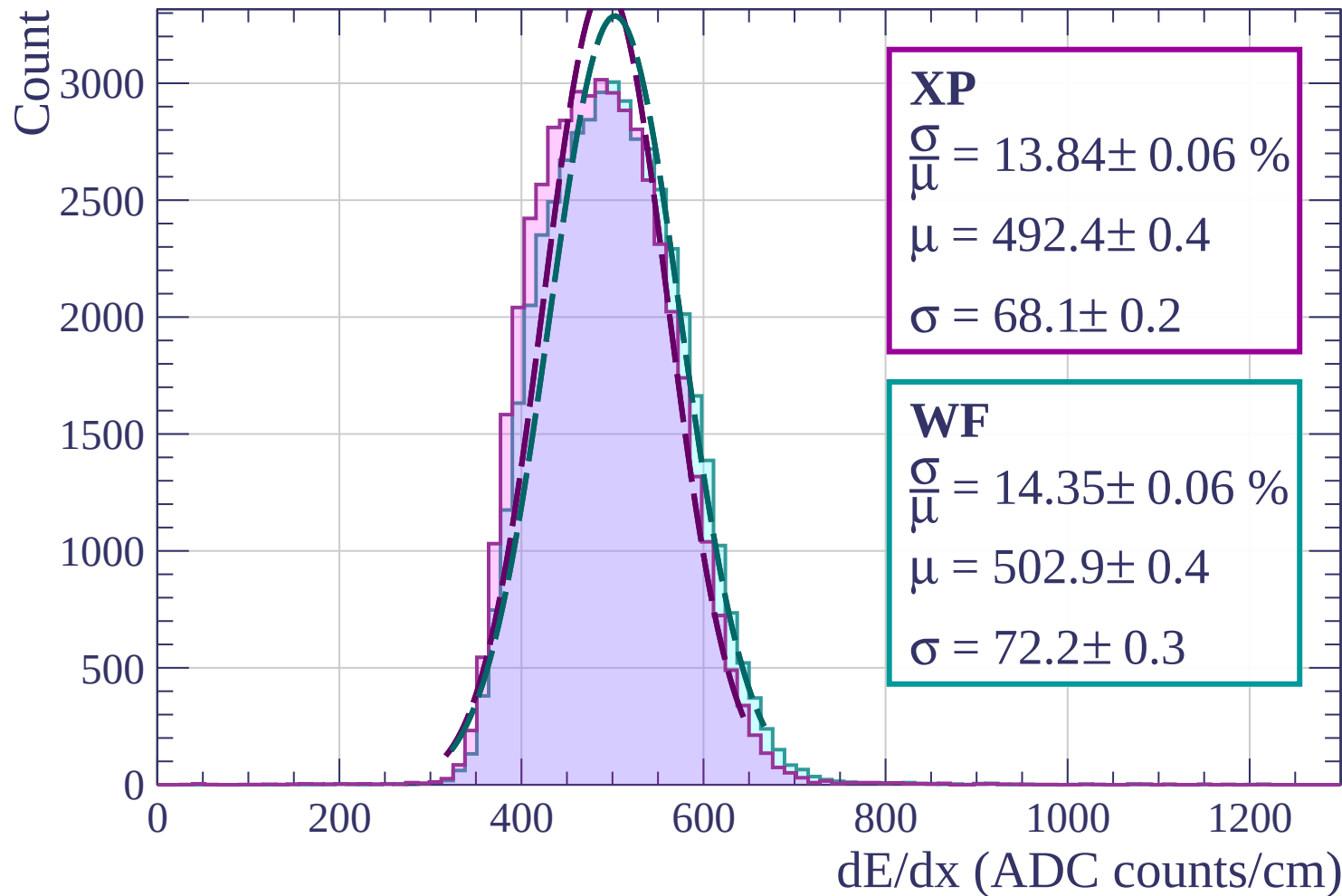
Count



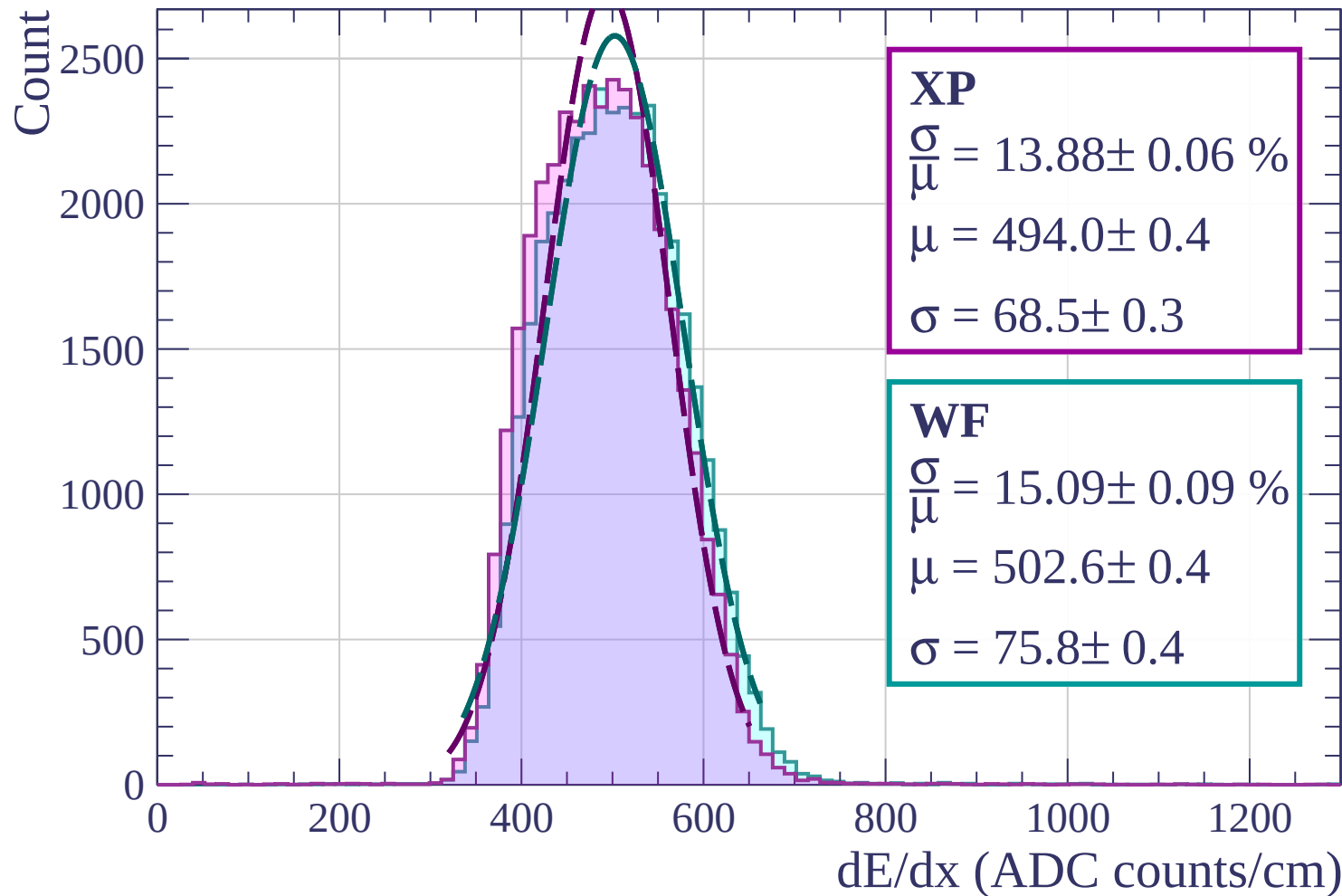
Energy loss | $18 < \theta < 21$



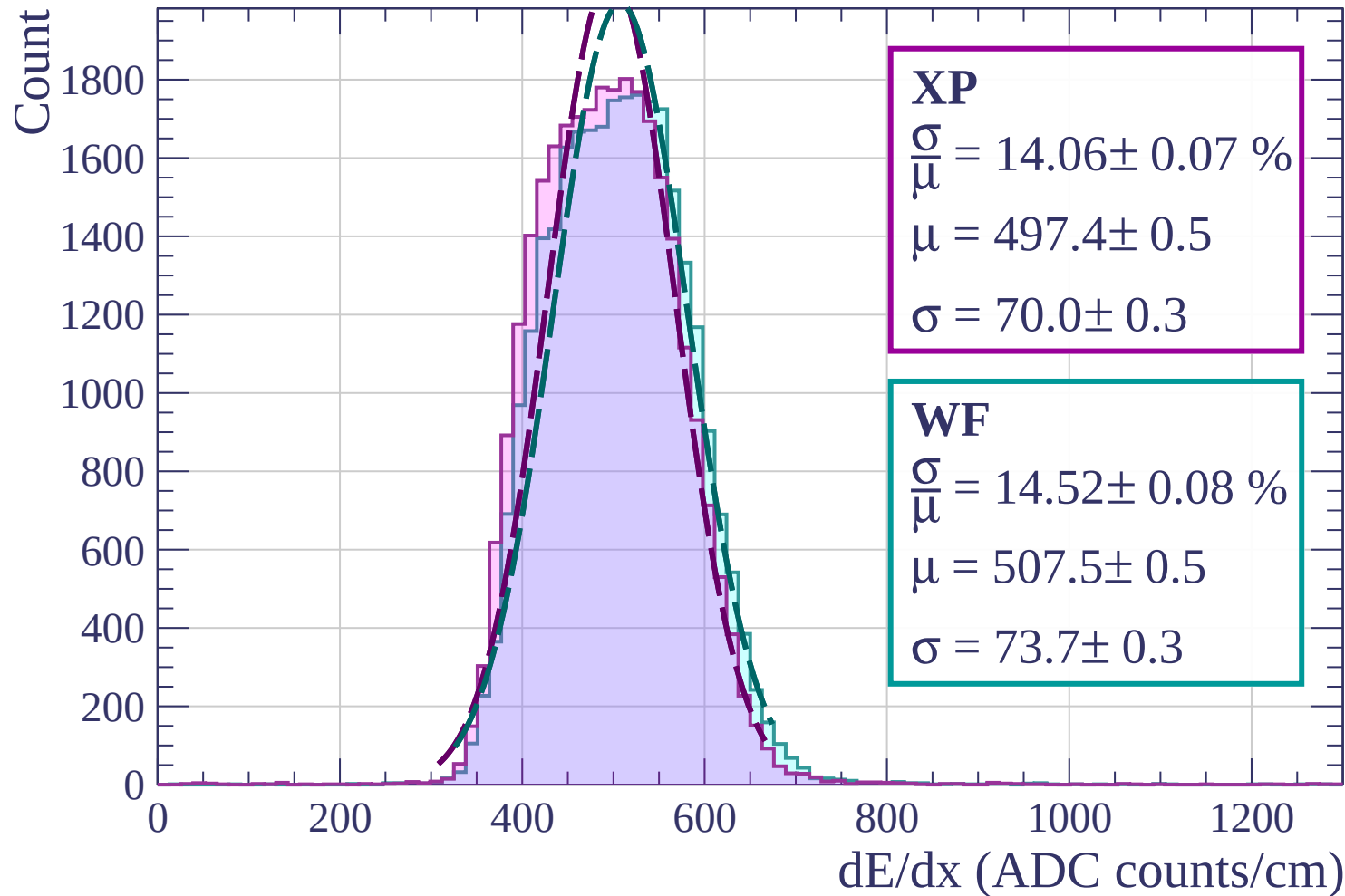
Energy loss | $21 < \theta < 24$



Energy loss | $24 < \theta < 27$

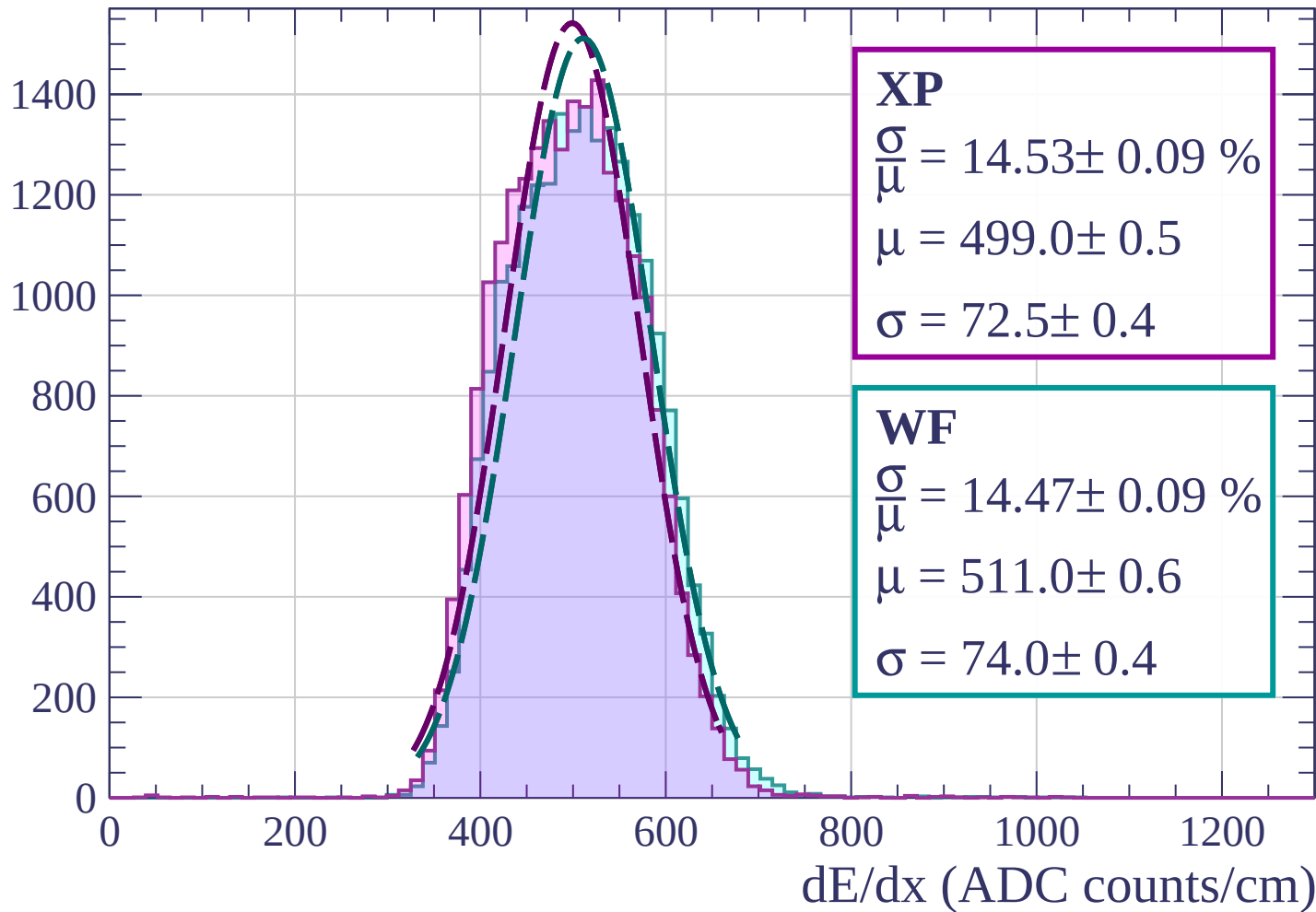


Energy loss | $27 < \theta < 30$

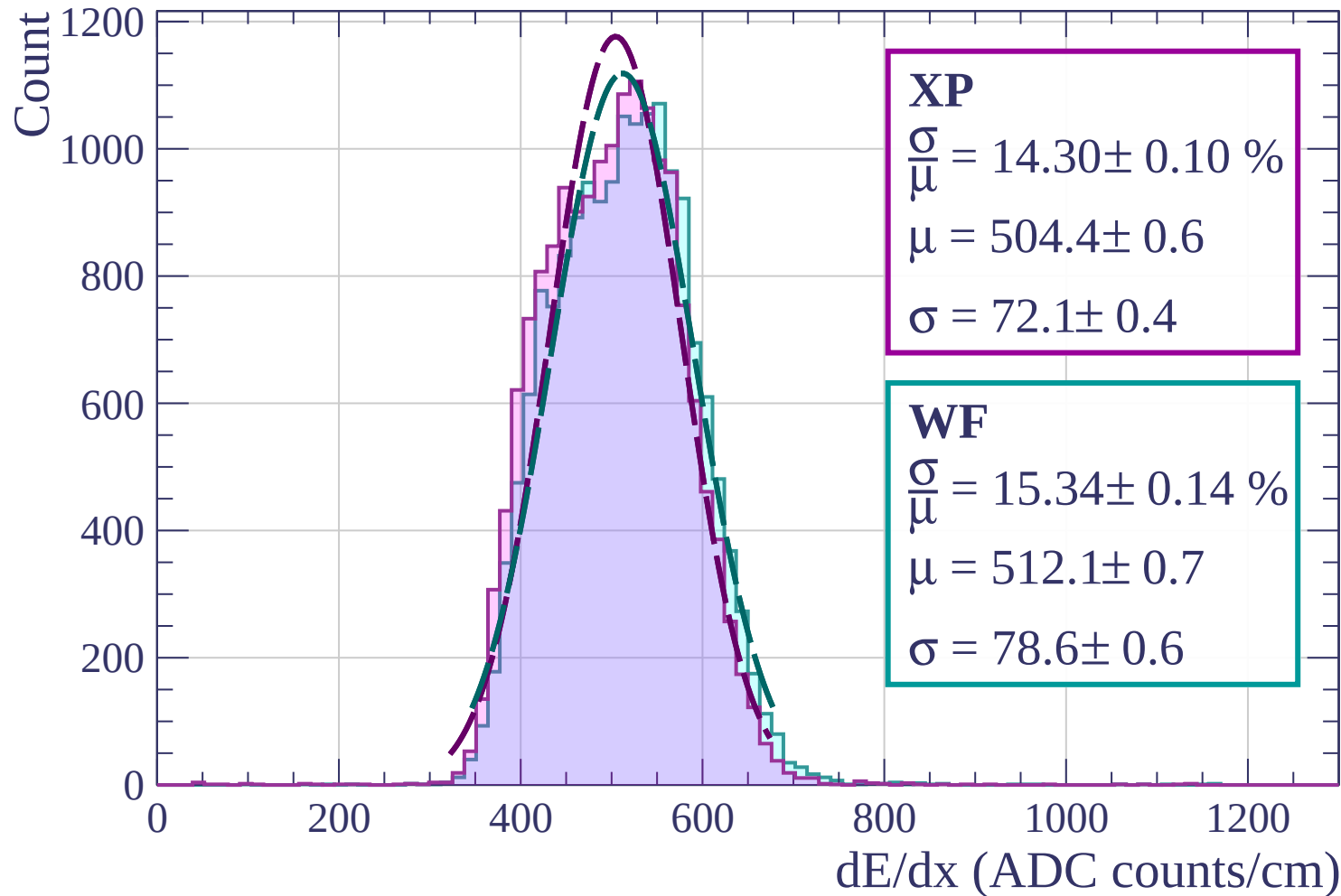


Energy loss | $30 < \theta < 33$

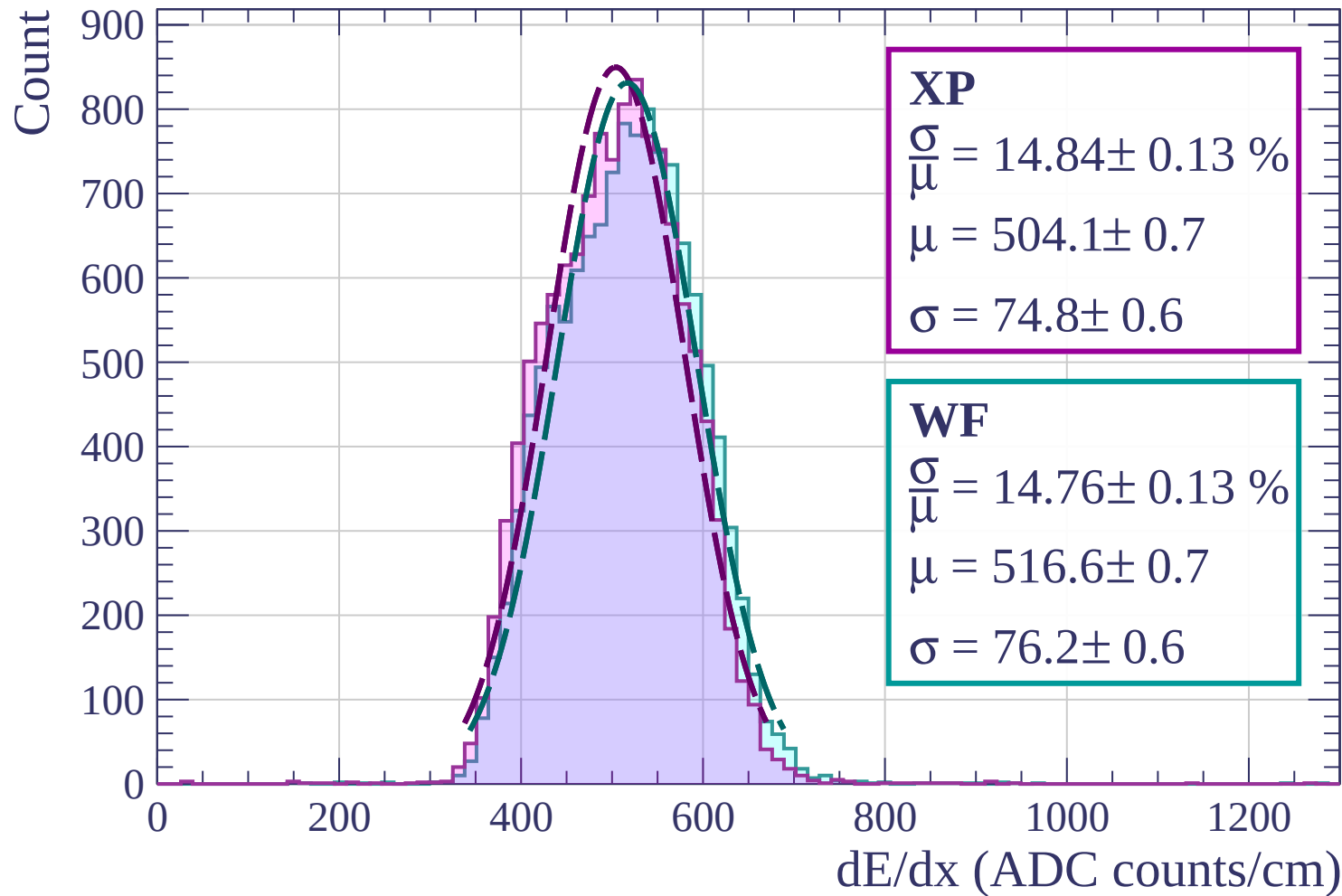
Count



Energy loss | $33 < \theta < 36$

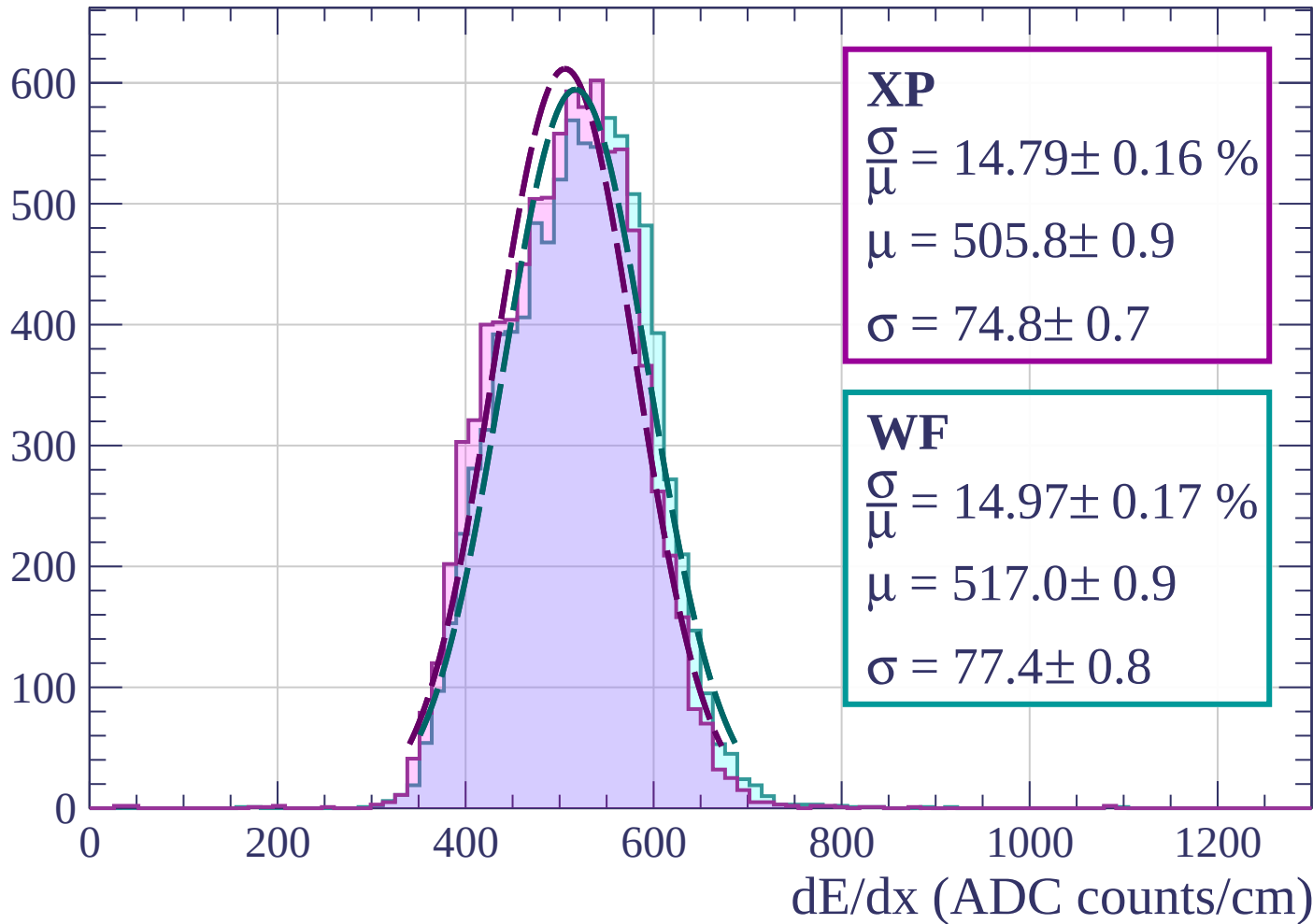


Energy loss | $36 < \theta < 39$



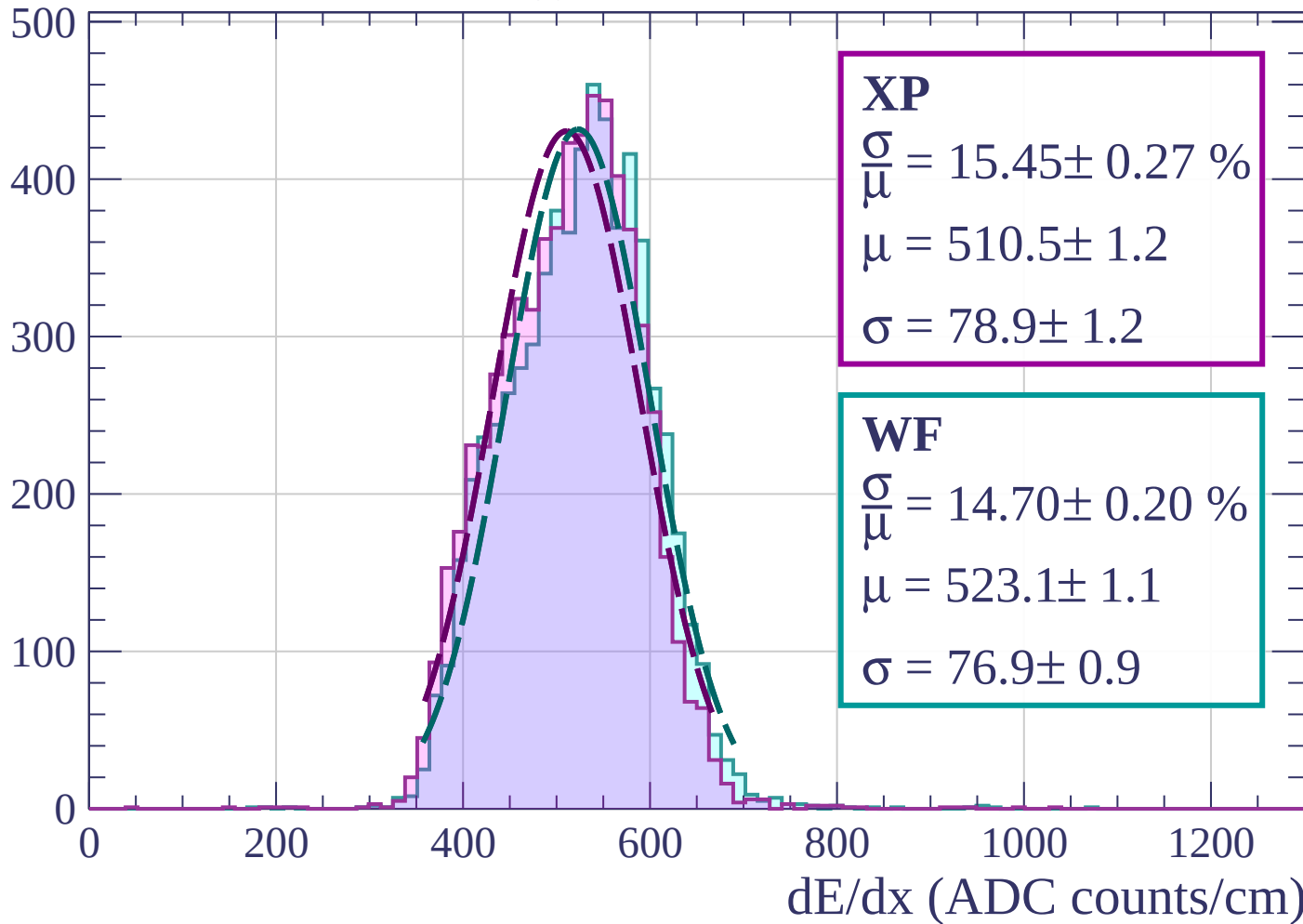
Energy loss | $39 < \theta < 42$

Count



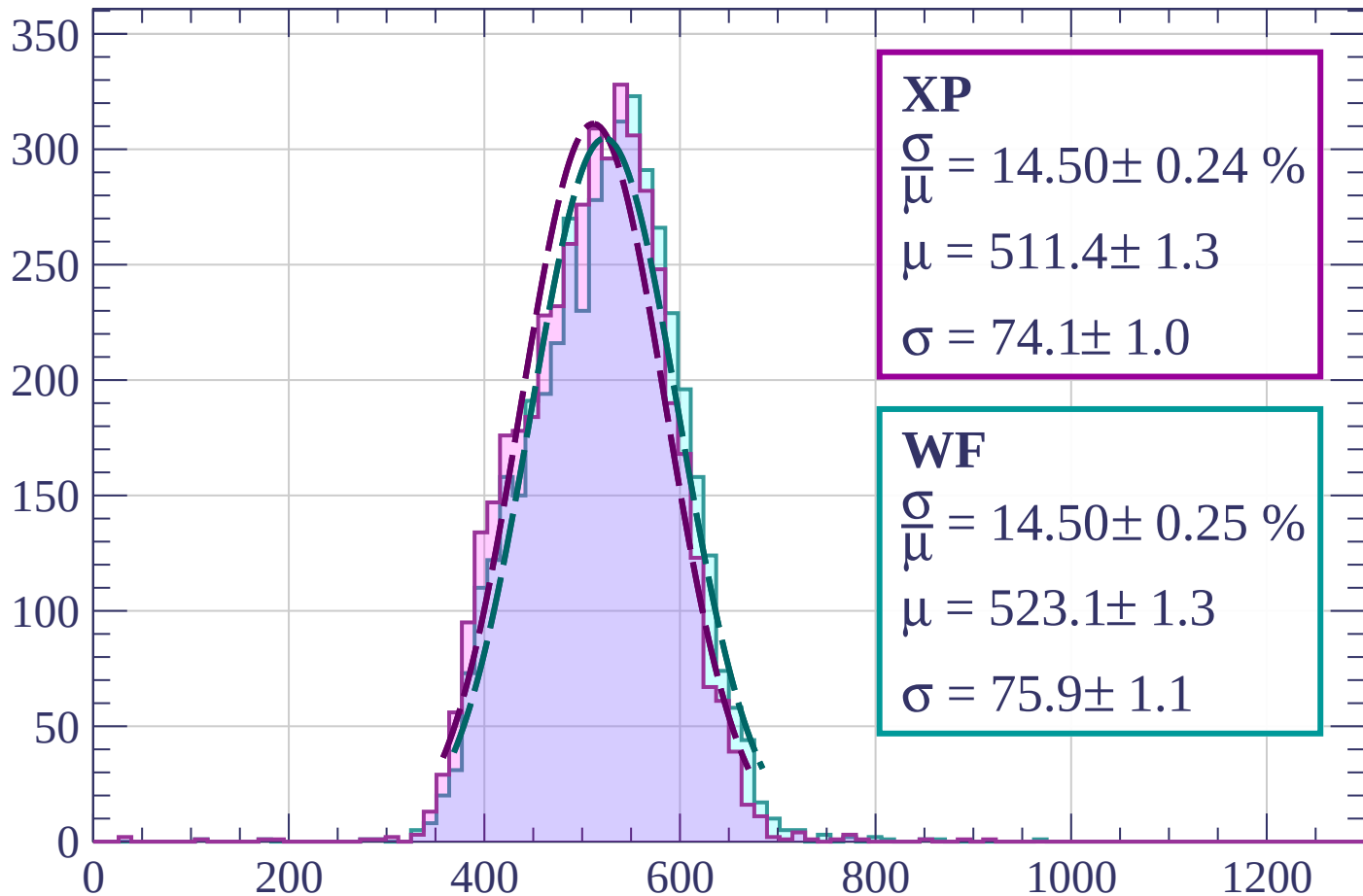
Energy loss | $42 < \theta < 45$

Count



Energy loss | $45 < \theta < 48$

Count



XP

$$\frac{\sigma}{\mu} = 14.50 \pm 0.24 \%$$

$$\mu = 511.4 \pm 1.3$$

$$\sigma = 74.1 \pm 1.0$$

WF

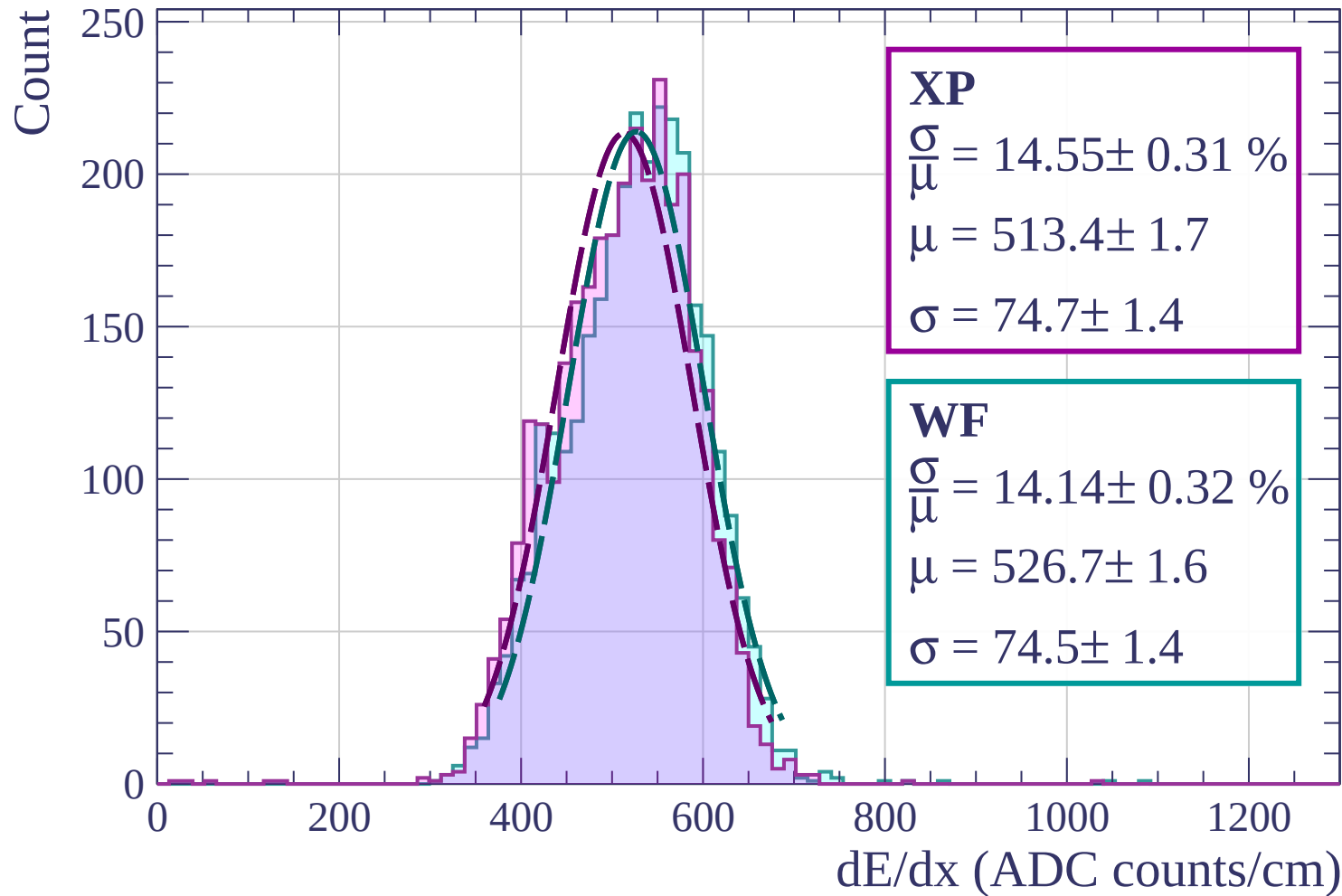
$$\frac{\sigma}{\mu} = 14.50 \pm 0.25 \%$$

$$\mu = 523.1 \pm 1.3$$

$$\sigma = 75.9 \pm 1.1$$

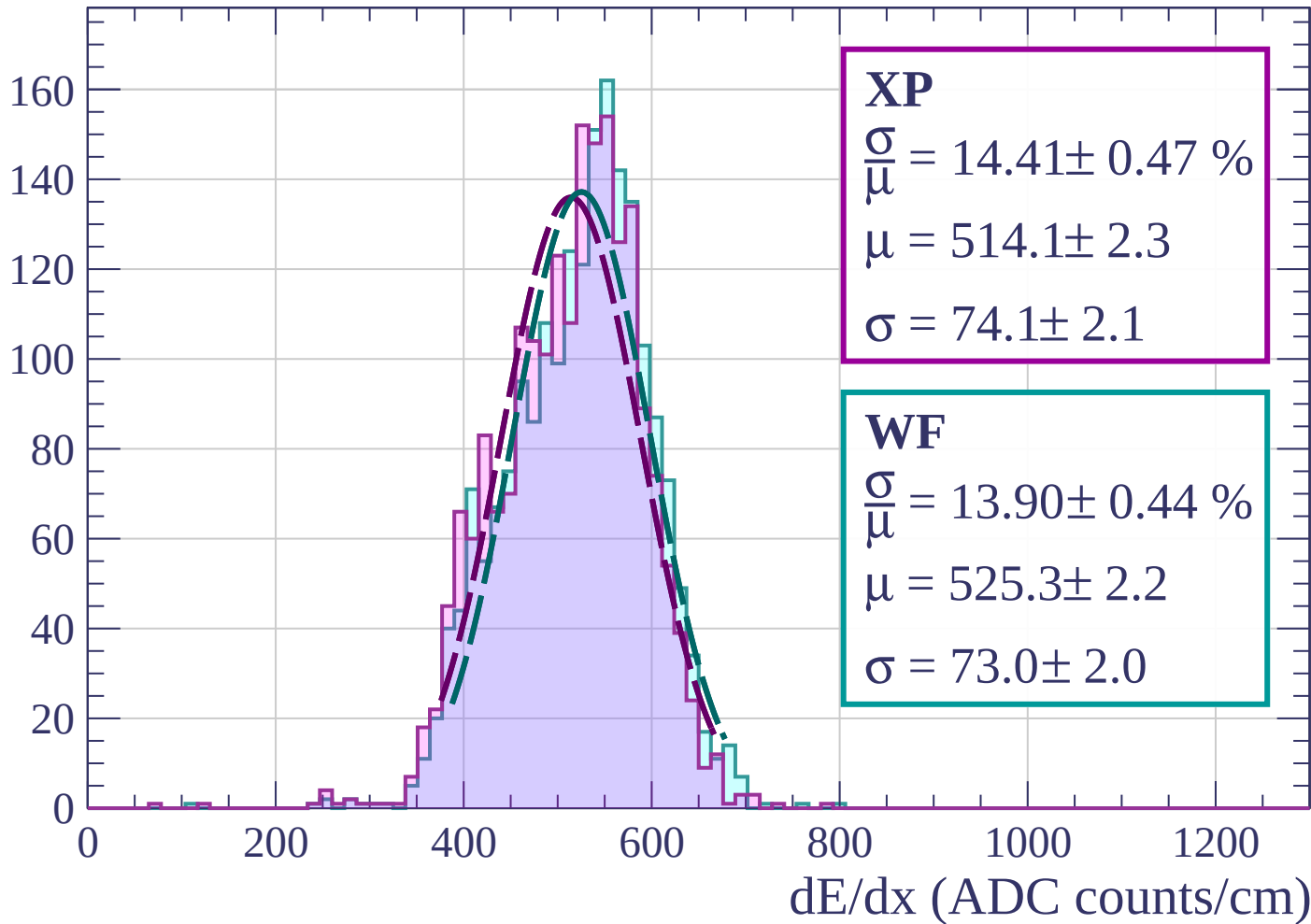
dE/dx (ADC counts/cm)

Energy loss | $48 < \theta < 51$



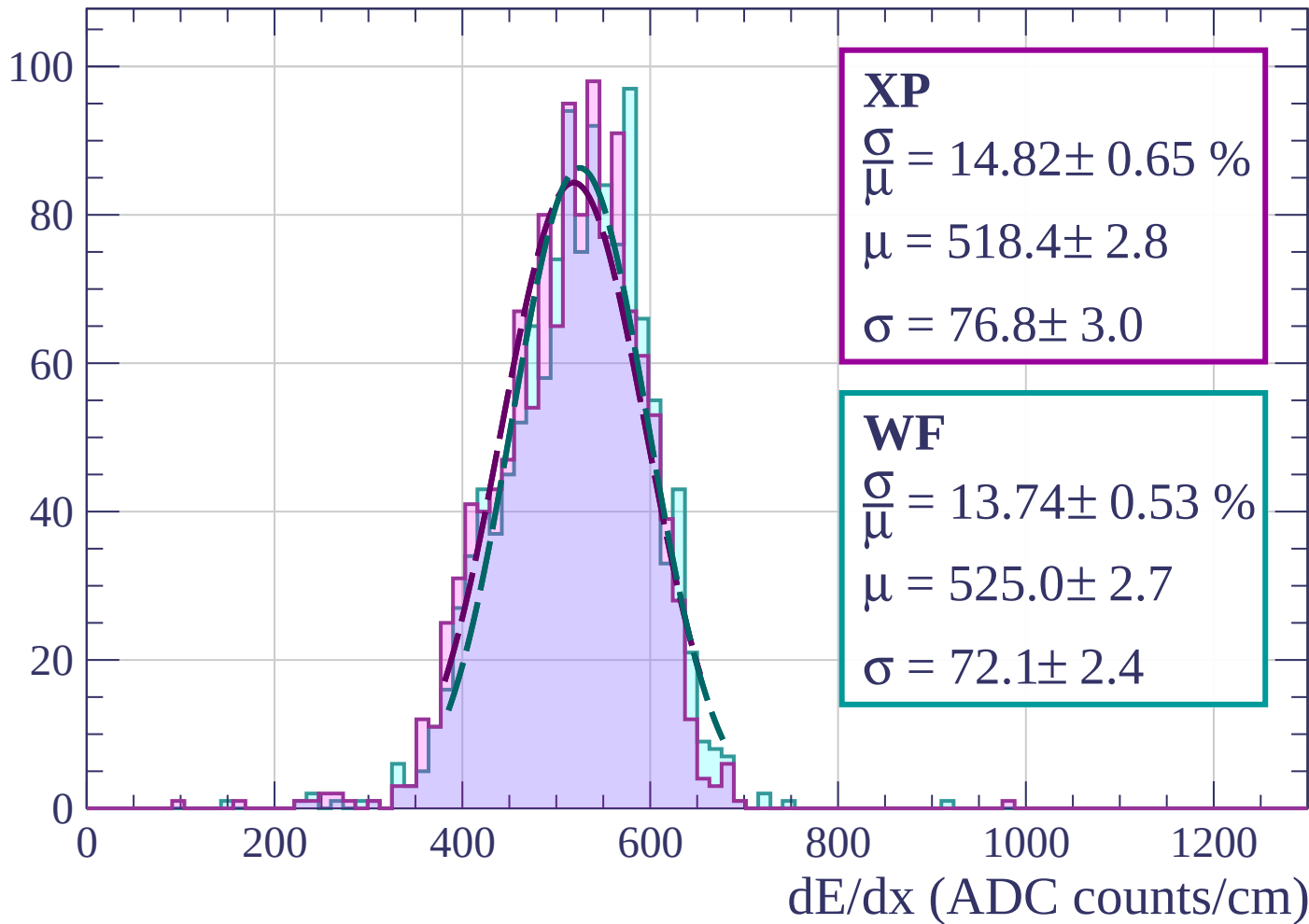
Energy loss | $51 < \theta < 54$

Count



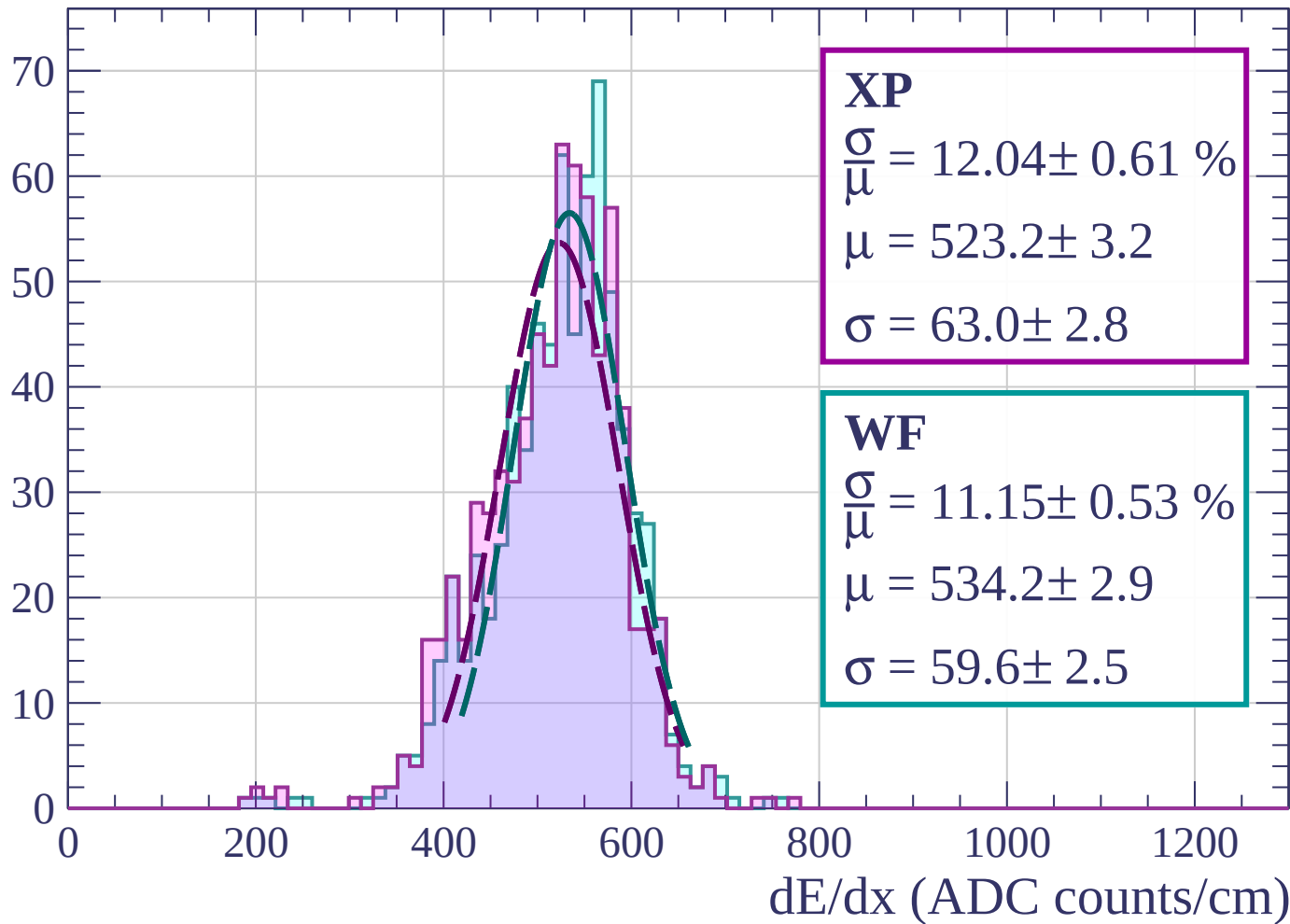
Energy loss | $54 < \theta < 57$

Count



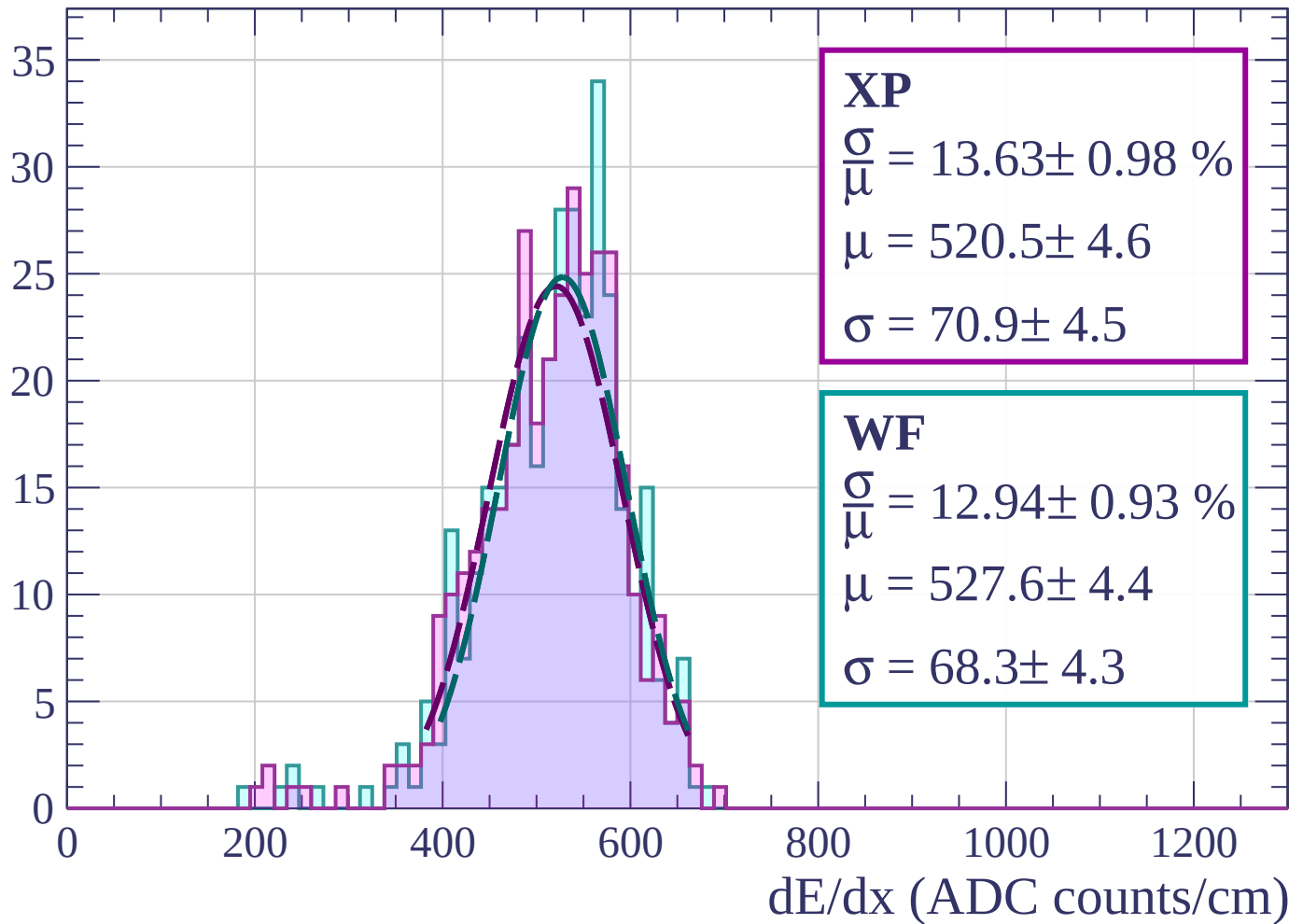
Energy loss | $57 < \theta < 60$

Count



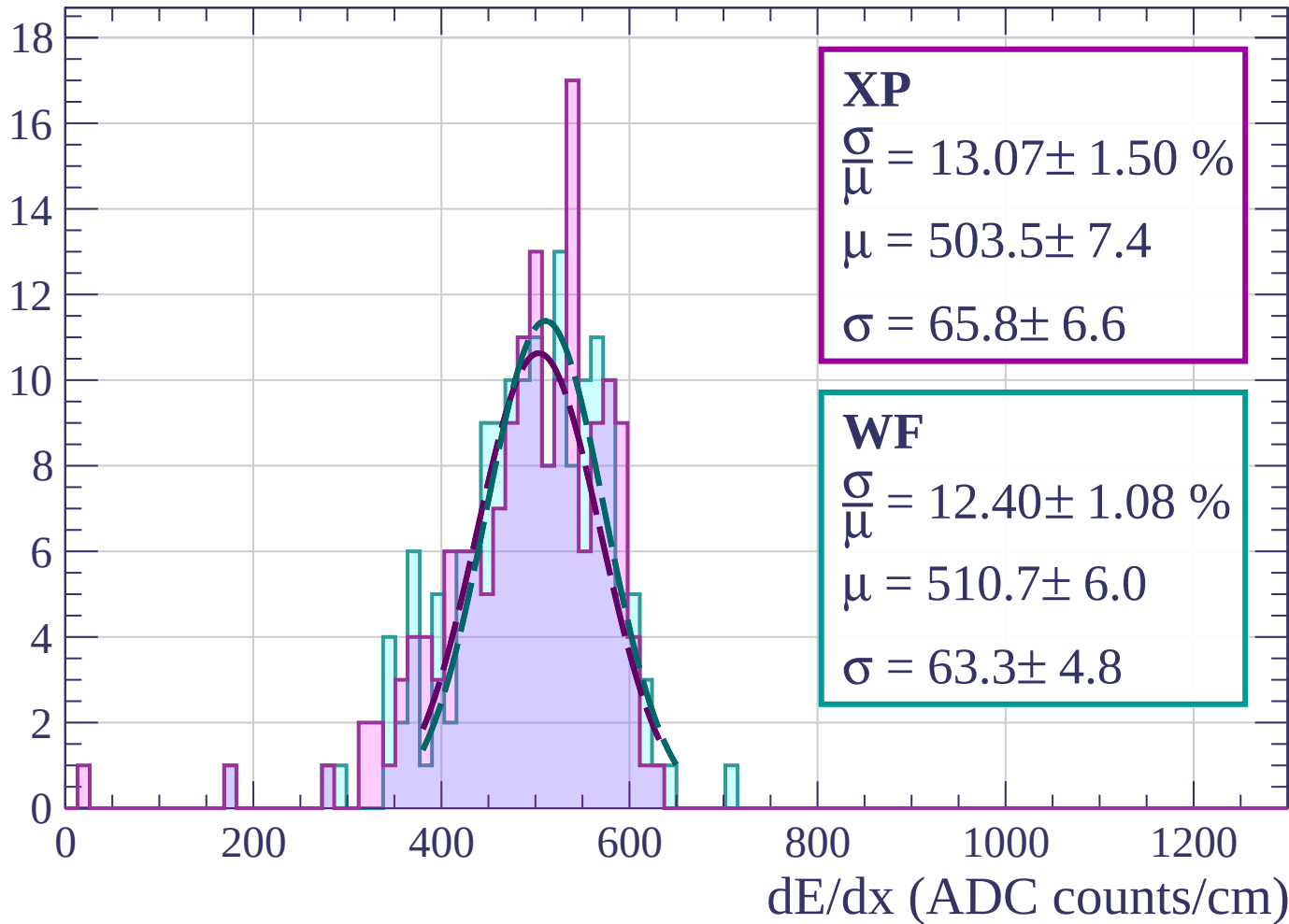
Energy loss | $60 < \theta < 63$

Count



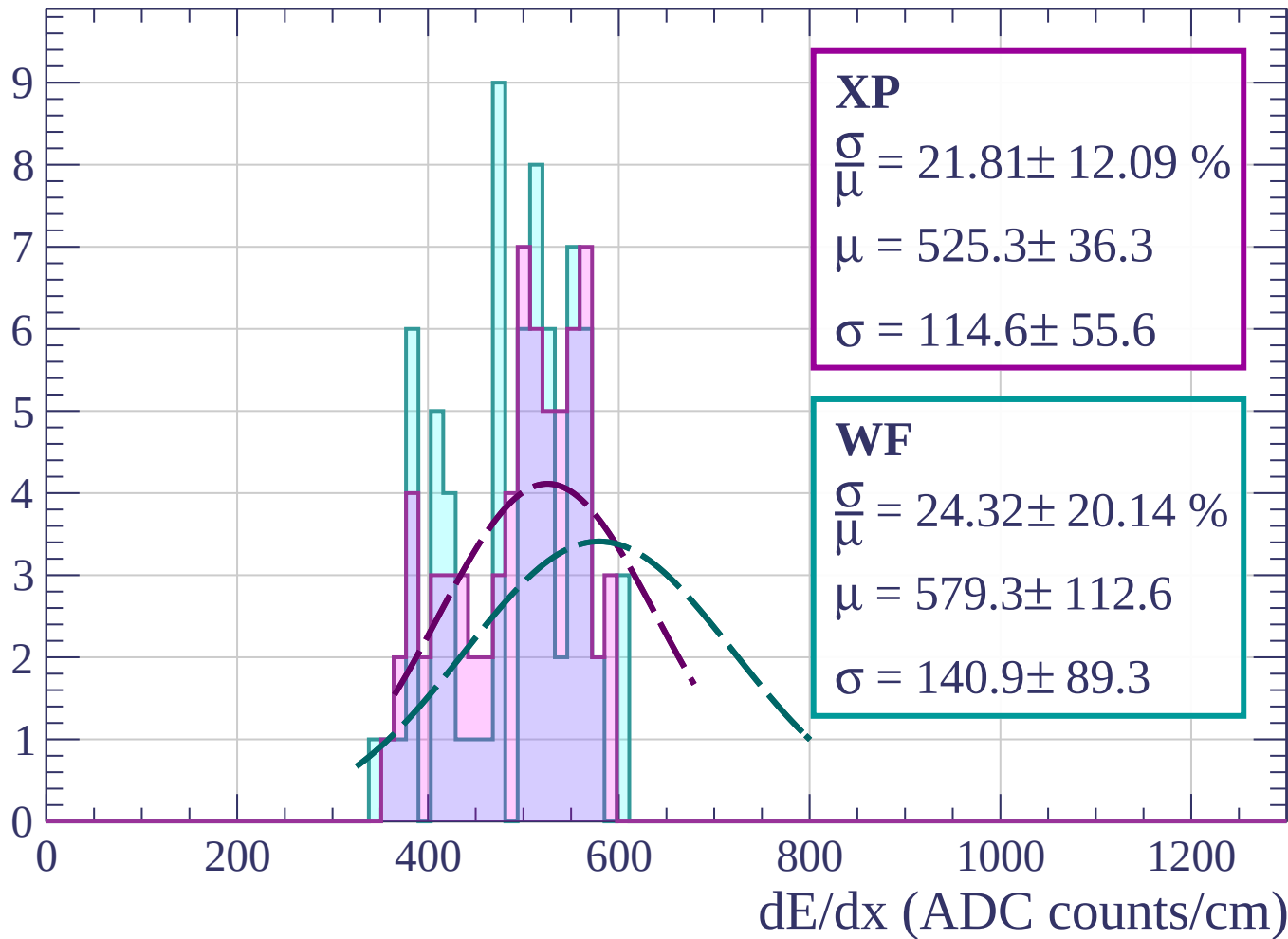
Energy loss | $63 < \theta < 66$

Count



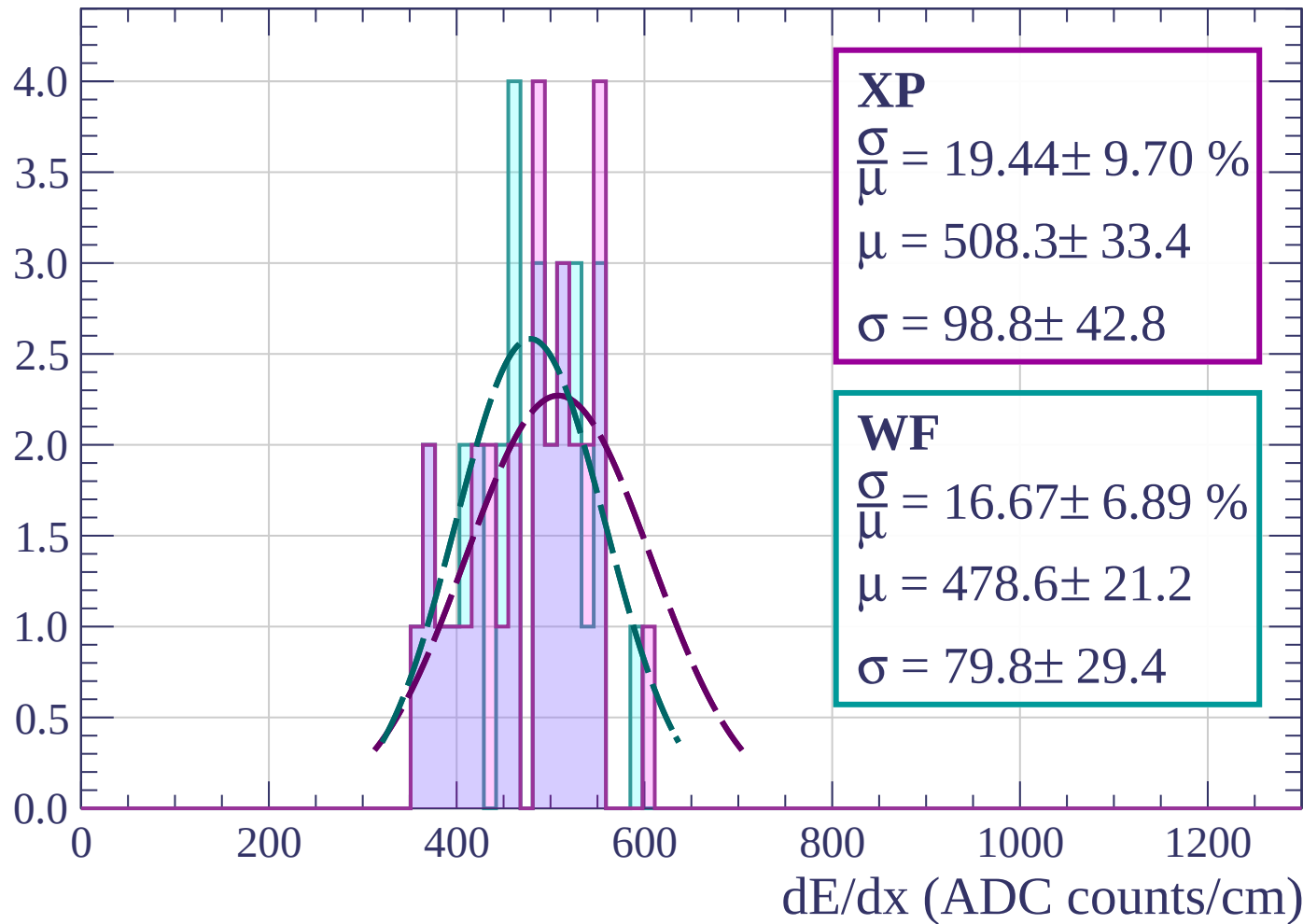
Energy loss | $66 < \theta < 69$

Count



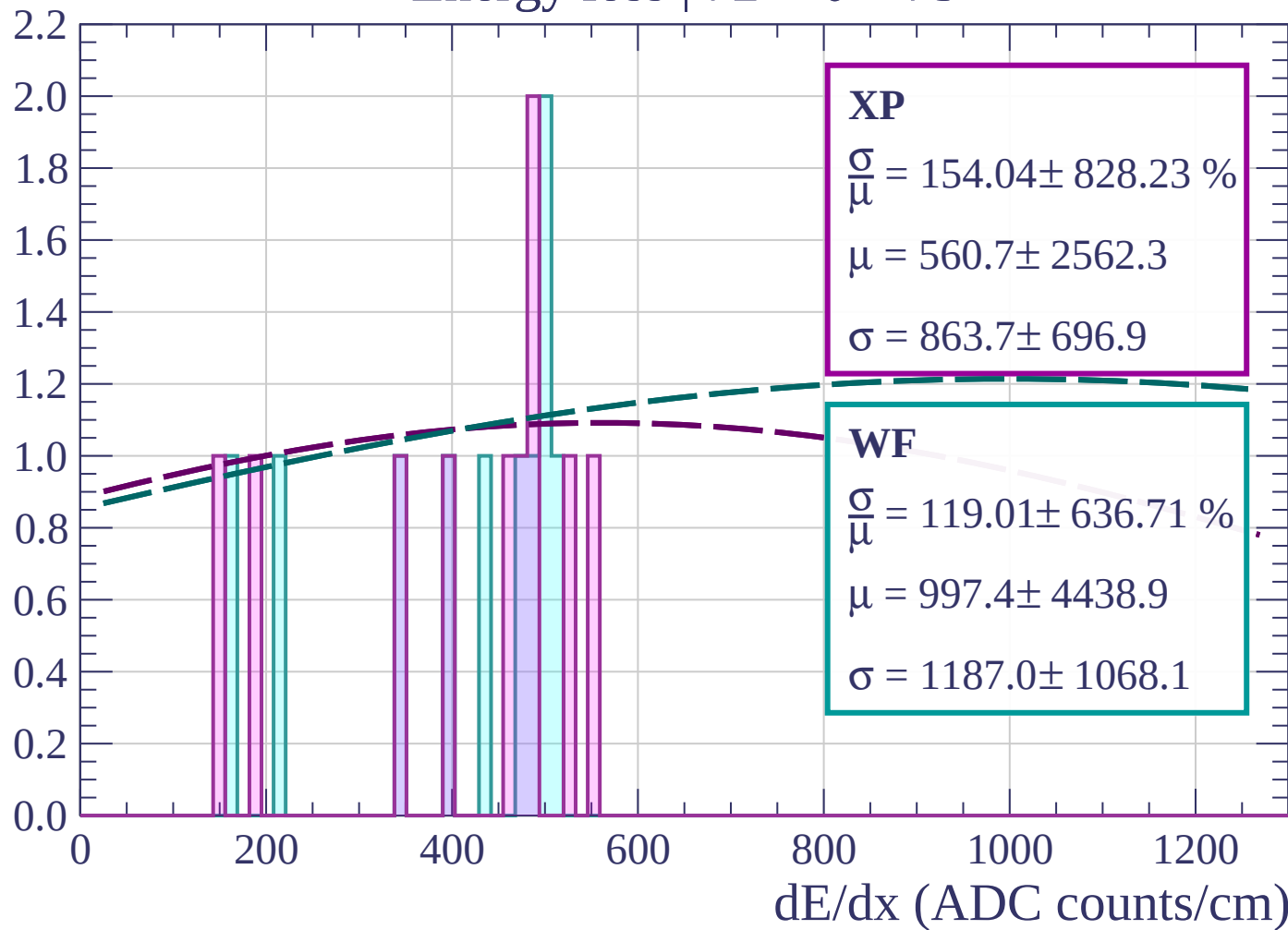
Energy loss | $69 < \theta < 72$

Count



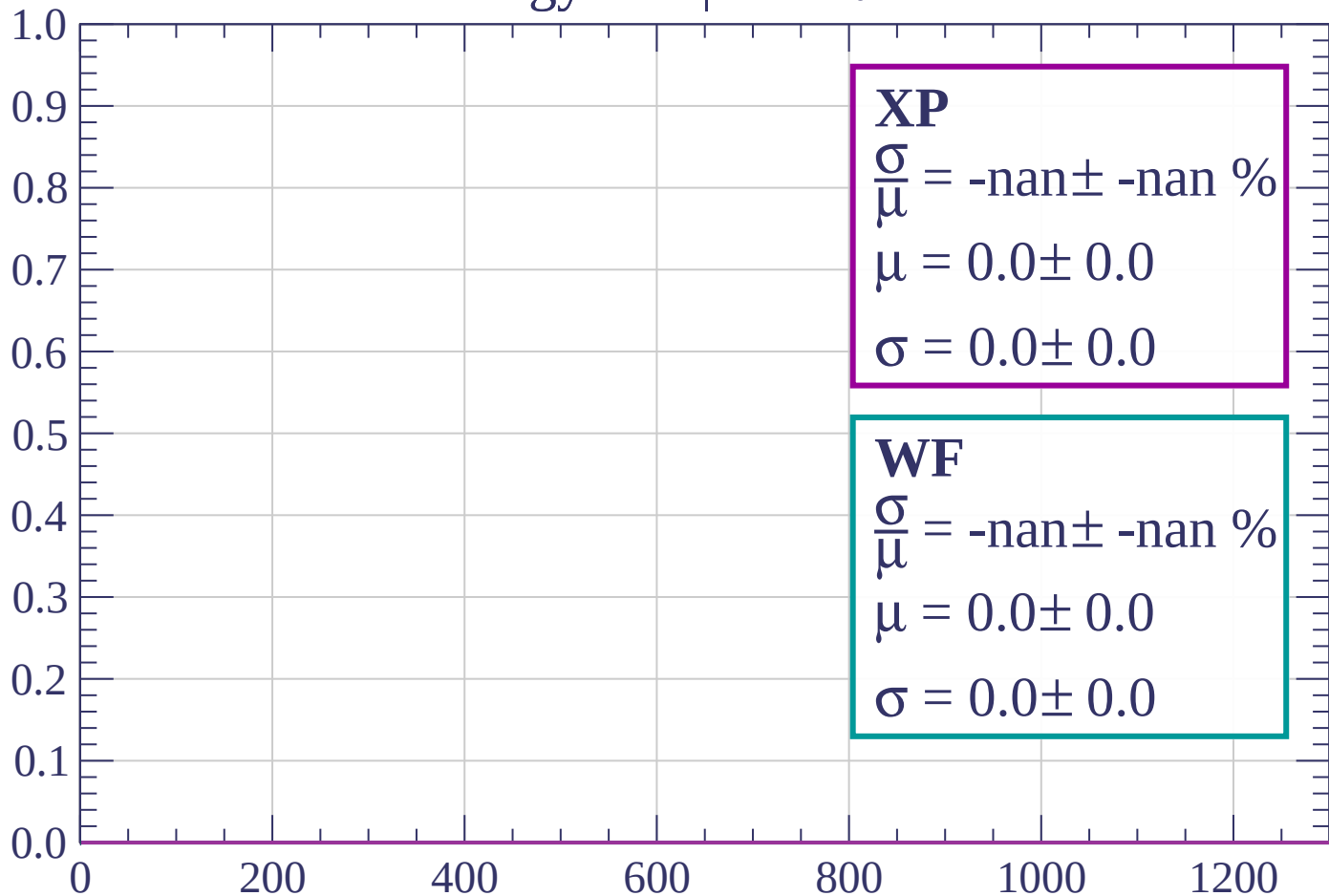
Energy loss | $72 < \theta < 75$

Count



Energy loss | $75 < \theta < 78$

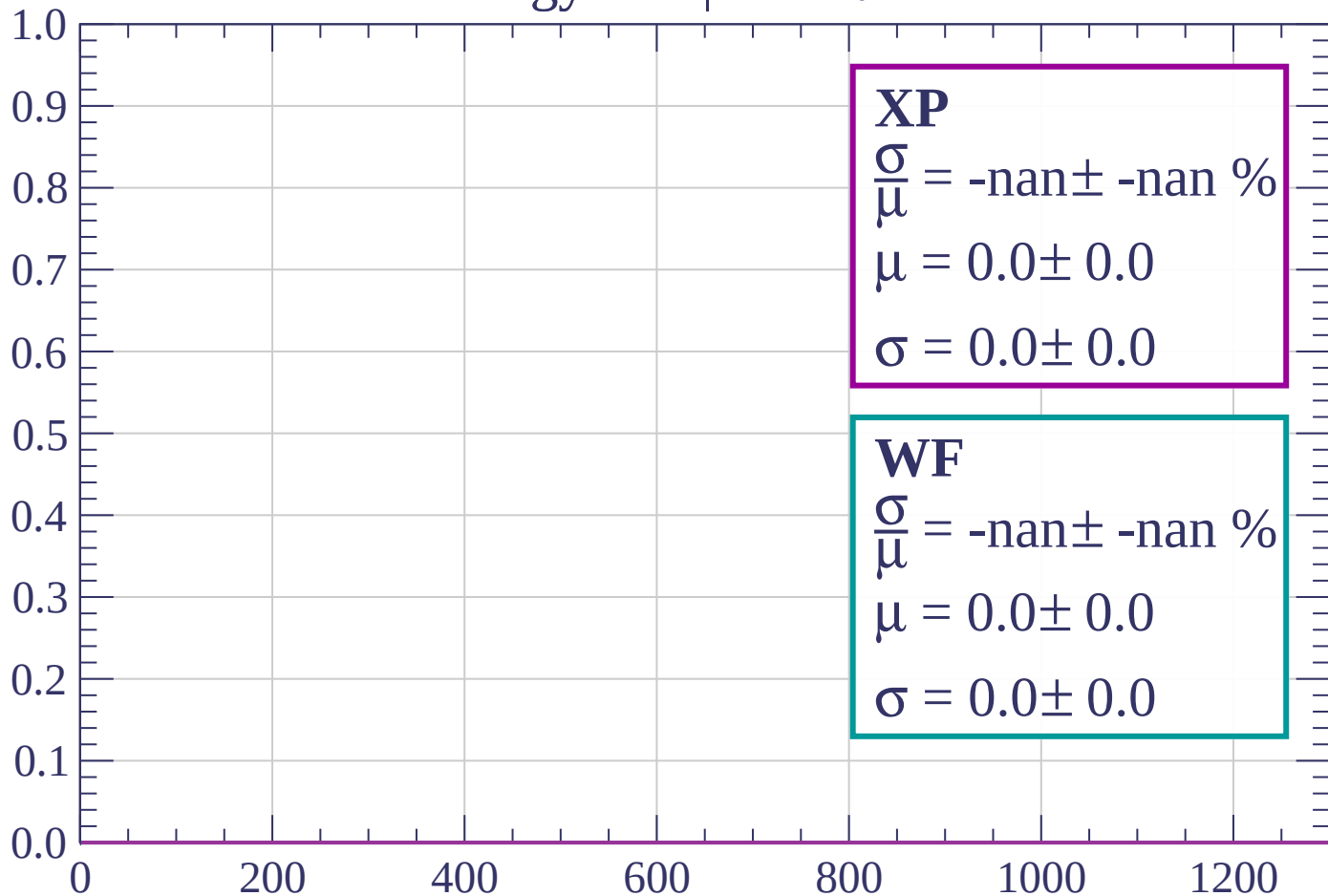
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 81$

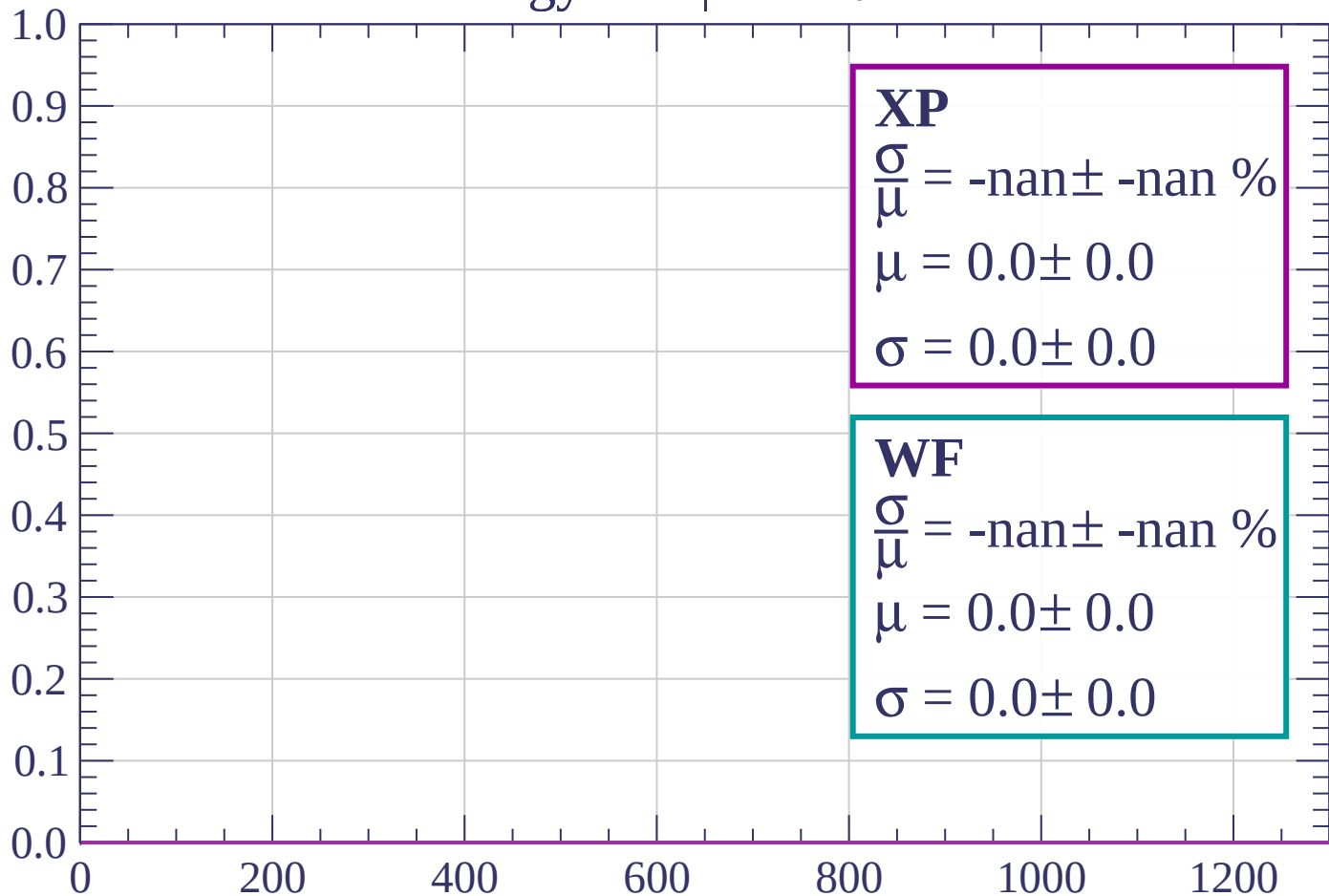
Count



dE/dx (ADC counts/cm)

Energy loss | $81 < \theta < 84$

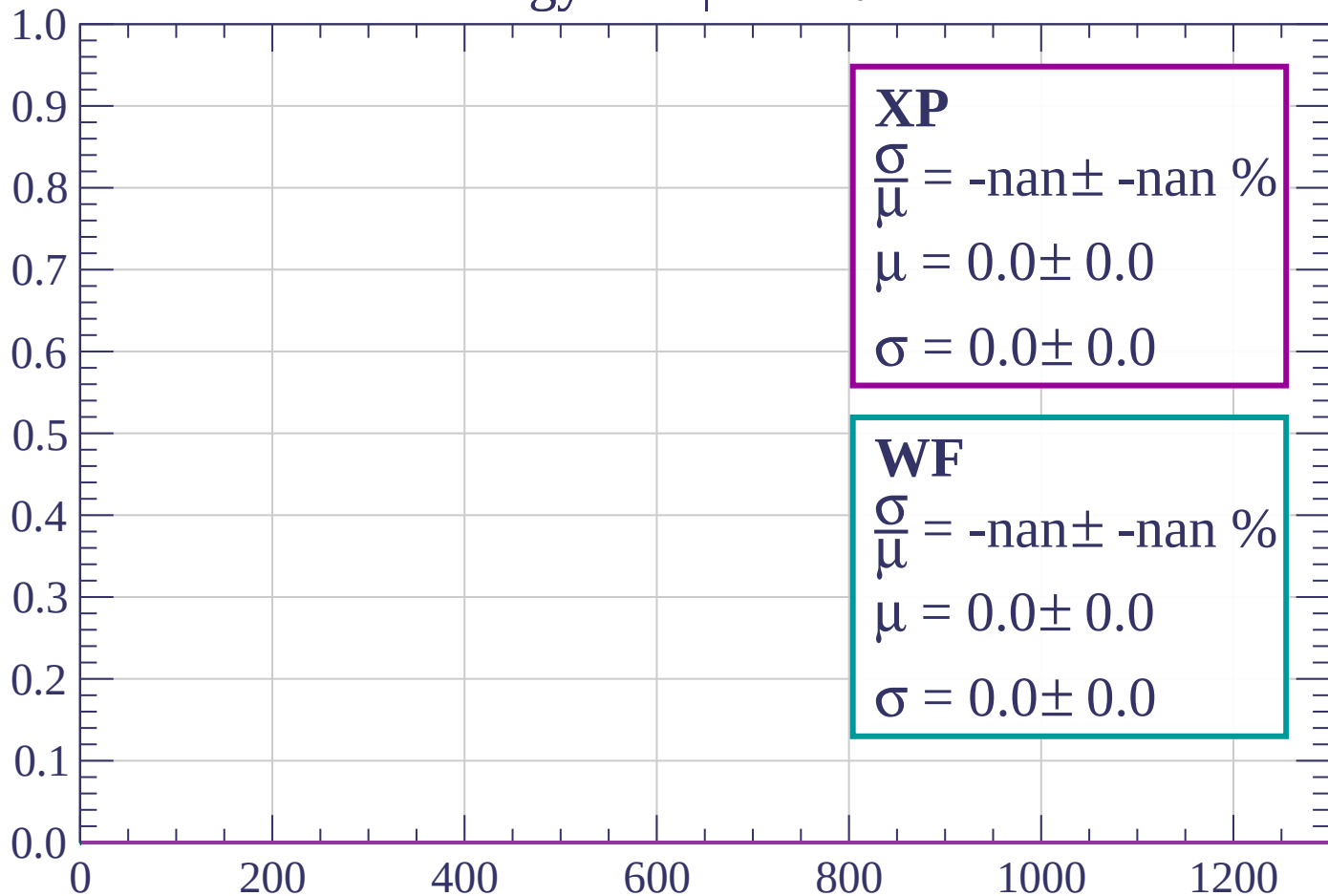
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 87$

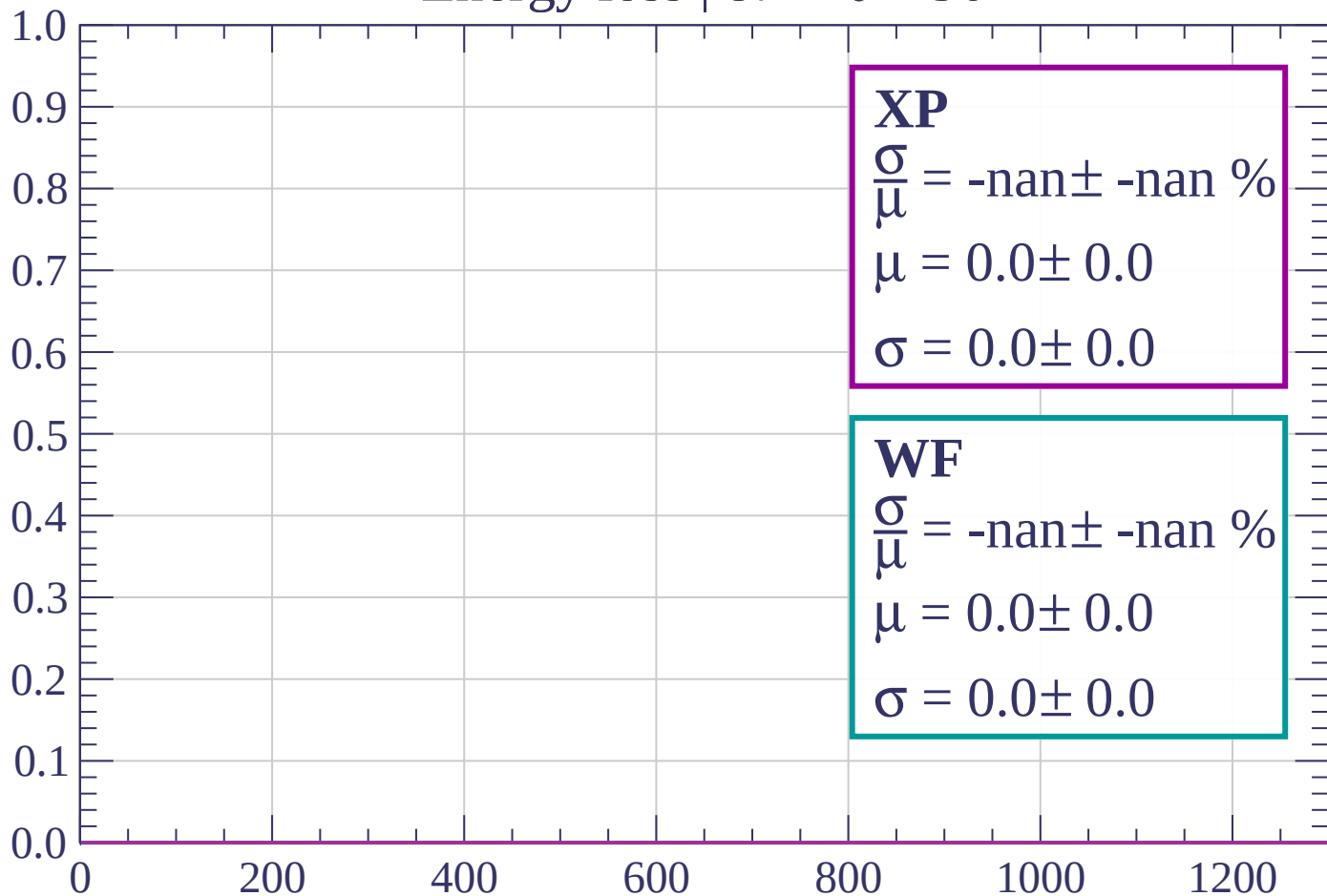
Count



dE/dx (ADC counts/cm)

Energy loss | $87 < \theta < 90$

Count



dE/dx (ADC counts/cm)

